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(54) **METHODS AND APPARATUS FOR TWO
DIMENSIONAL LOCATION BASED DIGITAL
CONTENT**

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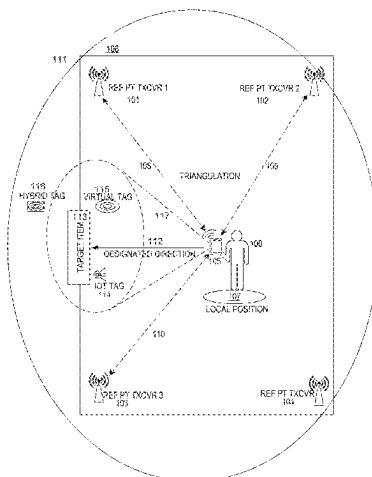
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(57) **ABSTRACT**

Augmented reality apparatus and methods of use are pro-
vided with secure persistent digital content linked to a
location coordinates. More specifically, the present inven-
tion links a physical location with digital content to enable
a user interface with augmented reality that combines
aspects of the physical area with location specific digital
content. According to the present invention, digital content
remains persistent with a location even if visual aspects of
the location change.

20 Claims, 57 Drawing Sheets



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which is a continuation-in-part of application No. 17/244,970, filed on Apr. 30, 2021, now Pat. No. 11,194,938, which is a continuation-in-part of application No. 17/196,146, filed on Mar. 9, 2021, now Pat. No. 11,514,207, said application No. 17/244,970 is a continuation-in-part of application No. 17/183,062, filed on Feb. 23, 2021, now Pat. No. 11,106,837, and a continuation-in-part of application No. 17/176,849, filed on Feb. 16, 2021, now Pat. No. 11,100,261, and a continuation-in-part of application No. 17/134,824, filed on Dec. 28, 2020, now Pat. No. 11,080,439, and a continuation-in-part of application No. 17/113,368, filed on Dec. 7, 2020, now Pat. No. 11,120,172, and a continuation-in-part of application No. 16/951,550, filed on Nov. 18, 2020, now Pat. No. 11,100,260, and a continuation-in-part of application No. 17/062,663, filed on Oct. 5, 2020, now Pat. No. 11,042,672, said application No. 17/134,824 is a continuation of application No. 16/943,750, filed on Jul. 30, 2020, now Pat. No. 10,902,160, said application No. 17/196,146 is a continuation of application No. 16/935,857, filed on Jul. 22, 2020, now Pat. No. 10,984,146, said application No. 17/176,849 is a continuation of application No. 16/915,155, filed on Jun. 29, 2020, now Pat. No. 10,943,034, said application No. 16/951,550 is a continuation of application No. 16/900,753, filed on Jun. 12, 2020, now Pat. No. 10,872,179, said application No. 17/183,062 is a continuation of application No. 16/898,602, filed on Jun. 11, 2020, now Pat. No. 10,949,579, said application No. 17/062,663 is a continuation of application No. 16/831,160, filed on Mar. 26, 2020, now Pat. No. 10,824,774, said application No. 16/915,155 is a continuation of application No. 16/775,223, filed on Jan. 28, 2020, now Pat. No. 10,740,503.

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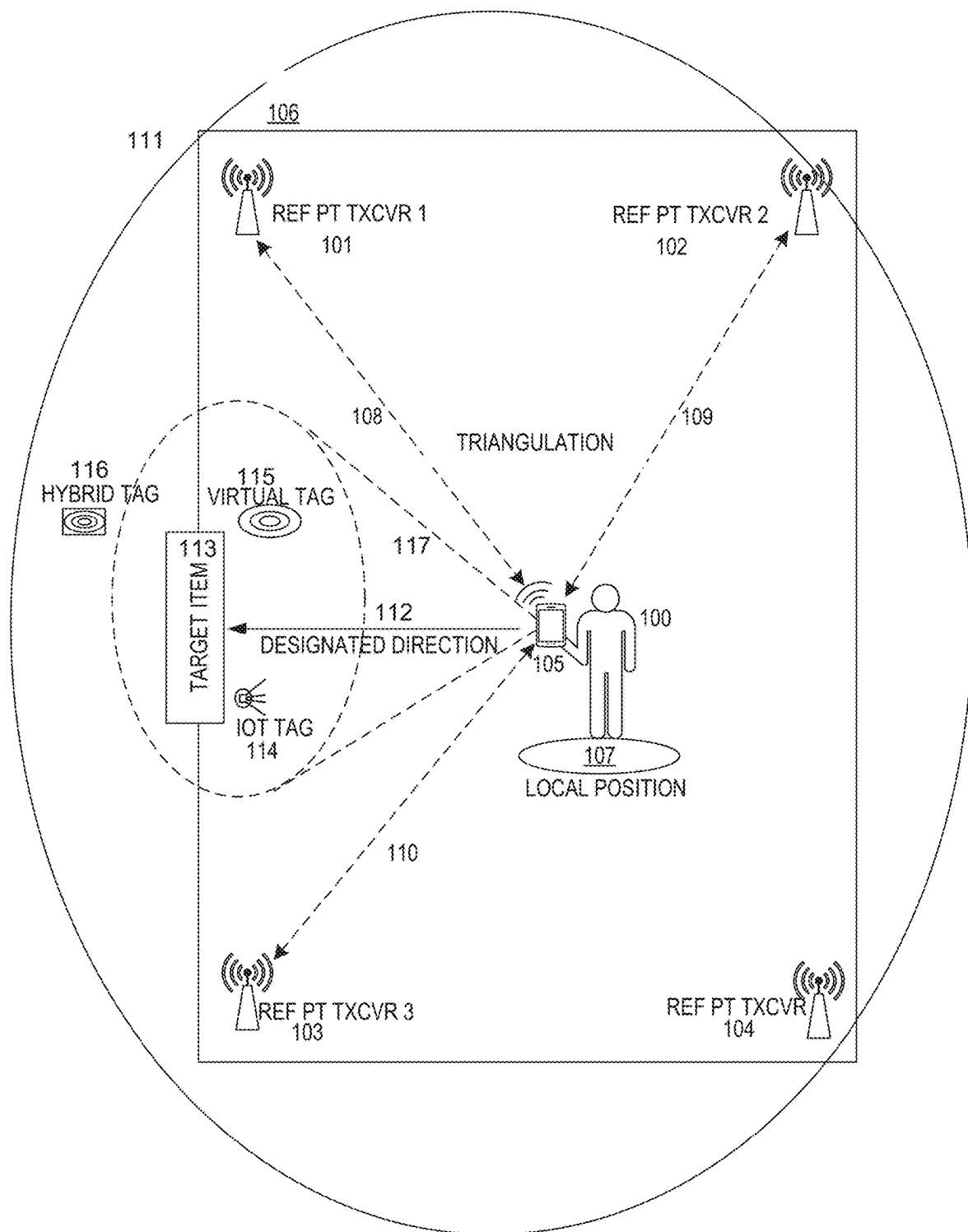


FIG. 1

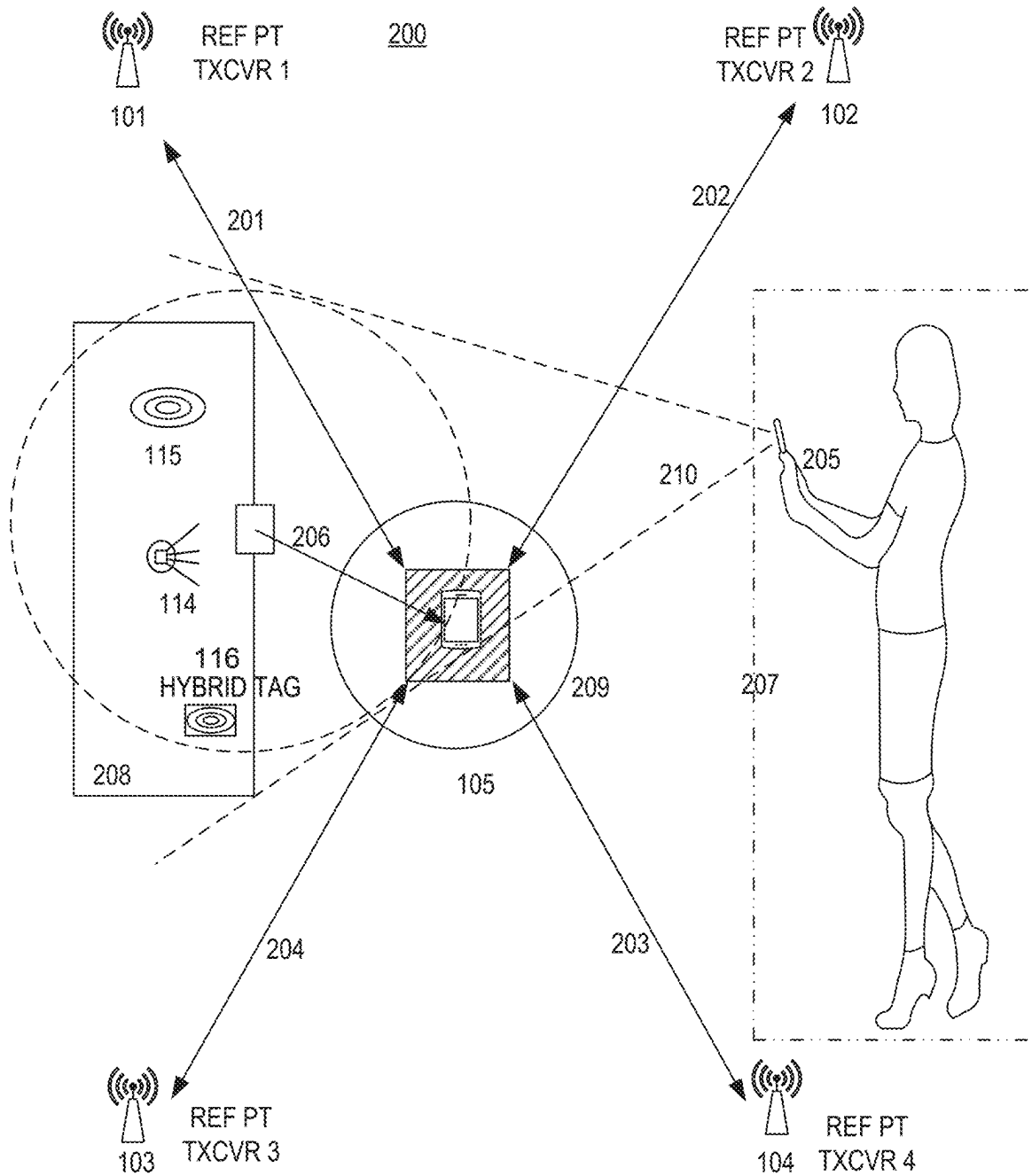


FIG. 2

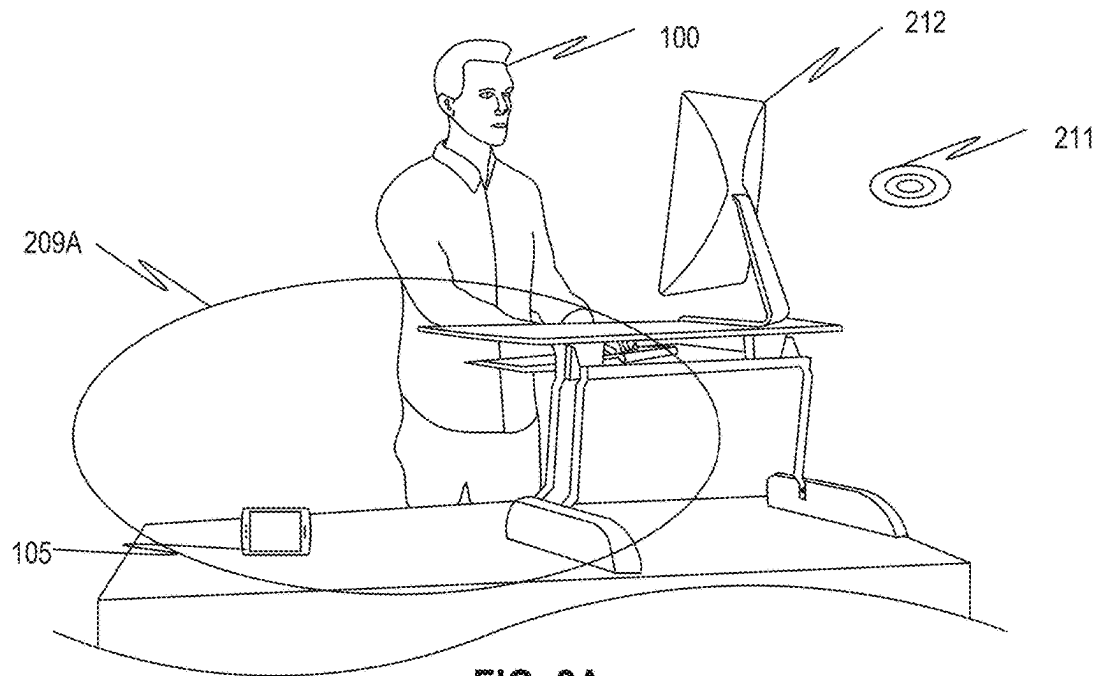


FIG. 2A

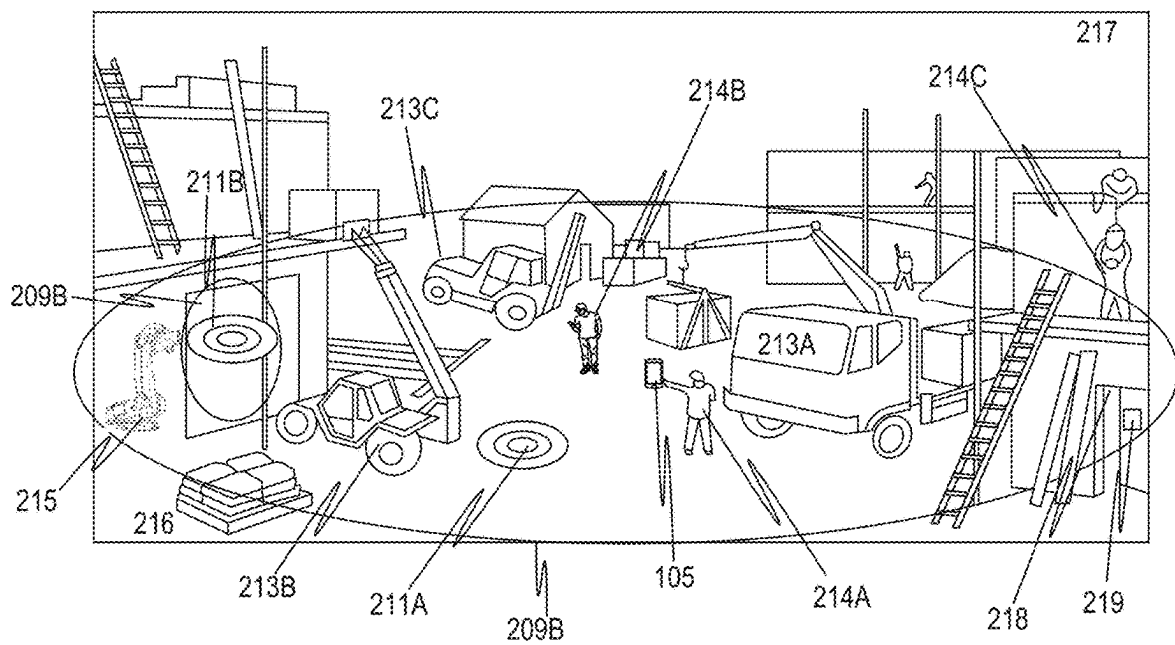


FIG. 2B

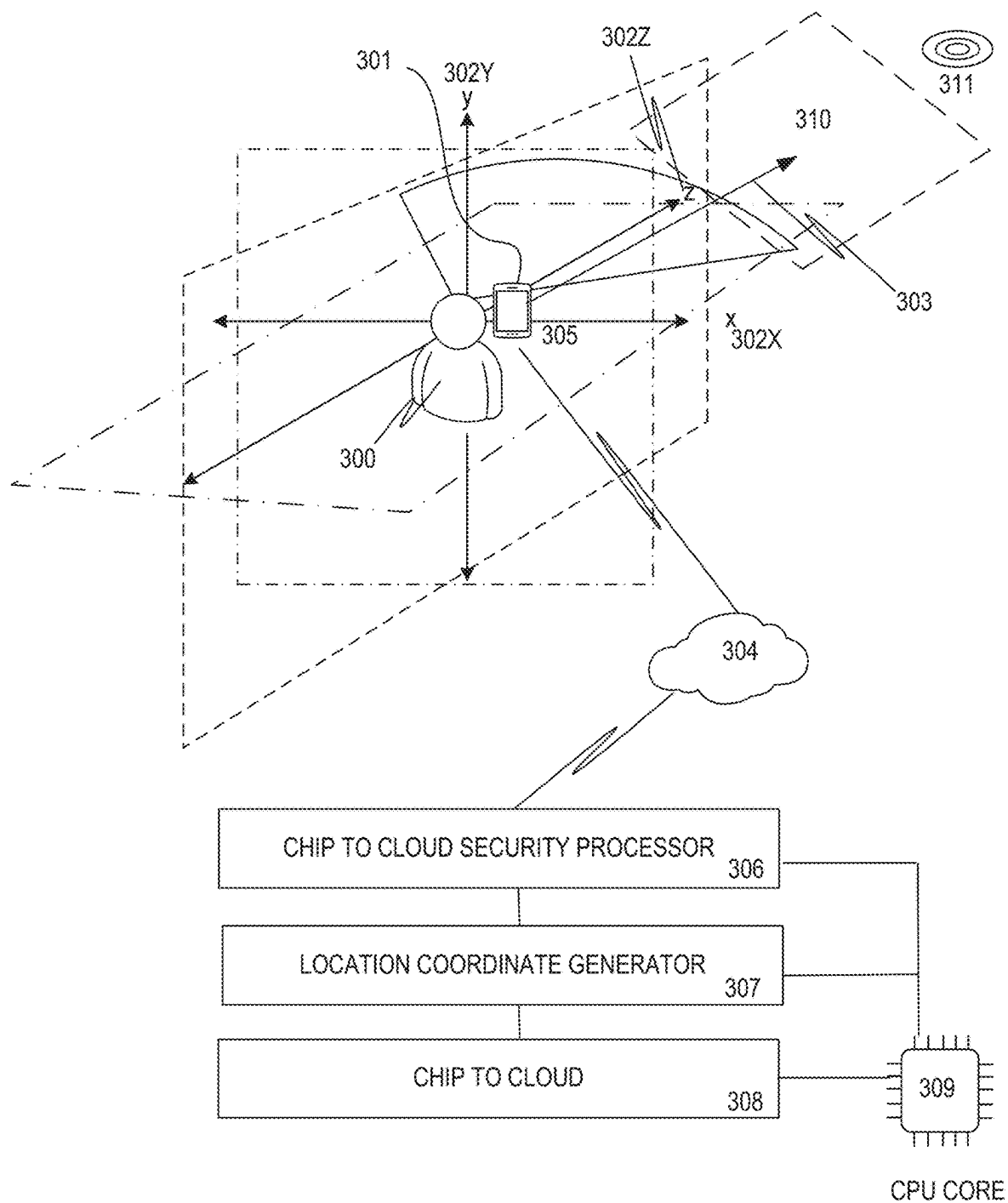


FIG. 3

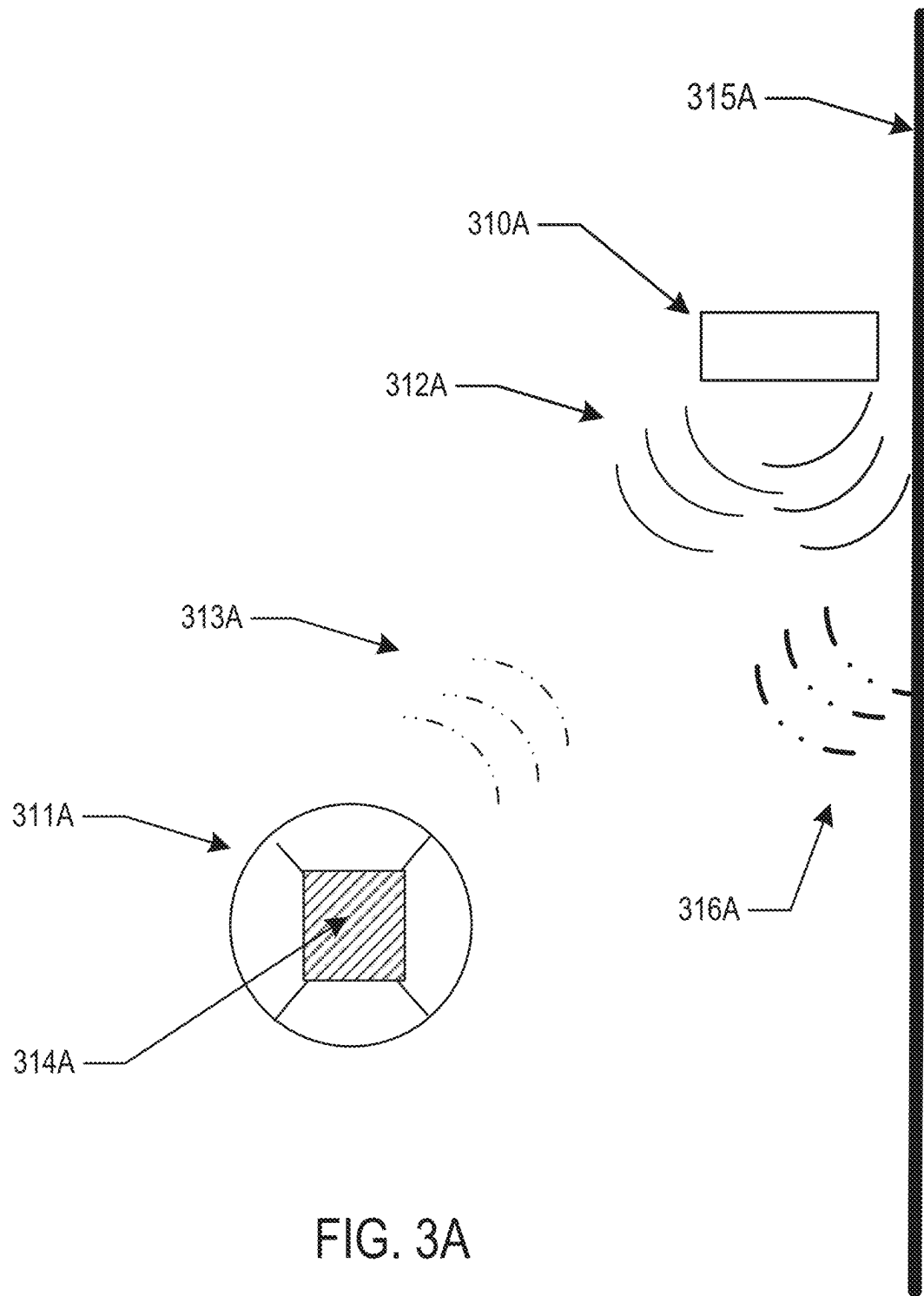


FIG. 3A

401

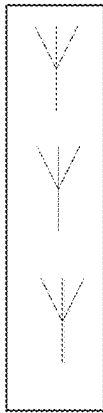


FIG. 4A

402

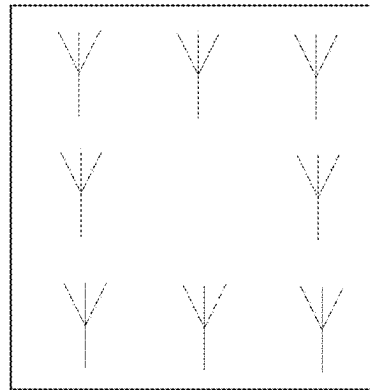


FIG. 4B

403

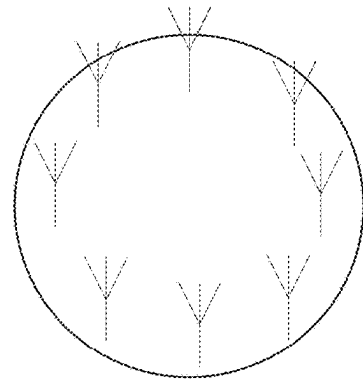


FIG. 4C

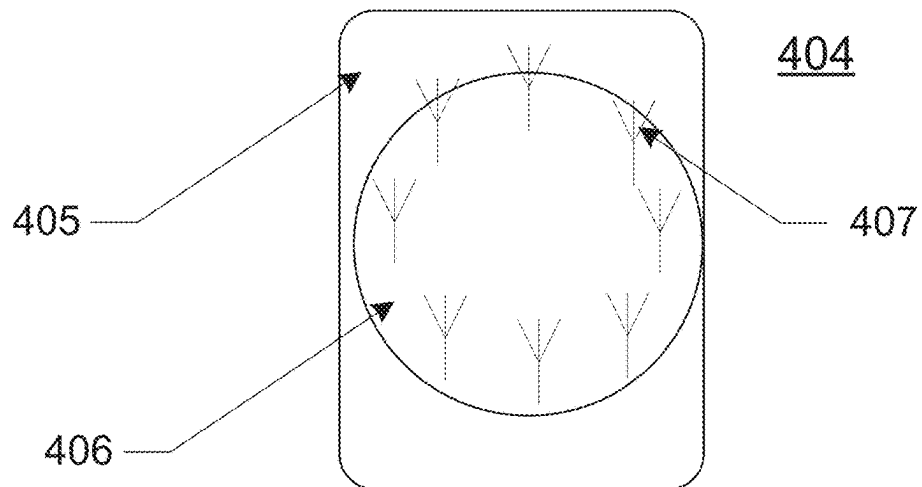


FIG. 4D

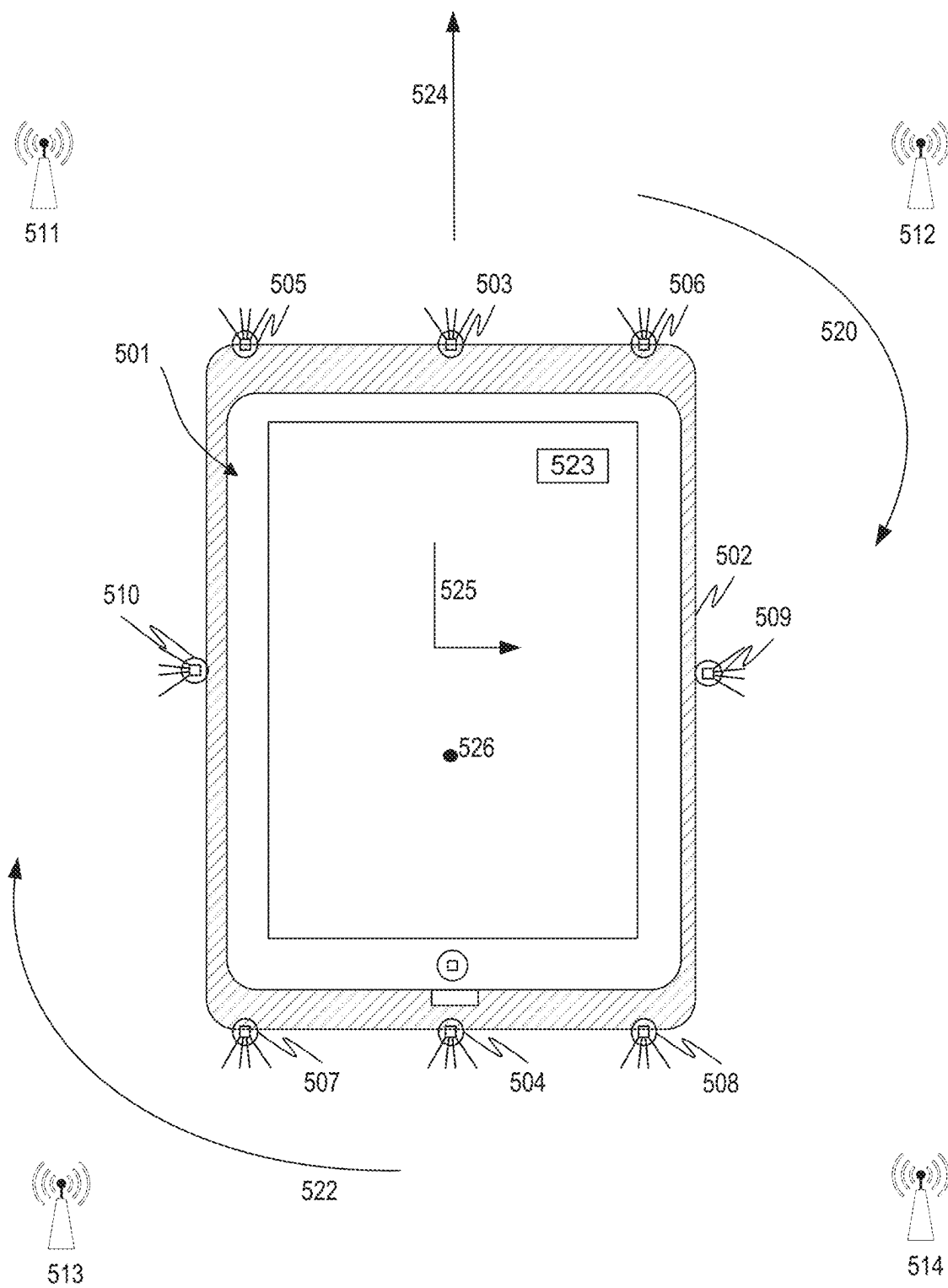


FIG. 5A

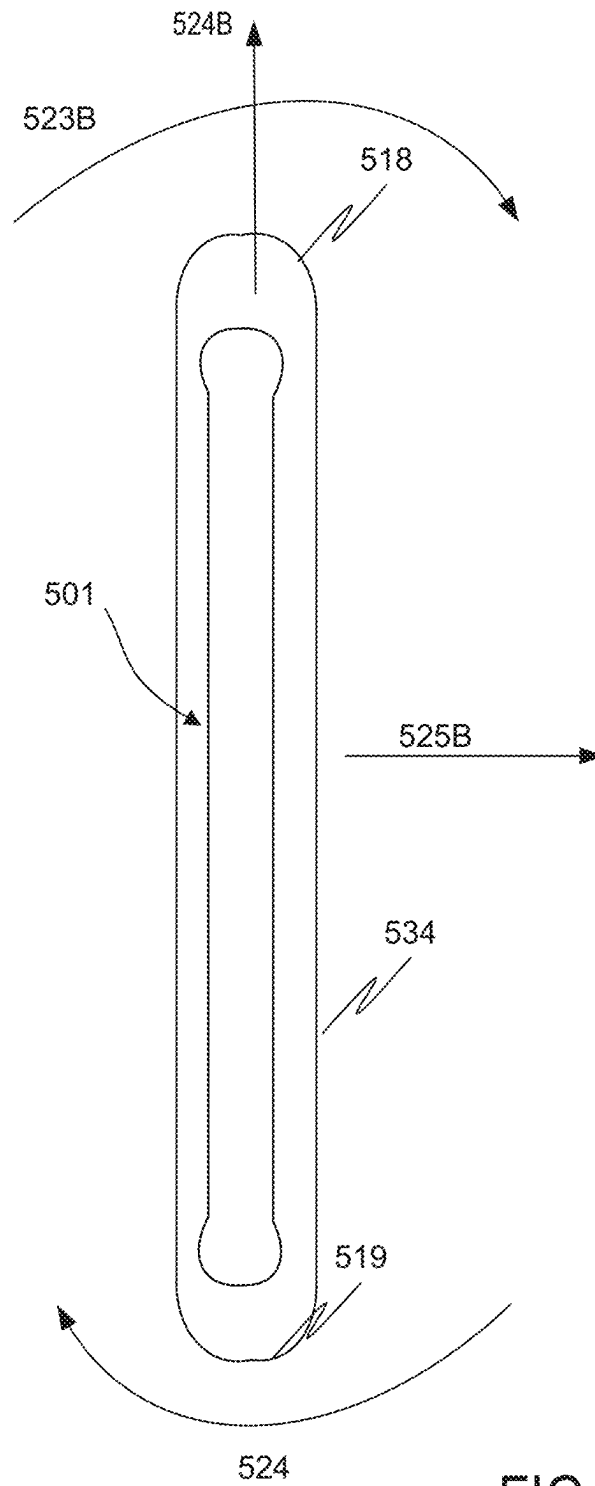


FIG. 5B

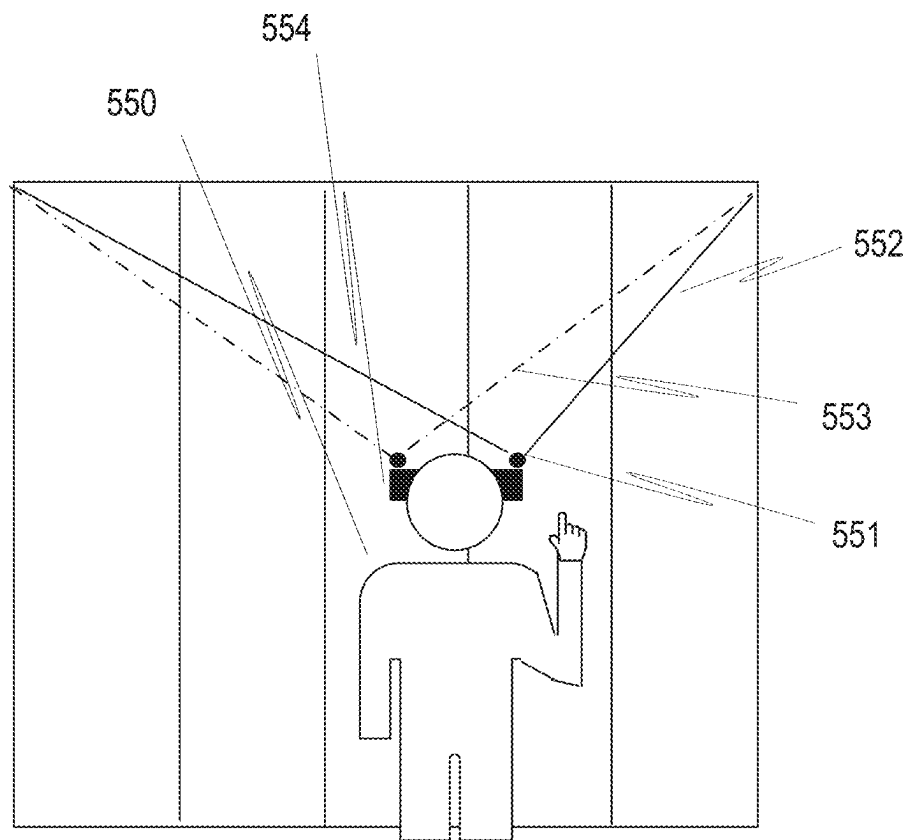


FIG. 5C

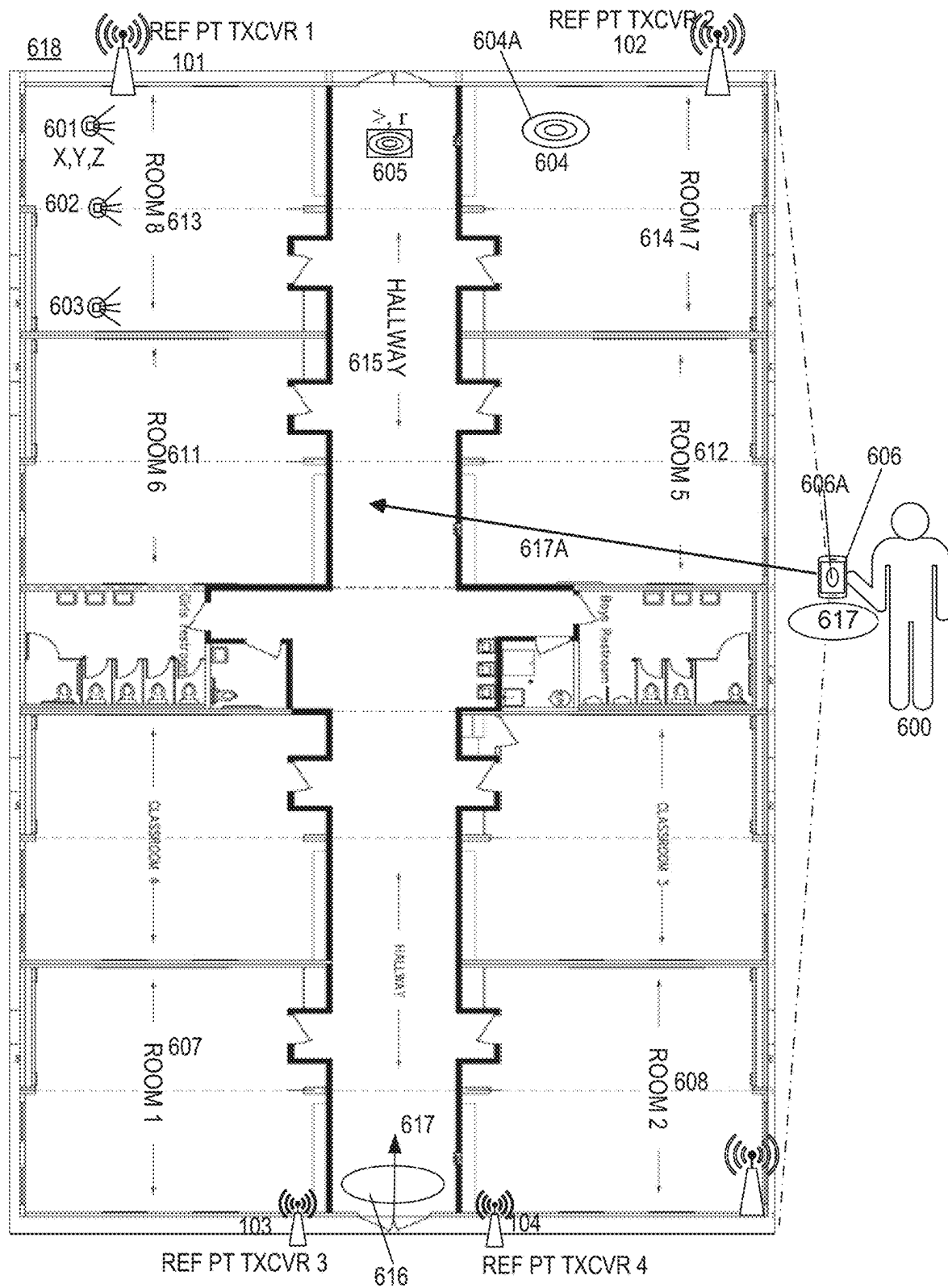


FIG. 6

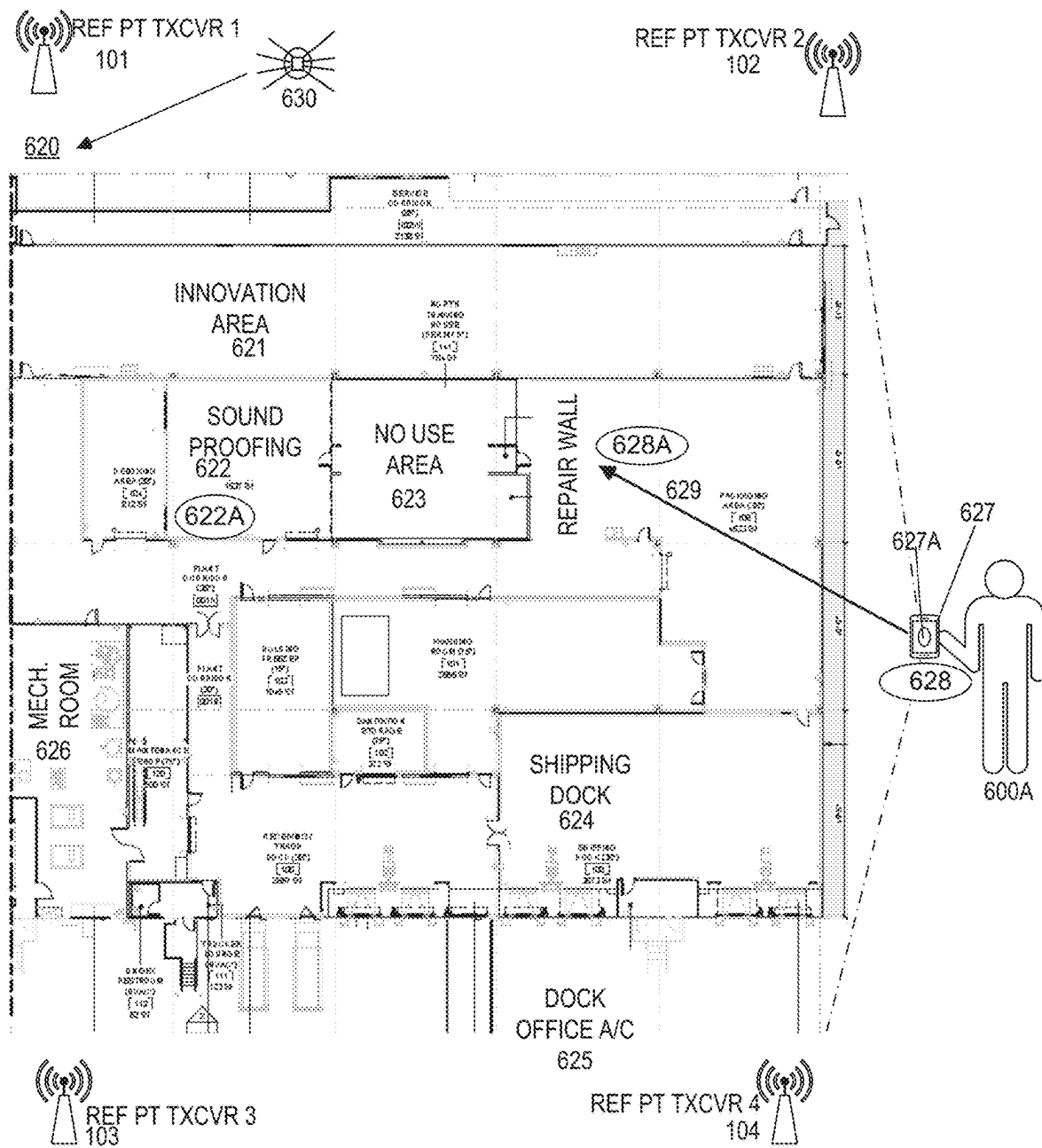


FIG. 6A

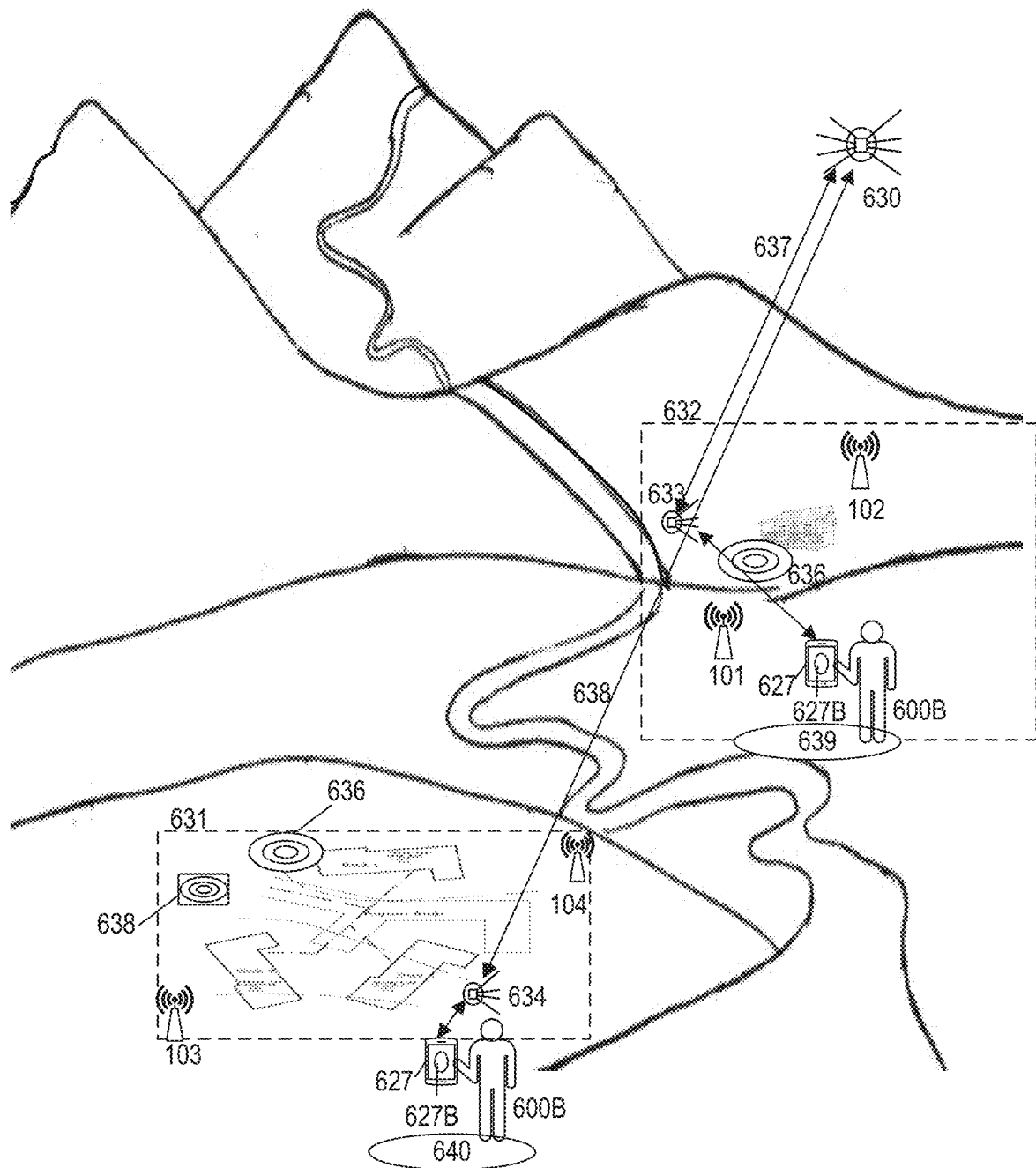


FIG. 6B

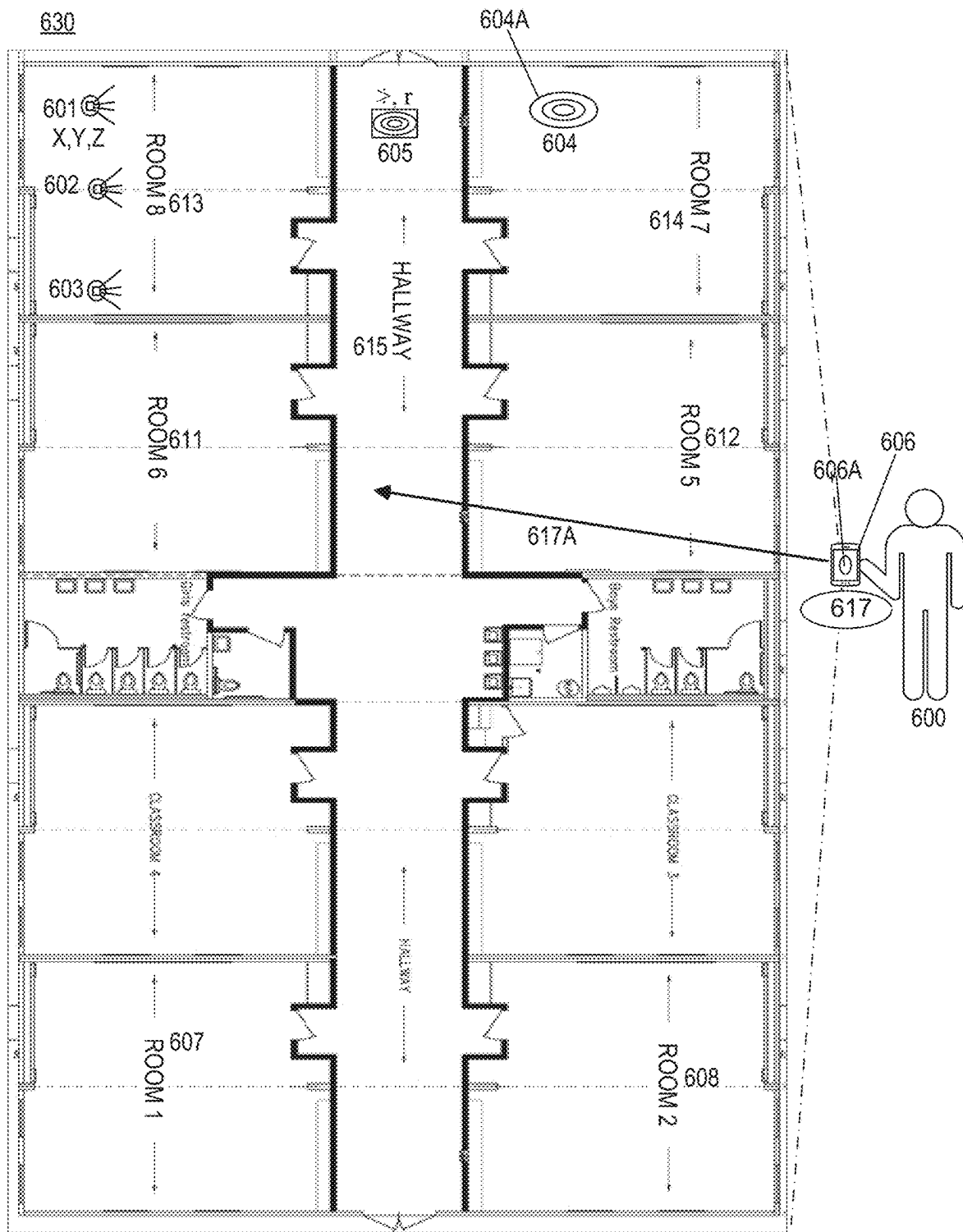


FIG. 6C

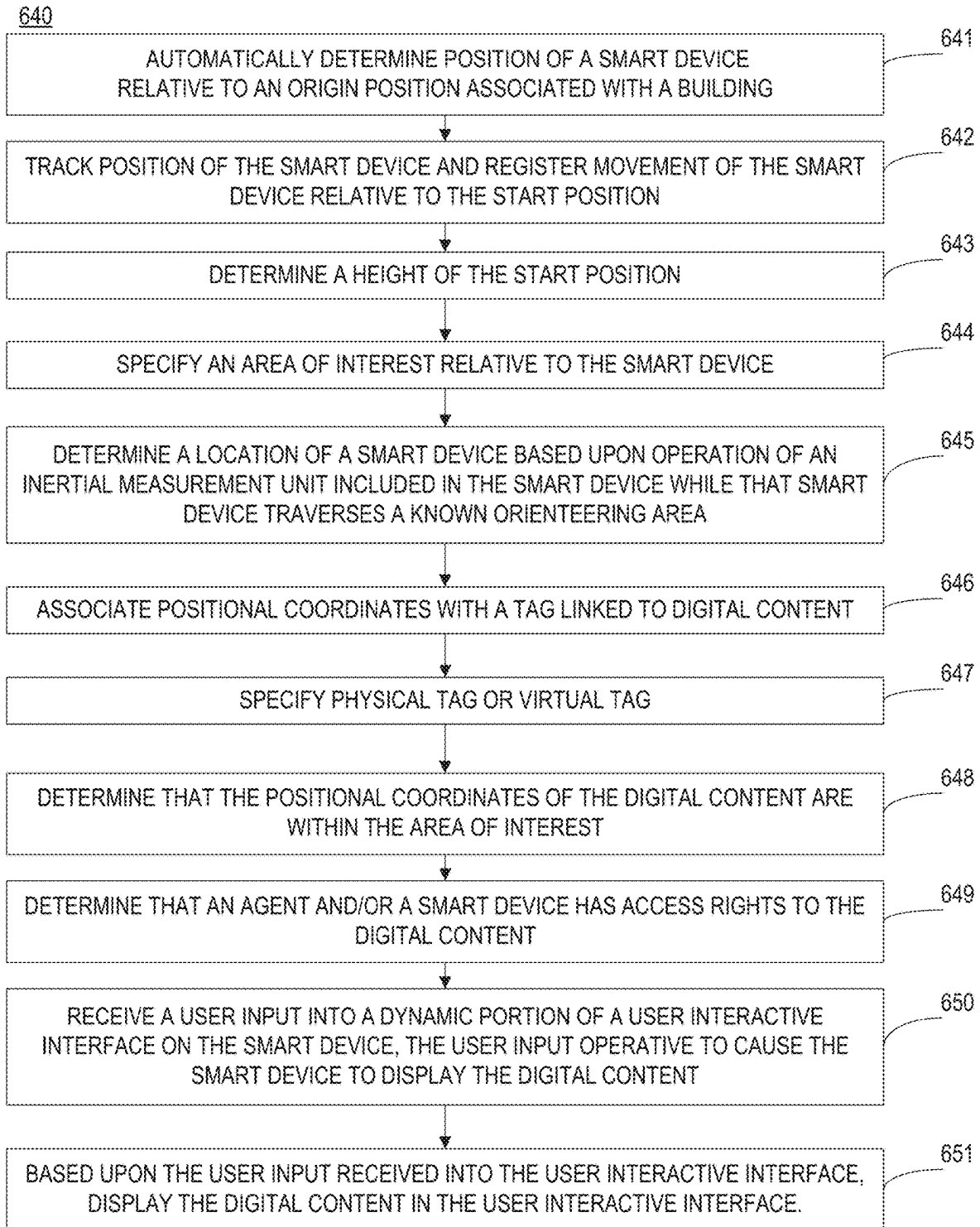


FIG. 6D



FIG. 7

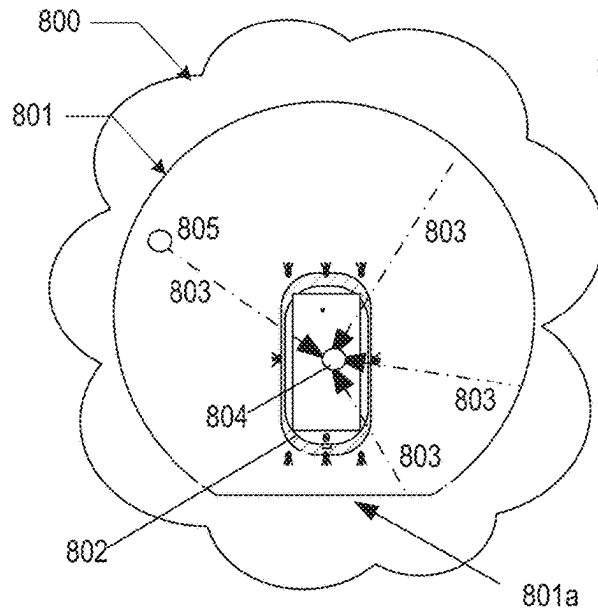


FIG. 8A

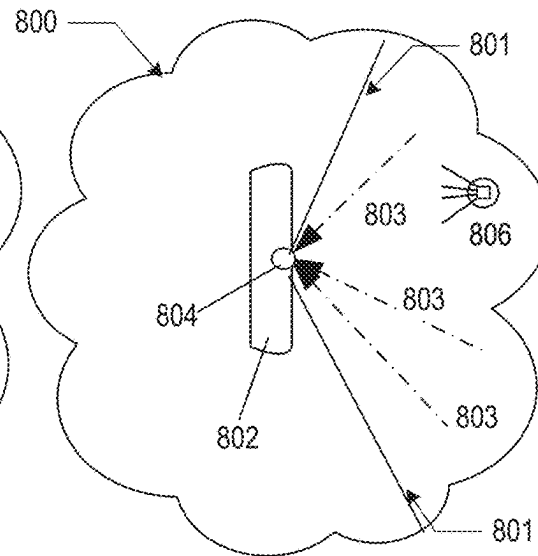


FIG. 8B

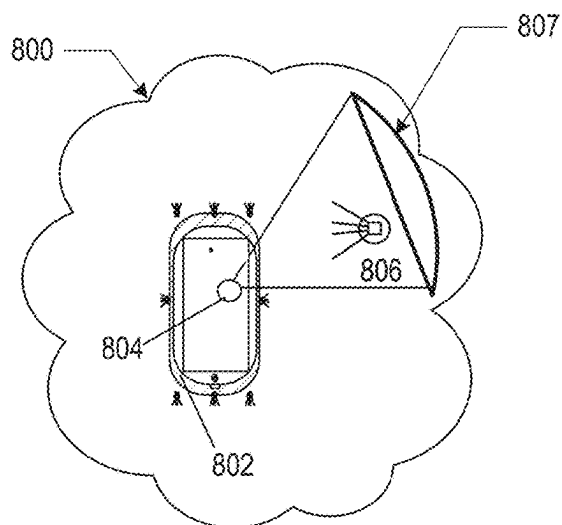


FIG. 8C

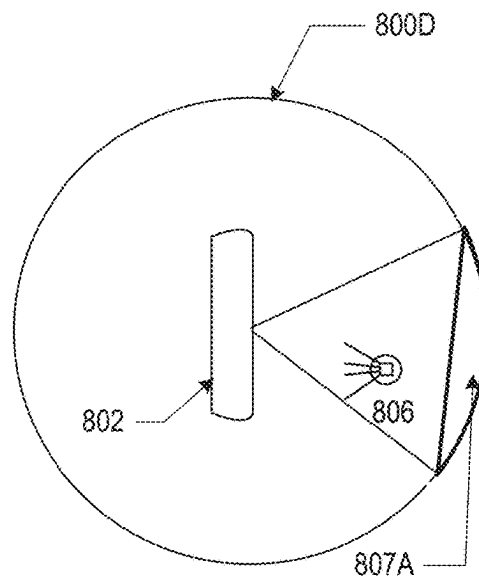


FIG. 8D

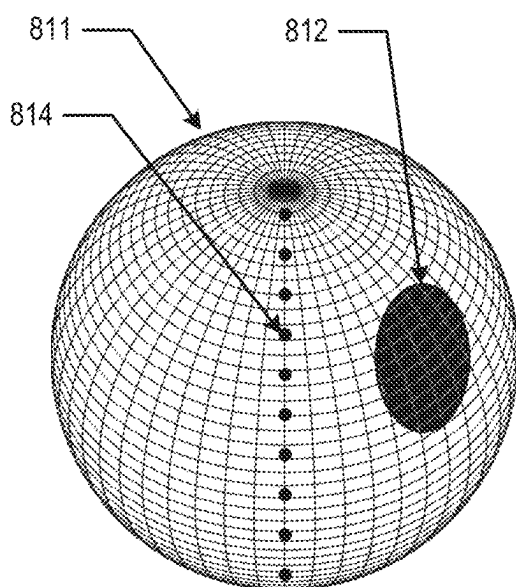


FIG. 8E

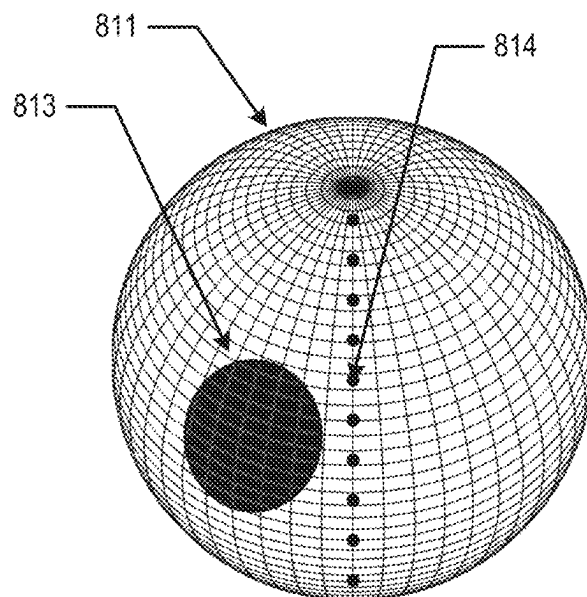


FIG. 8F

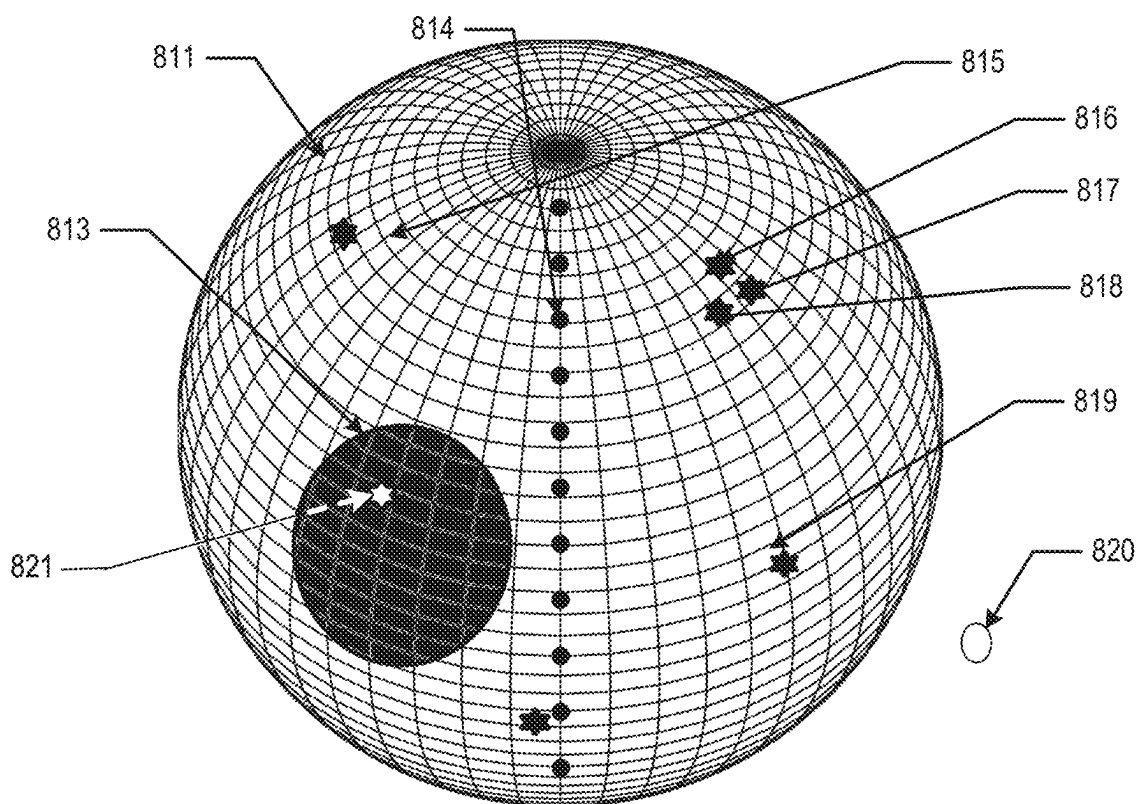


FIG. 8G

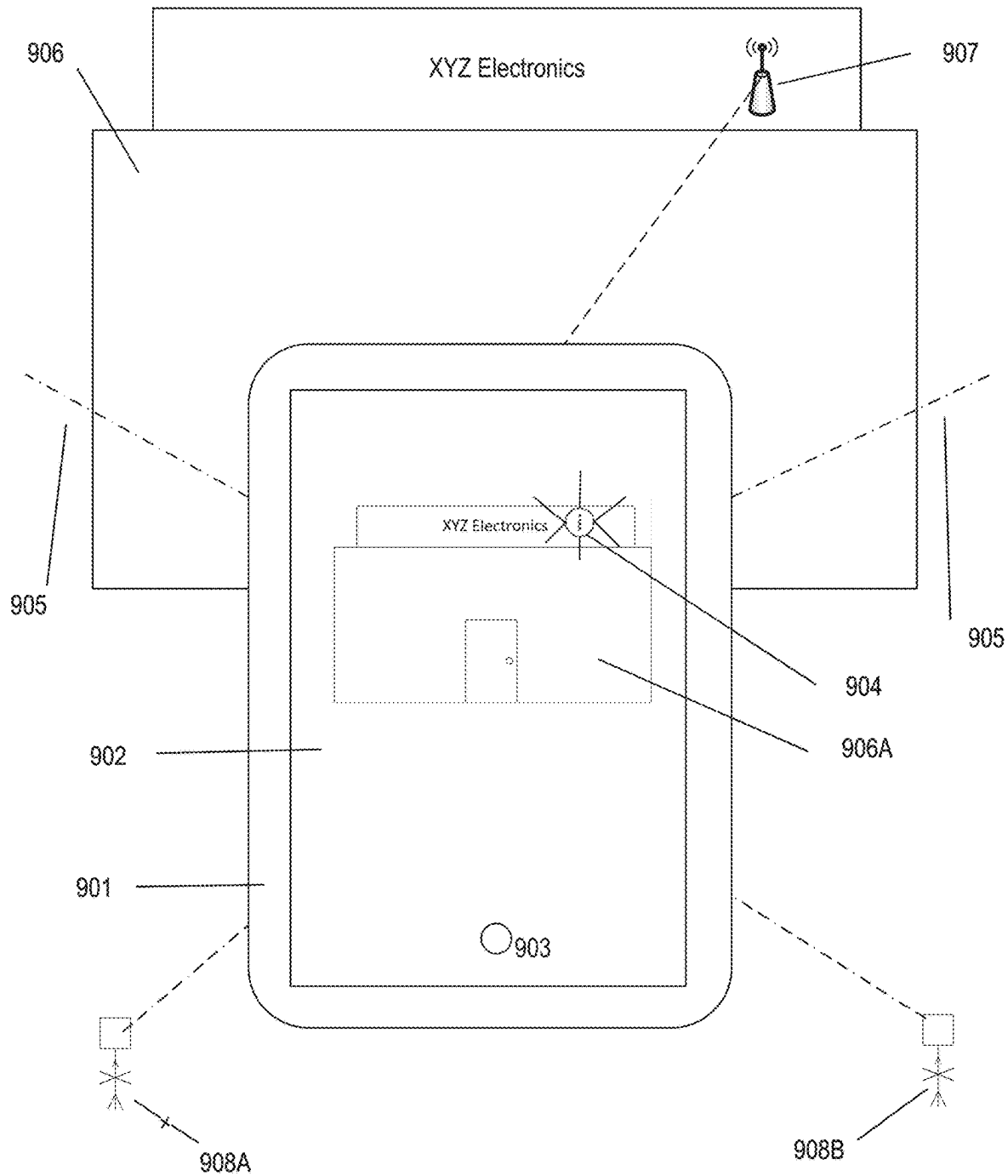


FIG. 9A

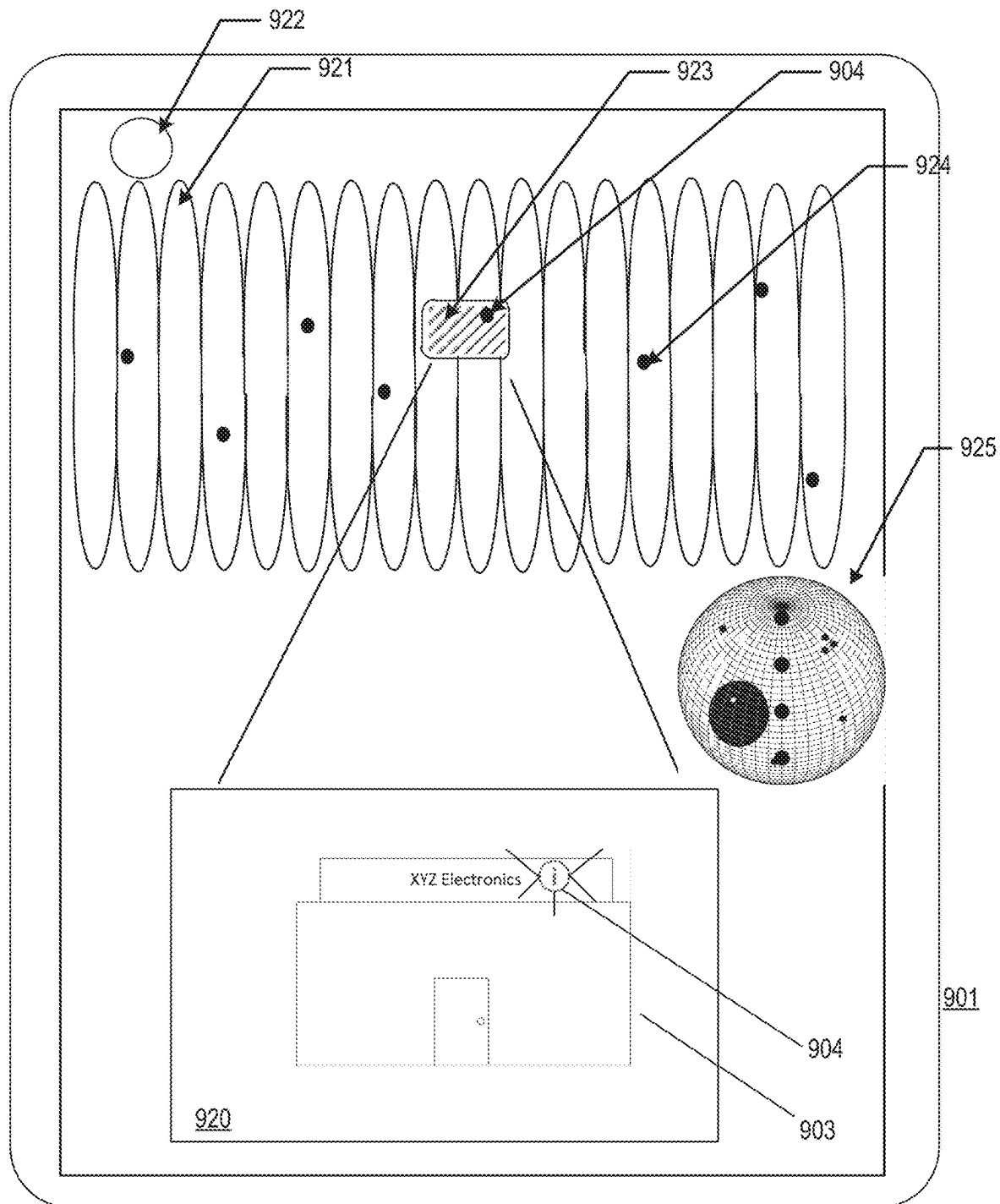


FIG. 9B

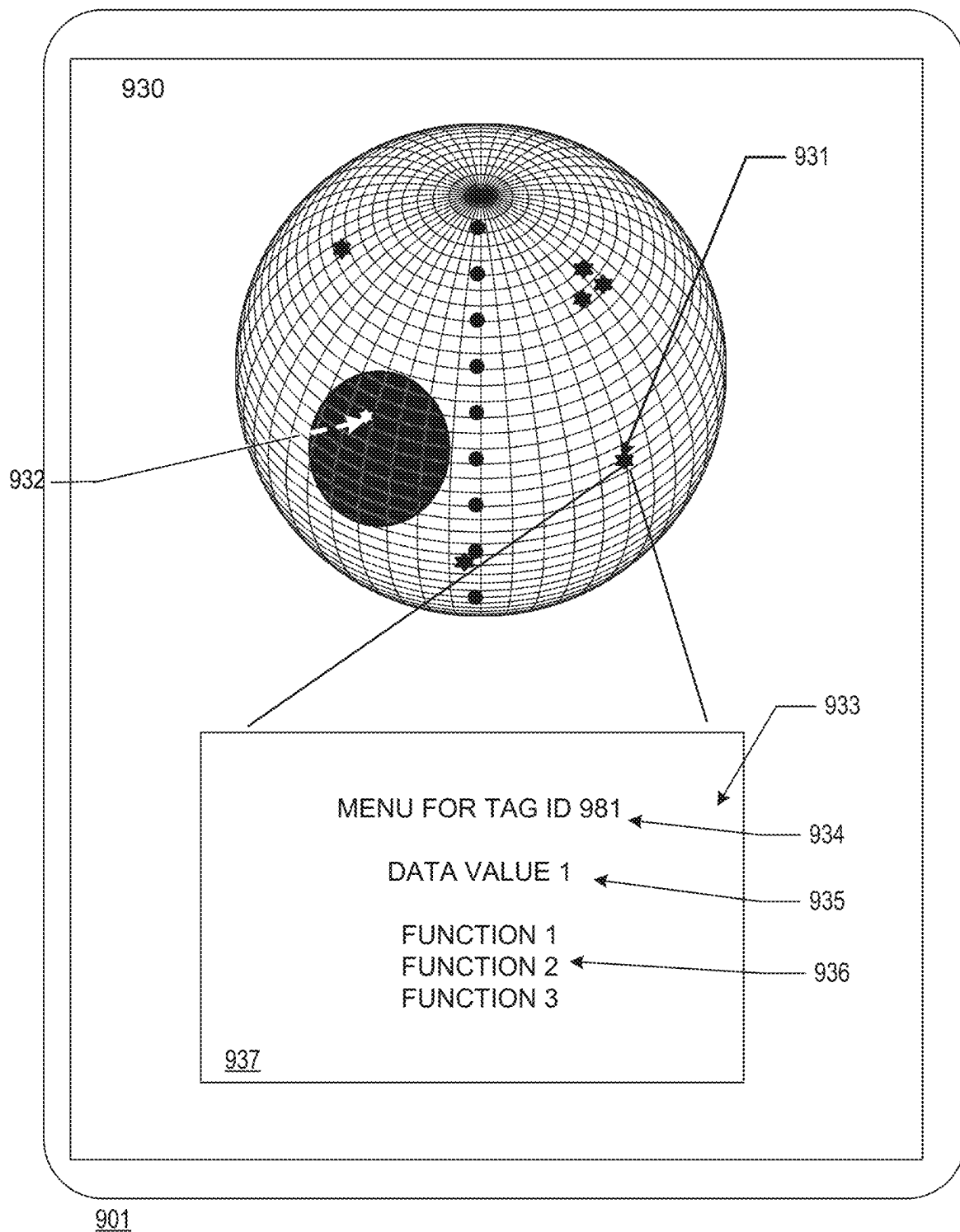


FIG. 9C

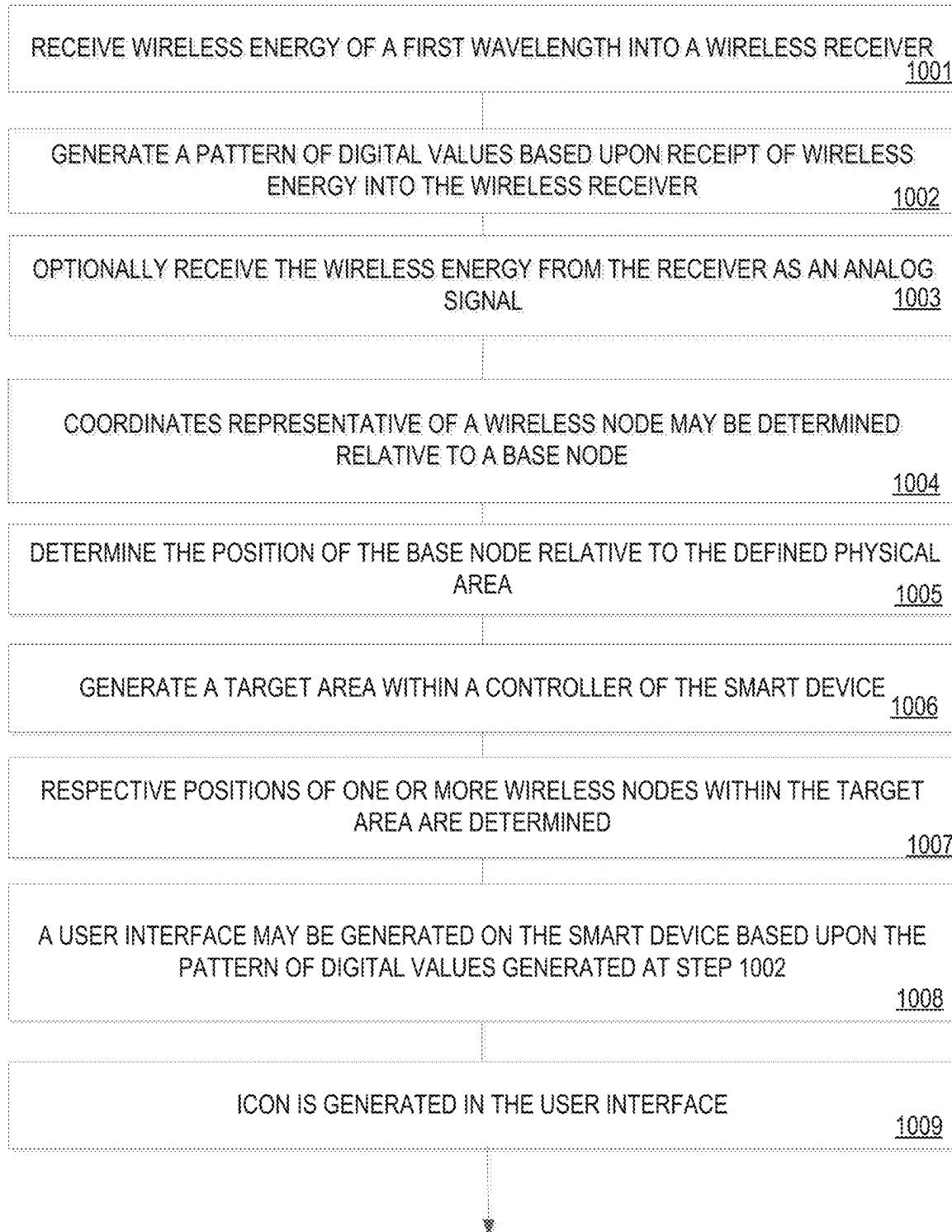


FIG. 10A

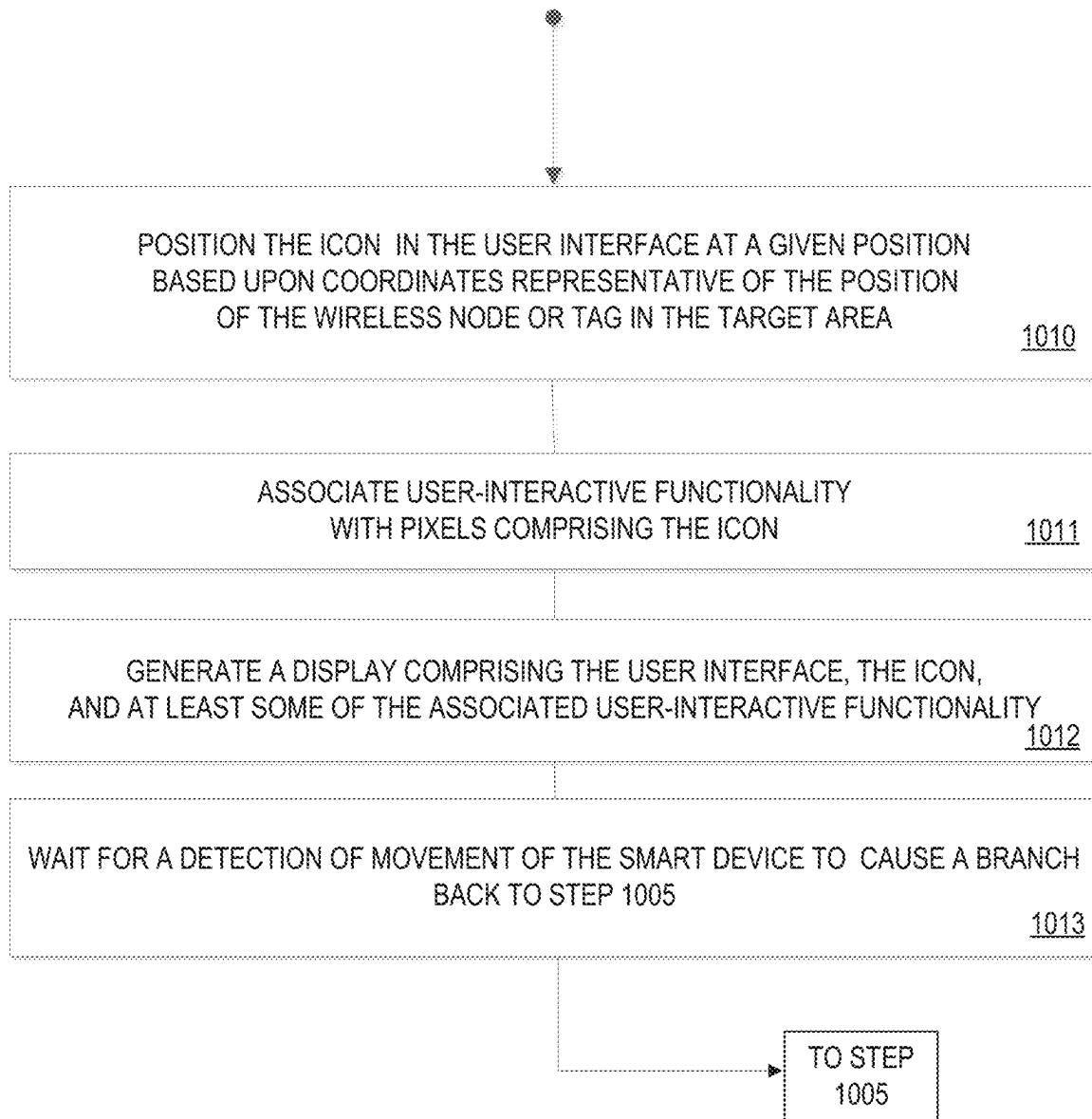
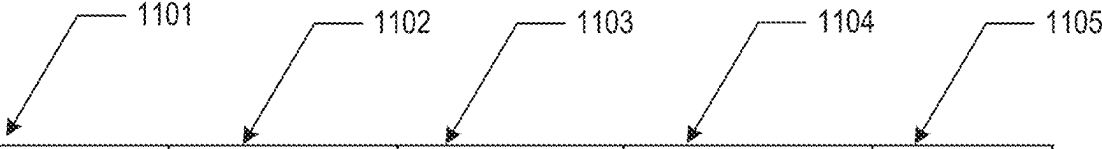


FIG. 10B



Coordinates	Permissions	ActionType	ActionAttributes	Notes
(1,1,1)	All	Information	"Store hours are 9:00-18:00"	Store hours
(1,1,1)	UserID = 333	Message	"Hi Jennifer!"	Message from store owner
(0,0,0)	RequestPassword()	Action	InitiateAlarm()	Emergency alarm
(2,2,2)	All	Directions	GreenArrows()	Display arrows leading to next destination

FIG. 11

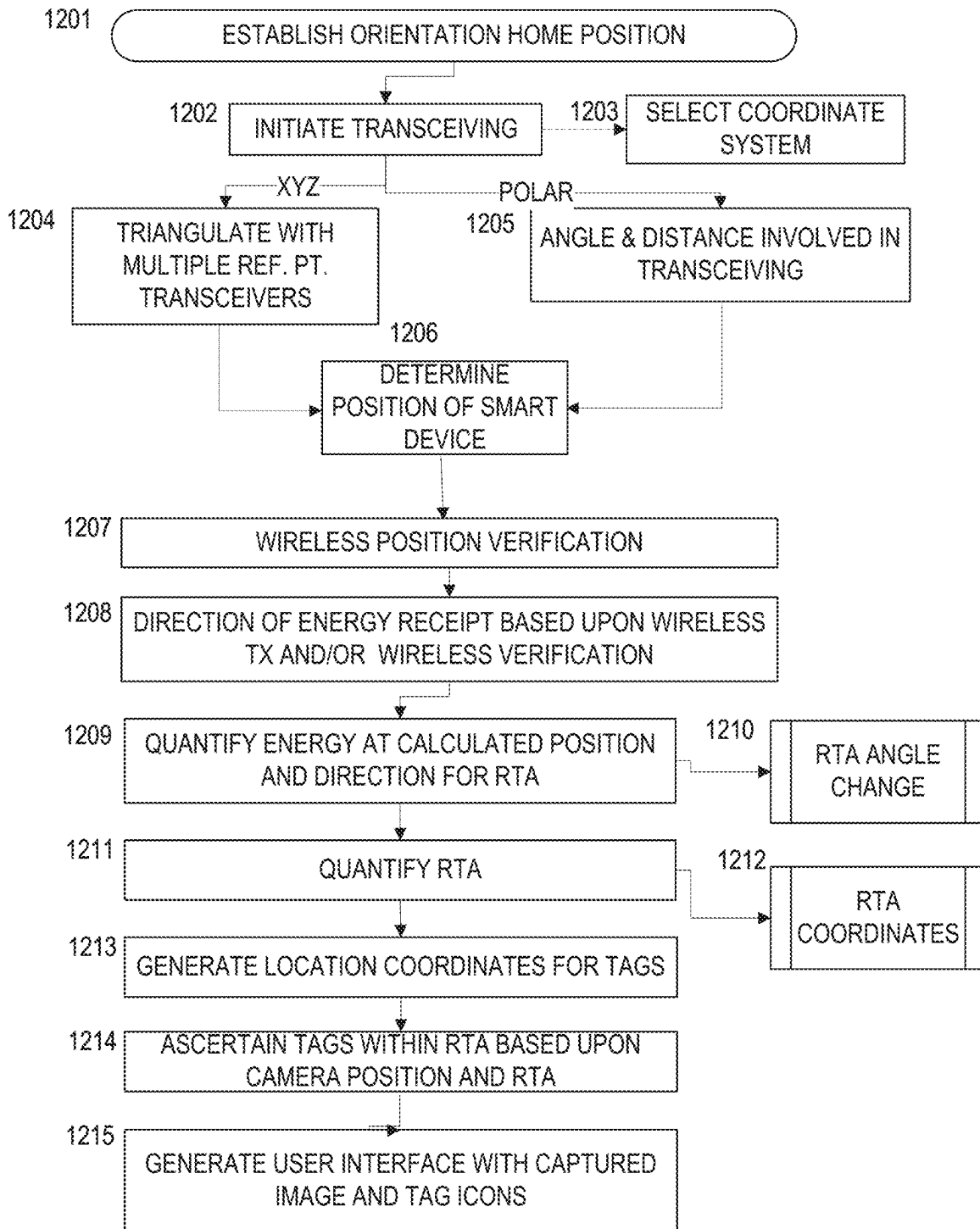


FIG. 12

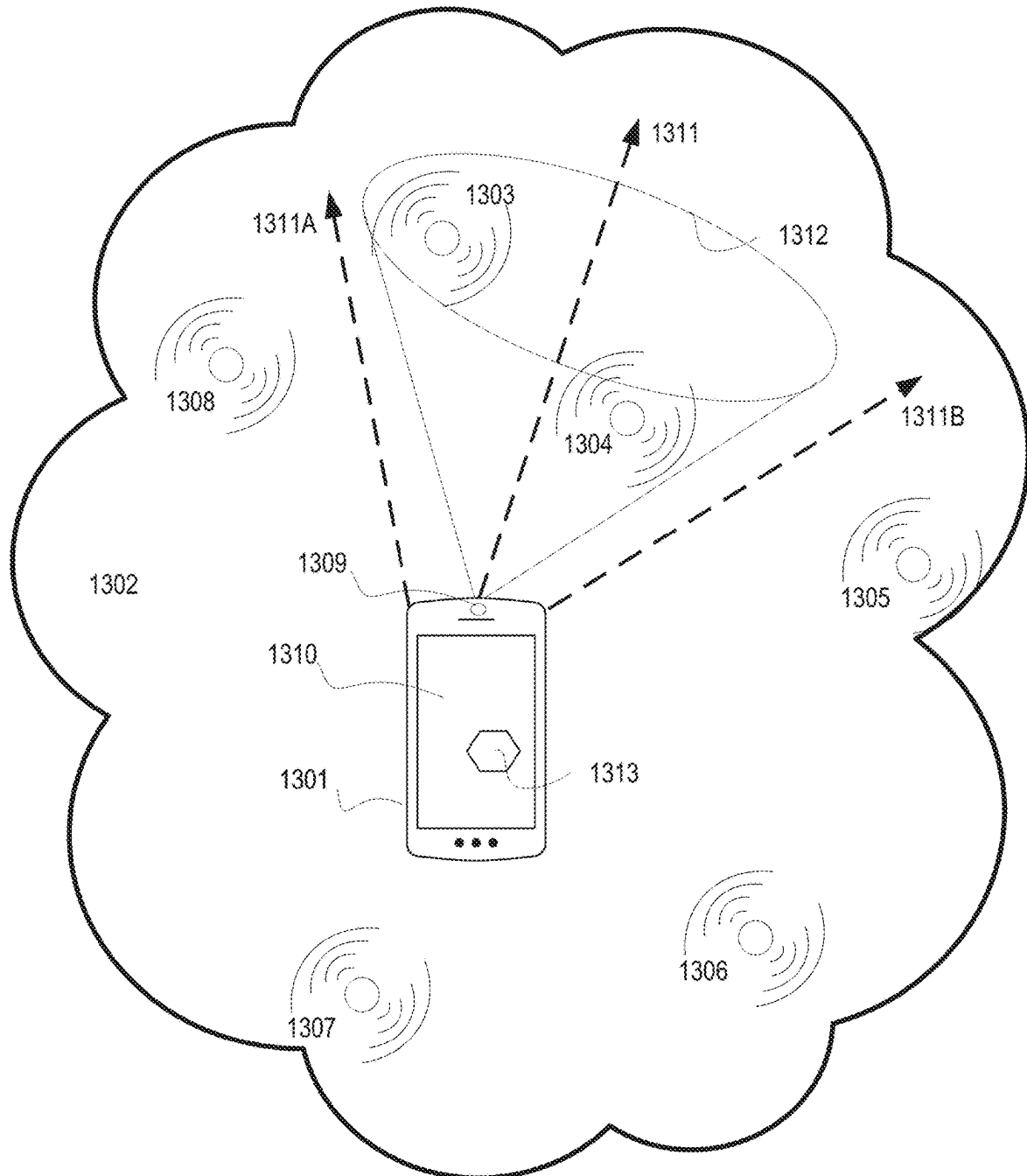


FIG. 13



FIG. 14

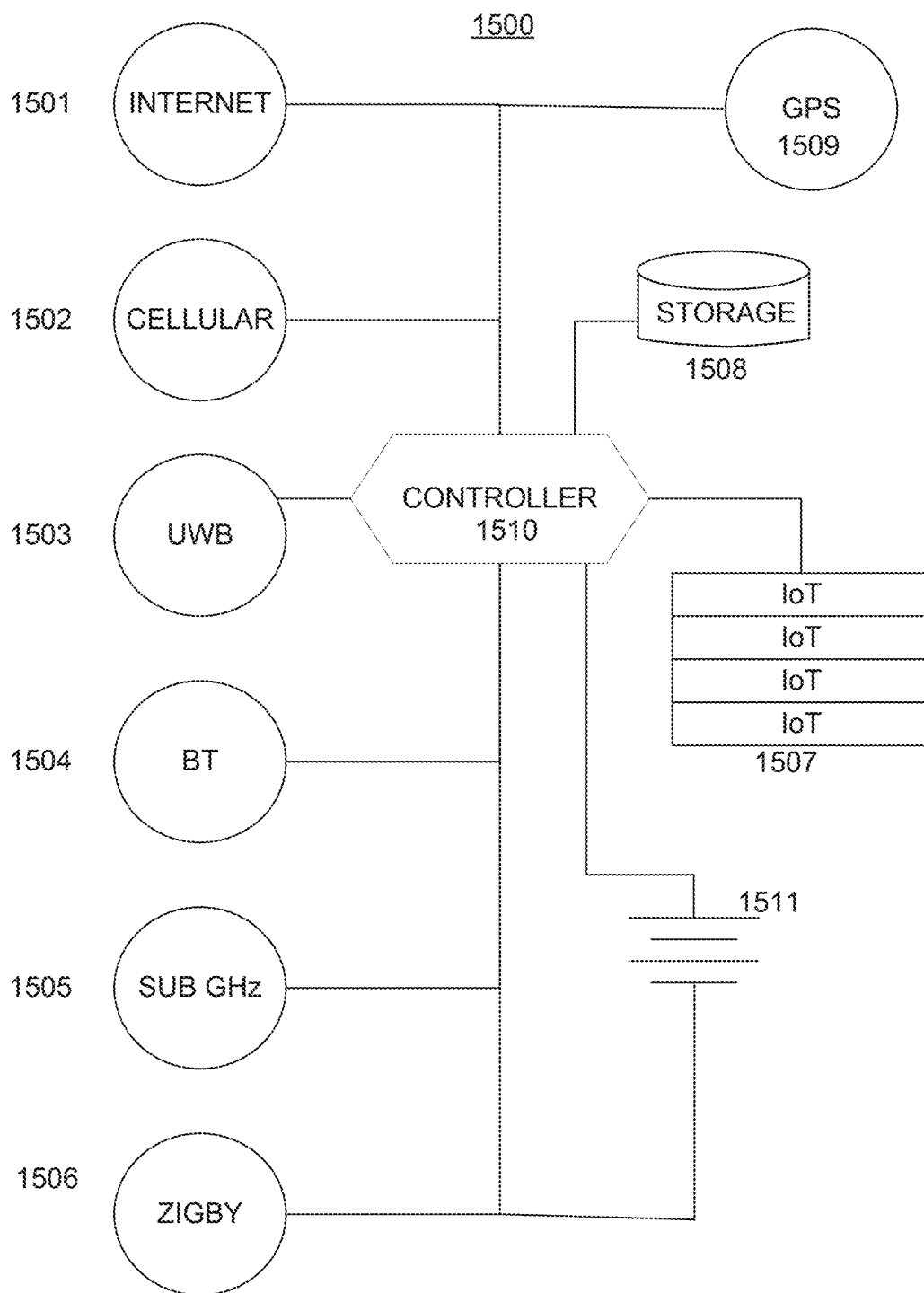


FIG. 15

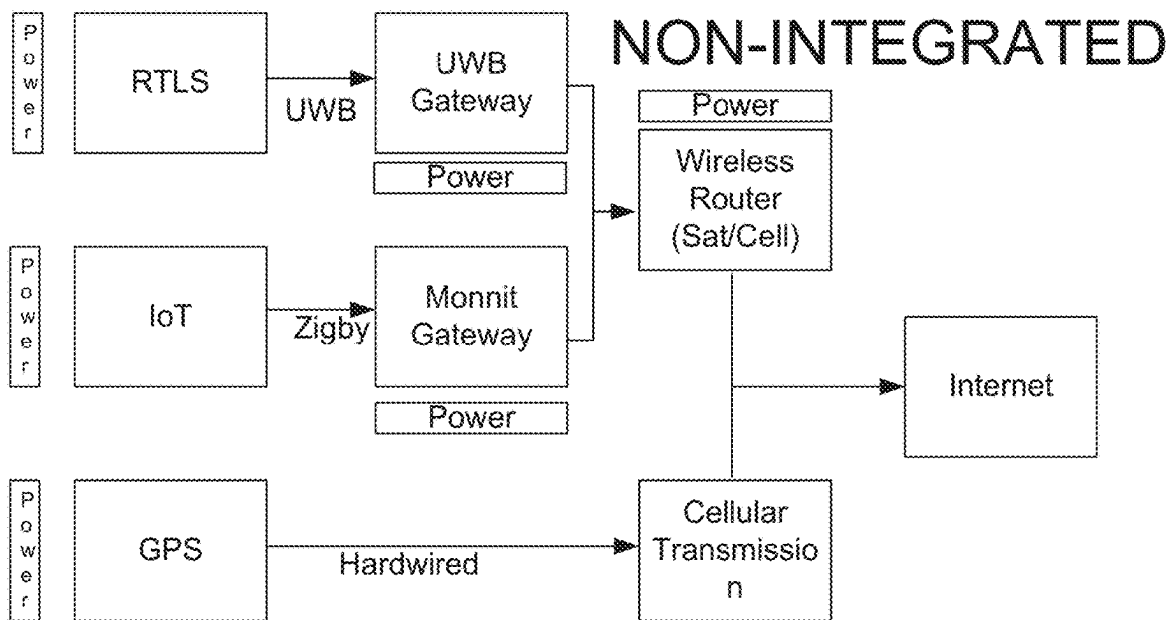


FIG. 16A

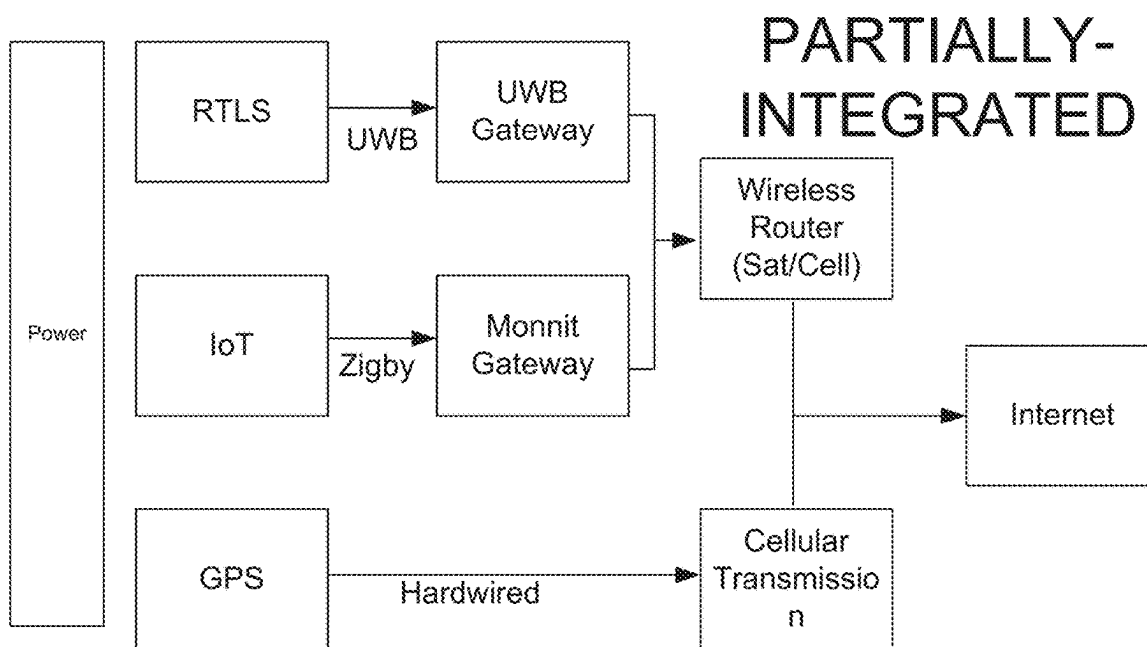


FIG. 16B

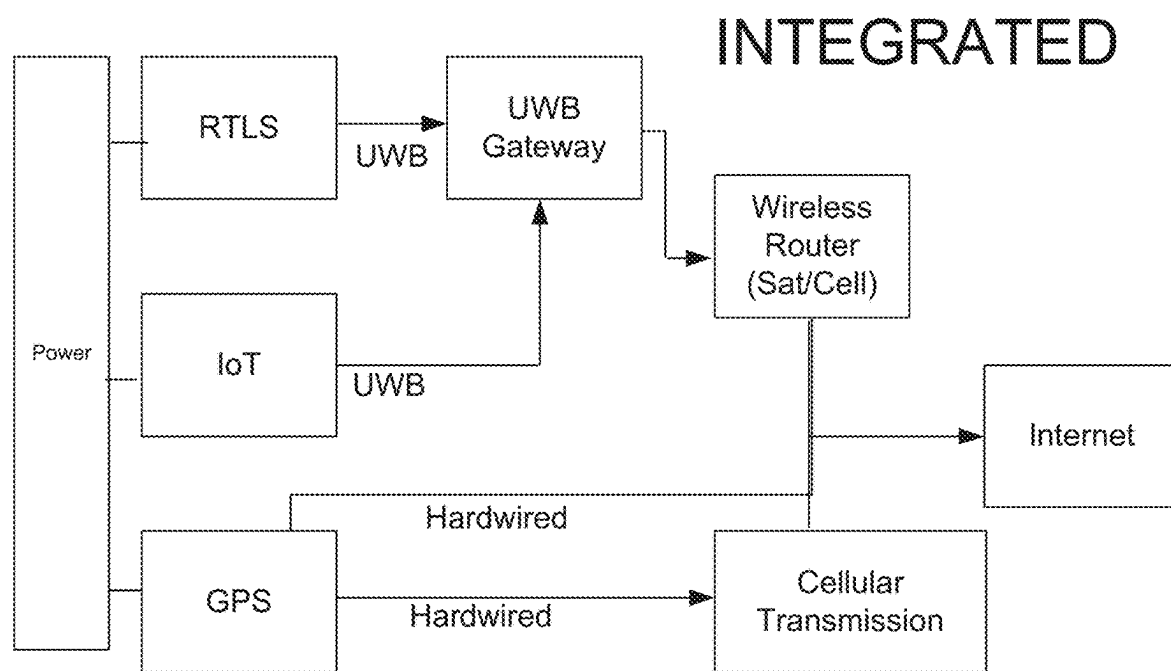


FIG. 16C

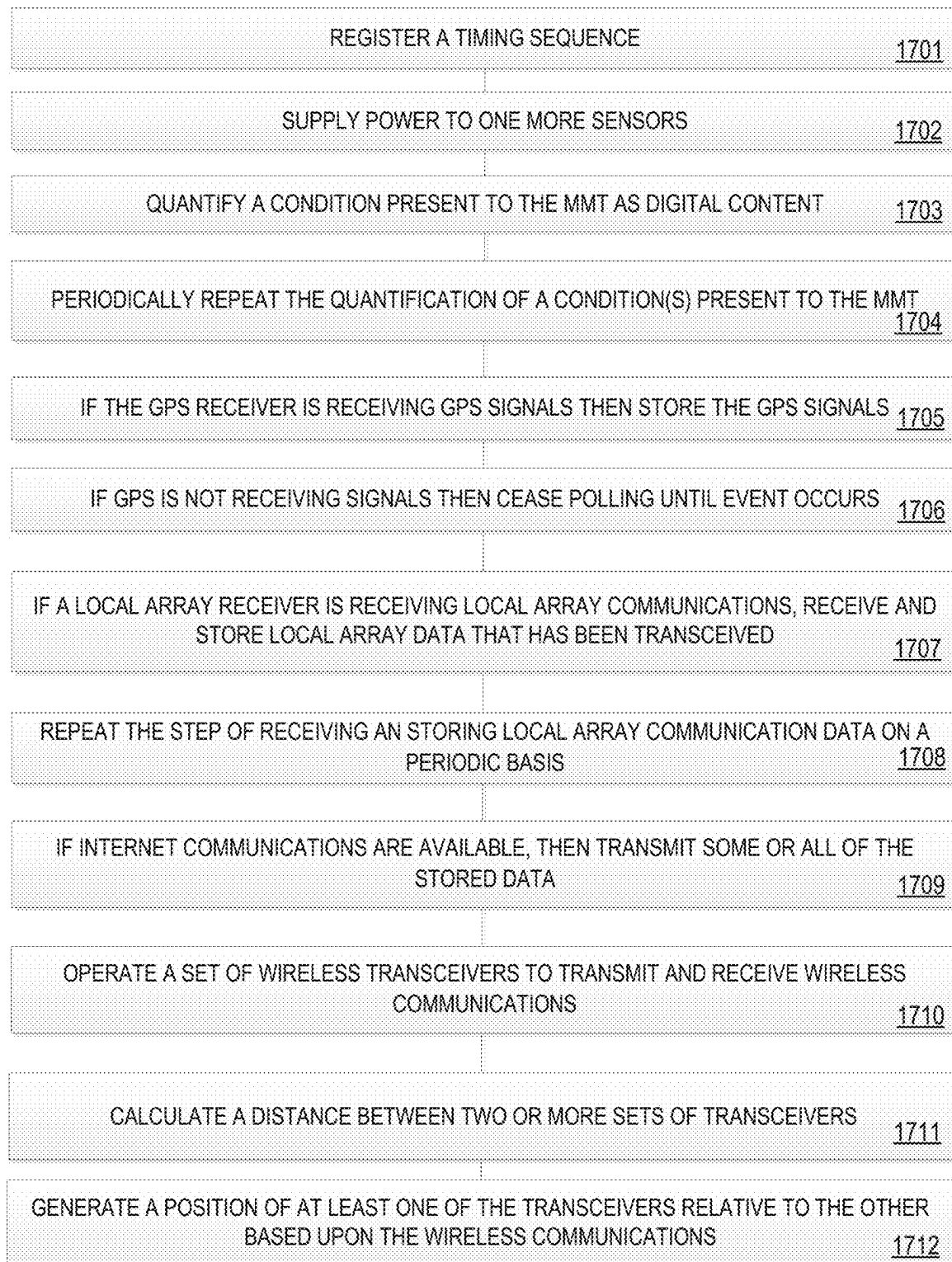


FIG. 17A

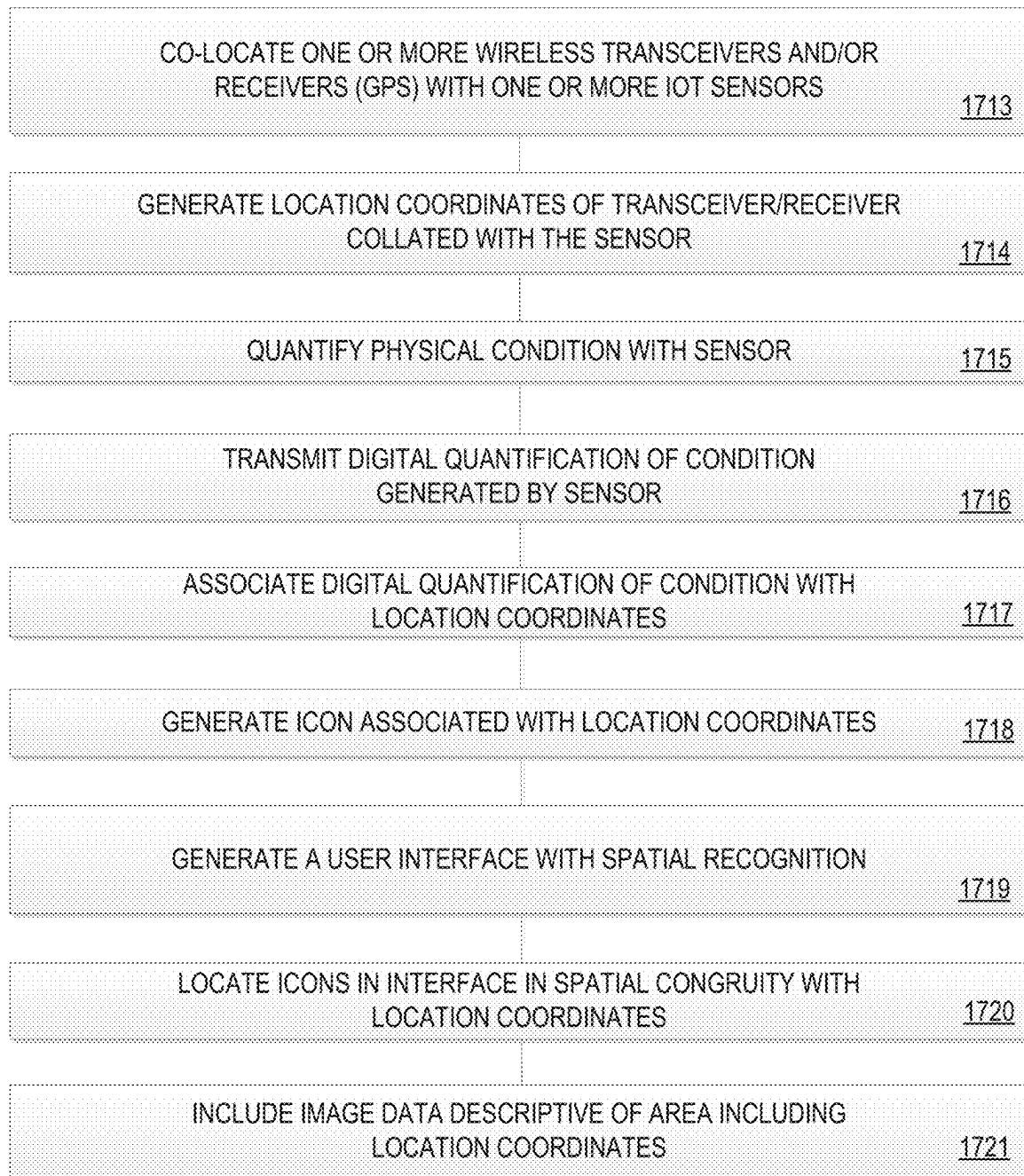


FIG. 17B

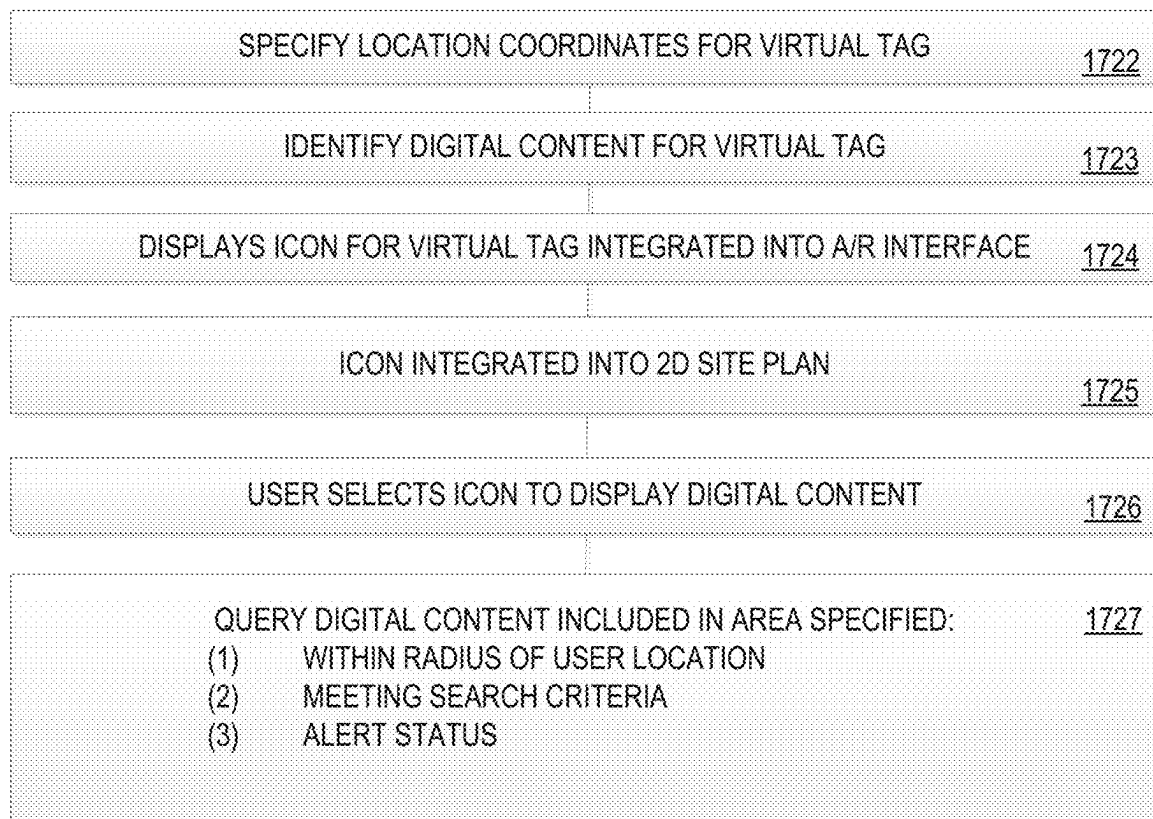


FIG. 17C

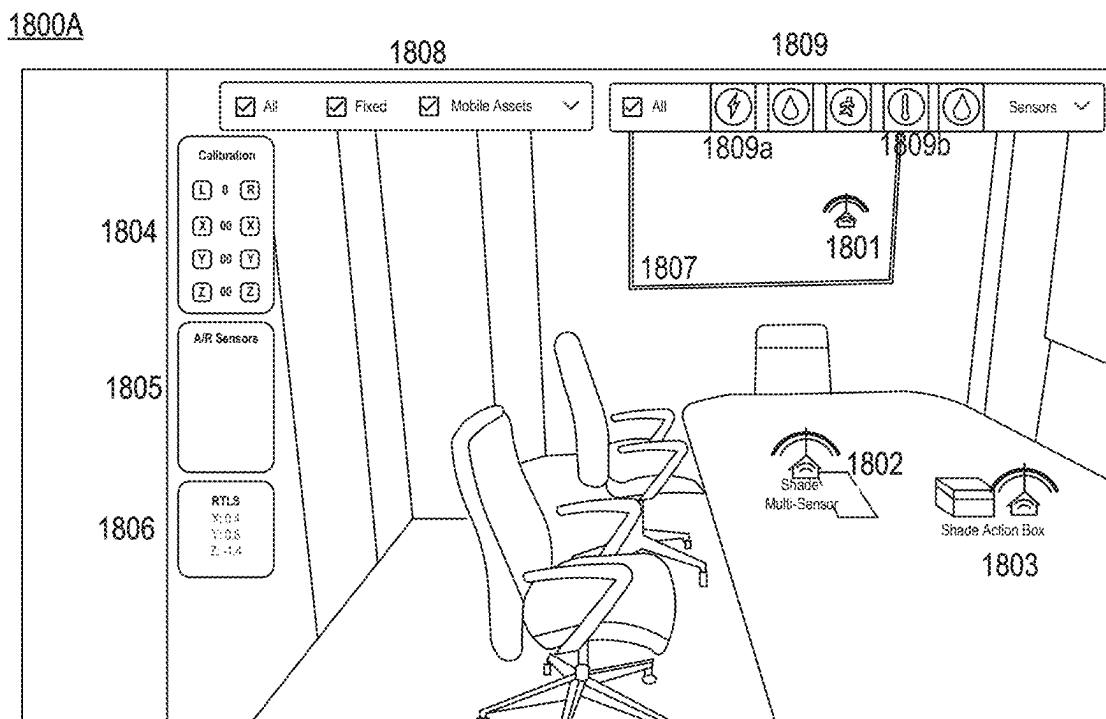


FIG. 18A

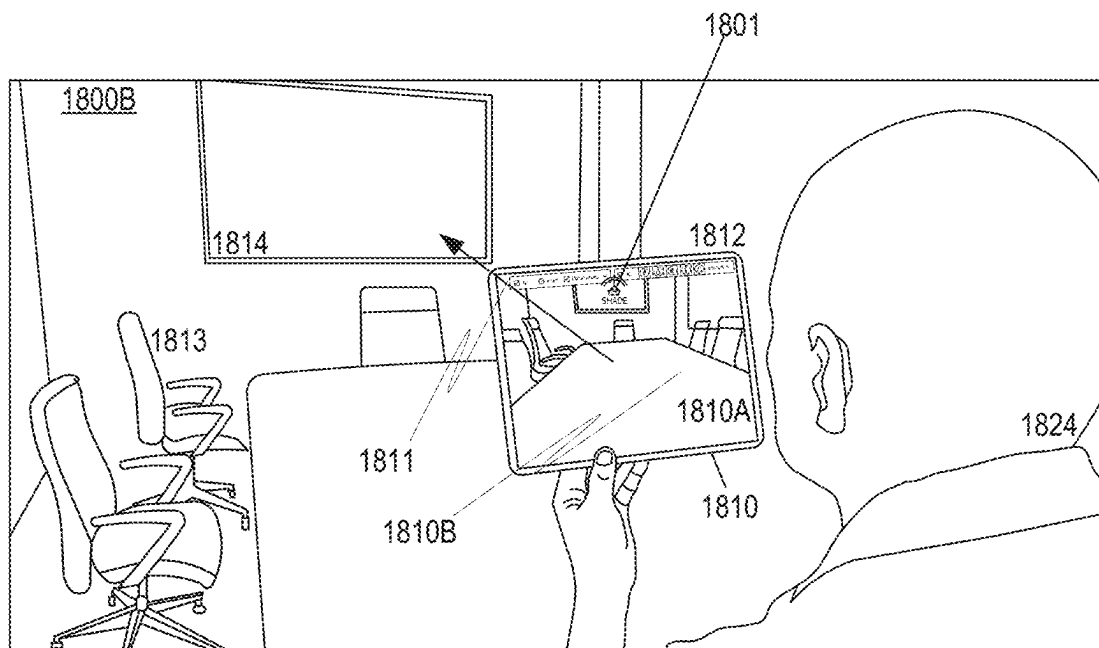


FIG. 18B

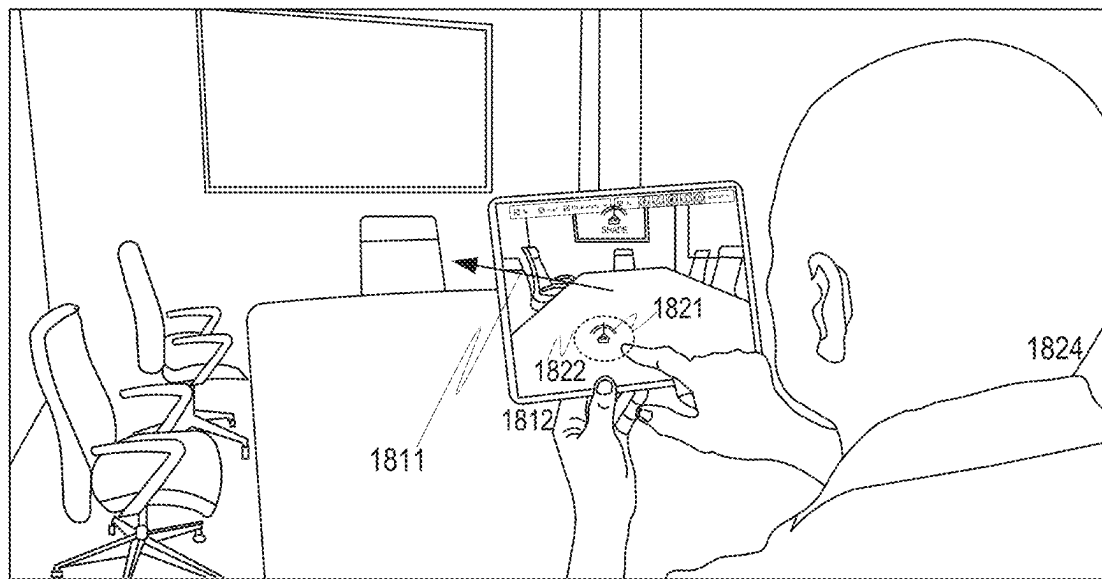


FIG. 18C

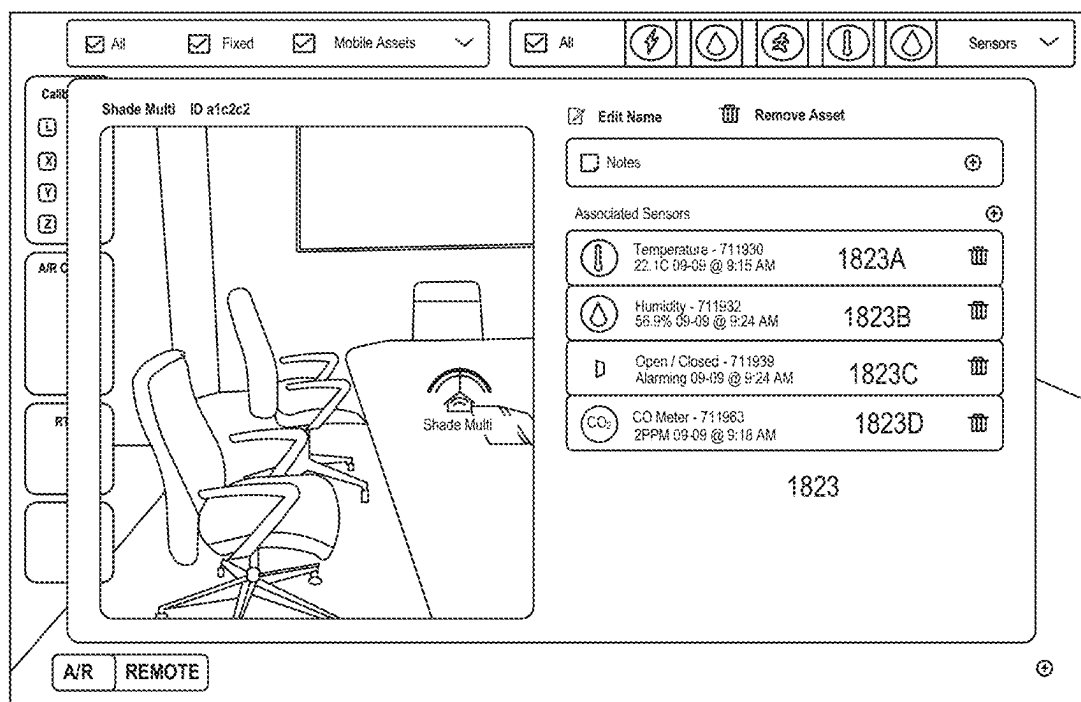


FIG. 18D

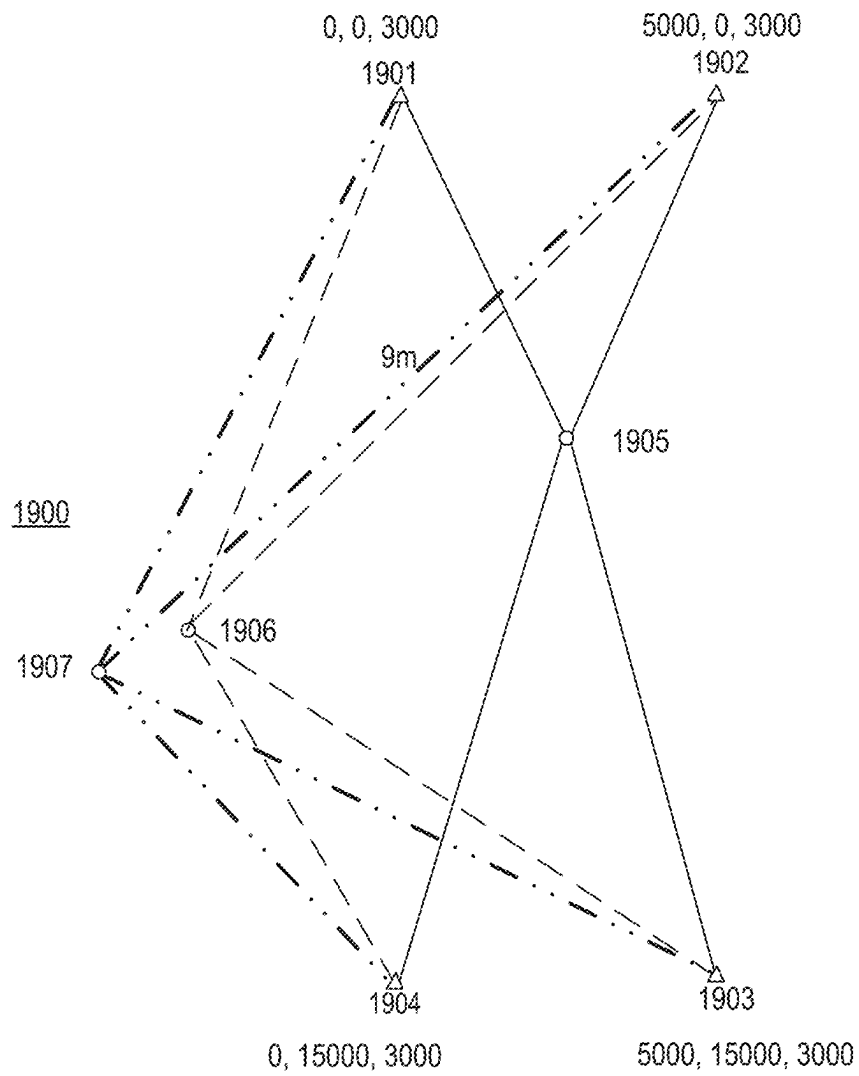


FIG. 19

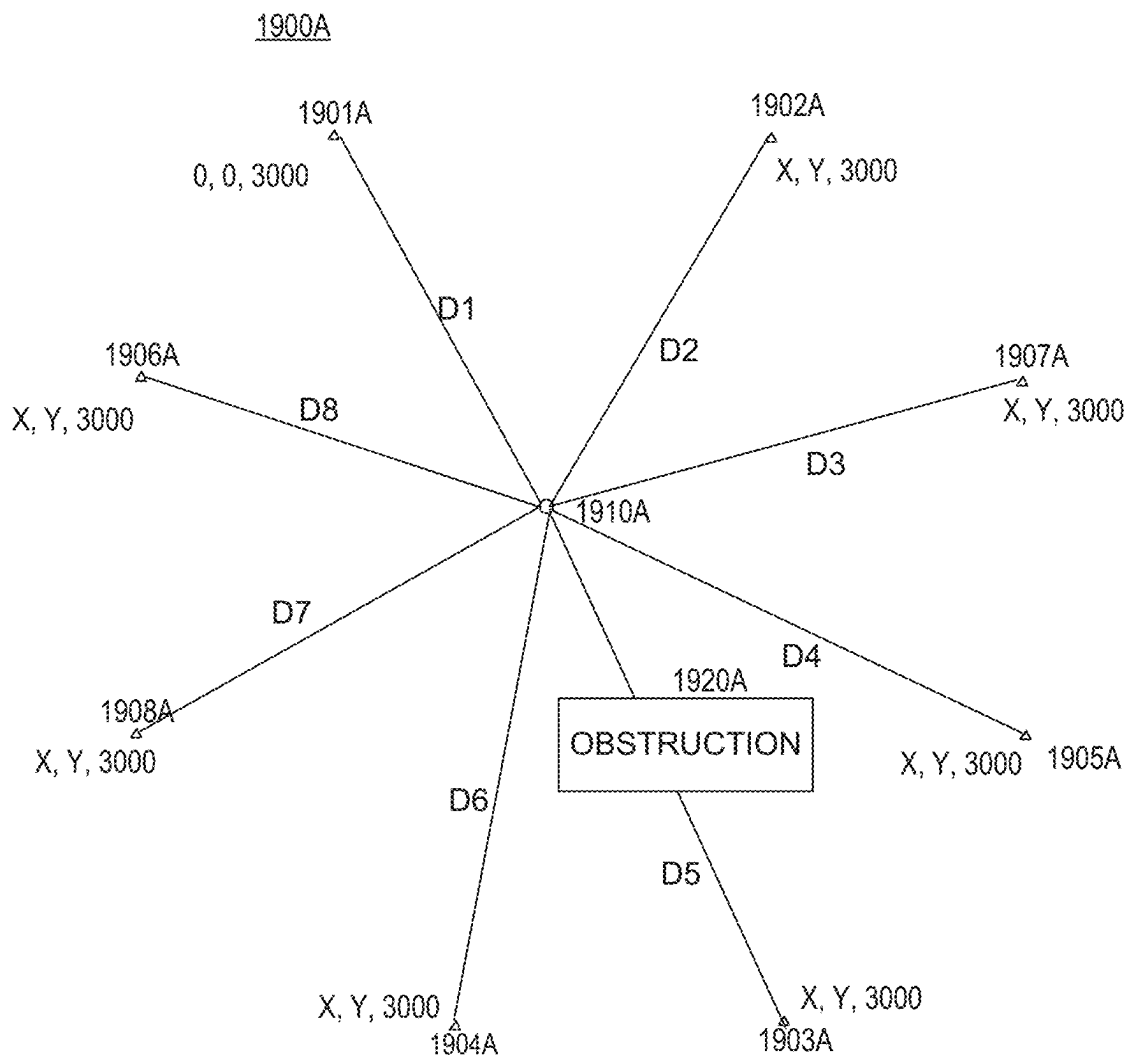


FIG. 19A

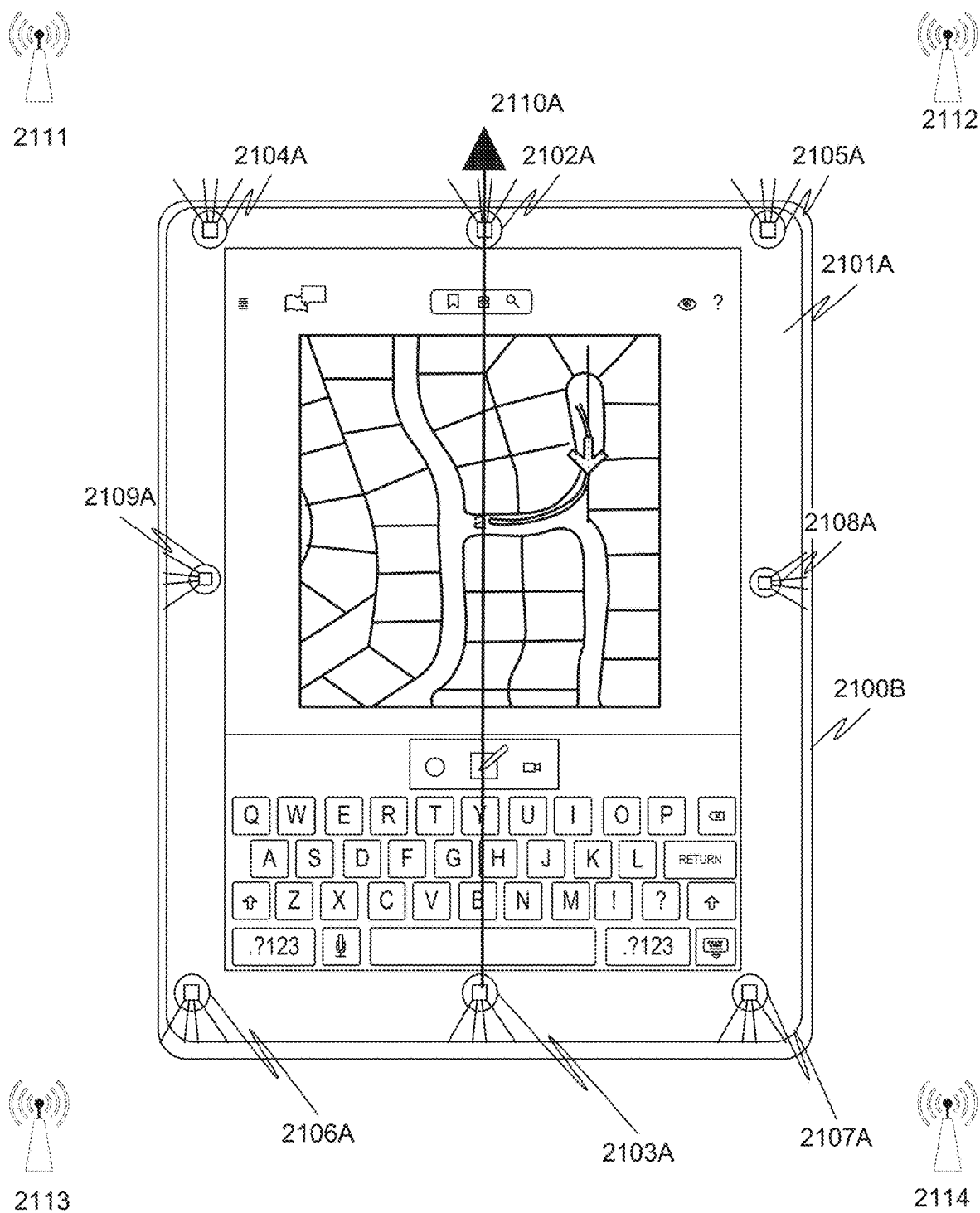


FIG. 20

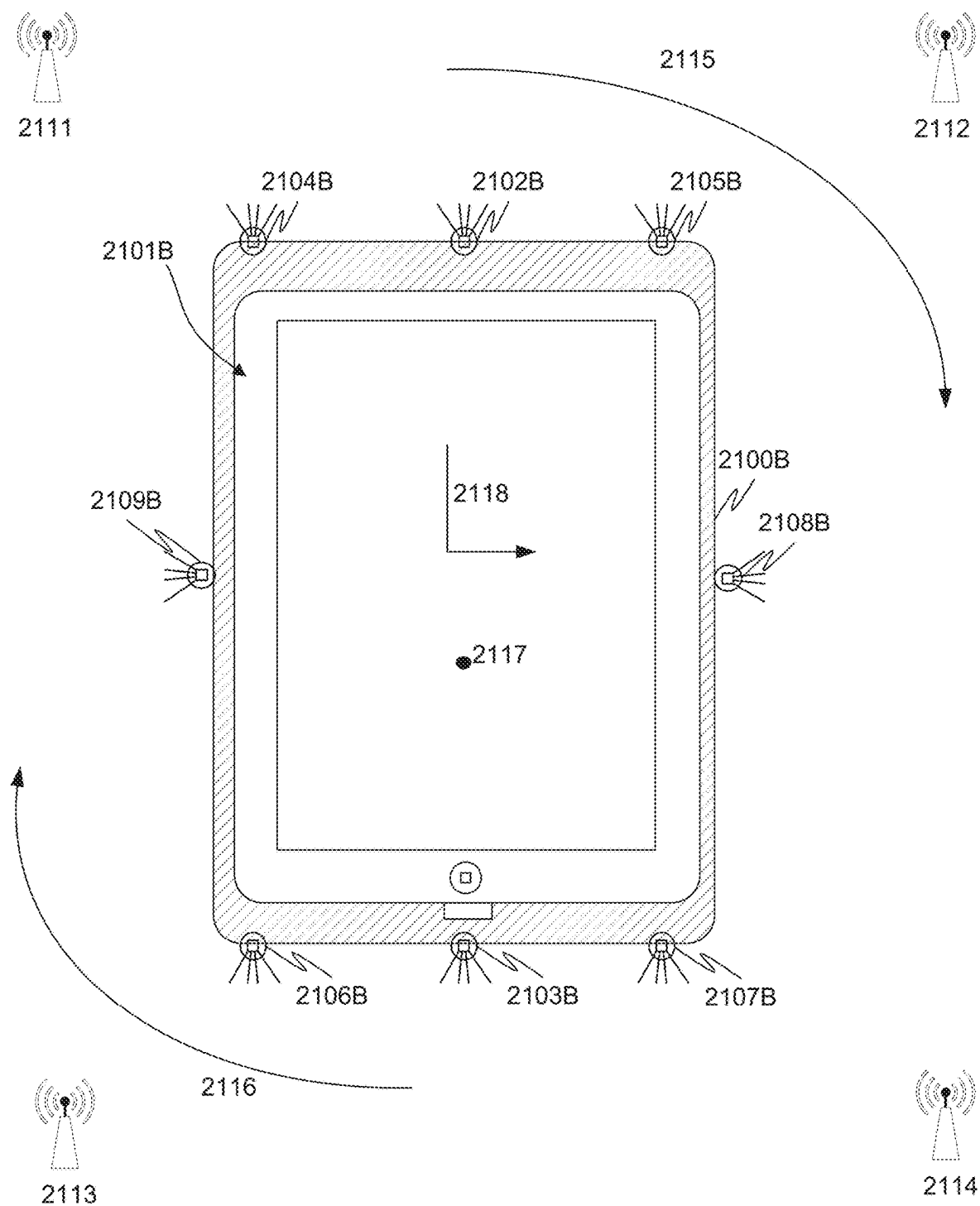


FIG. 21

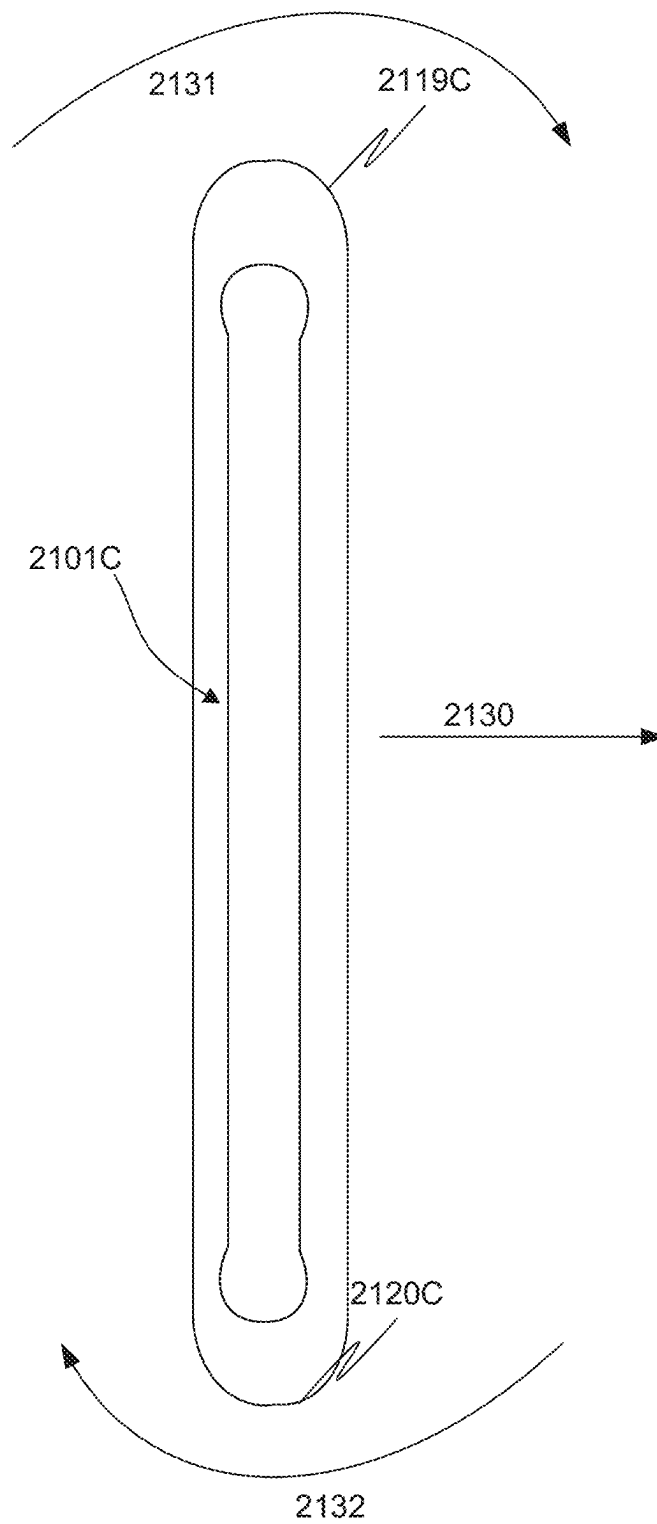


FIG. 21A

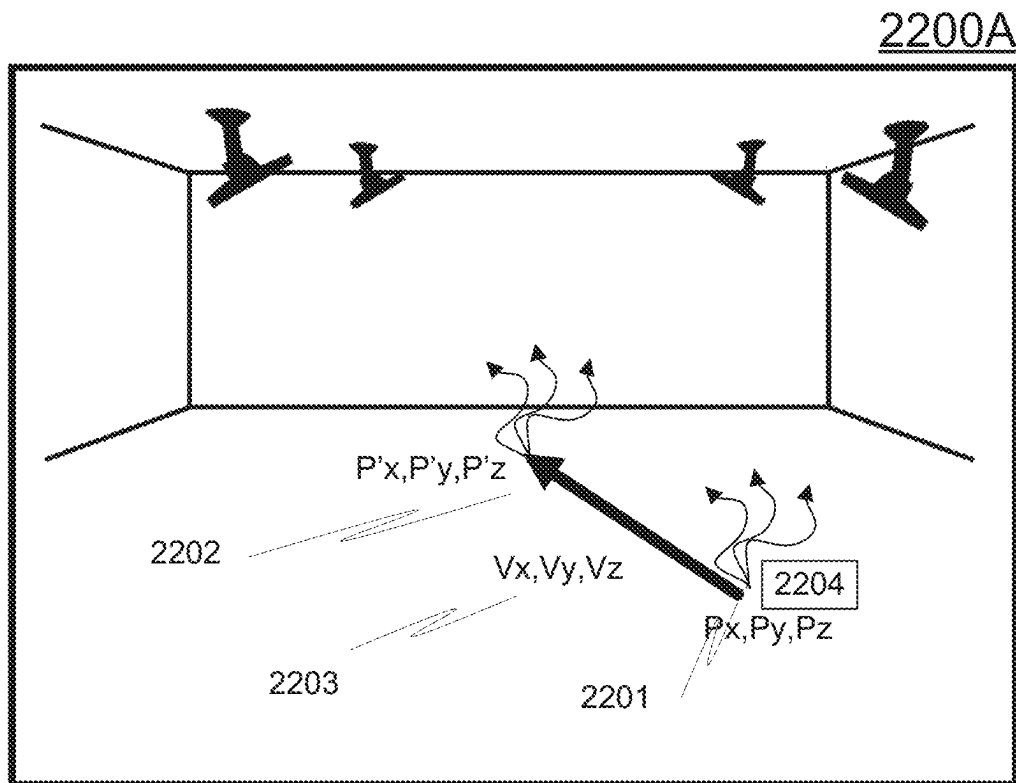


FIG. 22A

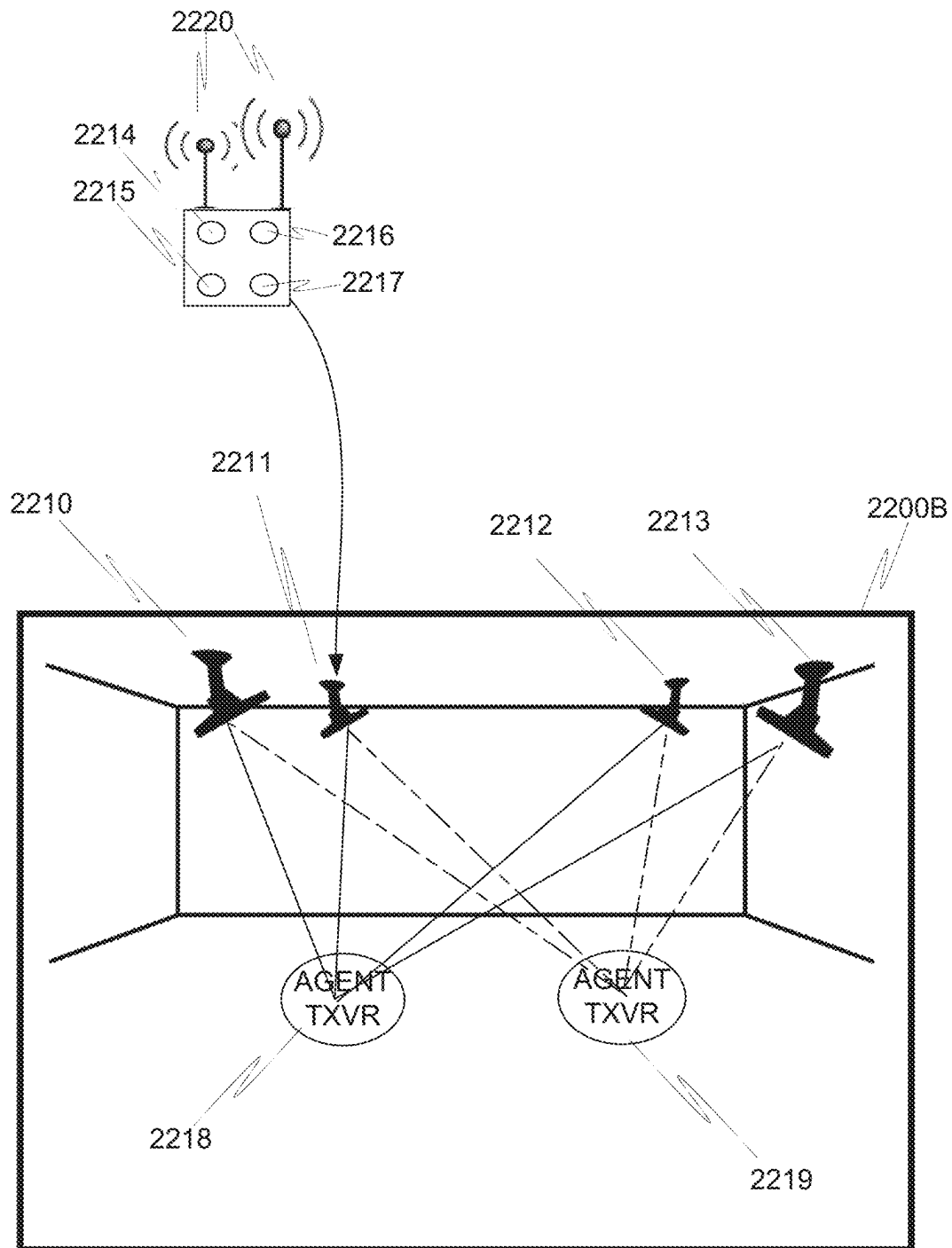


FIG. 22B

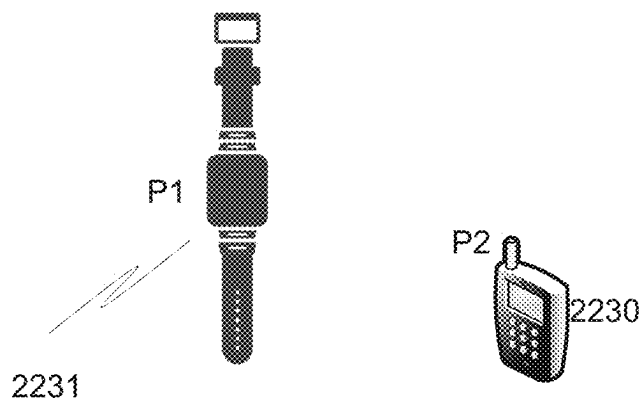


FIG. 22C

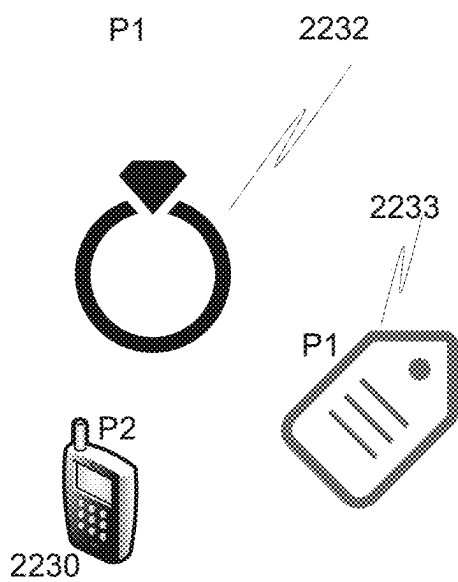


FIG. 22D

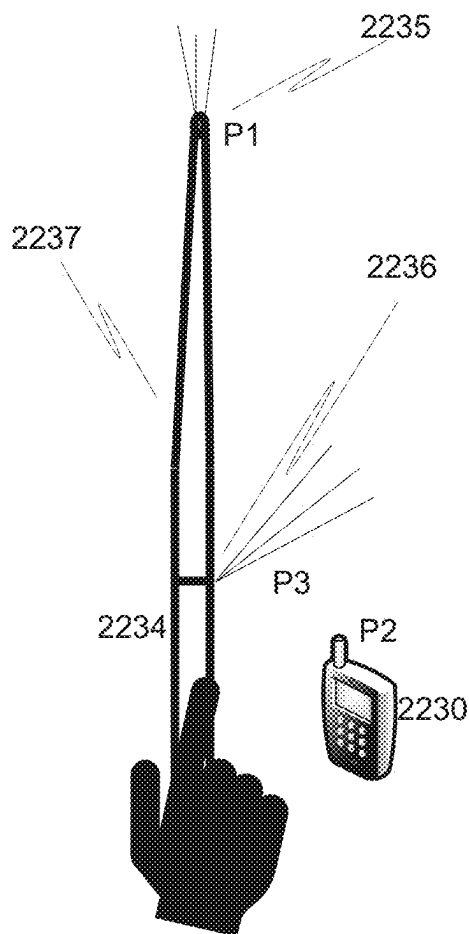


FIG. 22E

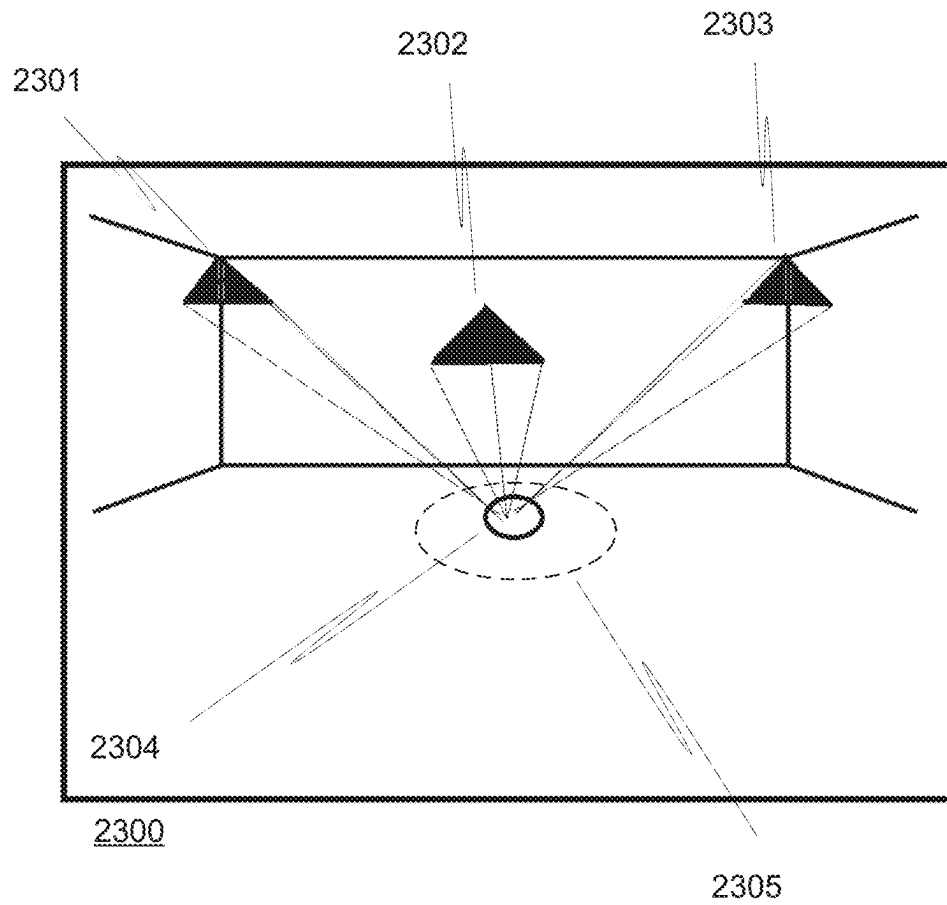


FIG. 23A

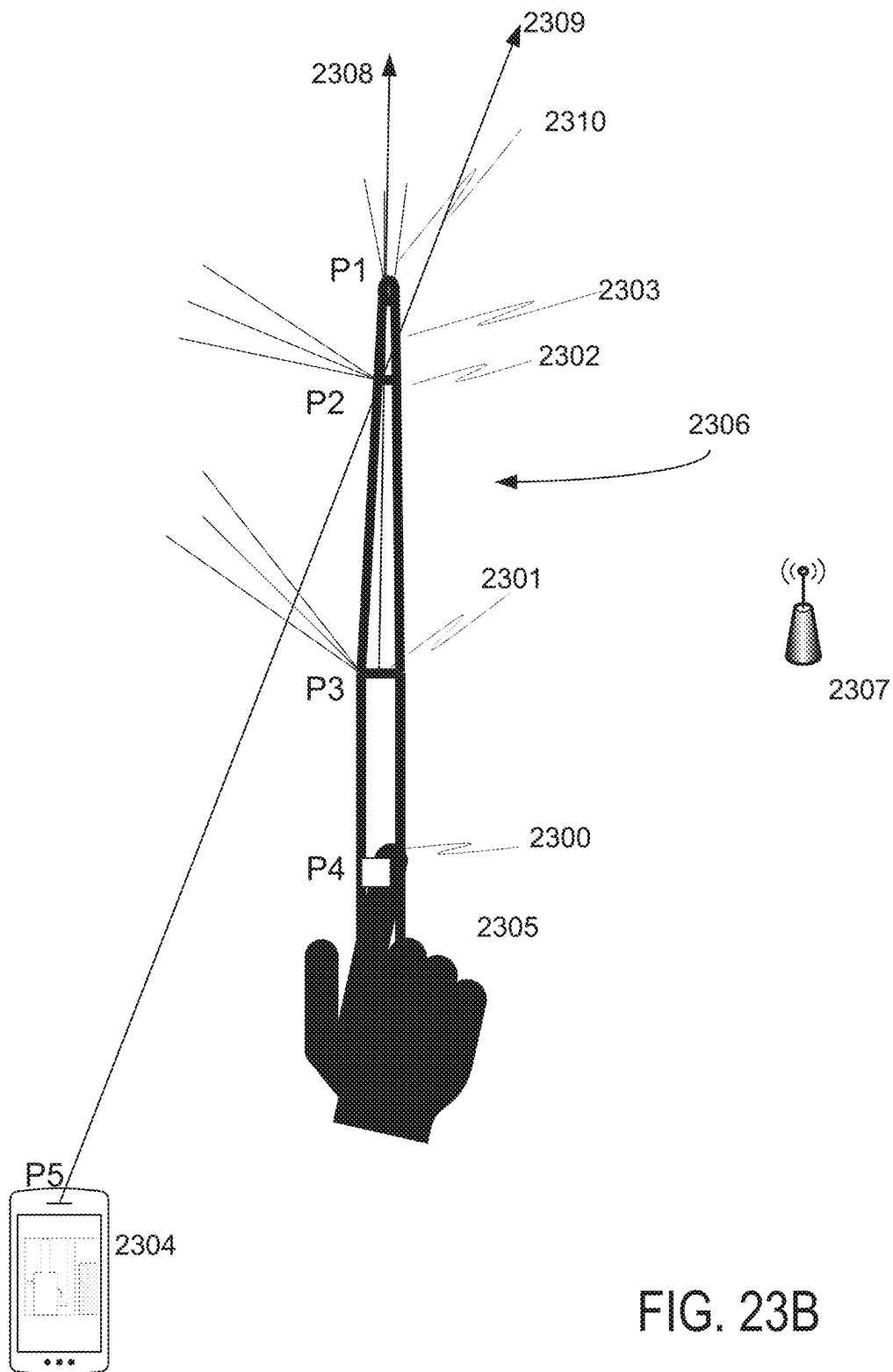


FIG. 23B

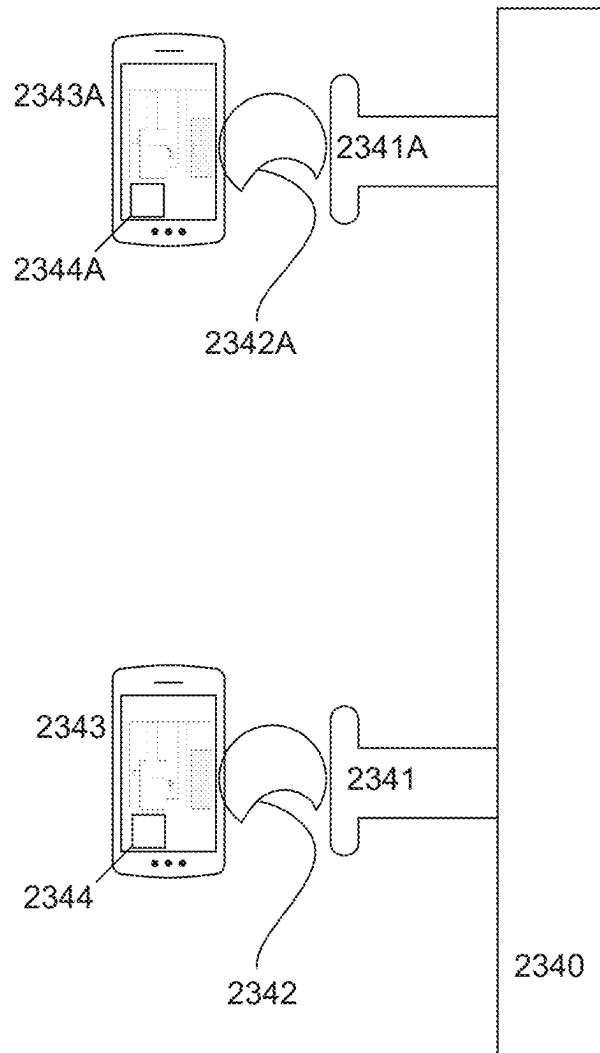


FIG. 23C

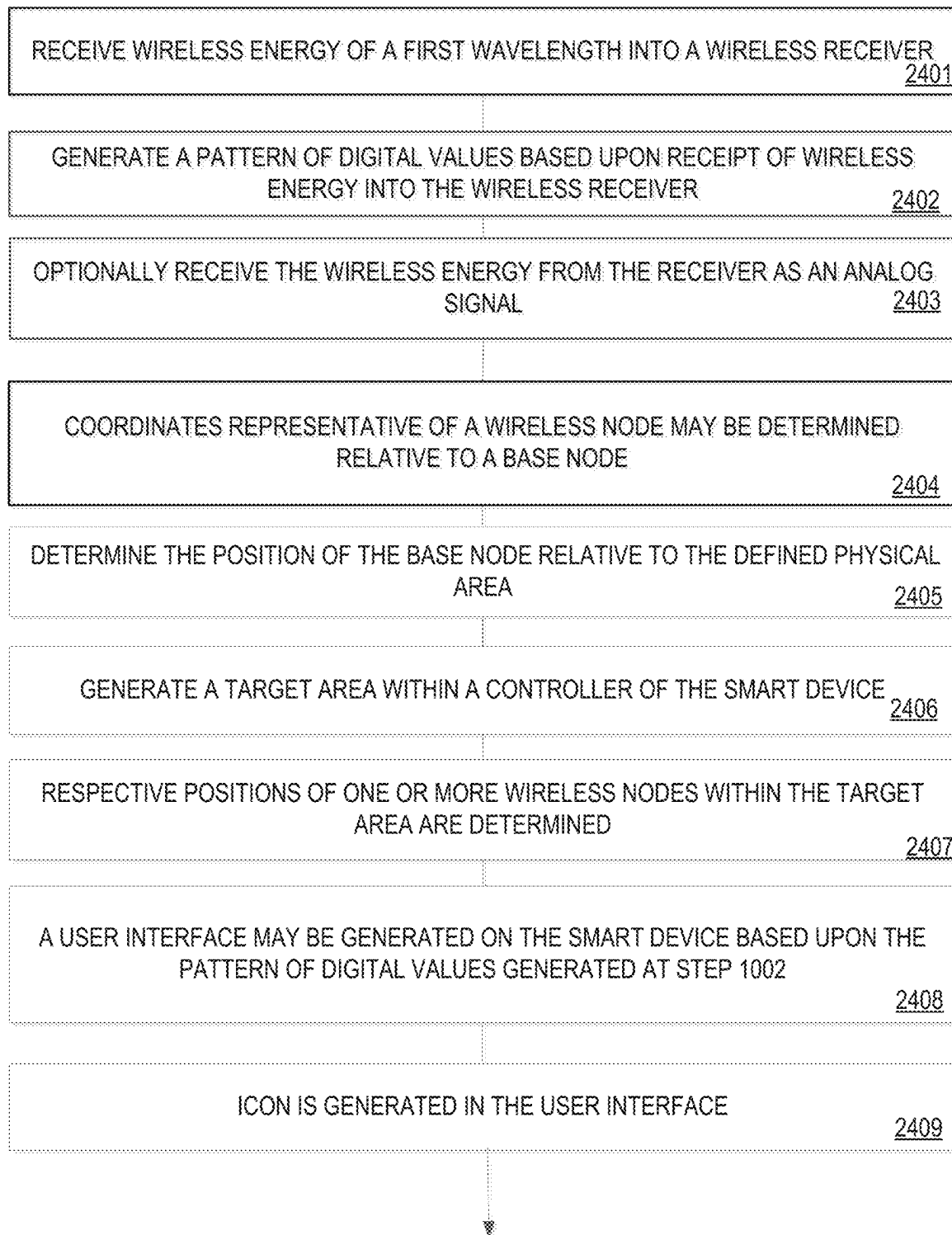


FIG. 24A

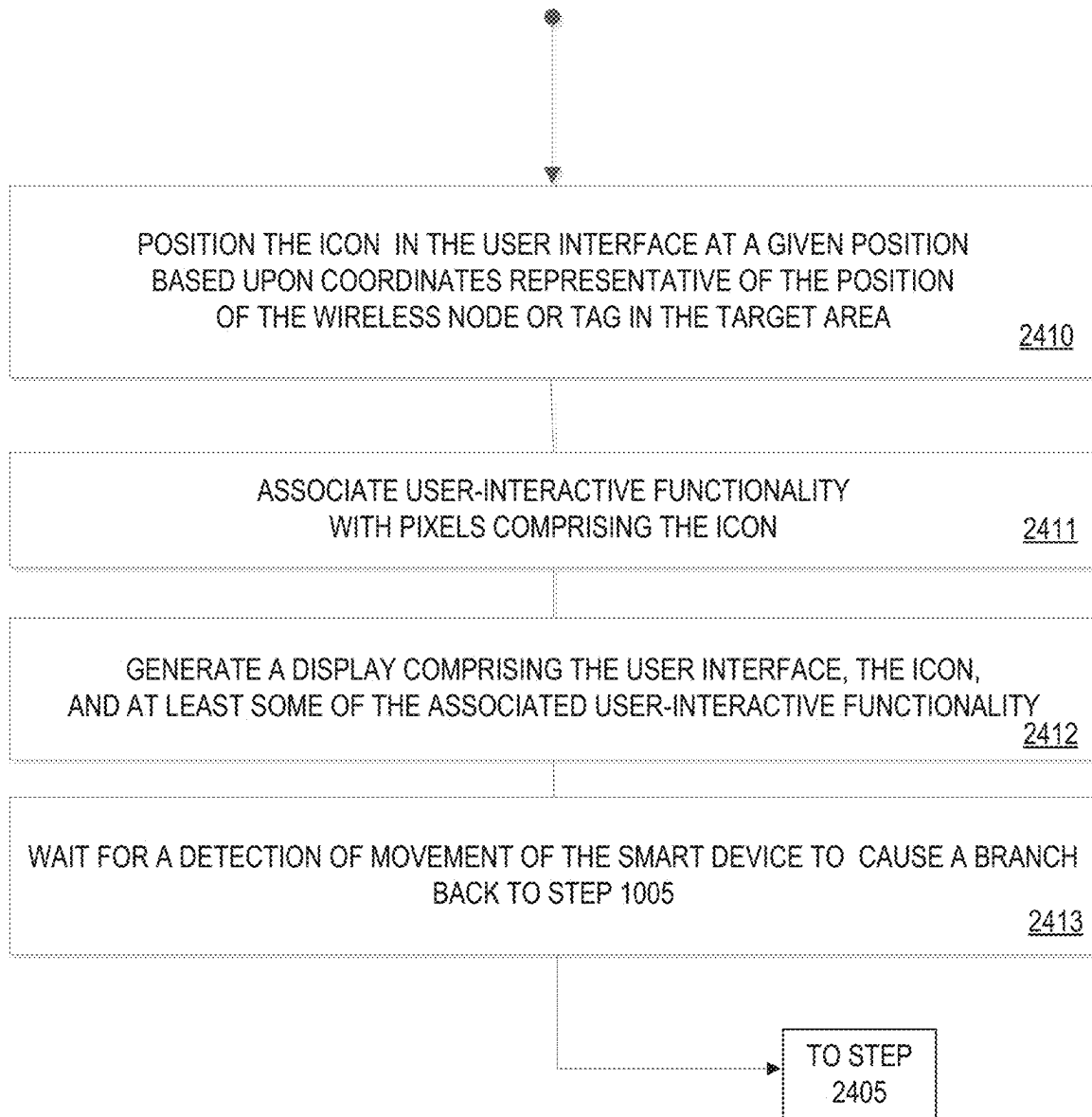


FIG. 24B

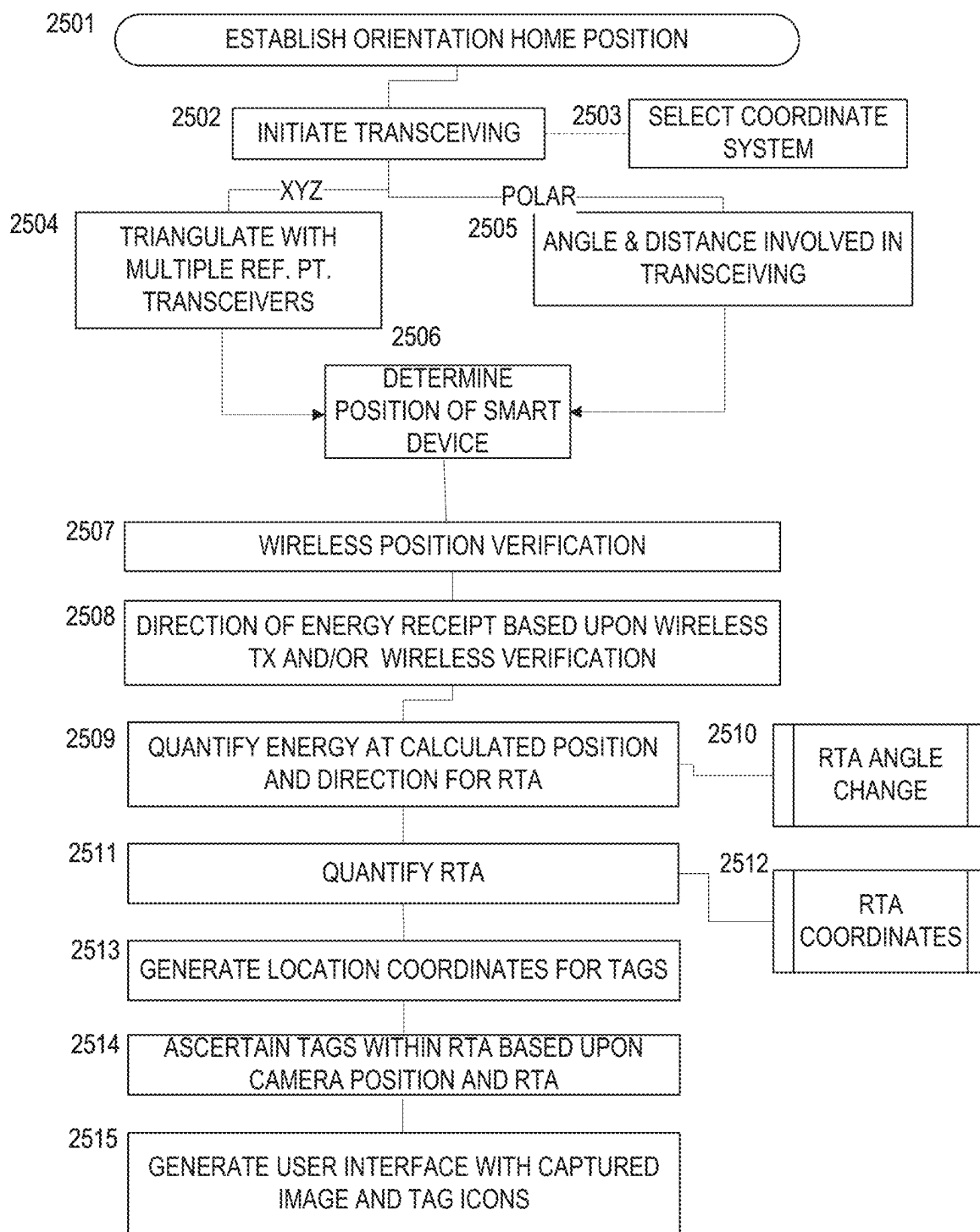


FIG. 25

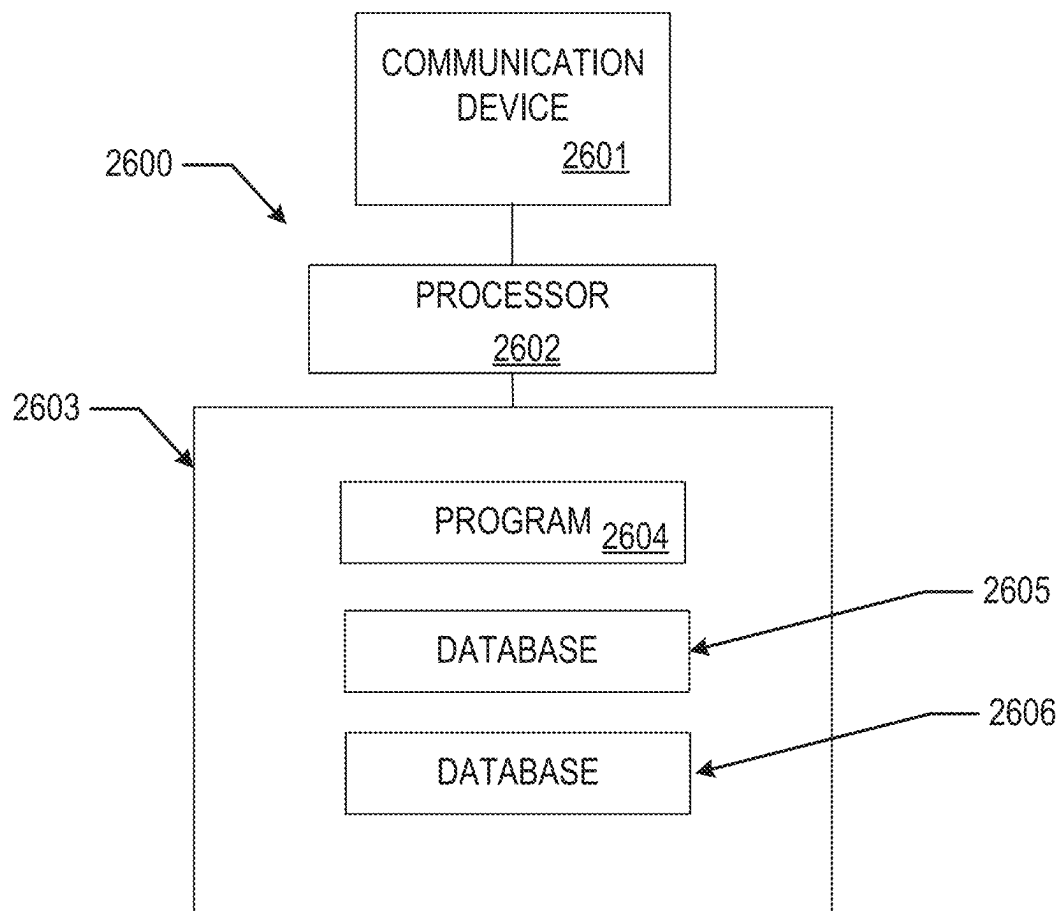


FIG. 26A

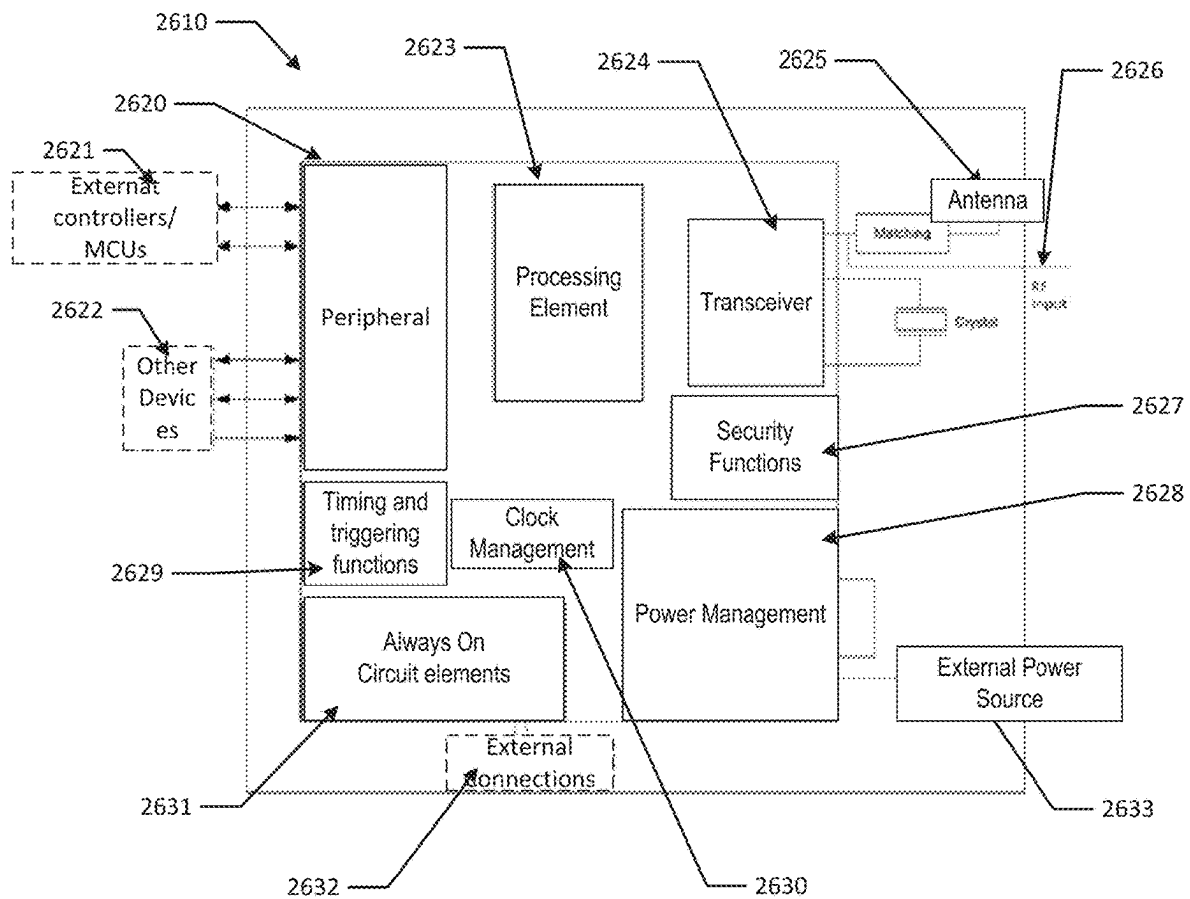


FIG. 26B

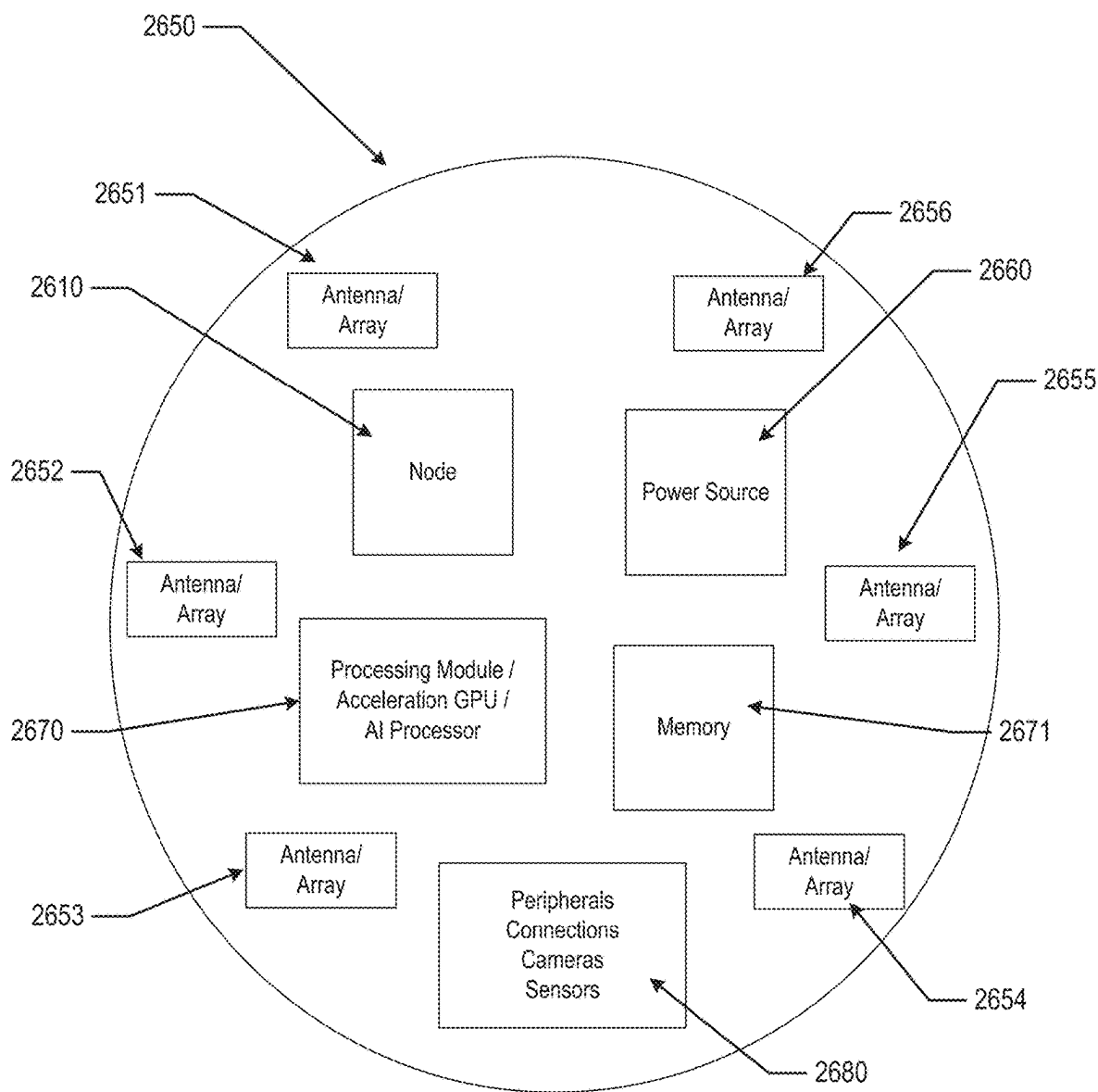


FIG. 26C

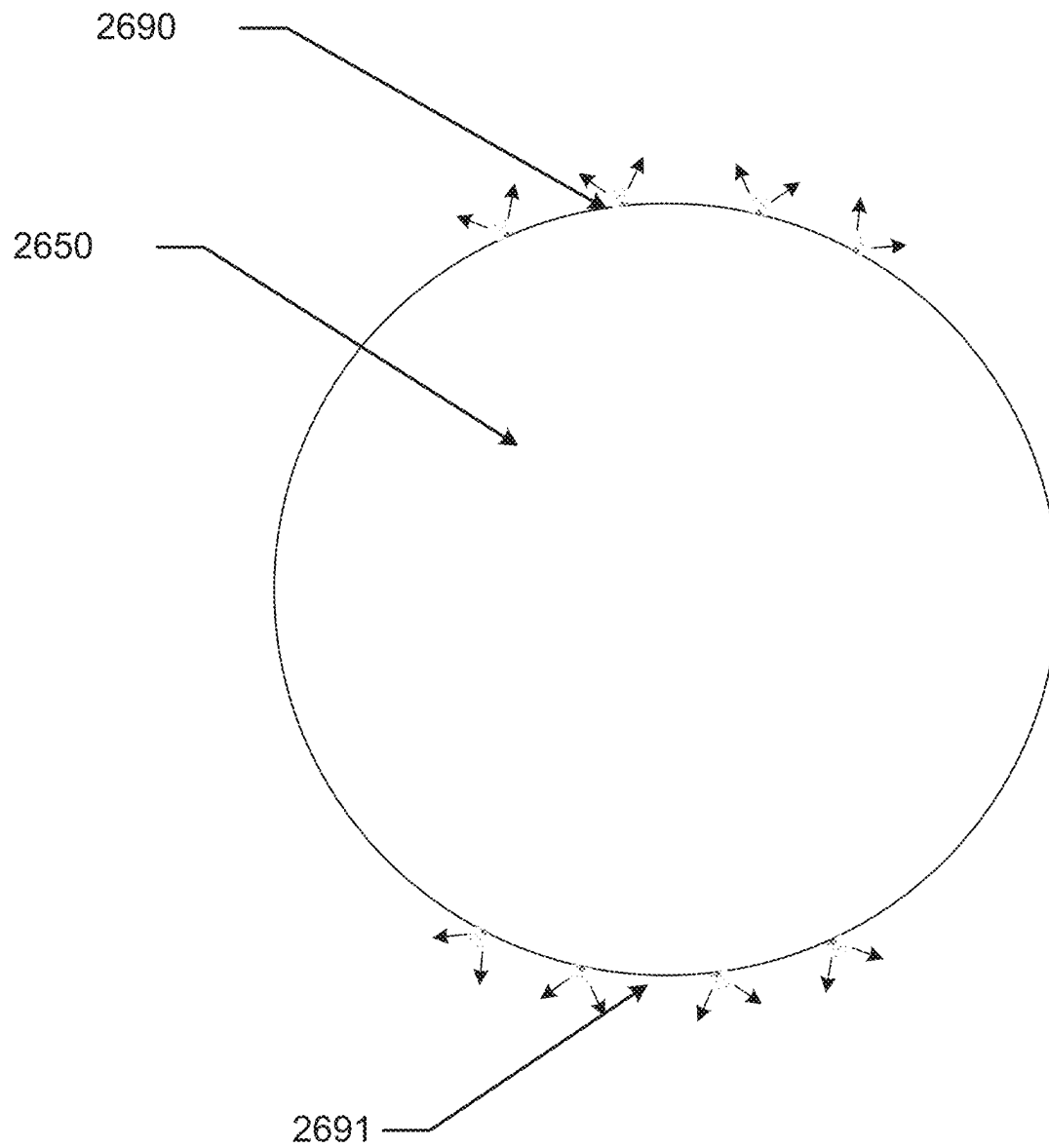


FIG. 26D

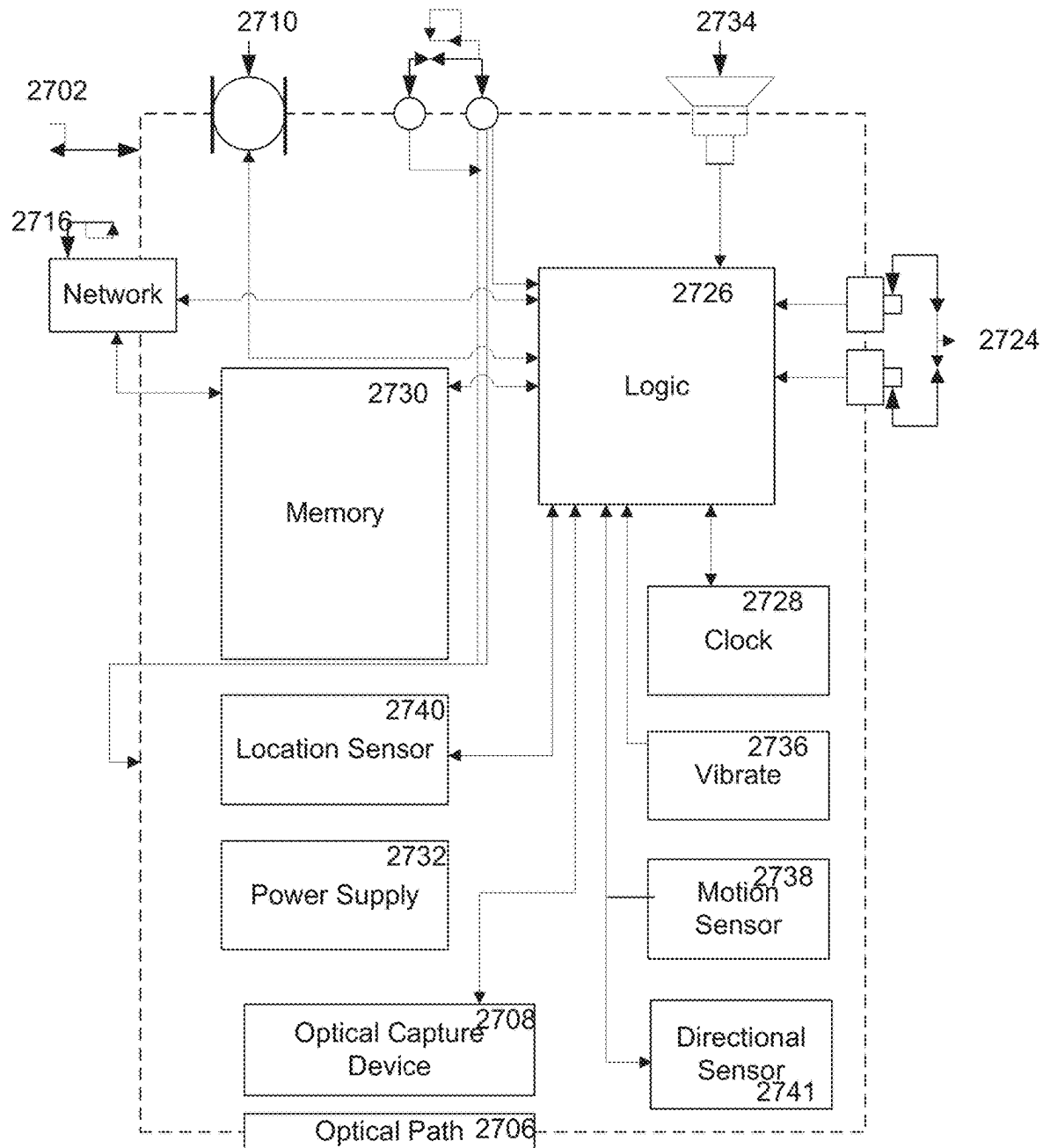


FIG. 27

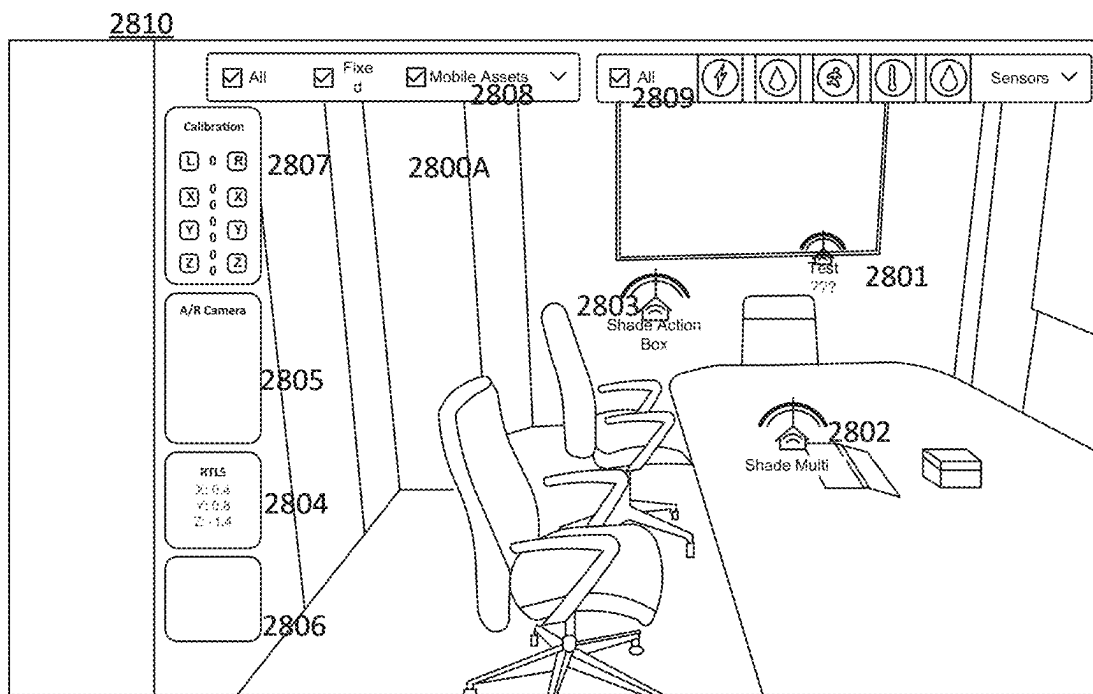


FIG. 28A

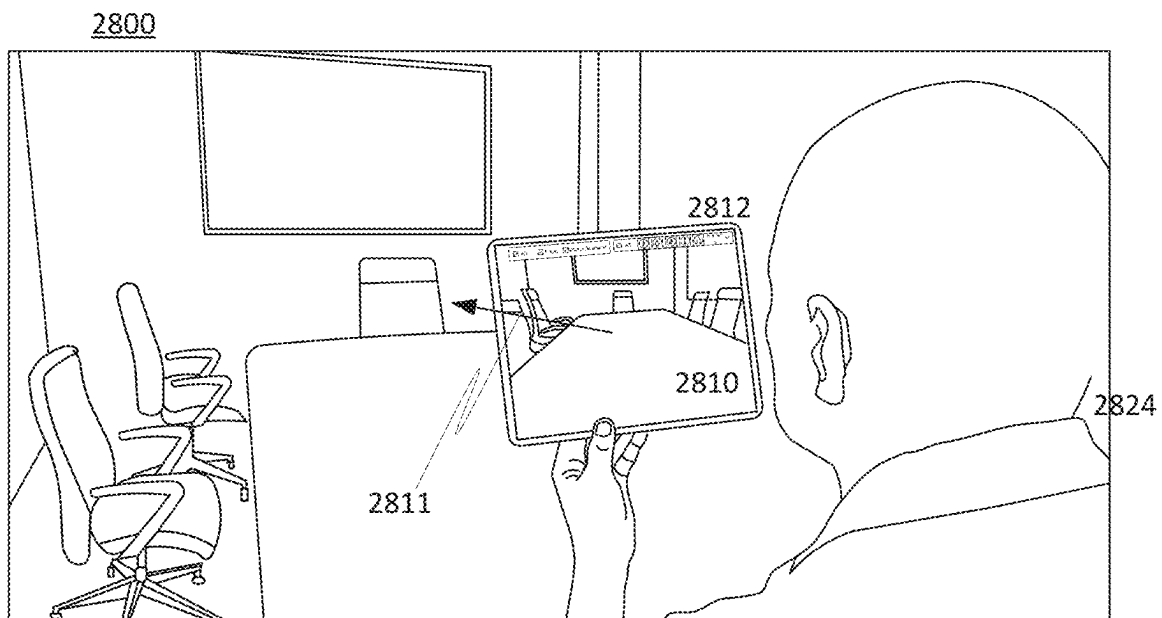


FIG. 28B

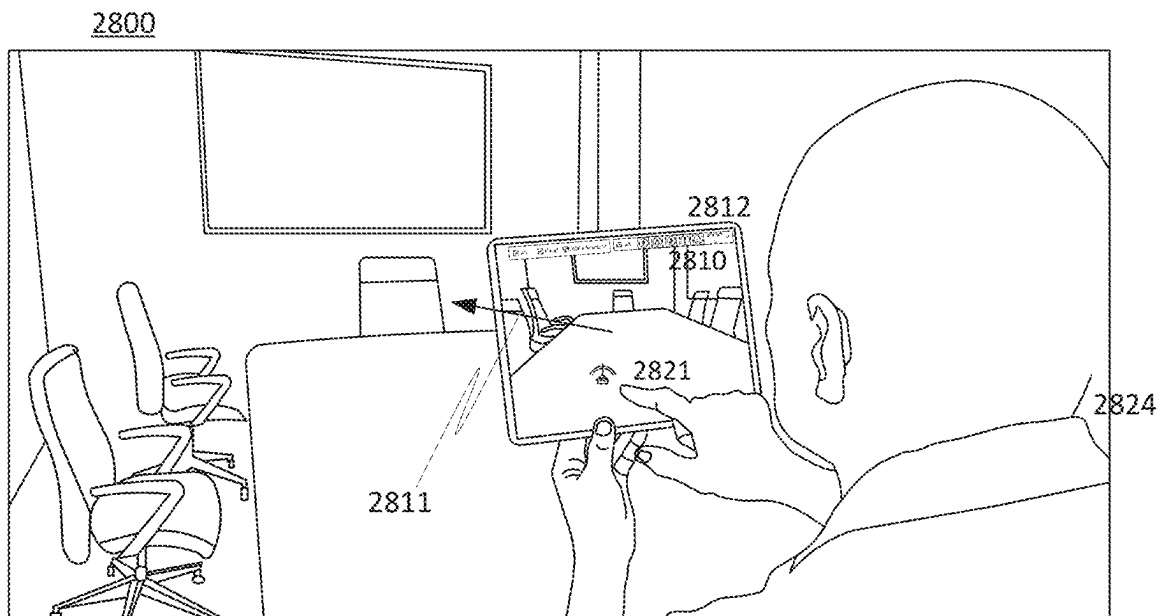


FIG. 28C

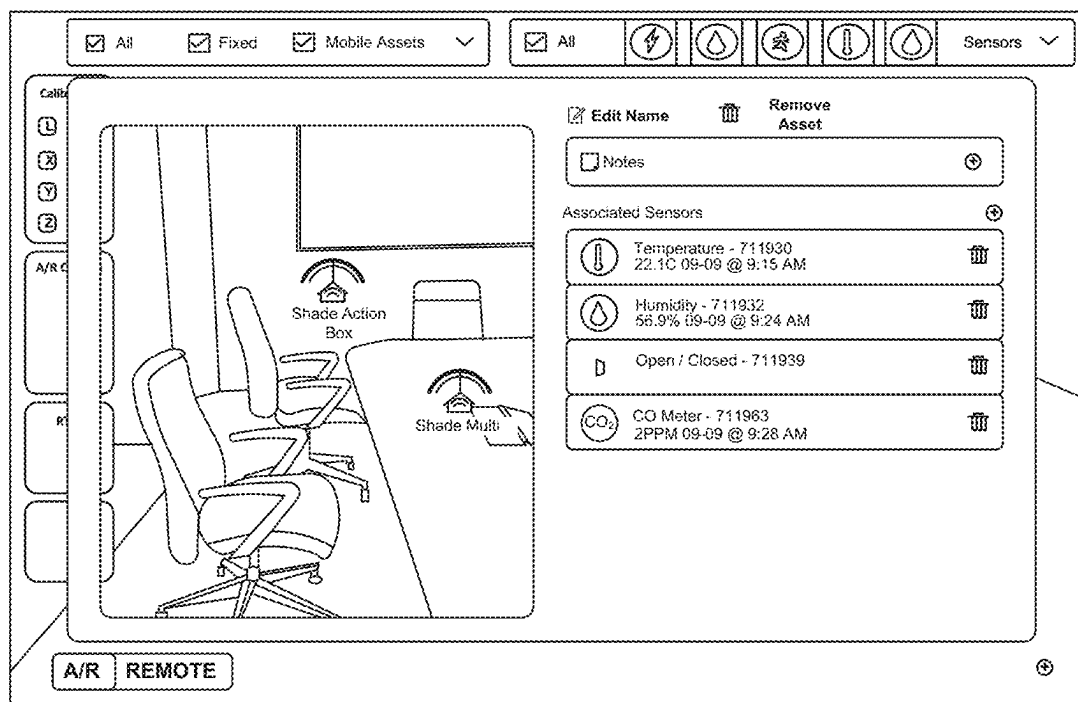


FIG. 28D

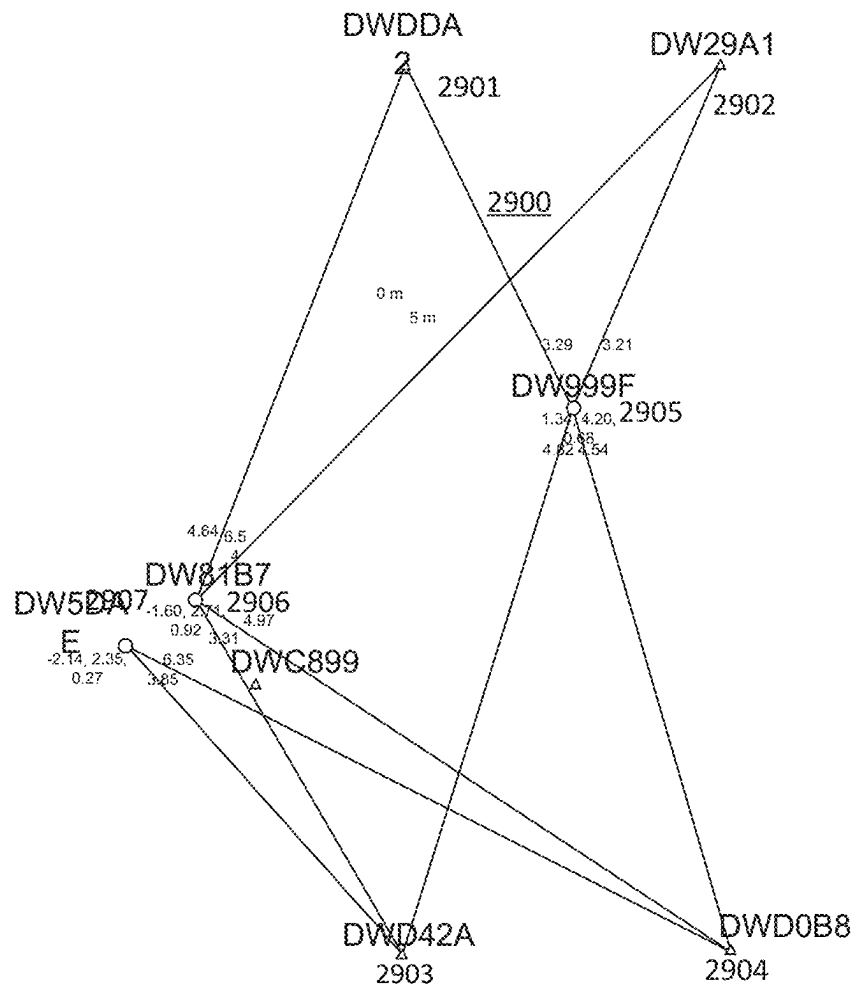
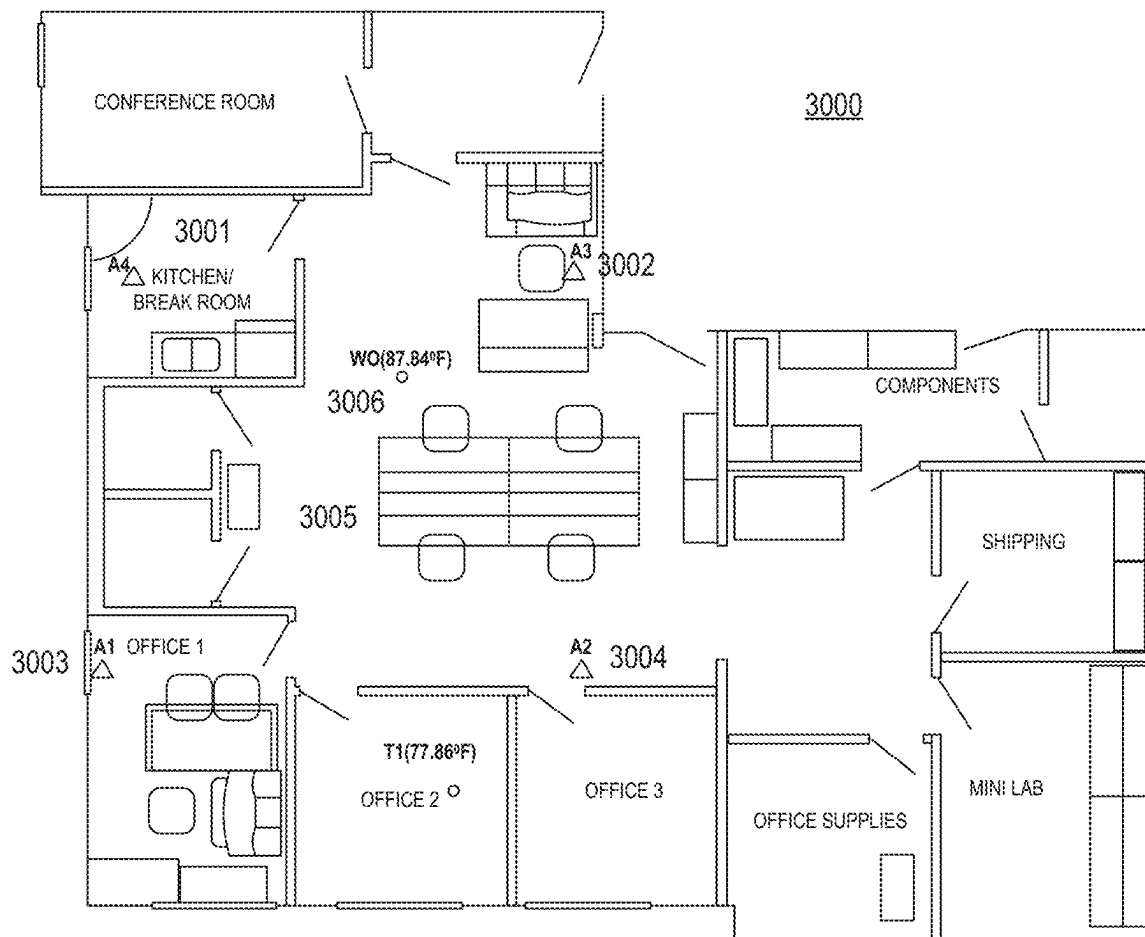


FIG. 29

**FIG. 30**

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METHODS AND APPARATUS FOR TWO DIMENSIONAL LOCATION BASED DIGITAL CONTENT

CROSS REFERENCE TO RELATED APPLICATIONS

The present application claims the benefit of Provisional Patent Application Ser. No. 63/118,231 filed Nov. 25, 2020; the entire contents of which are hereby incorporated by reference. The present application also claims the benefit of Provisional Patent Application Ser. No. 63/155,109 filed Mar. 1, 2021; the entire contents of which are hereby incorporated by reference. The present application also claims the benefit of Provisional Patent Application Ser. No. 63/213,782 filed Jun. 23, 2021; the entire contents of which are hereby incorporated by reference. The present application also claims the benefit of Provisional Patent Application Ser. No. 63/223,575 filed Jul. 20, 2021; the entire contents of which are hereby incorporated by reference. The present application also claims the benefit of Provisional Patent Application Ser. No. 63/277,334 filed Nov. 9, 2021; the entire contents of which are hereby incorporated by reference. The present application makes priority and benefit claims as outlined in the application data sheet, the present application also incorporates by reference in their entirety any matters included in the application data sheet as filed herewith.

INCORPORATION BY REFERENCE TO RELATED APPLICATIONS

The present application claims references Non Provisional patent application Ser. No. 16/935,857, filed Jul. 22, 2020 and entitled TRACKING SAFETY CONDITIONS OF AN AREA, the entire contents of which are hereby incorporated by reference. This application references the Non Provisional U.S. patent application Ser. No. 16/935,857, filed Jul. 22, 2020 and entitled TRACKING SAFETY CONDITIONS OF AN AREA, the entire contents of which are hereby incorporated by reference. This application also references the Non Provisional U.S. patent application Ser. No. 16/504,919, filed Jul. 8, 2019 and entitled METHOD AND APPARATUS FOR POSITION BASED QUERY WITH AUGMENTED REALITY HEADGEAR; and the Non Provisional patent application Ser. No. 16/688,775, filed Nov. 19, 2019 and entitled METHOD AND APPARATUS FOR WIRELESS DETERMINATION OF POSITION AND ORIENTATION OF A SMART DEVICE the entire contents of which are hereby incorporated by reference. This application references the Non Provisional patent application Ser. No. 16/503,878, filed Jul. 5, 2019 and entitled METHOD AND APPARATUS FOR ENHANCED AUTOMATED WIRELESS ORIENTEERING, the entire contents of which are hereby incorporated by reference. This application references the Non Provisional patent application Ser. No. 16/297,383, filed Mar. 8, 2019 and entitled SYSTEM FOR CONDUCTING A SERVICE CALL WITH ORIENTEERING, the entire contents of which are hereby incorporated by reference. This application references the Non Provisional patent application Ser. No. 16/249,574, filed Jan. 16, 2019 and entitled ORIENTEERING SYSTEM FOR RESPONDING TO AN EMERGENCY IN A STRUCTURE, the entire contents of which are hereby incorporated by reference. This application references the Non Provisional patent application Ser. No. 16/176,002, filed Oct. 31, 2018 and entitled SYSTEM FOR CONDUCTING A SER-

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VICE CALL WITH ORIENTEERING, the entire contents of which are hereby incorporated by reference. This application references the Non Provisional patent application Ser. No. 16/171,593, filed Oct. 26, 2018 and entitled SYSTEM FOR HIERARCHICAL ACTIONS BASED UPON MONITORED BUILDING CONDITIONS, the entire contents of which are hereby incorporated by reference. This application references the Non Provisional patent application Ser. No. 16/165,517, filed Oct. 19, 2018 and entitled BUILDING VITAL CONDITIONS MONITORING, the entire contents of which are hereby incorporated by reference. This application references the Non Provisional patent application Ser. No. 16/161,823, filed Oct. 16, 2018 and entitled BUILDING MODEL WITH CAPTURE OF AS BUILT FEATURES AND EXPERIENTIAL DATA, the entire contents of which are hereby incorporated by reference. This application references the Non Provisional patent application Ser. No. 16/142,275, filed Sep. 26, 2018 and entitled METHODS AND APPARATUS FOR ORIENTEERING, the entire contents of which are hereby incorporated by reference. This application references the Non Provisional patent application Ser. No. 15/887,637, filed Feb. 2, 2018 and entitled BUILDING MODEL WITH CAPTURE OF AS BUILT FEATURES AND EXPERIENTIAL DATA, the entire contents of which are hereby incorporated by reference. This application references the Non Provisional patent application Ser. No. 15/716,133, filed Sep. 26, 2017 and entitled BUILDING MODEL WITH VIRTUAL CAPTURE OF AS BUILT FEATURES AND OBJECTIVE PERFORMANCE TRACKING, the entire contents of which are hereby incorporated by reference. This application references the Non Provisional Patent application Ser. No. 15/703,310, filed Sep. 13, 2017 and entitled BUILDING MODEL WITH VIRTUAL CAPTURE OF AS BUILT FEATURES AND OBJECTIVE PERFORMANCE TRACKING, the entire contents of which are hereby incorporated by reference. This application references the Non Provisional patent application Ser. No. 16/528,104, filed Jul. 31, 2019 and entitled SMART CONSTRUCTION WITH AUTOMATED DETECTION OF ADVERSE STRUCTURE CONDITIONS AND REMEDIATION, the entire contents of which are hereby incorporated by reference. This application references the Non-Provisional U.S. patent application Ser. No. 16/657,660, filed Oct. 18, 2019 and entitled METHOD AND APPARATUS FOR CONSTRUCTION AND OPERATION OF CONNECTED INFRASTRUCTURE, the entire contents of which are hereby incorporated by reference. This application references the Non-Provisional U.S. patent application Ser. No. 16/721,906, filed Dec. 19, 2019 and entitled METHOD AND APPARATUS FOR WIRELESS DETERMINATION OF POSITION AND ORIENTATION OF A SMART DEVICE, the entire contents of which are hereby incorporated by reference. This application references the Non Provisional patent application Ser. No. 16/549,503, filed Aug. 23, 2019 and entitled METHOD AND APPARATUS FOR AUGMENTED VIRTUAL MODELS AND ORIENTEERING, the entire contents of which are hereby incorporated by reference. This application references the Non Provisional patent application Ser. No. 16/775,223, filed Jan. 28, 2020 and entitled SPATIAL SELF-VERIFYING ARRAY OF NODES, the entire contents of which are hereby incorporated by reference.

FIELD OF THE INVENTION

The present invention relates to augmented reality with persistent digital content linked to a two dimensional refer-

ence. More specifically, the present invention links an electronic interactive user interface descriptive of a physical location with digital content to enable an Agent interface with augmented reality that combines aspects of the physical area with location specific digital content. According to the present invention, digital content remains persistent with a location.

BACKGROUND OF THE INVENTION

The use of smart devices has become a daily part of life for many people. A smart device may be used as a communication tool and to receive information. Communication is generally point to point basis, and sometimes on a point to many points basis. Information dissemination to unknown recipients, or to recipients that are not ascertainable at a time of generation of the information, requires posting, or storing of the information in a centralized server where the information may be retrieved using traditional queries and search engines, such as Google or social media.

However, it is difficult, if not impossible to automatically receive information pertinent to a designated area. And it is especially difficult to receive information related to a subject area when a user does not know what to ask for, or what type of information may be available to the user.

SUMMARY OF THE INVENTION

Accordingly, the present invention combines methods and apparatus for providing digital content included in an Agent interface based upon an area at least partially defined by a geospatial position contemplated by the Agent interface. In some embodiments, an area defined by a geospatial position contemplated by an Agent is based upon a current position occupied by the Agent and a direction of interest provided by the Agent.

Digital content linked to a physical area is provided to an Agent based upon an identification of the Agent, where the Agent is, and which direction the Agent is oriented towards. In some embodiments, the present invention provides an interactive user interface with content derived from an immediate environment the Agent is located in and virtual content. For example, the present invention enables a user to direct a Smart Device, such as a smart phone, towards an area of interest to the user and a user interface is presented on the user's smart device (and/or a remote smart device) the interface includes an augmented reality environment that combines a rendition of the physical environment present to the user and location specific information in the form of digital content.

For example, a transceiver may be co-located with a sensor and engage in wireless communication. Based upon the wireless communication, location coordinates indicating where the sensor is located. The sensor quantifies a condition at a specific location. When a user views the physical area containing the sensor with the Smart Device, the Smart Device displays a rendition of the area monitored by the sensor and with the conditions quantified by the sensor.

Essentially, the present invention enables point and query (or ask and query) access to information or other content close to a Smart Device. The Smart Device may be used to generate an interface indicating what people, equipment, vehicles, or other items are viewable to the Smart Device and place those items into the context of the environment surrounding the Smart Device.

In general, the present invention associates digital content with Tags associated with location coordinates. A Tag may

include one or more of a Physical Tag, a Virtual Tag, and a Hybrid Tag (as described in the Glossary below). Tags provide persistent access to specified digital content based upon the Tag's association of the content with a set of location coordinates and/or an area of positional coordinates including the location coordinates. For example, a set of positional coordinates may be located near an architectural aspect of a structure, such as a point of intersection of two beams. A Tag may be associated with the set of positional coordinates. An Area of digital content interaction (e.g., retrieval of digital content and/or placement of digital content for subsequent retrieval) may include the set of positional coordinates and also include an area comprising additional sets of positional coordinates, surrounding the set of positional coordinates, adjacent to the set of positional coordinates, and/or proximate to the set of positional coordinates.

This functionality is accomplished by establishing a target area and determining which tags are present within the target area. Tags may be virtual; in which case the virtual tags are associated with positional coordinates and viewable whenever a target area is designated to encompass the coordinates the virtual tag.

Alternatively, the tags may be physical, such as a small disk adhered to an item of equipment, vehicle, or a person's employee badge. Tracking of a position and content associated a physical tag may be updated in real time or on a periodic basis. Physical tags may be moved into a target area, or the target area may be moved to encompass the physical tag. The present invention will automatically generate an interface indicating which tags contained in the interface, what those tags are associated with and where a tag is in relation to the Smart Device. It will also access any information that has been stored and associated with the tag and present it on the Smart Device.

By aligning real world and virtual world content, a real world site experience is enriched with content from a geospatially linked virtual world. The virtual world content is made available to an Agent based upon a position and a direction of a Radio Target Area ("RTA") specified by a Smart Device supported by the Agent. A geospatial position and direction of interest that is contained within the RTA is generated using wireless communication with reference point transmitters. Wireless communication capabilities of the Reference Point Transmitters determine parameters associated with a Wireless Communication Area ("WCA"). The RTA is a subset of the WCA.

The present invention provides for methods and apparatus for executing methods that augment a physical area, such as an area designate as a wireless communication area. The method may include the steps of transceiving a wireless communication between a Smart Device and multiple reference point transceivers fixedly located at a position within a wireless communication area; generating positional coordinates for the Smart Device based upon the wireless communication between the Smart Device and the multiple reference transceivers; establishing a radio target area for an energy receiving sensor; receiving energy into the energy receiving sensor from the radio target area; generating a digital representation of the energy received into the energy receiving sensor at an instance in time; generating positional coordinates for a tag at the instance in time, the tag comprising digital content and access rights to the digital content; determining the tag is located within the radio target area based upon the positional coordinates for the tag; generating a user interactive interface comprising static portions based upon the digital representation of the energy

received into the energy receiving sensor; generating a dynamic portion of the user interactive interface based upon the positional coordinates for the tag and the positional coordinates for the Smart Device; receiving a user input into the dynamic portion of the user interactive interface; and based upon the user input received into the dynamic portion of the user interactive interface, including the digital content in the user interactive interface.

In some embodiments, multiple disparate energy levels may be received into the energy receiving sensor at the instance in time, each disparate energy level received from a different geospatial location; associating positional coordinates with the disparate energy levels; and indicating the disparate energy levels and relative positions of the disparate energy levels in the user interactive interface. A tag may include a virtual tag with the digital content and a location identified via positional coordinates.

In another aspect, a physical tag may include a transceiver capable of wireless communication with the multiple reference transceivers and the method may include transceiving a wireless communication between a tag and multiple reference point transceivers; and generating positional coordinates for the tag based upon the wireless communication between the tag and the multiple reference transceivers. The wireless communication between the Smart Device and the multiple reference point transceivers may be accomplished by transceiving using an Ultra-Wideband modality; Bluetooth modality or another wireless modality, such as WiFi.

A wireless communication area may be identified as including a radio transmission area of the energy receiving sensor and the wireless communication area may be based upon a communication distance of the Ultra-Wideband modality in an area encompassing the energy receiving sensor.

Transceiving a wireless communication between a tag and multiple reference point transceivers may be accomplished using a wireless modality such as, for example, a UWB or Bluetooth modality; and generating positional coordinates for the tag based upon the wireless communication between the tag and the multiple reference transceivers may be accomplished using the same modalities. Positional coordinates may include one or more of: Cartesian Coordinates, an angle of arrival and an angle of departure and a distance.

In another aspect, access rights to tag content may be required and based upon an identifier of the Smart Device or a user operating the Smart Device. A dynamic portion of the user interactive interface may include an icon indicative of the digital content associated with the tag.

In some embodiments, the present invention combines methods and apparatus for providing multifactor security protection of digital content. Factors in addition to the typical two factor security are based upon location coordinates of Tags (Virtual, Physical or Hybrid) that are associated with specified digital content. For example, a third factor may be based upon knowledge of a location of a Virtual Tag and/or Physical Tag storing the digital content; and a fourth factor may be based upon a physical location of a user or other Agent requesting access to the digital content. A potential fifth factor may include location coordinates of the user's smart device for a time period prior to the request for access to the protected digital content. According to the present invention, knowledge of the location of the Tag may be provided via an Augmented Reality interface with icons specific to a User being presented in the context of an image of a physical environment. Determination of a physical location of the user may be accomplished via wireless communications.

Factors in security authentication according to the present invention may be based upon one or both of: location coordinates for a user's smart device at the time of request for access to the Virtual Tag; and location coordinates of the user's smart device at time intervals prior to the request for access to the Virtual Tag. In some embodiments, an Augmented Reality interface manifests interactive icons that may be selected to provide access to the digital content "in" or otherwise associated with the Virtual Tag. An AR view includes interactive icons incorporated into image data representative of a physical area, such as an image based upon energy levels received into a charged couple device (CCD). Image data may be live "video" type image data or a still image.

In some embodiments, access to protected digital content requires the presence of an Agent in a defined geospatial area and corresponding location coordinates. The A/R interface incorporates interactive icons representative of the Virtual Tags at the locations specified by the location coordinates within the defined geospatial area. For example, an authorized geospatial area may include a physical structure; such as an office; a manufacturing facility; a warehouse or other storage facility; a military base; an airport; a construction worksite; an energy substation; a cellular tower site; a distribution center; a residential facility; an infrastructure, such as a bridge or tunnel; or other definable area.

Digital content linked to a physical area is provided to an Agent based upon an identification of the Agent, where the Agent is, and which direction the Agent is oriented towards. In some embodiments, the present invention provides an interactive user interface with content derived from an immediate environment the Agent is located in and virtual content. For example, the present invention enables a user to direct a Smart Device, such as a smart phone, towards an area of interest to the user and a user interface is presented on the user's smart device (and/or a remote smart device) the interface includes an augmented reality environment that combines a rendition of the physical environment present to the user and location specific information in the form of digital content.

A Virtual Tag infrastructure (VTI) provides a cryptographic technique that enables secure access to digital content via one or both of: secure networks; and an insecure public networks, such as the Internet. Virtual Tag cryptography reliably verifies an authorized request for access to digital content by a user or other Agent. In various embodiments, location coordinates are used to verify an authorized request to access protected digital content. The location coordinates correspond with one or more of: a location of a Virtual Tag; a location of a Physical Tag; a location of a smart device; and a location of another type of Node.

In another aspect, Virtual Tag cryptography may be based upon location coordinates of designated areas of access to the digital content. Still further aspects may rely upon location coordinates of a Virtual Tag, a Physical Tag a smart device and/or a Node at intervals over a defined period of time prior to a request to access the digital content.

The VTI includes a system for the creation, storage, and distribution of digital content based upon presentation of authorizing credentials based, at least in part, upon location coordinates. The location coordinates may relate to a first security factor based upon one or both of a location of a Virtual Tag; and a Physical Tag; and a second factor based upon a location or sequence of locations of an Agent seeking to access the digital content to retrieve, modify, or deposit digital content.

In some embodiments, authorizing credentials may include an improved version of a PKI infrastructure, wherein the PKI public certificate is a value based upon location coordinates of at least one of a: Virtual Tag, Physical Tag, Smart Device, or other Node. The public certificate may be used to verify that a particular Virtual Tag is accessible to an entity requesting access. The VTI creates digital certificates that map Virtual Tags to locations, and maps authorized access areas to users requesting access (or who may request access). In this manner, the public key includes positional coordinates, or may be derived from positional coordinates. For example, derivation may be accomplished via an algorithmic processing of location coordinate values. Location coordinate values may be those determined at a time interval during a request to access the digital content, and/or during a time interval prior to a request to access digital content.

A VTI may include one or more of: a VTI certificate authority that stores, issues and processes digital certificates based upon location coordinates and time interval values; a central directory providing a secure location in which location values associated with Virtual Tags digital content are stored and indexed; and a Virtual Tag management system managing positional coordinates and associated digital content.

The present invention a multifactor authentication for protecting access to digital content that includes a mathematical scheme for verifying the authenticity of a request for access to the digital content and authenticity of a provider of digital content. Essentially, a valid digital signature or login credentials based upon location coordinate values, personal knowledge and possession of a known hardware device provides the VTI with extremely strong evidence to determine that an access request was submitted by a known sender, resulting in a location/user based authentication, and confidence in the integrity of the digital content.

In some embodiments, the present invention enables point and query (or ask and query); and point and deposit; access to information or other digital content proximate to a Smart Device. The Smart Device may be used to generate an interface indicating what people, equipment, vehicles, or other items are viewable to the Smart Device and place those items into the context of the environment surrounding the Smart Device.

The details of one or more examples of the invention are set forth in the accompanying drawings and the description below. The accompanying drawings that are incorporated in and constitute a part of this specification illustrate several examples of the invention and, together with the description, serve to explain the principles of the invention: other features, objects, and advantages of the invention will be apparent from the description, drawings, and claims herein.

DESCRIPTION OF THE DRAWINGS

The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate several embodiments of the invention and, together with the description, serve to explain the principles of the invention:

FIG. 1 illustrates location determination with wireless communication to reference points and Virtual Tags and IoT Tags.

FIG. 2 illustrates locations aspects with a location based transaction.

FIG. 2A-2B illustrate exemplary apparatus and methods that may involve Virtual Tags.

FIG. 3 illustrates methods of orienteering by device movement.

FIG. 3A illustrates methods of location determination with sonic systems.

FIGS. 4A-4D illustrate exemplary configurations of antenna arrays.

FIG. 5A illustrates an exemplary Smart Device with an array of antennas.

FIG. 5B illustrates exemplary methods of indicating directions with Smart Devices and antenna arrays.

FIG. 5C illustrates an exemplary method of a user utilizing an oriented stereoscopic sensor system to orient a direction of interest.

FIG. 6 illustrates a two dimensional reference that includes a floor plan view with Virtual Tags and IoT Tags.

FIG. 6A illustrates a two dimensional reference that includes a floor plan view with digital content from Virtual Tags and IoT Tags displayed.

FIG. 6B illustrates exemplary aspects related to operations in external and remote locations.

FIG. 6C illustrates a dynamic two dimensional interface according to some embodiments of the present invention.

FIG. 6D illustrates method steps that may be performed in practicing some embodiments of the present invention.

FIG. 7 illustrates an exemplary handheld device that may be used to implement aspects of the present disclosure including executable software.

FIGS. 8A-8G illustrate aspects of the determination of directions of interest and Fields of View and information display.

FIGS. 9A-9C illustrate additional aspects of information display.

FIGS. 10A-10B illustrates an exemplary method for generating an augmented-reality Radio Target Area for a Smart Device.

FIG. 11 illustrates an exemplary database structure according to the instant specification.

FIG. 12 illustrates additional exemplary method for displaying Radio Target Areas with Smart Devices.

FIG. 13 illustrate exemplary aspects of Wireless Communication Areas in Radio Target Area display.

FIG. 14 illustrates a set of polygons generated via LIDAR that may be used for geospatial recognition.

FIG. 15 illustrates a block diagram of a multi-modal tag according to some embodiments of the present invention.

FIGS. 16A-16C illustrate RTLS components and IoT sensor components in various degrees of integration.

FIGS. 17A-17C illustrate flowchart of method steps that may be implemented in various embodiments of the present invention.

FIGS. 18A-B illustrate exemplary user interfaces.

FIGS. 18C-D illustrate exemplary user interaction with user interfaces.

FIGS. 19, 19A illustrate examples of determining the location of a radio receiver in three dimensional space based on its interaction with multiple radio transceivers.

FIG. 20 illustrates an exemplary Smart Device with an array of antennas.

FIGS. 21 and 21A illustrate a device and vectors according to various embodiments of the present invention.

FIGS. 22A-22B illustrates a defined area with Transceivers.

FIGS. 22C-22E illustrate devices that may include a Transceiver.

FIG. 23A illustrates wireless communication, including directional Transceivers.

FIG. 23B illustrates an apparatus with Transceivers and generation of a vector.

FIG. 23C illustrates an exemplary apparatus for attaching an exemplary accelerometer to a component of a Structure.

FIGS. 24A-24B illustrates an exemplary method for generating an augmented-reality Radio Target Area for a Smart Device.

FIG. 25 illustrates additional exemplary method for displaying Radio Target Areas with Smart Devices.

FIG. 26A illustrates apparatus that may be used to implement aspects of the present invention including executable software.

FIG. 26B illustrates an exemplary block diagram of a controller with angle of arrival and angle of departure functionality.

FIG. 26C illustrates exemplary block diagram of an assembly with multiple antenna arrays such as a "puck"

FIG. 26D illustrates another view of a puck with directional antenna arrays.

FIG. 27 illustrates an exemplary handheld device that may be used to implement aspects of the present disclosure including executable software.

FIGS. 28A-28D illustrate a physical area and a augmented reality interface associated with the physical area.

FIG. 29 illustrates wireless determination of a Tag location within an authorized area.

FIG. 30 illustrates a 2D view of an authorized area and Tag location.

DETAILED DESCRIPTION

The present invention relates to the provision of digital content placed in a real world context, such as a human user interface, including for example an augmented reality interface, or a machine interface, with persistent digital content linked to a physical location. In some embodiments, the present invention links a physical area with digital content and thereby enables an augmented reality user interface with aspects of the physical area combined with location designated digital content. According to the present invention, digital content remains persistent with a location even if visual aspects of the location change.

Some embodiments include methods and apparatus for determining virtual world digital content linked to positional coordinates; and displaying real-world energy levels integrated with and aligned with the virtual-world digital content.

In some examples, a carefully placed reference point Node may function as a transceiver of signals. For example, a Node may receive and transmit signals in a radio frequency band of the electromagnetic spectrum. In a simple form, a Node may broadcast a radio frequency wireless communication. Nodes may also be capable of receiving a radio frequency on a same radio frequency and/or a different radio frequency as a received radio frequency. Frequencies utilized for wireless communication may include those within the electromagnetic spectrum radio frequencies used in one or more of: UWB, Wi-Fi, and Bluetooth modalities, as well as IR, visible and UV light as examples of transmission modalities. In some embodiments, sound emanations may also be used as a communication mechanism between a Smart Device and a reference point Node and/or between two Nodes. In various aspects of wireless communication, the Nodes may function to communicate a timing value via their electromagnetic or sonic transmissions or data other than timing signals, such as a digital value representing a condition quantified with an electronic sensor. Accordingly,

wireless communications may provide data identifying information unique to the Node, data related to the synchronization of timing at different well located reference points and may also function as general data communication Nodes.

A triangulation calculation and/or a distance and angle indicating a position of a Smart Device, or a Node may result from a system of multiple reference position Nodes communicating timing signals to or from the Smart Device or Node. Methods of calculating positions via wireless communications may include one or more of: RTT, RSSI, AoD, AoA, timing signal differential and the like. Triangulation or other mathematical techniques may also be employed in determining a location.

A process is disclosed for determination of a position based upon wireless communication between a Node and/or Smart Device and with reference point transceivers. The process may be accomplished, for example via executable software interacting with the controller, such as, for example via running an app on the Smart Device or as a service on a server accessible via the Internet.

In the following sections, detailed descriptions of examples and methods of the invention will be given. The description of both preferred and alternative examples though thorough are exemplary only, and it is understood that, to those skilled in the art, variations, modifications, and alterations may be apparent. It is therefore to be understood that the examples do not limit the broadness of the aspects of the underlying invention as defined by the claims.

In some embodiments, the location of a Node. may be determined via discernment of a physical artifact, such as, for example a visually discernable feature, shape, or printed aspect. A pattern on a surface may convey a reference point by a suitable shape such as a cross, Vernier or box structure as non-limiting examples. The printing may also include identification information, bar codes or lists of location coordinates directly. A Smart Device ascertaining a physical reference mark and a distance of the Smart Device to the mark may determine a relative location in space to a coordinate system of the marks. Reference points may be coded via identifiers, such as a UUID (Universally Unique Identifier), or other identification vehicle.

Marks tied to a geospatial coordinate system may be utilized to determine a relative location. A number of methods may be executed to determine a distance from the Smart Device to a mark such as, for example, a sensed reflection of light beams (preferably laser beams), electromagnetic beams of wavelength outside of the visible band such as IR, UV, radio and the like, or sound-based emanations. It may be important that the means of determining the distance can be focused into a relatively small size. It may be important that the means of determining the distance is reflected by the physical mark. For example, a light-source apparatus used to determine a distance may benefit from a mirror surface upon the physical mark. In addition, a reflected signal may emerge significantly towards a user. It may be desirable that physical reference points are placed with high accuracy at specific reference locations, or it may be desirable to be able to measure with high accuracy specific reference locations after placement.

Referring now to FIG. 1, aspects of a system for enhanced wireless position and orientation are illustrated. Reference Point Transceivers 101-104 are shown deployed within or proximate to (in this case within radio transceiving distance) a Structure 106, wireless communications between the reference point transceivers 101-104 and a Transceiver 105 supported an Agent 100 within or proximate to the Structure

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106 may be used to calculate a local position of the Transceiver 105 supported an Agent 100. In this discussion one or more of: a Transceiver 105 supported an Agent 100; and a reference point transceivers 101-104 may be embodied as a Node; a Smart Device; or a dedicated transceiver.

A designated direction 112 may be generated based upon wireless communications and may be used to determine a targeted item 113 and/or a targeted direction. A designated direction may be determined via the wireless modalities discussed herein. Some embodiments will include one or more of: radio communications involving multiple antennas; radio communications with one or more antennas at two or more locations at different instances of time; a magnetometer; a compass; LiDAR; laser; sonic; accelerometer; gyroscope or other electronic mechanism.

A Radio Target Area 117 (sometime abbreviated as “RTA”) may be identified with an area of from which energy is received from a direction congruent with and/or overlapping an area indicated with the designated direction 112. In some embodiments, an RTA may also be congruent and/or overlap a field of view (FOV) of a camera device included in the Smart Device at the location of transceiver 105.

In another aspect, in some embodiments, one or more of a physical transceiver, such as one or more of: an IoT Tag 114; a Virtual Tag 115; and a Hybrid Tag 116. In general, an IoT Tag 114 includes a physical tag with a transceiver that transceives logical data that may include a timing signal from which a position may be calculated. In some embodiments, an IoT Tag 114 may also include an electronic sensor that generates digital content that may also be transceived and/or stored in the IoT Tag 114. A Virtual Tag includes digital content associated with a set of coordinates but does not include a physical transceiver. A Hybrid Tag 116 includes digital content associated with a set of coordinates similar to a Virtual Tag 115 and may be a subset type of Virtual Tag 115 that includes digital content generated via an IoT Tag 114, while the IoT Tag 114 was located at the coordinates associated with the Hybrid Tag 116.

In some embodiments, one or more of the IoT Tag (Virtual Tag 115), the Smart Device at the location of transceiver 105, and the Reference Point Transceivers 101-104 may comprise a Node with a transceiver, a controller, and a memory that may store digital content.

Still further, in some embodiments, a Node deployed as one or more of the IoT Tag (Virtual Tag 115), the Smart Device at the location of transceiver 105, and the Reference Point Transceivers 101-104 may receive a digital content download via a wireless modality other than the modality used to determine a geospatial location 107 of the Smart Device at the location of transceiver 105, such as, for example via a cellular transmission or a satellite download in a remote area, such as a remote construction work site, a wilderness location, a hiking trail, a remote section of an aqueduct or utility or a pipeline (e.g. a gas or oil pipeline). Digital content downloaded and/or stored in a Node deployed as one or more of the IoT Tag (Virtual Tag 115), the Smart Device at the location of transceiver 105, and the Reference Point Transceivers 101-104 may be communicated during a transceiving process engaged in between Nods and/or device described herein.

Reference Point Transceivers 101-104 may be fixed in a certain location within or proximate to the Structure 106 and define a wireless communication area 111. The Reference Point Transceivers 101-104 may transceive in a manner suitable for determination of a position of one or more Transceivers 105 supported by an Agent 100. Transceiving may be conducted via one or more wireless transmission

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modalities between the Transceiver 105 supported by the Agent 100 and one or more Reference Point Transceivers 101-104.

By way of non-limiting example, Transceivers 105 supported by the Agent 100 may be included in, and/or be in logical communication with, a Smart Device, such as, one or more of: a smart phone, tablet, headgear, ring, watch, wand, pointer, badge, Tag, Node; a Smart Receptacle; headgear; or Tag supported by the Agent 100 or other Agent supportable device with Transceiver 105 operable to transceive with the Reference Point Transceivers 101-104.

The Reference Point Transceivers 101-104 may include devices, such as, for example, a radio transmitter, radio receiver, a light generator, a light receiver, a pattern recognizing device, a sonic or ultrasonic device. A radio frequency transceiver may transmitters and receivers operative to communicate via wireless modalities such as, for example: Wi-Fi, Bluetooth, Ultra-wideband (“UWB”), ultrasonic, infrared, or other communication modality capable of logical communication between Transceivers 101-105.

In some embodiments, a Reference Point Transceivers 101-104 may include a multi-modality transceiver, that communicates more locally via a first communication modality and a second communication modality. Each of the first modality and the second modality, may include for example apparatus and software code for wireless communications via one of: UWB, Bluetooth, Wi-Fi, ANT, Zigbee, BLE, Z Wave, 6LoWPAN, Thread, Wi-Fi, Wi-Fi-ah, NFC (near field communications), Dash 7, Wireless HART, or similar modality. In some embodiments, a first communication modality may engage in wireless communications with a first set of distance constraints and a second communication modality may engage in wireless communication with a second set of distance constraints. For example, a first communication modality may involve UWB communications modalities that travel a shorter distance than a second communication modality but may provide a more accurate location determination. In this example, a second communication modality may include one or more of: satellite communications (e.g., GPS); cellular communication modalities (e.g., 3G, 4G 5G and the like), sub GHz communications, or another modality.

In some embodiments, satellite communications (e.g., global positioning system “GPS”) and/or cellular communications may be used to determine an identification of a structure 106 within which transceiving with the Reference Point Transceivers 101-104 may be established.

Wireless communications between Transceivers 101-105 may engage in logical communications to provide data capable of generating one or more of: Cartesian coordinates, polar coordinates, vector values, AoA, AoD, RTT, RSS, a GPS position, or other data that may be utilized for one or more of: locating one or both of an Agent 100; indicating a direction of interest; and identify a Structure or defined area of structure 106. Logical communications may include digital data indicative or a value, which may be used, for example, to determine a distance via processes involving one or more of: a time difference of arrival (TDOA); frequency difference in arrival (FDOA); and time of flight (TOF).

A precise geospatial location 107 may be determined via triangulation based upon a measured distance from three or more Reference Point Transceivers 101-104. For example, a radio transmission or light signal may be measured and compared from the three reference point transceivers 101-103. Other embodiments may include a device recognizable

via image analysis and a sensor or other Image Capture Device, such as a CCD device, may capture an image of three or more Reference Point Transceivers **101-104**. Image analysis may recognize the identification of each of three or more of the Reference Point Transceivers **101-104** and a size ratio of the respective image captured with Reference Point Transceivers **101-104** may be utilized to calculate a precise position. Similarly, a height designation may be made via triangulation using the reference point transceivers as reference to a known height or a reference height.

Transceivers **101-105** may include circuitry and logic capable of transceiving in a single modality, or multiple disparate modalities. Similarly, a Reference Point Transceivers **101-104** and/or an Agent-supported device at the location of Transceiver **105** may include multiple transceiver device, including, transmitters and receivers.

A modality, as used in conjunction with a Transceiver, transmitter and/or receiver refers to one or both of a bandwidth of wireless communication and a protocol associated with a bandwidth. By way of non-limiting example, a modality, as used in relation to a Transceiver, transmitter and/or receiver may include Wi-Fi; Wi-Fi RTT; Bluetooth; UWB; Ultrasonic, sonic, infrared; ANT, Zigbee, BLE, Z Wave, 6LoWPAN, Thread, Wi-Fi, Wi-Fi 33-ah, NFC (near field communications), sub-GHz, Dash 7, Wireless HART, or other logical communication medium.

Triangulation essentially includes determining an intersection of three distances **108-110**, each of the three distances **108-110** calculated from a reference point transceivers **101-104** to an Agent-supported device at the location of transceiver **105**. The present invention allows for a first distance **108**, to be determined based upon a wireless communication in a first modality; and a second distance **109** and a third distance **110** determined based upon a wireless communication in a same or different modality as the first modality. For example, a first distance **108** may be determined based upon a wireless communication using UWB; a second distance **109** may be determined based upon a wireless communication using Bluetooth; and a third communication may be determined based upon a wireless communication using ultrasonic communication (other combinations of same and/or different communication modalities are also within the scope of the present invention and may include, for example, infrared communications, image recognition, RFID, accelerometer readings or other data generated by an electronic device or mechanical mechanism).

A geospatial location **107** may be determined via triangulation based upon a measured distance from three or more reference point transceivers **101-103** to the transceiver **105** supported by the Agent **100**. For example, timing associated with a radio transmission or light signal may be measured and compared from the three reference point transceivers **101-103**. Other embodiments may include a device recognizable via image analysis and a sensor or other Image Capture Device, such as a CCD device, may capture an image of three or more reference point transceivers **101-104**.

Additional embodiments may include image analysis of image data captured via a CCD included in a Smart Device to recognize the identification of each of three or more of the reference point transceivers **101-104** and a size ratio of the respective image captured where reference point transceivers **101-104** may be utilized to calculate a precise position. Similarly, a height designation may be made via triangulation using the reference point transceivers as reference to a known height or a reference height. In a similar fashion,

triangulation may be utilized to determine a relative elevation of the Smart Device as compared to a reference elevation of the reference points.

In some embodiments, the geospatial location **107** of the Agent-supported device at the location of Transceiver **105** may be ascertained via one or more of: triangulation; trilateration; and multilateration (MLT) techniques.

In some embodiments, a geospatial location based upon triangulation may be generated based upon a controller receiving a measurement of angles between the position and known points at either end of a fixed baseline. By way of non-limiting example, a point of a geospatial location may be determined based upon generation of a triangle with one known side and two known angles. Moreover, a geospatial location based upon multilateration may be generated based upon a controller receiving measurement of a difference in distance to two reference positions, each reference position being associated with a known location. Wireless signals may be available at one or more of: periodically, within determined timespans and continually. The determination of the difference in distance between two reference positions provides multiple potential locations at the determined distance. A controller may be used to generate a plot of potential locations. In some embodiments, the potential determinations generally form a curve. Specific embodiments will generate a hyperbolic curve.

The controller may be programmed to execute code to locate a relatively exact position along a generated curve, which is used to generate a geospatial location. The multilateration system thereby receives as input multiple measurements of distance to reference points, wherein a second measurement taken to a second set of stations (which may include one station of a first set of stations) is used to generate a second curve. A point of intersection of the first curve and the second curve may be used to indicate a specific location.

In another aspect, in some embodiments, the location of a Transceiver **101-105** may be determined and/or aided via discernment of data based upon a physical artifact, such as, for example a visually discernable feature, shape or printed aspect located within the Structure **106**. Discernment of the physical artifact may, for example, be based upon topographical renditions of physical aspects included in the Structure, such as those measured using LIDAR, a magnetic force, image data (or a point cloud derived from image data). A pattern on a surface may convey a reference point by a recognizable pattern (which may be unique to the setting), Vernier or three-dimensional structure as non-limiting examples. A Smart Device ascertaining a physical reference mark and a distance of the Smart Device to the mark may determine a relative location in space to a coordinate system of the marks.

Marks and/or markers with a known location indicated via coordinate values tied to a geospatial coordinate system may be utilized to determine a relative location. Recognition of the marker may be accomplished for example, via one or more of image recognition, laser reflection, RFID activation and reading, sonic reading, LiDAR, or other recognition technology. Similarly, number of methods and apparatus may be executed in various embodiments, to determine a distance from the Smart Device to a mark such as, for example, a sense reflection of light beams (preferably laser beams), electromagnetic beams of wavelength outside of the visible band such as IR, UV, Radio and the like, or sound-based emanations.

In some examples, a carefully placed reference point Node may function as a transceiver of signals. For example,

a Node may receive and transmit signals in a radio frequency band of the electromagnetic spectrum. In a simple form, a Node may detect an incoming signal and coincidentally broadcast a radio frequency wireless communication. Frequencies utilized for wireless communication may include those within the electromagnetic spectrum radio frequencies used in UWB, Wi-Fi, and Bluetooth modalities, as well as IR, Visible and UV light as examples.

In some embodiments, sound emanations may also be used as a communication mechanism between a smart device and a Reference Point Transceivers **101-104**. In some examples, the Reference Point Transceivers **101-104** may function to communicate data with their electromagnetic or sonic transmissions. Such communications may provide identifying information unique to the Node, data related to the synchronization of timing at different well located reference points and may also function as general data communication nodes. A triangulation calculation of the position of a Smart Device or a Node may result from a system of multiple reference position Nodes communicating timing signals to or from the Smart Device or Node. Methods of calculating positions via wireless communications may include one or more of: RTT, RSSI, AoD, AoA, timing signal differential and the like, Triangulation or other mathematical techniques may also be employed in determining a location.

The process of determination of a position based upon triangulation with the reference points may be accomplished, for example via executable software interacting with the controller in a Smart Device, such as, for example via running an app on the Smart Device.

Reference points may be coded via identifiers, such as a UUID (Universally Unique Identifier), or other identification vehicle.

Reference point transceivers **101-104** may be deployed in a defined area of structure **106** to determine a geospatial location **107** of an Agent **100** within or proximate to the defined wireless communication area **111**. Reference point transceivers **101-104** may be fixed in a certain location and transceive in a manner suitable for a triangulation determination of the position of the Agent. Transceiving may occur via wireless transmission to one or more Transceivers **105** supported by the Agent **100**. By way of non-limiting example, Transceivers **105** supported by the Agent **100** may be included in, or be in logical communication with, a Smart Device with Transceivers **105** able to transceive with the reference point transceivers **101-104**.

The reference point transceivers **101-104** may include devices such as a radio transmitter, radio receiver, a light generator, or an image-recognizable device (i.e., an apparatus set out in a distinctive pattern recognizable by a sensor). A radio transmitter may include a UWB Node, Wi-Fi, Bluetooth, or other communication device for entering into logical communication between Transceivers **101-105**. In some embodiments, Reference Point Transceivers **101-104** may include a Wi-Fi router that additionally provides access to a distributed network, such as the Internet. Cartesian coordinates (including Cartesian coordinates generated relative to a GPS or other reference point), or any other coordinate system, may be used as data that may be utilized for one or more of: locating one or both of an Agent **100**; indicating a direction of interest; and identifying a Structure **106** or defined area of structure **106**. A radio transmitter may include a router or other Wi-Fi device. The radio transmitter may include transmissions via an Ultra-Wideband ("UWB") frequencies including, for example, 3.5-6.5 GHz; on Wi-Fi frequencies (300 MHz-60 GHz), sub GHz frequencies or

another modality. A light generator may distribute light at human-safe intensities and at virtually any frequency known in the art. Such frequencies include, without limitation, infrared, ultraviolet, visible, or nonvisible light. Further, the light beacon may comprise a laser, which may transmit light at any of the aforementioned frequencies in a coherent beam.

This plurality of modalities allows for increased accuracy because each modality may have a different degree of reliability. For example, a Smart Device **101** and/or Smart Receptacle may measure a timing signal transmitted by a Reference Point Transceivers **101-104** within a different error tolerance than it may measure the receipt into a photodetector of infrared laser light. This has at least two principal benefits. First, a location calculation may, in some embodiments, be a weighted average of the location calculated from each modality. Second, outliers may be shed. For example, if the standard location calculation comprises a weighted average of the location as calculated by five modalities, but one modality yields a location greater than two standard deviations from the average computed location, then that modality may not be considered for future weighted location calculations.

Additionally, the radio transmitters and/or transceiver in the Smart Device may comprise multiple antennas that transmit signals in a staggered fashion to reduce noise. By way of non-limiting example, if there are three antennas, then they may transmit a signal in intervals of 20 milliseconds. Given this rate of transmission, a detected time of arrival may be used to determine the distance between the transmitter and the antenna (i.e., the Smart Device). Moreover, the antennas may comprise varying lengths to accommodate desirable wavelengths. Further, dead reckoning may be used to measure location, using traditional methods of numerical integration.

A precise location may be determined based upon wireless transmissions between Nodes, such as between a Smart Device and the Reference Position Transceivers. Timing determinations—as well as signal qualities like angle of arrival, angle of departure, transmission strength, transmission noise, and transmission interruptions—may be considered in generating relative positions of Nodes. Additional considerations may include AI and unstructured queries of transmissions between Nodes and triangulation logic based upon a measured distance from three or more Reference point transceivers **101-104**. For example, a radio transmission or light emission may be measured, and timing associated with the radio transmission or light to determine a distance between Nodes. Distances from three reference point transceivers **101-103** may be used to generate a position of a Node in consideration. Other methodologies include determination of a distance from one or more Nodes and a respective angle of arrival and/or angle of departure of a radio or light transmission between the Node in consideration and another Node (Reference Point Node or dynamic position Node).

In some embodiments of the invention, position determination in a Structure or on a Property contemplates determination of a geospatial location using triangulation, trilateration, or multilateration techniques. In some embodiments, a geospatial location relative to one or more known reference points is generated. The geospatial location in space may be referred to as having an X, Y position indicating a planar designation (e.g., a position on a flat floor), and a Z position (e.g., a level within a Structure, such as a second floor) may be generated based upon indicators of distance from reference points. Indicators of distance may include a comparison of timing signals received from wire-

less references. A geospatial location may be generated relative to the reference points. In some embodiments, a geospatial location with reference to a larger geographic area is associated with the reference points, however, in many embodiments, a controller will generate a geospatial location relative to the reference point(s) and it is not relevant where the position is located in relation to a greater geospatial area. In addition to these Cartesian coordinates, polar coordinates may be used, as further described below.

A geospatial location based upon triangulation may be generated based upon a controller receiving a measurement of angles between the position and known points at either end of a fixed baseline. A point of a geospatial location may be determined based upon generation of a triangle with one known side and two known angles.

Triangulation essentially includes determining an intersection of three distances **108-110**, each of the three distances **108-110** calculated from a reference point transceivers **101-104** to an Agent-supported device at the location of transceiver **105**. The present invention allows for a first distance **108** to be determined based upon a wireless communication in a first modality; and a second distance **109** and a third distance **110** determined based upon a wireless communication in a same or different modality as the first modality. For example, a first distance **108** may be determined based upon a wireless communication using UWB; a second distance **109** may be determined based upon a wireless communication using Bluetooth; and a third communication may be determined based upon a wireless communication using ultrasonic communication (other combinations of same and/or different communication modalities are also within the scope of the present invention).

Geospatial location **107** may be based upon triangulation generated based upon a controller receiving a measurement of angles between the position and known points at either end of a fixed baseline. A point of a geospatial location may be determined based upon generation of a triangle with one known side and two known angles.

In some embodiments, a geospatial location **107** may be based upon multilateration and generated based on a controller receiving a measurement of a difference in distance to two reference positions, each reference position being associated with a known location. Wireless signals may be available at one or more of: periodically, within determined time spans, and continually. The determination of the difference in distance between two reference positions provides multiple potential locations at the determined distance. A controller (such as one in the Smart Device) may be used to generate a plot of potential locations. In some embodiments, the potential determinations generally form a curve. Specific embodiments will generate a hyperbolic curve.

A controller may be programmed to execute code to locate an exact position along a generated curve, which is used to generate a geospatial location. The multilateration thereby receives as input multiple measurements of distance to reference points, wherein a second measurement taken to a second set of stations (which may include one station of a first set of stations) is used to generate a second curve. A point of intersection of the first curve and the second curve is used to indicate a specific location.

In exemplary embodiments, as described herein, the distances may be triangulated based on measurements of UWB, Wi-Fi or sub GHz strength at two points. Transceiver signals propagate outward as a wave, ideally according to an inverse square law. Ultimately, a crucial feature of the present invention relies on measuring relative distances between two points. In light of the speed of Wi-Fi waves and the real-time

computations involved in orienteering; these computations need to be as computationally simple as possible. Thus, depending upon a specific application and mechanism for quantifying a condition or location, such as a measurement, various coordinate systems may be desirable. In particular, if the Smart Device moves only in a planar direction while the elevation is constant, or only at an angle relative to the ground, the computation less complicated.

One exemplary coordinate system includes a polar coordinate system. One example of a three-dimensional polar coordinate system is a spherical coordinate system. A spherical coordinate system typically comprises three coordinates: a radial coordinate, a polar angle, and an azimuthal angle (r , θ , and ϕ , respectively, though θ and ϕ are occasionally swapped conventionally).

By way of non-limiting example, suppose Point 1 is considered the origin for a spherical coordinate system (i.e., the point $(0, 0, 0)$). For each Transceiver **101-105**, emitter e_1 , e_2 , e_3 , . . . can be described as points (r_1, θ_1, ϕ_1) , (r_2, θ_2, ϕ_2) , and (r_3, θ_3, ϕ_3) . . . respectively. Each of the r_i 's ($1 \leq i \leq 3$ or more) represent the distance between the Transceiver **101-105** emitter and the Transceiver **101-105** receiver on the Smart Device **101** or Smart Receptacle (see FIG. 5A).

It is understood that in some embodiments, an azimuth may include an angle, such as a horizontal angle determined in an arcuate manner from a reference plane or other base direction line, such as an angle formed between a reference point or reference direction; and line (ray or vector) such as a ray or vector generated from or continuing to a Smart Device. In preferred embodiments, the ray or vector may be generally directed from a Reference Position Transceiver towards, and/or intersect one or more of: an item of interest; a point of interest; an architectural aspect (such as a wall, beam, header, corner, arch, doorway, window, etc.); an installed component that may act as a reference in an augmented virtual model (AVM) (such as, for example, an electrical outlet, a light fixture, a plumbing fixture, an architectural aspect; an item of equipment; an appliance; a multimedia device, etc.); another Reference Position Transceiver or other identifiable destination.

Accordingly, in some embodiments, a spherical coordinate system may include Reference Position Transceiver that is capable of determining an angle of departure of a location signal and a Transceiver that is capable of determining an angle of arrival of the location signal; one or both of which may be used to facilitate determination of an applicable azimuth.

According to various embodiments of the present invention, one or both of an angle of departure and an angle of arrival may therefore be registered by a Transceiver that is transmitting and/or receiving wireless signals (e.g., radio frequency, UWB, Bluetooth 5.1, sonic frequency, or light frequency).

In some embodiments, locating an Agent occurs in or proximate to a Structure in which Reference point transceivers, (including, for example, one or more of: Wi-Fi Transceivers, UWB Transceivers, Bluetooth Transceivers, infrared Transceivers, and ultrasonic Transceivers) may be located above and/or below an Agent. In these embodiments, a cylindrical coordinate system may be more appropriate. A cylindrical coordinate system typically comprises three coordinates: a radial coordinate, an angular coordinate, and an elevation (r , θ , and z , respectively). A cylindrical coordinate system may be desirable where, for example, all Wi-Fi emitters have the same elevation. Angles may be determined as described above.

In some embodiments, transceivers **101-105** including arrays of antennas may be used to measure an angle of radio communication (e.g., angle of arrival and/or angle of departure). Various configurations of transmitting antennas and receiving antennas may be used. For example, a radio transmission may be transmitted with a single antenna and received with a receiver with an array of antennas, the phase or timing difference of arriving signals can be used to calculate the angle at which the signals emerged. In angle of departure schemes, a transmitter may contain an array of antennas and may send a pattern of signals through the array that arrive at a receiver with a single antenna where the angle of departure (AoD) is communicated.

Measurement of angle of arrival may be performed as mentioned by calculation of time difference of arrival at the antennas in an array or alternatively can be performed by rotation of antenna elements.

Some modalities, such as those modalities that adhere to the Bluetooth 5.1 or BLE5.1 standards, allow a Smart Device **101**, Smart Receptacle, or other Node to determine an angle of arrival (AoA) or an angle of departure (AoD) for a wireless transmission. An array of antennas may be used to measure aspects of the Bluetooth signaling that may be useful to calculate these AoA and AoD parameters. By calibrating an antenna system, the system may be used to determine angles in one or two dimensions depending on the design of the antenna. The result may be significant improvement in pinpointing the location of origin of a signal.

An array of antennas may be positioned relative to each other and a transmitting transceiver to allow for extraction of an AoA/AoD. Such an array may include a rectangular array; a polar or circular array; a linear array; and a patterned array, where a number of antennas are deployed in a pattern conducive to a particular environment for transceiving. Antennas may be separated by characterized distances from each other, and in some examples, a training protocol for the antenna array results in antenna positioning incorporating superior angle and location precision. Some transceivers may transceive in 2.4-2.482 GHz frequency bands, and thus the radiofrequency transmissions may have wavelengths in the roughly 125 mm length scale. A collection of antennas separated by significantly less than the wavelength may function by comparing a phase of RF transmissions arriving at the antennas. An accurate extraction of phase differences can yield a difference in path length that, when accumulated, can lead to a solution for the angles involved. In some embodiments, Transceivers **101-105** may include antenna arrays combined with batteries and circuitry to form complete self-contained devices.

In an example, an Agent-supported device at the location of transceiver **105** may be located at a position and may transmit a signal of the various types as have been described. Nodes, such as Reference Point Transceivers **101-104** located at reference points in the wireless communication area around the position of the Agent **100** may receive the transmission and determine the angle of arrival of that transmission. Similarly, transmission associated with other reference point transceivers **101-103** may also determine an angle of arrival of transmissions. In some embodiments, reference point transceivers **101-103** may communicate with one or more of: each other, a smart device, a controller, or other processor to mathematically process multiple angles and locations of the transceivers and calculate a position of a transmission emanation. In examples where calculations

are not performed at a smart device, a communication to the smart device of the calculated position may be communicated.

In certain embodiments of the invention, a direction of interest indicated by Smart Device **101**, or a Smart Receptacle (see FIG. 5A) may be determined based upon a movement of Smart Device **101** or a Smart Receptacle **502**. For example, a device with a controller and an accelerometer, such as mobile Smart Device, may include an Agent display that allows a direction to be indicated by movement of the device from a determined location acting as a base position towards an As Built feature in an extended position. In some implementations, the Smart Device may first determine a first position based upon triangulation with the reference points and a second position (extended position) also based upon triangulation with the reference points. These position determinations may proceed as described above. The process of determination of a position based upon triangulation with the reference points may be accomplished, for example via executable software interacting with the controller in the Smart Device, such as, for example via running an app on the Smart Device. Logical communications relevant to location determination may include, for example, one or more of: timing signals; SIM information; received signal strength; GPS data; raw radio measurements; Cell-ID; round trip time of a signal; phase; and angle of received/transmitted signal; time of arrival of a signal; a time difference of arrival; and other data useful in determining a location.

In another aspect, captured data may be compared to a library of stored data using image recognition software to ascertain and/or affirm a specific location, elevation and direction of an image capture location and proper alignment with the virtual model. Similarly, a Smart Device may capture image data and/or audio data from a determined position and in some embodiments, a determined direction. The captured image data may be stored in a library of image data and referenced in subsequent analysis of where a location is.

In various embodiments, a position of an Agent Smart Device may be determined by any of the methods described herein. A user or other Agent may position a sensor of an associated smart device to be pointing in a direction of interest and obtain an image. The image may be passed on to a server with access to database of images containing stored images of the space around the user. Algorithms on the server may compare the stored images to the image captured by the user and may calculate adjustments to the comparative image based on where the reference image was taken in relationship to the location of the user. Based on the determination that the calculated adjusted image compared to the image obtained by the user in the direction of interest, a direction may be inferred with known location of objects in the reference image. In some variations, the differences in features of the user obtained image compared to a reference image may be used to calculate a direction of interest based upon a location at which the reference image was obtained.

In some examples, stored images may be obtained at multiple angles to improve accuracy of orienteering. These examples may include sensor arrays, audio capture arrays and sensor arrays with multiple data collection angles. In some examples a full 360-degree sensor perspective may be obtained by such arrays. In some directional arrays, a Sensor array (including image capture sensors) may include at least 120 degrees of data capture. By collecting such image collections as the Sensor/Sensor systems are moved, a

database of image perspectives may be formed and utilized to assist in orienteering as described.

Non-limiting examples may include image-based identification where a device with some imaging means, including but not limited to a mobile device sensor, tablet device sensor, computer sensor, security sensor, or A/R headset sensor, may image points of interest in a direction of interest. These points of interest may be identified. Image recognition software may be used to identify the visualized landscape by its identifying features. Machine learning may be used to train systems using this software to identify specific features of the environment in question.

To create a supplemental topographic part of a model of the environment of a user, laser scanning and/or LiDAR may be performed during the acquisition of imagery for a reference database. A resulting three-dimensional shape model may be modelled with the captured imagery to help in the comparison to user data. Three-dimensional shapes can be used to infer comparative imagery at different angles of acquisition than exist in a database. In another example, a device of a user may have the means of performing laser scanning or LiDAR scanning of the environment as well as obtaining images. The resultant three-dimensional data or a composite of the three-dimensional data, and the imagery may be used to recognize features and determine a direction that the user was facing when they collected the image.

The results of scanning may be stored and presented in different manners. In some examples, the scanned data may be represented by a point cloud representation; in other examples an estimated topographic surface representation may be used to visualize the three-dimensional shape data obtained. In some examples, outward facing planes of the surface topography may have the captured imagery superimposed upon them. The resulting image and three-dimensional models may be used to calculate a direction of interest or a device field of view in a dynamic sense or alternatively upon user request.

In some examples other methods of capturing spatially accurate information may include the use of drones and optical scanning techniques which may include high resolution imagery obtained from multiple viewpoints. Scanning may be performed with light based methods such as a CCD sensor. Other methods may include infrared, ultraviolet, acoustic, and magnetic and electric field mapping techniques.

In other embodiments, a single distance to a point of interest in an image, which may be obtained by a laser, other collimated light source, sound source, or the like, may be used with models of the environment of the user. A comparison of the imagery and the measurement of the distance of the user to a prominent feature in the image may allow for an orientation of the user to be determined algorithmically.

In some exemplary embodiments, radio frequencies used for wireless communication include sound waves used to perform one or more of: location determination; movement tracking in interior or exterior areas; and distance calculation. Sound wave transmissions include a number of significant attributes, which may translate into a benefit for a given set of circumstances when used for RF based location determination.

According to the present invention, sonic waves may be deployed independently, or in combination with electromagnetic transmissions and reception of logical communications utilizing other bandwidths, such as bandwidths associated with Ultrawideband, Wi-Fi, Bluetooth, Ultrawideband,

ANT, infrared or almost any wavelength in the Industrial, Scientific, and Medical bands (sometimes referred to as "ISM Bands").

For example, sound waves travel through an ambient atmosphere at a significantly slower speed than electromagnetic radiation (6×10^2 m/sec versus 3×10^8 m/sec). Therefore, a relative accuracy for measurements related to travel times may be improved in some environments by orders of magnitude using sonic-based location tracking as compared to electromagnetic-based measurements. Therefore, using sonic communications may result in increased accuracy of location determination, in some environments.

The present invention also provides for sonic wave emanations may be used to complement electromagnetic emanations based upon a tendency that sonic waves generally do not efficiently penetrate walls other physical items or structures. Sonic transceivers may be particularly advantageous in a defined area where location can be unambiguously determined to be within a particular room (the use of multiple bandwidth transmitting and receiving for different purposes is further discussed below). Sound wave interaction with a solid surface, such as a wall, may require that for optimal performance, transceiver/transmitters/receivers to be located in each room where location detection is desired. In some embodiments, a reflected sonic transmission may be received and analyzed to determine an effect of the reflected nature of the transmission.

Accordingly, methods may be employed using sonic emanations and reception for location determination. In general, frequencies of effective indoor sonic location detection may be at ultrasonic bandwidths (commonly used bandwidths include frequencies of between 1 MHz and 10 MHz, although frequencies of less than 50 kHz to greater than 200 MHz are possible). The utilized frequencies may be either below or above audible detection by people or other animals in the location, such as at frequencies above 20 kHz.

Sound waves may be used to perform one or more of: location determination, movement tracking in interior or exterior locations, and distance calculation from a position to a Smart Device at the location of transceiver 105, which may be accomplished based upon transmission and receipt of sonic transmission. Sound wave transmissions include several significant attributes, which may be beneficial for a given set of circumstances when used for radiofrequency-based location determination. According to the present invention, sonic waves may be deployed independently, or in combination with, transmissions and reception of logical communications utilizing other bandwidths, such as bandwidths associated with Wi-Fi, Bluetooth, ANT, infrared or almost any wavelength in the ISM Bands.

In some examples, as may be used in orienteering herein, an Agent-supported device at the location of transceiver 105 may support receivers, transmitters or transceivers that interact with ultrasonic transceivers fixedly secured to a reference point position, such as via mechanical mounting within a room environment. An ultrasonic positioning system may have indoor positioning accuracy at centimeter, millimeter, and even sub-millimeter accuracy. Multiple ultrasonic Transceivers may transceive from an Agent-supported device to communicate with fixed reference point transceivers may transmit signals. Arrival of the sound transmissions may be accurately timed and converted to distances. In some embodiments, distance determinations may be improved with knowledge of temperatures in the environment containing the sound transceiving. For example, temperature

may be measured at one or more of: an Agent-supported Smart Device, a Reference Point position, and an ambient environment.

In some examples, synced timing apparatus is able to generate a location of a stationary Agent to within centimeter accuracy using sonic wave transmissions and reception and preferably within several millimeters of accuracy. In addition, in some embodiments sensors are able to detect frequency shifts within the sonic emanations which may add information about the relative rate of movement of the Agent, which may then in turn allow for correction to the timing signals.

In some examples, a combination of radio frequency emissions and ultrasonic emissions may be used. For example, a complement of radio frequency emissions/receptions and ultrasonic of radio frequency emissions and ultrasonic emissions may be reconciled to generate more accurate location determination. In another aspect, a radio frequency signal may be used to transmit syncing signals that establish a time that ultrasonic signals are transmitted. Since, the electromagnetic transmissions may be orders of magnitude faster than sound transmissions, the electromagnetic transmissions relatively small time of travel from the Transceivers to the Agent may be negligible and therefore used as “zero-time” setpoints as received at the Agent-supported Transceiver. In such embodiments, a controller determining a location may use not only relative arrival times, but also a delta time between a radiofrequency transmission and ultrasonic transmission to determine a distance from a transmitting Transceiver. An array of such ultrasonic and/or radiofrequency transceivers provide increased accuracy in triangulating a location of the Agent.

In still further examples, RF communications may transmit a syncing pulse and also transmit digital data about various aspects of a defined area, such as the defined area’s identification, its relative and/or absolute location in space and other refinements. In some examples, data related to improved distance calculation may also be transmitted by RF communication such as temperature of the environment, humidity and the like which may influence the speed of sound in the environment as a non-limiting example. In some examples, such a system may result in millimeter level accuracy of position determination.

In some examples, the process may be iterated to refine the direction of each of the ultrasonic transmitters and maximize signal levels of the transmission which may provide additional information in the calculation of a position. UWB, Bluetooth, Wi-Fi, ANT, Zigbee, BLE, Z Wave, 6LoWPAN, Thread, Wi-Fi, Wi-Fi-ah, NFC (near field communications), Dash 7, and/or Wireless HART transmissions may be used for data communications and syncing. In other examples, an Agent-supported device at the location of transceiver **105** may be moved and an iterative process may be used to track the Agent-supported device at the location of transceiver **105** as it moves through space. Stationary Agents may be tracked with submillimeter accuracy in some embodiments.

A direction dimension may be based upon multiple transceivers included in a Smart Device or a Smart Receptacle or via a movement of a Smart Device or Smart Receptacle while an Agent supporting the Smart Device or Smart Receptacle remains in a stationary position in relation to reference, such as a ground plane position. For example, a device with a controller and an accelerometer, such as mobile Smart Device, may include a user display that allows a direction to be indicated by movement of the device from a determined location acting as a base position towards a

feature in the intended direction where the movement results in an extended position. In some implementations, the Smart Device may first determine a first position based upon triangulation with the reference points and a second position (extended position) also based upon triangulation with the reference points.

As described above, facing a mobile device towards an area in a Structure and movement of the mobile device in a particular pattern may be used to ascertain a specific area in space to be interpreted by various methods. In some examples, the leading edge of a smart device, or the top portion of the user screen (in either portrait or landscape mode of display) may be the reference for the direction pointed in by the user. If the smart device is held at an angle relative to the ground, in some examples, the angle formed by the perpendicular to the top portion of the user screen may be projected upon the ground and that projection taken as the indication of direction.

Referring now to FIG. 2, a secure transaction system **200** with apparatus and methods of completing a transaction with multifactor security is illustrated. A Smart Device **205** is illustrated proximate to a Transaction Apparatus **208**. The methods and apparatus discussed herein may be used to determine that the Smart Device **205** is located within a Smart Device Position Area **207**. In some embodiments, a Smart Device Position Area **207** is generally defined as an area in front of (or to a particular side of) a Transaction Apparatus **208** within a margin of accuracy for wireless communication modalities and environmental conditions (including, for example, environmental interference with wireless communication).

An Authorized Transaction Area (ATA) **209** may be specified to indicate an area from which a Smart device **205** is authorized to perform transactions with the Transaction Apparatus **208**. In some embodiments, an ATA **209** may be exclusive to a particular Transaction Apparatus **208**. For example, as discussed further below, an ATA **209** may be exclusive to transactions with a specified Transaction Apparatus **208**, such as, for example, a particular workstation; dispensing apparatus; building delivery; utility station; point of sale device; and construction site under management.

Inclusion of a Smart device **205** within an ATA **209** may be determined according to any of the methods and apparatus discussed herein, in addition a Transaction Apparatus Transceiver (TAT) **210** may be positioned with the Transaction Apparatus **208** (such as, co-located with, or incorporated into the Transaction Apparatus **208**). The TAT **210** may include a relatively shorter range and more accurate location modality than GPS, such as one or more of: UWB, Bluetooth, Wi-Fi, ANT, Zigbee, BLE, Z Wave, 6LoWPAN, Thread, Wi-Fi, Wi-Fi-ah, NFC (near field communications), Dash 7, and/or Wireless HART transmissions. The TAT **210** may provide data sufficient to calculate an ATA/Smart Device distance **206** indicative of how far the Smart Device at the location of transceiver **105** is from the TAT **210** based upon a timing signal included in transceiving between the TAT **210** and the Smart Device at the location of transceiver **105**.

In general, a TA/Smart Device distance **206** may not be enough to determine that the Smart Device at the location of transceiver **105** is located within a transaction Apparatus **208** since a radio distance is often a radius. Therefore, the present invention provides for a TAT **210** to provide for directional transmissions. For example, a directional transmission may be limited to transmission from a front side of the transaction Apparatus **208** in a forward direction (as indicated by the arrow).

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In addition, one or more Reference Point Transceivers **201-204** may engage wireless communications with the Smart Device **205** and a respective distance between the respective reference point transceivers **101-104** and the Smart Device **205** may be calculated. Since only one respec- 5
tive distance **201-202** per reference point transceivers **101-104** will intersect a radius of a distance from a TAT **210**, a determination made be made as to whether a SD is within an ATA **209**.

With a Smart Device **205** positioned within an ATA **209**, a transaction with improved security may be entered into and completed. A number of factors may be ascertained before a transaction with the transaction Apparatus **208** is executed. In some preferred embodiments, the completion of the transaction with the transaction Apparatus **208** will include one or more of: a) designating a user via a unique user identification; b) exchange of a user password, private key, synchronous dynamic password or other security code; c) location verification via wireless communication, which may include determination that the Smart Device at the location of transceiver **105** is within a ATA **209**; identifying a transaction Apparatus **208**; d) present transaction Apparatus **208** credentials to a transaction process; f) present credentials to a transaction process; g) designate actions to be completed during transaction execution; h) execute the actions; generation of an augmented reality icon viewable form the ATA **209**; and interaction with the A/R icon.

According to the present invention, presentation of credentials from one or both of the transaction apparatus **208** and Agent **100** may only be made once location determinations indicate that the Agent **100** with the User Smart Device at the location of transceiver **105** is within the ATA **209**.

In addition, in some embodiments, credentials may be withheld until an A/R Virtual Tag icon has been generated and made available to the unique identifier of the user. The Virtual Tag icon will be located in virtual space at a geospatial position that is viewable via an A/R interface generated with input from geospatial indicators generated by the Smart Device. The viewable area may include a Radio Target Area for an energy receiving device in (or attached to) the Smart device. Geospatial indicators may include, for example, location coordinates for the Smart Device that are generated as discussed herein. Generation of the icon in a virtual space that is aligned with a specific location creates a deterrent to unauthorized access to the Transaction Apparatus since a person that is not present to the TA will not be able to view the Virtual Tag icon and therefore not be able to interact with the icon in order to complete a process for gaining access to the secure transaction.

Referring now to FIGS. 2A-2B exemplary embodiments of apparatus and methods that may be involved in various embodiments involving Virtual Tags **211**, **211A-211B** according to the present invention are illustrated. These illustrations are non-limiting and depict only some specific examples of how the present invention may be implemented to provide enhanced security and data sources to memorializing events and/or for analysis by machine learning apparatus, such as via those apparatus capable of artificial intelligence processes.

Referring now to FIG. 2A, a workstation **212** is illustrated as a Transaction Apparatus for which a geospatial component is required to login to the workstation **212** or to otherwise operate the workstation. For example, a Smart Device at the location of transceiver **105** may be required to be within an ATA **209A** located in an area proximate to the workstation **212**. In this scenario, the term "proximate to" may be a defined radius distance from the workstation, such

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as for example, within 3 feet, 6 feet, 12 feet, or other defined distance. "Proximate to" may also include a directional component, such as, for example, in front of the workstation **212** in order for an Agent **100** to login to the workstation **212** and operate the workstation. An ATA **209A** may also be a larger area such as an area within a structure in which the workstation **212** resides, or a property in which the workstation **212** resides.

In some embodiments, a presence of the Smart Device at the location of transceiver **105** within an ATA **209A** permits an Agent **100** to enter User credentials. In other embodiments, the locating of the Smart Device at the location of transceiver **105** within the ATA **209A** causes a first controller (such as a CPU in a workstation **212**) to execute a security related process that presents User credentials associated with the Smart Device at the location of transceiver **105** to a related login process. Presentation of credentials may also be a part of other various secure transactions that require different credentials to be presented. In addition, credentials from one or both of a TA, such as the workstation **212**, and an Agent **100** may be presented to a process that will transacted.

In some embodiments, a Virtual Tag **211A** is presented at location coordinates within an RTA (not illustrated). The Virtual Tag **211A** is preferably viewable from within the ATA **209A** with the Smart Device at the location of transceiver **105**, when the Smart Device at the location of transceiver **105** is supported by the Agent **100**.

Referring now to FIG. 2B, an exemplary worksite is illustrated with various items of equipment **213A-213C**, personnel **214A-214C**, robotic Agents **215**, materials **216**, authorized areas of work **217**, and locations of architectural aspects **218**, such as a building, walkway, stairs etc.

According to the present invention, a personnel, which in some embodiments may also be a User **214A** may locate a Smart Device at the location of transceiver **105** with an ATA **209B** and interact with an App (or other executable code executed by a controller) that identifies a particular authorized area of work **217** which may also represent a TA based upon the Smart Device at the location of transceiver **105** being located in the ATA **209B**. The App may present Agent credentials for a controller, software, automation, or other provider of access to Virtual Tag(s) **211**, **211A-211B**.

The App may also present payor credentials for the User **214A** (or other entity that authorizes payment for sale of materials, energy, services, or other saleable item or quantity to the User **214A** or an entity associated with the User **214A**).

In some preferred embodiments, the credentials are virtually presented to a known payment processor, such as a bank, a credit card company, online payment company (e.g., Venmo, PayPal, Zelle etc.) via a secure Internet connection or device specific app, thereby adding further security features. In this manner, the present invention provides for a payee for a transaction from within a specific TA for one or more of: a) a finite list of potential purchases; b) a finite list of potential vendors to purchase from; c) a capped amount of purchase contingent upon the Agent identification and the TA the Agent is in; d) unlimited purchase conditions with each purchase associated with the particular project or other accounting designation, wherein the accounting designation is associated with a particular TA; and f) unlimited purchase conditions and/or accounting variables based upon the TA and Agent credentials.

For example, a User **214A** in an ATA **209B** may execute an app that provides a user interface with user interactive devices enabling one or more of: the purchase, shipping,

supply, lease and rental of one or more of: materials; services; utilities; equipment; supplies; food; space; real estate; or other purchasable item and have the purchase linked with a jobsite associated with one or more of: an ATA **209B**; an architectural aspect **218**; and a User **214A-214B**. No card swipe, chip scan, near field communication or other locally interceptable transfer of data takes place; and the point of sale may be anywhere within an ATA **209B**.

In some embodiments, an additional step may be included in a secure transaction according to the present invention. The additional step may include generation of a Virtual Tag **211** in RTA viewable from the ATA **209B**. Use of the Virtual Tag **211B** verification requires that a User be able to see the Virtual Tag **211B** and interact with the Virtual Tag **211B**. In some embodiments, a Virtual Tag **211B** may occupy a congruent space in a user interface as an item in a visual representation (e.g., image data reproducing a physical area). The item may be, by way of non-limiting example, an item of equipment **213A-213C**; personnel **214A-C**; robotic agent **215**; materials **216**, authorized area **216-219**; or other ascertainable item in a visual representation included in a user interface presented on a Smart Device at the location of transceiver **105**.

A remote hacker or a hacker that was not in or very close to the ATA **209B** would not be able to view and interact with the Virtual Tag **211B** since the Virtual Tag **211B** is assigned to a set of location coordinates that are unpublished to the User **214A** except as an icon in an A/R interface. Similar processes may be utilized in other secure transactions that may, for example, involve access to protected digital content and submission of protected digital content.

In some embodiments, an action taken, or quantification of a condition on the worksite may be accomplished via a robotic Agent **215**, such as an UAV or UGV, in such embodiments an alternate ATA **209B** may be designated as an area for a specific action and an icon may be locatable by the robotic Agent **215**, as well as by a User **214A** overseeing the robotic Agent **215**. The icon may include digital content related to an action to be performed by the robotic device **215** at a particular location. For example, the icon may include a type of material to be used; a fastener type; parameters of installation, time of installation; sequence of installation; values for variables related to installation (e.g., force, pressure, torque, thickness of material, rate of process used, etc.).

In another aspect, in some embodiments, a secure transaction may include one or more of: operation of various items of equipment **213A-213C**; operation of a device **219**, such as an automated lock that provides entry into a designated area or structure (such as a delivery area, equipment corral, commercial building, garage, parking area, recreation facility or fenced in area); association of digital content with a position of a design model, site plan, floorplan or other AVM; and access to a utility area. Utility areas may include, for example, an electrical power substation, a gas line control area, water supply area, wastewater area, Internet or other distributed network area, watershed area, or other utility area. A secure transaction may include a delivery, or simple access to the area via an ATA **209B**. As with other secure transactions according to the present invention, a Virtual Tag Icon may be generated proximate to the ATA **209B** (in this case viewable from the ATA **209B**) such that User **214A** interaction with the generated icon may be used to further verify an authorized User **214A** is present at the designated area allocated as the ATA **209B**. A utility area may also be equipped with an ATA **209C** accessible via UAV and/or UGV.

While a construction site is illustrated, other outdoor, or combined outdoor and structure interior environments are within the scope of the present invention. Accordingly, a definable geospatial area, such as a recreation area; hunt club area; private land; Federal Bureau Land Management area, National Park, State, or local park, including land areas and/or water body areas may be designated to contain one or more ATA's **209B**. One or more ATA's **209B** may be associated with one or more Virtual Tag Icons to verify a User **214A** and a location that may be referenced in an automated process for authorizing a transaction, which may include authorization to be present on the land included as a managed area, a delivery, a movement of items within the area, or other action. Areas that contain wildlife and/or human activity may also require authorization for activities monitoring movement of the wildlife and/or humans and/or equipment.

Referring now to FIG. 3, in some embodiments, aspects of location determination and verification may be integrated into other security mechanisms that may include processes that involve security specific protocols and hardware which may involve communications with cloud storage **304**. For example, one or more of the wireless communications described herein may also involve the use of a security processor, such as a chip to cloud security processor **306** and associated software and processes. According to the present invention, the chip to cloud security processor **306** and one or more additional apparatus involving chip to cloud technology may interact with a location coordinate generator **307** to further ascertain that a requested process involving digital content is being requested by an authorized Agent, from an authorized geospatial area and for an authorized purpose.

In some embodiments, an Agent **300** may position a smart device **305** including one or more transceivers and antenna arrays in a first position **301** proximate to a portion in a space of interest **310**. The first position **301** of the Smart Device **305** may be determined and recorded, for example in terms of cartesian coordinates X, Y and Z, representative of three planes **302X**, **302Y**, **302Z**. The Agent **300** may orient the smart device **305** in a general direction of the portion in space of interest **310**. Positional coordinates may also include other known coordinate values, such as: polar coordinates and/or cylindrical coordinates.

A controller in an external system or in the smart device **305** may generate a directional indicator **303** (e.g., one or both of a ray and a vector). The directional indicator **303** may be directed towards a portion of a space of interest in which the Agent **300** would like to execute a secure transaction, such as interaction with a Virtual Tag **311** (e.g.: generation of a new virtual tag; modification an existing virtual tag; or receipt of digital content included in an existing virtual tag). Interaction with a Virtual Tag **311** may include one or more of: receipt of digital content associated with the Virtual Tag **311**; associating digital content with the Virtual Tag **311**; and conditions for access of the Virtual Tag **311**.

In some embodiments, the vector may have a length determined by a controller that is based upon a distance to a feature of interest in space as represented in a model on the controller in the direction of the generated vector. The vector may represent a distance from the second position to the space of interest **310** along the axis defined by a line between the first position **301** and the second position. In contrast, a ray may include just a starting point and a direction.

In still other embodiments, a device with a controller and an accelerometer, such as mobile phone, tablet or other

Smart Device **305** that includes a Transceiver, may include a user display that allows a direction to be indicated by movement of the device from a determined location acting as a base position towards a point in a direction of interest or representing a center of an RTA of the device. The movement may occur to a second location in an extended position. In some implementations, the Smart Device **305** determines a first position **301** based upon triangulation with the reference points. The process of determination of a position based upon triangulation with the reference points may be accomplished, for example via executable software interacting with a controller in the Smart Device **305**, such as, for example by running an app on the Smart Device **305**.

An array of antennas positioned at a user reference point may allow for the accurate receipt of orientation information from a transmitter. As discussed earlier a combination devices with arrays of antennas may be used to calculation a position. A single Node with an array of antennas can be used for orienteering and determining a direction of interest. Each of the antennas in such an array receiving a signal from a source may have different phase aspects of the received signal at the antennas due to different distances that the emitted signal passes through. The phase differences can be turned into a computed angle that the source makes with the antenna array.

In some embodiments, aspects of location determination and verification may be integrated with other security mechanisms **306-309** that may implement processes that involve security specific protocols and hardware. For example, any of the wireless communications may also involve the use of a security processor, such as a cloud security processor **306** and associated software and processes. According to the present invention, the security processor **306** and other chip to cloud technology **308** may interact with a location coordinate generator **307** to further ascertain that a requested process involving digital content is being requested by an authorized Agent **300**, from an authorized geospatial area and for an authorized purpose.

In some embodiments, methods, and devices for determining a direction that may be referenced for one or both of data capture and data presentation of a particular portion of the virtual representation of surroundings of a user. An Agent **300** may position a Smart Device **305** in a first position **301** proximate to a portion in a space of interest **310**. The first position **301** of the Smart Device **305** may be determined and recorded. The Agent **300** may then relocate the Smart Device **305** to a second position in a general direction of the portion in space of interest. With associated position information obtained for the first and second positions a controller in an external system or in an associated Smart Device **305** may generate one or both of a ray and a vector towards the portion of a space of interest.

In some embodiments, the vector may have a length determined by a controller that is based upon a distance to a feature of interest in space as represented in a model on the controller in the direction of the generated vector. The vector may represent a distance from the second position to the space of interest **310** along the axis defined by a line between the first position **301** and the second position. In contrast, a ray may include just a starting point and a direction. In various embodiments, a vector may be referenced to define a volume of space that has a perimeter of a defined shape, such as a rectangle, circle, oval, square or other perimeter shape with a volume defined by the perimeter being extended through space in a direction and for a distance indicated by the vector.

In still other embodiments, a device with a controller and an accelerometer, such as mobile phone, tablet or other Smart Device **305** that includes a Transceiver, may include a user display that allows a direction to be indicated by movement of the device from a determined location acting as a base position towards a point in a direction of interest or representing a center of an RTA of the device. The movement may occur to a second location in an extended position. In some implementations, the Smart Device determines a first position **301** based upon triangulation with the reference points. The process of determination of a position based upon triangulation with the reference points may be accomplished, for example via executable software interacting with a controller in the Smart Device, such as, for example by running an app on the Smart Device **305**.

An array of antennas positioned at a user reference point may allow for the accurate receipt of orientation information from a transmitter. As discussed earlier a combination devices with arrays of antennas may be used to calculation a position. A single Node with an array of antennas can be used for orienteering and determining a direction of interest. Each of the antennas in such an array receiving a signal from a source may have different phase aspects of the received signal at the antennas due to different distances that the emitted signal passes through. The phase differences can be turned into a computed angle that the source makes with the antenna array.

As illustrated in FIG. 3A, in some embodiments, a sonic Transceiver **310A** may transmit a sonic transmission **312A** and determine a location **311A** based upon receiving an echo **313A** back from an Agent-supported device **314A**. Walls **315A** may also generate reflected sonic emanations **316A**.

In some examples, as may be used in orienteering herein, an Agent-supported device **314A** may support receivers, transmitters or transceivers that interact with ultrasonic transceivers fixedly secured to a reference point position, such as via mechanical mounting within a room environment. An ultrasonic positioning system may have indoor positioning accuracy at centimeter, millimeter, and even sub-millimeter accuracy. Multiple ultrasonic Transceivers may transceive from an Agent-supported device to communicate with fixed reference point transceivers may transmit signals. Arrival of the sound transmissions may be accurately timed and converted to distances. In some embodiments, distance determinations may be improved with knowledge of temperatures in the environment containing the sound transceiving. For example, temperature may be measured at one or more of: an Agent-supported Smart Device, a Reference Point position, and an ambient environment.

In some examples, synced timing apparatus is able to generate a location of a stationary Agent to within centimeter accuracy using sonic wave transmissions and reception and preferably within several millimeters of accuracy. In addition, in some embodiments sensors are able to detect frequency shifts within the sonic emanations which may add information about the relative rate of movement of the Agent, which may then in turn allow for correction to the timing signals.

In some examples, a combination of radio frequency emissions and ultrasonic emissions may be used. For example, a complement of radio frequency emissions/receptions and ultrasonic of radio frequency emissions and ultrasonic emissions may be reconciled to generate more accurate location determination. In another aspect, a radio frequency signal may be used to transmit syncing signals that establish a time that ultrasonic signals are transmitted.

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Since, the electromagnetic transmissions may be orders of magnitude faster than sound transmissions, the electromagnetic transmissions relatively small time of travel from the Transceivers to the Agent may be negligible and therefore used as “zero-time” setpoints as received at the Agent-supported Transceiver. In such embodiments, a controller determining a location may use not only relative arrival times, but also a delta time between a radiofrequency transmission and ultrasonic transmission to determine a distance from a transmitting Transceiver. An array of such ultrasonic and/or radiofrequency transceivers provide increased accuracy in triangulating a location of the Agent.

In still further examples, RF communications may not only transmit a syncing pulse, but also transmit digital data about various aspects of a defined area, such as the defined area’s identification, its relative and/or absolute location in space and other refinements. In some examples, data related to improved distance calculation may also be transmitted by RF communication such as temperature of the environment, humidity and the like which may influence the speed of sound in the environment as a non-limiting example. In some examples, such a system may result in millimeter level accuracy of position determination.

In some examples, the process may be iterated to refine the direction of each of the ultrasonic transmitters and maximize signal levels of the transmission which may provide additional information in the calculation of a position. RF and Wi-Fi transmissions may be used for data communications and syncing as have been described. In other examples, as an Agent-supported device **314A** is moving, an iterative process may be used to track the Agent-supported device **314A** moves through space. Stationary Agents may be tracked with submillimeter accuracy in some embodiments.

A direction dimension may be based upon multiple transceivers included in a Smart Device or a Smart Receptacle or via a movement of a Smart Device or Smart Receptacle while an Agent supporting the Smart Device or Smart Receptacle remains in a stationary position in relation to reference, such as a ground plane position. For example, a device with a controller and an accelerometer, such as mobile Smart Device, may include a user display that allows a direction to be indicated by movement of the device from a determined location acting as a base position towards a feature in the intended direction where the movement results in an extended position. In some implementations, the Smart Device may first determine a first position based upon triangulation with the reference points and a second position (extended position) also based upon triangulation with the reference points.

As described above, facing a mobile device towards an area in a Structure and movement of the mobile device in a particular pattern may be used to ascertain a specific area in space to be interpreted by various methods. In some examples, the leading edge of a smart device, or the top portion of the user screen (in either portrait or landscape mode of display) may be the reference for the direction pointed in by the user. If the smart device is held at an angle relative to the ground, in some examples, the angle formed by the perpendicular to the top portion of the user screen may be projected upon the ground and that projection taken as the indication of direction.

Referring to FIGS. 4A-D a series of exemplary devices employing matrices (or arrays) of antennas for use with Nodes that communicate wirelessly, such as via exemplary UWB, Sonic, Bluetooth, a Wi-Fi, or other modality, is illustrated. Linear antenna arrays **401** are illustrated in FIG.

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4A. Rectangular antenna arrays **402** are illustrated in FIG. **4B**. Circular antenna arrays **403** are illustrated in FIG. **4C**, other shapes for arrays are within the scope of the invention. In addition, an antenna array may be omni-directional or directional.

In some embodiments, see FIG. **4D** item **404**, a Smart Device **405** may include one or more Nodes **406** internal to the Smart Device **405** or fixedly attached or removably attached to the Smart Device **405**. Each Node **406** may include antenna **407** arrays combined with a power source and circuitry to form complete self-contained devices. The Nodes **406** or a controller may determine an RTT, time of arrival, AoA and/or AoD or other related angular determinations based upon values for variables involved in wireless communications. For example, a composite device may be formed when a Node **406** with a configuration of antennas, such as the illustrated exemplary circular configuration of antennas **407**, is attached to a Smart Device **405**. The Node **406** attached to the Smart Device **405** may communicate information from and to the Smart Device **405** including calculated results received from or about another Node **406**, such as a Node **406** fixed as a Reference Point Transceiver or a Node with dynamic locations, wherein the wireless communications are conducive to generation of data useful for determination of a position (e.g., timing data, angles of departure and/or arrival, amplitude, strength, phase change, etc.). A combination of angles from multiple fixed reference points to the antenna array can allow for a location of a user in space. However, with even a single wireless source able to communicate with the antenna array, it may be possible to determine a direction of interest or a device related field of view.

An array of antennas positioned at a reference point may allow for the accurate receipt of orientation information from a transmitter. As discussed earlier a combination devices with arrays of antennas may be used to calculation a position. A single Node with an array of antennas can be used for orienteering and determining a direction of interest. Each of the antennas in such an array receiving a signal from a source will have different phase aspects of the received signal at the antennas due to different distances that the emitted signal passes through. The phase differences can be turned into a computed angle that the source makes with the antenna array.

Referring now to FIG. **5A**, in some embodiments, one or both of a Smart Device **501** and a Smart Receptacle **502** may incorporate multiple Transceivers **503-510** and a direction of interest may be ascertained by generating a vector **526** passing through a respective position of each of at least two of the transceivers (as illustrated through transceiver **505** and **507**). The respective positions of each of the transceivers **503-510** supported by the Smart Device **501** and/or Smart Receptacle **502** may be ascertained according to the methods presented herein, including for example via triangulation, trilateration, signal strength analysis, RTT, AoD, AoA, topography recognition, and the like. Reference point transceivers **511-514** may be fixed in a certain location.

In some embodiments, one or both of the Smart Device **501** and the Smart Receptacle **502** incorporating Transceivers **503-510** may be rotated in a manner (such as, for example in a clockwise or counterclockwise movement **520**, **522** relative to a display screen) that repositions one or more Transceivers **503-510** from a first position to a second position. A vector **526** may be generated at an angle that is zero degrees with a plane of a display screen **523** or perpendicular **525** or some other designated angle in relation to the smart device **501** and an associated display screen

523. In some embodiments, an angle in relation to the smart device is perpendicular **525** to a display screen **523** and thereby viewable via a forward-looking sensor (or other CCD or LIDAR device) on the smart device. In addition, a mirror or other angle-altering device may be used in conjunction with a CCD, LIDAR or other energy receiving device.

Movements of a Smart Device **501** equipped with an antenna array can be determined via relative positions of the antenna and/or via operation of an accelerometer within the Smart Device **501** or Smart Receptacle **502**. Rough movement sense may be inferred with a single source to the antenna array. However, with multiple sources, the positional movement of each of the antennas can be used to sense many types of movements including translations and rotations.

A user may position the smart device **501** such that an object in a direction of interest is within in the CCD view. The smart device may then be moved to reposition one or more of the transceivers **503-510** from a first position to a second position and thereby capture the direction of interest via a generation of a vector in the direction of interest.

In some embodiments, movement of the Smart Device **501** and/or the Smart Receptacle **502** may not be desired and another method of indicating direction may be utilized, such as a value generated based upon a magnetic force detection device, such as a magnetometer included in the Smart Device **501** and/or Smart Receptacle **502**. Registration of a magnetic force may be determined in relation to a particular direction of interest **524** and a determination of magnetic force referenced or provide a subsequent orientation of the Smart Device **501** or Smart Receptacle **502**. In some embodiments therefore, multiple modalities of wireless input may be used to determine a position of a Smart Device and direction of interest **524**.

For example, a first wireless modality may be used to determine a structure in which a Smart Device **501** is located. Accordingly, the first modality may include a cellular transmission or GPS signals. A second modality of wireless communication may be used to determine a position within the structure. Appropriate second modalities may therefore include, one or more of UWB, Bluetooth, sonic, ultrasonic, LiDAR, or infrared modalities (or other modalities discussed herein). A third modality, such as a magnetic force, a movement of a device, a relative position of two or more antennas may be used to determine a direction of interest.

In some embodiments, the magnetic force detection device may be used combination with, or in place of directional movement of the Smart Device transceivers **503-510** to quantify a direction of interest to a user. Embodiments may include an electronic and/or magnetic sensor to indicate a direction of interest when a Smart Device **501** and/or Smart Receptacle **502** is aligned in a direction of interest. Alignment may include, for example, pointing a specified side of a Smart Device **501** and/or Smart Receptacle **502**, or pointing an arrow or other symbol displayed upon a user interface on the Smart Device **501** towards a direction of interest.

A magnetic force detection device (sometimes referred to as a magnetic sensor) may detect a magnetic field particular to a setting that a Smart Device is located. For example, in some embodiments, a particular structure or other setting may have a magnetic force that is primarily subject to the earth's magnetic field or may be primarily subject to electromagnetic forces from equipment, power lines, or some other magnetic influence or disturbance. An initial quanti-

fication of a magnetic influence at a first instance in time may be completed and may be compared to a subsequent quantification of magnetic influence at a later instance in time. In this manner an initial direction of interest indicated by a position, orientation, pitch, and yaw of a Node, such as a Smart Device may be compared to a subsequent position, orientation, pitch, and yaw of the Smart Device.

In some embodiments, an initial position, pitch, and yaw of a Smart Device **501** may be described as a relative angle to a presiding magnetic force. Examples of presiding magnetic forces include, magnetic influences of electrical charges, Earth's magnetic field, magnetized materials, permanent magnetic material, strong magnetic fields, ferromagnetism, ferrimagnetism, antiferromagnetism, paramagnetism, and diamagnetism, or electric fields that are generated at reference nodes at known positions which may be locally used to indicate a direction of interest.

Smart devices may include electronic magnetic sensors as part of their device infrastructure. The magnetic sensors may typically include sensing elements deployed along three axis. In some examples, the magnetic sensors may be supplemented with electronic accelerometers, such as MEMS accelerometers.

In some examples the magnetic sensor may include a Hall effect sensor that is operative to measure a sensed magnetic field perpendicular to the body of the sensor via operation of the Hall effect sensor. In some examples, a Hall effect sensor may be built into silicon to generate a relatively sensitive sensing capability for magnetic fields. In some Hall effect sensors, electrons and holes flowing in a region of the silicon may interact with the regional magnetic field and build up on the fringes of the conduction region, thus generating a measurable voltage potential. In other examples, anisotropic magnetoresistance sensors may sensitively detect the magnetic field at the device as a significant change in resistance of structure elements in the device.

In still further examples, the magnetic sensor may include one or more giant magnetoresistance (GMR) sensors may detect the magnetic field. In some of these examples, the GMR sensors may detect a magnetic field with a current-perpendicular-to-plane (CPP) GMR configuration. In other examples, a current-in-plane (CIP) GMR sensor configuration may be used. The resulting three-axis magnetic sensors may perform a sensitive compass function to determine a direction of a specified portion of the Smart Device and/or an edge of the smart device relative to the local magnetic field environment. A specified portion of the Smart Device may be indicated via a user interface presented on a screen of the Smart Device.

Referring now to FIG. **5B**, as illustrated, a vector in a direction of interest **525B** may be based upon a forward arcuate movement **523B** or a complementary arcuate movement **524B** of the smart device **501**, such as a movement of an upper edge **518** in a forward arcuate movement **523B**. The lower edge **519** may also be moved in a complementary arcuate movement **524B** or remain stationary. The movement of one or both the upper edge **518**- and lower edge **519** also results in movement of one or more transceivers **503-510** (Shown in FIG. **5A**) and/or registration in an onboard accelerometer **534**. The movement of the transceivers **503-510** (Shown in FIG. **5A**) will preferably be a sufficient distance to register disparate geospatial positions based upon wireless transmissions and/or sufficient to register movement via the onboard accelerometer **534**.

As presented herein, a direction dimension may be based upon one or more of: a wireless position of two or more transceivers, a movement of a device, a magnetic force

determination, a LIDAR transmission and receiving, CCD energy determinations and other assessments of an environment containing the Smart Device and/or Smart Receptacle. For example, a device with a controller and an accelerometer, such as a mobile Smart Device, may include a user display that allows a direction to be indicated by movement of the device from a determined location acting as a base position towards a feature in the intended direction where the movement results in an extended position. In some implementations, the Smart Device may first determine a first position based upon triangulation with the reference points and a second position (extended position) also based upon triangulation with the reference points.

As described above, facing a mobile device towards an area in a Structure and movement of the mobile device in a particular pattern may be used to ascertain a specific area in space to be interpreted by various methods.

Referring to FIG. 5C, an illustration of an Agent 550 utilizing an oriented stereoscopic sensor system 551 and 554 to orient a direction of interest is shown. The stereoscopic sensor systems 551 and 554 may obtain two different images from different viewpoints 552-553 which may be used to create topographical shape profiles algorithmically. A controller may obtain the image and topographic data and algorithmically compare them to previously stored images and topographic data in the general environment of the user. The resulting comparison of the imagery and topography may determine an orientation in space of the Agent 550 and in some examples determine a device field of view. The controller may utilize this determined field of view for various functionality as described herein.

Referring now to FIG. 6 an exemplary two dimensional (sometimes designated as "2D") representation of a structure 618 is illustrated as it may be generated in an user interface 606A on a Smart Device 606 operated by an Agent 600, such as a human user or an automation. The form of the interface may include and digital or electronic format that is able to be interpreted by an Agent. According to the present invention, a representation of a structure 618 may include digital content associated with one or more of IoT Tags 601-603; Virtual Tags 604, 607-615; and Hybrid Tags 605. A representation of a Virtual Tag 604 may include a Virtual Tag Icon 604A that symbolizes an interactive area on a user interface 606A that an Agent 600 may activate to access digital content associated with the Virtual Tag 604. Other Virtual Tags 607-615 may be integrated into image data presented in the user interface, as illustrated, such image data may include a 2D representation of a portion of a structure 618.

In some embodiments, a portion of the 2D representation will be displayed in a user interface 606A based upon a position 617 of the smart device 606 and a direction of interest 617A generated via wireless mechanisms and methodologies discussed herein. The wireless mechanisms and methodologies may include for example, one or more of: radio communications; magnetic readings (e.g., a compass reading); accelerometer readings; sonic readings; image recognition and the like.

In some embodiments, an Agent 600 may select one or both of: an area to be included in a user interface 606A; and a direction of interest via interaction with the user interface 606A. Interaction to indicate a direction of interest and/or an area to be included in a user interface display, may include, for example, dragging over an area of the user interface 606A; creation of a geometric shape, which may be a polygon or arcuate shape or combination of both; free hand movement; and selection of areas designation by a pattern of Virtual Tags 604 and/or IoT Tags 601-603 and/or Hybrid

Tags 605. In some embodiments, for example, one or both of a direction of interest 617A and an area or a representation of a structure 618 may be defined in whole or in part via a tag (virtual, hybrid and/or virtual) that define different zones, quadrants, rooms, architectural aspects, structure features, or even flood zones, seismic zones, animal sanctuaries, dangerous conditions, or other subject.

Digital content associated with one or more of IoT Tags 601-603, Virtual Tag 604, Hybrid Tag 605; may include almost any content representable via a digital value or pattern, such as, by way of non-limiting example, one or more of: narrative, text, video, image, audio, uniform resource locator, IP address alone or in combination with a port and/or socket; and ecommerce vehicle.

In some non-limiting examples that relate to construction site management, a worker may be on site and instead of having a series of physical documents that must be carried and updated and deciphered as to their meaning and revision level and applicability to a particular worker, the worker or other Agent 600 may simply carry a Smart Device 606 and based upon a position of the Smart Device and/or a direction of interest 617A, a user interface 606A may display information specific to the Agent 600 via one or more of IoT tags, Virtual Tags 604, 607-615, and Hybrid Tags 605.

For example, an interface 606A for an HVAC worker may include data specific to HVAC systems present in a building in which the HVAC worker is present (or which the HVAC worker selects) the HVAC data specific to HVAC systems may include data generated by an IoT Tag 601-603 with an electronic sensor that provides readings on one or more of: heat, humidity, water presence, air flow and the like. A plumber, electrician and/or a sheet rock worker may or may not be shown similar or same data on a user interface presented to them.

A Hybrid Tag 605 may include a time and position of an IoT tag 601 at a time previous to a present time period and digital content generated by the IoT Tag 601 at that time and position. As discussed herein, a position of the IoT Tag 601 may include a set of Cartesian Coordinates (e.g., X, Y, Z) and/or polar coordinates or cylindrical coordinates (including θ angle and radius). A position of a tag, including for example, one or more of: an IoT Tag 601; Virtual Tag 607-615; and a hybrid tag 605 may also be determined via positioning techniques, such as position within a point cloud, visual recognition, position relative to a known orienteering point 616 using one or more of: inertial measurement mechanisms and software in a smart device 600; magnetometers, magnetic positioning, dead reckoning, optical, sonic or other acoustic technologies.

In some embodiments, a smart device 600 may generate a known location via one or more of: triangulation, distance and angle determination, and proximity to or being located within a known location area 616. One or more of inertial movement tracking, augmented reality processing steps, magnetic positioning, Lidar, point cloud positioning, optical recognition positioning, received signal strength, sonic reckoning, acoustic reckoning, dead reckoning, or other tacking technology and method steps may be used to track movement of the smart device 600 as it traverses away from the known location area 616. In some embodiments, a trajectory 617 of a smart device 600 as it traverses through a known location area 616 may be tracked and the trajectory may be continued using the above techniques in order to further track a position of a smart device 600 in relation to a content tag, including one or more of: an IoT Tag 601; Virtual Tag 607-615; and a hybrid tag 605.

Accordingly, an augmented reality positional reference of a smart device **600** may be determined via one of, or a combination of multiple of: wireless communications, inertial measurement unit operation, magnetic position device operation, image location techniques, lidar, sonic device operation, laser location techniques in sequence or concurrent execution,

For example, in some embodiments, a smart device **600** may communicate wirelessly with one or more of a Bluetooth transceiver, a UWB transceiver, a RFID device or an optical or sonic transceiver as the smart device traverses a known location area **616** in order to reset a starting point for a device locating technology and method. In some embodiments, an Agent or other remote user may execute software that is operative with a controller to determine a location for the smart device **600** and thereby generate a known location for the smart device **600** which may act a start point for locating methodologies and mechanisms that do not rely upon triangulation based upon wireless communications or an angle of arrival and distance based upon wireless communications. Some embodiments may also include one or both of: periodic determination of a known location, and determination of a known location based upon an occurrence of a reset event. A reset event may include, by way of non-limiting example, an IMU reading, or other sensor measurement.

Some additional embodiments may include a known location via approach and traversing through a known orienteering area **616**. An approach may be determined via any of the methodologies and devices described herein, and/or traversing a path enabling communication via a series of wireless transmissions, such as, by way of non-limiting example a sequence of RFID communications or Bluetooth, UWB or light based communications.

Similarly, an electrician may be shown wiring and electrical fixtures on the electricians User Interface **606A** and a plumber may be shown piping and plumbing fixtures on the plumbers User Interface **606A**. Accordingly, a user interface **606A** may include content based upon an Agent that will interact with the user interface **606A** and the position **617** and direction of interest **617A** for the Smart Device **606**.

In some embodiments, digital content included in a user interface **606A** may be stored in a data receptacle (e.g., a data storage vehicle) and linked, such as, via a data file, relational database, index, ledger, blockchain, or other data association mechanism enabling specific digital content to be associated with coordinates that indicate a particular location, which may be referred to as a Tag location. In addition, the specific digital content may be associated with one or more Agents **600**. A Smart Device **606** may also be associated with the Agent **600**, and a position **617** and direction of interest **617A**. An Area of Interest, and/or Radio Target Area may be based upon the position **617** and direction of interest **617A** and an area shape and thereby include the location of the Tag. If the Area of Interest, and/or Radio Target Area does include the location of the Tag, then the digital content associated with the Agent may be included in a user interface **606A** presented on the Smart Device **606**.

Referring now to FIG. **6A**, a 2D user interface representation of a structure **620** is illustrated as it may be generated in a user interface **627A** on a Smart Device **627** operated by an Agent **600A**. Virtual Tags **621-626** are presented on the user interface at locations correlated with locations in the physical world that are represented via the 2D graphic, such as, in FIG. **6A**, an illustration of a floor plan of a structure **620**. As presented, the digital content includes narrative or

text, such as innovation area **621**, sound proofing **622B**, no use area **623**, shipping dock **624**, dock office A/C **625**, and mechanical room **626**. Such text may also act as an interactive device that may be selected to link out to additional content, such as video, audio, ecommerce and/or a URL link.

In addition, in some embodiments, activation of an initial interactive area represented as a Virtual Tag **621-626** may bring a user from a first user interface **627A** to a second or subsequent user interface. For example, a subsequent user interface may include information derived from a BIM model, AVM or other sophisticated resource normally requiring a proprietary interface.

In some specific embodiments, one or more of a Virtual Tag, IoT Tag and Hybrid Tag may be used to guide a robot or other unmanned vehicle (including, for example, one or both of an unmanned aerial vehicle and/or an unmanned ground vehicle or machine) to a location for performing an action; and/or confirm a location for performing an action. The action may be almost any procedure that the robot is capable of performing while deployed on a location. By way of illustration and non-limiting example, one or more of a Virtual Tag, IoT Tag and Hybrid Tag may be used to guide a robot or other unmanned vehicle may be guided to a person, such as a patient in an assisted living situation, rehabilitation, or medical care facility and assist the patient in a manner that the robot is designed. Wireless location and direction of interest as described herein may be combined with Virtual Tags and/or IoT Tag data to provide input to the robot as to a function to perform at a particular location for a particular patient.

In some embodiments, user interface representation of a structure **620** may additionally include an indicator **628A** of a current location of a smart device **627**. Preferably the indicator **628A** of a current location of a smart device **627** will be relative to a position **622** of one or more Virtual Tags **622**.

In another example, a construction site robot may be guided to a position **622A** within a structure **620** that corresponds with a Virtual Tag **622** and be provided with an instruction for an action to be conducted at the Virtual Tag position **622A**. The action may be an action achievable by a robot or other automation engaged to receive instruction from the Virtual Tag **622**. As illustrated, the instruction may include, for example to install sound proofing on the walls, or some subset of the walls the robot may access from the indicated Virtual Tag position **622A**.

In some embodiments, a Virtual Tag may be created remotely by associating coordinates with digital content. On location, the Virtual Tag may be accessible as would any other tag based upon the location coordinates, such as via an interactive user interface.

According to the present invention, multiple modalities of wireless position determination may be utilized to provide an increasingly accurate location of the smart Device position **628** and direction of interest **629**. For example, a satellite **630** communication modality, such as GPS may provide the smart Device position **628** determination accurate enough to determine an identification of a structure **620**; and a radio transceiving such as an UWB and/or Bluetooth modality may provide a Smart Device position **628** and direction of interest **629** and also a Virtual Tag position **622A**. Image recognition, LiDAR, laser, sonic, ultrasonic or other highly accurate modality may further guide a precise location for an action to be performed. By way of non-limiting illustration, in the construction site example, a radio transceiving such as an UWB and/or Bluetooth modality may provide a Smart Device position **628** and direction of

interest **629** may be used to indicate a particular wall to be worked on by a robot, and one or more of Image recognition, LiDAR, laser, sonic, ultrasonic may be used to specify a precise location of an action, such as where to drill a hole, or place a beam or board.

In some embodiments, the 2D user interface representation of a structure **620** may be generated in a user interface **627A** on a Smart Device **627** operated by an Agent **600A** and/or referenced to generate a dynamic two dimensional interface that includes a semblance of the floorplan and interactive areas at locations on the semblance of the floorplan. The links may be activated by a user to bring the user to one or more resources that make the digital content included in the Virtual Tag and/or IoT Tag and/or Hybrid Tag available to the user. By way of non-limiting example, a dynamic two dimensional interface may include a PDF document, word processing document, image document, html document, a URL page or other website, touchscreen or other user interactive artifact that align interactive areas with locations designated by the coordinates of one or more Virtual Tags and/or IoT Tags and/or Hybrid Tags.

Referring now to FIG. 6C, in some embodiments, the dynamic two dimensional interface **630** may be generated by capturing a screen image of the user interface at an instance in time and integrating into the captured image the user interactive areas with links to the digital content included in the one or more Virtual Tags and/or IoT Tags and/or Hybrid Tags. In some embodiments, the user interactive areas may be designated by an icon, highlight, boundary, color, pattern or other visual indicator, or portions of the image itself may be made interactive. For example, certain pixels included in an image may be interactive so that when selected they link the user to the digital content included in the one or more Virtual Tags and/or IoT Tags and/or Hybrid Tags.

In this manner, some embodiments may memorialize what one or more Virtual Tags and/or IoT Tags and/or Hybrid Tags are available as being associated with a particular physical setting and/or area at a designated time-frame.

In addition, the dynamic two dimensional user interface may be communicated from a first user with access to apparatus and software normally used to access the one or more Virtual Tags and/or IoT Tags and/or Hybrid Tags, to second user who does not have access to a particular app or other apparatus and software normally used to access the one or more Virtual Tags and/or IoT Tags and/or Hybrid Tags. The first and second users may be remote or co-located.

In some embodiments, the dynamic two dimensional interface **630** may be customized for a particular user, and/or a particular purpose. For example, a punch list for a trades worker may include items to be repaired or constructed within a particular trade (electrician, plumber, carpenter, metal worker etc.). Another example includes a dynamic two dimensional interface **630** with path or other directions for travel through a structure represented by the dynamic two dimensional interface **630** to bring a user to a particular destination. Accordingly, the dynamic two dimensional interface **630** may include a representation of a structure with additional content to support the functions described herein and in the related patent documents to which this specification claims priority to and incorporates by reference.

Referring now to FIG. 6B, in remote location areas, such as, for example, a remote construction site **631** and/or a remote wilderness location **632**; continuous communications with a distributed network, such as the Internet or a

VPN may be difficult. In such scenarios, a long distance communication, such as a transceiving communication **637-638** involving a satellite **630** and a physical Node, such as an IoT Tag **633-634**, may be conducted on a periodic basis.

In this manner, cost associated with such communications with remote locations may be controlled based upon a frequency of communications on a periodic basis.

In some embodiments, digital content may be transceived to an IoT Tag **633-634** and stored. Subsequently, an Agent **600B** may transceive between a Smart Device **627** and the IoT Tag **633-634** to retrieve the information stored in the IoT Tag **633-634** that was downloaded from the satellite **630**.

In another aspect, digital content generated on sites of remote locations **631-632** may be stored in a Node such as an IoT Tag **633-634** and transceived via a satellite uplink communication and from the satellite **630** conveyed to a distributed network, such as the Internet and/or a virtual private network.

Embodiments illustrated in FIG. 6B enable digital content to be generated and transceived and be waiting for an Agent **600B** to receive the digital content when the Agent **600B** and associated Smart device **627** are within transceiving proximity to a Node, such as an IoT Tag **633-634**.

As with other settings, an Agent **600B** in a remote wilderness location **632** may generate a Virtual Tag **636** by placing a Smart device **627** at a location **639-640** and associating a set of location coordinates with the location **639-640** of the Smart Device **627**. The agent **600B** may interact with a user interface **627B** on the Smart Device **627** to acknowledge the set of location coordinates and associate Agent specified digital content with the set of location coordinates. Similarly, an IoT Tag **633-634** may generate data, or other IoT Tag **633-634** digital content and associate the digital content with set of location coordinates.

In a case where the IoT Tag **633-634** subsequently changes its location, a hybrid tag **638** may be left as an artifact from the previous presence of the IoT Tag **633-634** at that location. While such a capability may be useful in many scenarios and locations, it may be particularly important at remote locations **631-632** to track a presence of an Agent **600B** at the remote locations **631-632** and information in the form of digital content that the Agent may be able to input into a system utilizing one or more of location based: Virtual Tags, IoT Tags, and Hybrid Tags.

While a Smart Device **627** is within remote locations **631-632** or proximate to such area of remote locations **631-632** (within transceiving distance to Reference point transceivers located therein); an Agent **600B** may transceive with Reference Point Transceivers **101-104** to generate location coordinates and perform the actions described herein, including accessing digital content via IoT tags **633-634**, or virtual tags **636**, and generating Virtual Tags **636, 638**.

Referring now to FIG. 6D, a method **640** presented as method steps **641-651** that may be executed in some embodiments of the present invention. The steps listed are exemplary and any, or all, of the method steps may be completed and may be practiced in the sequence presented or any other sequence that is appropriate for a given circumstance.

At step **641**, an initial position of a smart device may be automatically determined, or manually entered by an Agent. The smart device is preferably determined relative to an origin position associated with a building. For example, a position may be a Cartesian Coordinate relative to a coordinate of 0, 0, 0; or a position that is specified as a distance at an angle relative to an origin position. The initial position

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may be determined, for example via one or more of: triangulation based upon multiple distance calculations based upon wireless communications (e.g. ultrawideband, Bluetooth, including Bluetooth Low Energy “BLE”, WiFi, cellular or other communications); radio frequency (RFID) communication; near field communication (NFC); hash code; barcode; sonic or ultrasonic communication; infrared (or other light based) registration of a smart device and/or communication with a smart device.

For example, an Agent may enter an initiation area, such as passage through a doorway and a smart device may communicate via a wireless modality (including for example, one or more of: BT, UWB, RFID, sonic, or light transmission). The initiation area may be used as a start position for position tracking that is used to calculate a position relative to an origin position specified in building.

In some embodiments, an automated position determination process may execute an initial position determination. In other embodiments, an Agent may initiate an initial position determination via a user interface on the Smart Device.

At step 642, a position of the smart device may be tracked. Position tracking processes may be executed by the Smart Device commencing with the initial position determination. Some embodiments may also include an initial position determination be conducted on a periodic basis or called as a routine in the event of a smart device becoming disoriented and position tracking may resume following the determination of the new “initial” position. Disorientation may be determined, by way of non-limiting example, by incongruity between captured imagery and stored image based records of an environment.

For example, a point cloud, or other pattern record, may be associated with a location. A smart device software routine may access an internal IMU or other device to predict a location of the Smart Device. However, if captured image data from a Smart Device position is not validated by a point cloud associated with the IMU location, then a software routine may access a process to determine an initial position and begin position tracking again from the determined initial position.

Position tracking processes may include, for example, one or more indoor positioning routines that may access one or more of: inertial measurement unit (sometimes referred to as “IMU”) sensor-based systems, accelerometers, magnetometers, LiDAR, image sensors (including smart device cameras); wireless communication systems employing time difference of arrival (TDOA); frequency difference of arrival (FDOA), two way ranging (TWR); time of flight (TOF) or other combination of electronic devices and software processes.

At step 643, in some embodiments, one or both of an Agent and a smart device automated process may determine a height designation (e.g., a floor of a building, an elevation above sea level, a height above a ground level) of a start position. The height designation may be determined, by way of non-limiting example, one or more of: barometric pressure, wireless triangulation, angle and distance of arrival or departure of a wireless transmission; Agent input, RFID, NFC, hash mark, barcode, sonic, infrared registering movement of the smart device relative to the start position.

At step 644, an Agent may specify an area of interest relative to the smart device. For example, an area of interest may include an area in front of an Agent as the Agent traverses along a path. Other areas of interest may include an area at an angle to the Smart Device.

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At step 645, a location of a smart device may be determined based upon operation of an inertial measurement unit (IMU) included in the smart device. Preferably, the IMU will be accessed to determine a position of the Smart Device while the Smart Device traverses a known orienteering area, such as, for example a floor of a building, a building campus, or craft.

At step 646, positional coordinates may be associated with a tag linked to digital content. Digital content may be active content, such as a link, code, video, or other content that leads to an action. An action may include, by way of non-limiting example, one or more of: bringing a user interface to another resource or virtual location, execution of software code, ecommerce, or playing an audio or video segment.

Some specific examples of digital content associated with a building may include: Primary Systems such as:

Construction Site Data, such as, for example, one or more of:

- a. Gas;
 - b. Electrical (switchgear, transformers, motors, etc.);
 - c. Fire Protection (pumps, tanks, hydrants, PRVs, etc.); and
 - d. Wastewater (pumps, tanks, equipment, etc.).
- Structural Data, such as, for example, one or more of:
- a. Footings/Foundations (location, size, depth, isolated foundations, equipment pads, etc.);
 - b. Slab (thickness, etc.);
 - c. X-bracing/Wind Girts (location, size, etc.);
 - d. Hardened Rooms (i.e., Tornado Shelter, etc.); and
 - e. Cranes (location, type, rating, etc.).
 - f. Architectural Data, such as, for example, one or more of: Doors (location, type, rating, size, hardware, etc.); Finishes (floor topping, curbs, wall type/color, ceiling type/color, etc.); and safety equipment (eye wash, emergency showers, fire extinguishers, rated walls, fire penetrations, etc.).

Mechanical Data, such as, for example, one or more of:

- a. Plumbing (floor drains, door foamers, grease traps, neutralization pits, etc.)
- b. Fire Protection (risers, valves, etc.)
- c. HVAC (RTUs, EFs, supply/returns, systems as a whole, etc.)
- d. Utilities (boilers plus, air compressors plus, HW sets, valves, etc.)
- e. Refrigeration (compressors plus, condensers plus, chilled water pumps, valves, etc.)
- f. Safety (fire dampers, etc.)
- g. Electrical Data, such as, for example, one or more of: Electrical Rooms (MCCs, etc.); Lighting (fixtures, controls, breakers, etc.); motors; battery chargers; outlets (120V, 240V, Welding, equipment, etc.); safety (emergency lighting/wall packs, etc.).

Availability of digital content may be indicated on a user interface on a Smart Device via an icon, symbol, highlight, border, outline, change of color, or another Agent ascertainable indication.

Links to the digital content may be used to archive documentation related to equipment, architectural aspects, change orders, construction takeoffs or other material lists, change orders, as built versus design specification, manufacturer manuals or other documentation, warranty information, operation videos, service videos, maintenance videos, or other instructional videos, checklists, procedural checklists, maintenance and/or operation records, as built imagery,

as built videos, as built audio record, records of proximity of an Agent (e.g. a particular Agent came within a particular distance to a location).

At step 647, in some embodiments, one or both of an Agent and software may specify whether a physical tag or virtual tag is associated with the digital content.

At step 648, in some embodiments, it may be determined that the positional coordinates of the digital content are within an area of interest. As discussed herein, an area of interest may be indicated via one or more of: interaction with a user interface; a position of a Smart Device; a position and direction of a Smart Device; and a record (e.g., a database) of digital content and positions associated with specific portions of the digital content.

At step 649, in some embodiments, it may be determined that an Agent and/or an associated Smart Device has access rights to the digital content. Access rights may include, by way of non-limiting example, multi-factor authorization, including whether an Agent is positioned within an area authorized for receipt of the digital content.

At step 650, a software routine may receive a user input via a dynamic portion of a user interactive interface on the Smart Device, the user input may cause a controller to be operative to cause the Smart Device to display the digital content.

At step 651, based upon the user input received into the dynamic portion of the user interactive interface display the digital content in the user interactive interface.

By way of non-limiting example, illustrative example, a process of designing and conveying information to on site workers may be improved as compared to presently available methods by associating portions of a design with an onsite area in which the design is to be manifested as a constructed building. For example, details describing architectural aspects, such as walls, doorways, windows, archways, beams, trusses, electrical requirement, equipment, plumbing, fixtures, hardware, and the like may be quantified as digital content and associated with positional coordinates. Each positional coordinate will represent an onsite position. Design features associated with the position may be made available via one or more of: an onsite position of an Agent; an onsite position of a smart device; a position selected on a user interactive interface, such as a replication of a floorplan; or other area descriptor.

During construction a worker, such as a tradesperson, may provide as built records, such as one or more of: imagery, documentation, sensor readings, audio recordings, or other records indicating what was actually built, or what transpired during construction, who was present, who competed work, certification of operation, or other data quantifying events that occurred on site.

In installed available according to the position of the equipment. Documentation for the equipment may include one or more of: operators manual; technical documentation; warranty information, commissioning details, including time and date of placing equipment in operation; sensor information, such as, equipment sensors and/or IoT sensors.

Relatedly, sensor data, such as, equipment sensor data, and/or IoT sensor data may be made available based upon location of the sensors and/or an area or item of equipment monitored by the sensor.

A two dimensional user interface such as those illustrated in FIGS. 6A-6C may indicate the presence of available digital content and be available to one or both of an on site Agent 600; and on remote or offsite Agent (not illustrated in FIGS. 6-6D). In this manner, the on site Agent 600 and an offsite Agent may exchange digital content and also access

historical records that are congruent. Work orders may be created and issued that are placed in positional context and accessed via that positional context or via traditional search queries. Likewise reporting and record keeping may be accessed in positional context and/or aggregated, summarized, tallied, and referenced in context of other dealings.

Accordingly, in some embodiments, digital content placed in positional context of a two dimensional reference, such as a design plan or floor plan may be accessed as a construction effort is being concluded and change orders that were completed during the course of the construction effort may be catalogued in the digital content such that each change order may show its positional context, which workers were involved, what timeframes the workers were involved, materials involved in the change orders, as built records of the change orders and any other relevant digital content involved in the change orders.

Additionally, some embodiments may include digital content that is associated with maintenance and operations. Work orders may be quantified in manners similar to change orders, with onsite personal and remote users accessing congruent data about the work order. Ecommerce may include links to consumables that are placed in positional context. For example, one or more of: filters; consumables; toners; inks; lights, lubricants; matching paints; matching floor coverings; matching wall coverings; matching hardware; matching building components; and/or complimentary items in similar or same categories may all be made available with links to vendors that sell the items.

In addition, in another aspect, qualitative indicators and/or ratings made be made available in positional context, such as, by way of non-limiting example, a consumable, such as a filter may be rated based upon user satisfaction, particulate size that is filtered, suggested filter life, environmental impact and the like, and/or simply rated via alphanumerical value or good, better best rating. Accordingly, a user may order a new filter via selecting a position of an equipment item and indicating that they want to order the best available filter from a selected vendor. The user will only need to indicate a position of the equipment item and not need to be aware of a make, model, year of manufacture, or other equipment details traditionally required to order replacement parts and/or consumables. A position may be as simple as one or more of: indicating that it is the equipment in a particular corner of a room portrayed on a floor print; or the equipment in front of a worker standing in the room on site.

Similarly, a user may access service records based upon a position on a floor plan, or via a position relative to an onsite Agent.

Referring now to FIG. 7 a flowchart illustrates some exemplary process steps that may be executed according to the present invention. At step 701, a structure may be identified, such as, for example via triangulation using one or both of satellite and cell tower communications. Other methods of identifying a structure may include RFID as an Agent enters the structure or is within radio range of the RFID circuit. Still other methods of determining an identification are to establish a known location within the structure (see step 702) and thereby determine that the Agent is within the structure associated with the know location.

At step 702 a location is designated via the wireless processes and using the apparatus discussed herein.

At step 703, a set of coordinates is associated with the designated location. Coordinates may include any grouping of alphanumerical values that is capable of indicating within a required accuracy range a location of the Agent. A required

accuracy range may be contingent upon a modality of wireless communication used to generate the coordinates.

At step **704**, a direction of interest from the designated location is generated. The direction of interest may be generated, for example, via generation of a vector intersecting a position of a Smart Device supported by the Agent. Generation of the direction of interest may reference, by way of non-limiting example, locations of reference point transceiver, angle of arrival of radio spectrum energy; angle of departure of radio spectrum energy; values generated by a magnetic sensor, image data recognition processes, LiDAR, ultrasonic measurements, and the like.

At step **705**, a shape of a directional area associated with the direction of interest may be designated. The shape of a directional area may include for example an area congruent with a radio a target area or other geometric area. Non-limiting examples of geometric areas may include areas definable as a frustum; conical area; an area with a square profile; an area with a rectangular profile, an area with another polygonal profile; an area with a circular profile, an area with an oval profile; and an area with a complex geometric shape profile, combining linear, arcuate and angular aspects.

At step **706**, digital content is designated that is accessible from the designated location in the direction of interest, and in some embodiments within the area of interest.

At step **707**, an interactive user interface is generated that is accessible via a smart device or other automated device having a controller and a display.

At step **708**, user interactive areas within the interactive user interface are associated with coordinates, and at step **709**, the coordinates are associated with specific digital content.

At step **710**, the Agent will activate a user interactive area associated with specific coordinates such that at step **711**, the digital content associated with the coordinates is presented on the user interface. Other actions may also be triggered by the Agents activation of an interactive area, such as control of an item of equipment, activation (or deactivation) of an alarms state, generation of an electronic message, or almost any action activatable via an electronic console.

Referring now to FIGS. **8A-8G**, exemplary Wireless Communication Areas (WCA) and Radio Target Areas (RTAs) are illustrated. In general, a WCA is an area through which wireless communication may be completed. A size of a WCA may be dependent upon a specified modality of wireless communication and an environment through which the wireless communication takes place. In this disclosure (and as illustrated), a WCA may be portrayed in a representative spherical shape; however, in a physical environment, a shape of a WCA may be amorphous or of changing shape and more resemble a cloud of thinning density around the edges. In general, an RTA is an area from which an energy-receiving Sensor will receive energy of a type and bandwidth that may be quantified by the energy-receiving Sensor. The RTA shape and size may be affected by an environment through which the energy must be conveyed and further effected by obstructions.

Referring now to FIG. **8A**, a side view illustrates a WCA **800** surrounding a Node, such as a Smart Device **802**. Energy **803**, which is illustrated as rays, is received by one or more energy-receiving Sensors **804** in the Smart Device **802** (energy-receiving Sensors may also be in a Smart Receptacle associated with the Smart Device, though this is not illustrated in FIG. **8A**). An exemplary ray may proceed from a position **805** within RTA **801** boundary to the energy-receiving Sensor **804**.

As illustrated, a portion of the RTA **801** may flatten out **801a** in response to a ground plane, wall, partition, or other obstruction encountered. A Node **806** may be located on or within a surface that makes up a relevant obstruction and the Node **806** may appear to be along a perimeter of the RTA **801**. Similarly, a Virtual Tag may be associated with location coordinates that appear on or within a floor, wall, partition, or other article acting as a radio frequency obstruction and thereby appear to be a part of the obstruction, however, since it is virtual, the Virtual Tag will not affect the physical properties of the obstruction. Essentially, a Virtual Tag may have location coordinates that correspond to anywhere in the physical real-world. In some examples, a software limit or setting may limit location coordinates of Virtual Tags to some distance from a base position or a distance from a designated position, such as a location of a designated Physical Tag, Reference Point Transceiver, or other definable position.

In addition to obstructions, a topography of an environment within an RTA **801** may also limit wireless conveyance of energy within an RTA **801** to an energy-receiving Sensor **804**. Topography artifacts may include, for example, a terrain, buildings, infrastructure, machinery, shelving, or other items and/or other structures that may create impediments to the receipt of wireless energy.

Energy **803** received into the energy-receiving Sensor **804** may be used to create aspects of a user interface that is descriptive of the environment within the RTA **801**. According to the present invention, environmental aspects, Nodes **806**, Tags (both physical Tags and Virtual Tags) and the like may be combined with user interactive mechanisms, such as switches, or other control devices built into a user interactive device and included in a user interactive interface. For example, energy levels received into an energy-receiving Sensor **804** may be combined with location coordinates of Physical Tags and/or Virtual Tags and a user interactive device may be positioned in an interactive user interface at a position correlating with the position coordinates and be surrounded with a visual indicator or the received energy levels.

In this manner, a single user interface will include a static image representative of received energy levels at an instance in time; a visual representation of a location(s) of Physical and/or Virtual Tag(s), and devices with user interactive functionality. In some embodiments, the devices with user interactive functionality may be positioned at a location in the user interactive interface correlating with the position(s) of the Physical and/or Virtual Tag(s).

This disclosure will discuss RTAs **801** that are frustums of a generally conical shape, however, RTAs **801** of other volume shapes are within the scope of the invention. For example, if an energy-receiving Sensor **804** included a receiving surface that was a shape other than round, or had multiple receiving surfaces, each of a round or other shape, the RTA **801** associated with such an energy-receiving Sensor may have a shape other than a frustum of generally conical shape.

Referring now to FIG. **8B**, a top-down view of an RTA **801** is depicted. An RTA **801** will include some portion of a WCA **800**. As illustrated, the WCA **800** includes a space with irregular boundaries encompassing 360 degrees around the Smart Device **802**. Aspects such as topography, strength of signals and atmospheric conditions (or other medium through which a wireless communication will travel) may affect and/or limit a perimeter of the WCA **800**. A location of the RTA **801** may be referenced to determine which Tags (Physical and/or Virtual) such as Node **806** are included

within an interactive user interface. Generally, preferred embodiments may only include Tags with location coordinates with the RTA **801** in the interactive user interface. However, embodiments may include Tags external to the RTA **801** in a particular interactive user interface.

Referring now to FIG. **8C**, a side view of a WCA **800** is presented where an energy-receiving Sensor **804** is capable of quantifying a particular form of energy, such as a particular bandwidth of energy received from a user selected RTA **807**. A Smart Device **802** may incorporate or be in logical communication with multiple energy-receiving Sensors **804**, each energy-receiving Sensor capable of quantifying a limited energy spectrum in an environment defined by the RTA **807** selected by the user.

Some embodiments include an RTA **807** that varies according to a type of energy-receiving Sensor **804** receiving a corresponding type of energy. For example, an energy-receiving Sensor **804** that receives energy in a lower bandwidth may have an RTA **807** that extends a greater distance than an energy-receiving Sensor **804** that receives energy in a higher bandwidth. Similarly, some energy-receiving Sensors **804** may be affected by forces outside of the RTA **807**, such as a magnetometer which may be sensitive to signal interactions around all of the WCA **800**, and an RTA **807** associated with a magnetometer may accordingly be the same as the WCA **800**.

By way of non-limiting example, an RTA **807** for a CCD-type energy receiver may be represented as a frustum with an expansion angle of approximately 60 degrees in shape. Accordingly, the RTA **807** subtends only a portion of the universal WCA **800**.

Referring now to FIG. **8D**, a top view of a WCA **800D** is illustrated with an RTA **807A** comprising a frustum with an expansion angle of approximately 60 degrees. A Smart Device **802** with an energy receiver that quantifies a specified bandwidth of energy from the RTA **807A** may generate a user interface with an image based upon energy quantified from RTA **807A**.

In FIG. **8D**, the WCA **800D** is represented as a spherical area. A WCA **800D** may be designated that is less than an entire area of possible radio communication using a specific designated wireless communication modality. For example, WCA **800D** may be spherical and stay within boundaries of a modality based upon a UWB wireless communication protocol.

A user interface based upon quantified energy in an RTA **807A**, may present a representation of energy within the respective RTA **807A** as quantified by an energy-receiving Sensor in a Smart Device **802**. Energy levels of other three-dimensional spaces within the WCA **800D** may be quantified by energy receivers and presented in a user interface by directing energy from a selected three-dimensional space into the energy receivers and thereby defining a different RTA **807A**. In this manner, energy levels may be quantified from essentially any area within the WCA **800D** and represented as part of a user interface. Quantified energy levels may vary based upon a receiving Sensor. For example, a CCD Sensor may quantify visible light spectrum energy, and a LIDAR receiver a broad spectrum, an infrared receiver may quantify infrared energy levels, and energy-receiving Sensors. A particular feature present in a particular portion of the electromagnetic spectrum quantified by an energy-receiving Sensor may have a unique physical shape which characterizes it, and which may be associated with a corresponding virtual-world aspect and Tag associated with the location.

In some examples, as has been described, quantification of energy levels associated with aspects of the physical world may be for one or more of: characterizing an RTA **807A** by quantifying energy levels and patterns existing at an instance in time; determining a location and/or orientation of a Smart Device **802** or other Node, such as Node **806**; and verifying a location and/or orientation of a Smart Device **802**. Various embodiments include energy levels associated with aspects of the physical world may be communicated by the Smart Device **802** to a remote controller for further processing, and the remote controller may communicate information back to the Smart Device or to another user interface. Information communicated from the controller may include, for example: an orientation of physical and/or virtual aspects located within the universal RTA in relation to the Smart Device; and quantified energy indicating of or more of a topographical feature, a surface temperature, a vibration level, information associated with a Virtual Tag, information associated with a physical Tag, sensor data, or other information associated with the RTA **807A**.

A view of an RTA **807A** (Radio Target Area) may be a relatively small portion of the entire wireless communication area (WCA) that surrounds a Smart Device. An area of energy to be quantified by a sensor (sometimes referred to herein as the Radio Target Area) may be displayed surrounded by the WCA **800D**.

Referring now to FIG. **8E**, an exemplary presentation of an RTA **812** superimposed upon a representation of a WCA **811** is illustrated. The WCA **811** is illustrated with a perspective view of a spheroid with an alignment feature **814** such as a spheroid dividing arc, or a line. A blackened ellipsoid feature is a representation of the RTA **812** associated with a particular Smart Device which would be located at a center of the spheroid WCA **811**. If desired, one or more energy receiving devices associated with or incorporated into a Smart Device may be repositioned or have a changed orientation in space to ultimately scan all of the accessible universal Radio Target Area space.

Referring to FIG. **8F**, an illustration of how moving the one or more energy receiving devices around in space may alter an area defined as the RTA **813**. The same orientation of the universal WCA **811** may be noted by a same location of the alignment feature **814**. Relative movement of the ellipsoid feature illustrates a change in an area designated as RTA **813**.

Referring to FIG. **8G**, an illustration of adding Tag locations (which may be Physical Tags or Virtual Tags) to a mapping of the WCA **811** is provided. A Tag may be represented in the WCA, for example, as an icon (two- or three-dimensional) positioned in space according to a coordinate system, such as Cartesian coordinates, polar coordinates, spherical coordinates, or other mechanism for designating a position. Coordinates may specify one or both of physical real-world Tags and Virtual Tags.

A location of a real-world Tag or Virtual Tag may be in either RTA **813**, the WCA **811** or external to both the RTA **813** and the WCA **811**. Examples of Tags outside the RTA **813** and within the WCA **811** include Tags **815-819**. An example of a Tag in the device RTA is Tag **820**. A Tag located external to of the WCA **811**, and the RTA **813** includes Tag **821**.

In some examples, a display on the user's Smart Device may illustrate image data captured via a CCD included in a Smart Device. Portions of the image data captured via a CCD may be removed and replaced with an icon at position correlating to a position in space within the RTA **813**. The icon may indicate of a Tag **821** located within the RTA **813**,

or at least the direction in the RTA **813** along which the Tag **821** may be located at an instance in time. In addition, an area of a user interface portraying the Icon may user interactive device such that when the device is activated, the Smart Device is operative to perform an action.

The actual positions of the Tags in real-world space (or the digital equivalent in the real-world space) may be stored and maintained in a database. Positions of physical Tags may be determined via techniques based upon wireless communication and be updated periodically. A period of update may be contingent upon variables including, user preference, Tag movement, change in environmental conditions, User query or other variable that may be converted into a programmable command. In another example of some embodiment, an Agent may interact with a user interface and understand the presence of Tags that are outside of the RTA **813** and adjust one or both of a position and direction that the Smart Device to cause the Smart Device to be positioned such that the RTA **813** encompasses a position of the Tag of interest.

Referring to illustration FIG. **9A**, an exemplary apparatus for effectuating the methods described herein is shown, wherein Smart Device **901** has within its Radio Target Area **905** a Structure **906**. Smart Device **901** may display a user interface **902** based upon data generated by an energy-receiving Sensor **903** incorporated into the Smart Device **901** or operative in conjunction with the Smart Device **901**. The energy-receiving Sensor **903** may produce data representative of an area from which the energy-receiving Sensor **903** received energy. A user interface **902** may be generated that is based upon relative values of some or all of values for data variables produced by the energy-receiving Sensor **903**.

Smart Device **901** may have its position and direction of orientation determined using the orienteering methods described herein, with reference to one or more Reference Point Transceivers **908 A-B**. The position may be determined relative to a Base Node, such as Reference Point Transceiver **908A**. The Base Node may operate as an origin in a coordinate system associated with Structure **906** and its surroundings. A position-determination process may be aided with reference to transmitter **907**, which in some embodiments, may be a Reference Point Transceiver. In this example, transmitter **907** is positioned proximate to the Structure **906**.

A receiver on Smart Device **901** may be operative to receive a wireless logical communication from transmitter **907**. This communication may be in one of a variety of modalities, such as Bluetooth, ultra-wideband (UWB), radiofrequency, infrared, ultrasound, etc. Based upon the signal, Smart Device **901** may transmit a database query based upon a determined set of coordinates of transmitter **907**, a set of coordinates of the Smart Device **901**, the RTA **905**, or other position and direction relevant variable.

If the database contains an entry comprising a set of coordinates (as a data structure) and the set of coordinates define a point within displayable distance to the set of coordinates of transmitter **907**, then the user interface **902** may display an interface that includes an interactive area **904** that manifests a Virtual Tag (such as an icon or other definable area) in context to a virtual structure **906A** representative of the physical structure **906**. In this way, a user of Smart Device **901** may be alerted to the presence of information associated with structure **906** in which the user may be interested.

In some embodiments, inclusion of an interactive area **904** on the user interface **902** may be contingent upon an Agent operating the Smart Device **901** presenting appropriate

credentials and/or permissions to access digital content made accessible via the interactive area. Still further appropriate credentials and/or permission may be required to ascertain that an interactive area exists. For example, an image displayed on a user interface may include imagery descriptive of item of equipment or a person. A user with proper credentials may be presented with a user interface that includes an interactive area **904** that manifests a Virtual Tag, such as an icon or outline of imagery descriptive of an item of equipment or a person; while a user who has not presented proper credentials may not be made aware of the existence of such an interactive area, nor the content included in the Virtual Tag associated with the interactive area **904**.

In another aspect, in some embodiments, an interactive area **904** may only display if Smart Device **901** is in active communication with a specified Wi-Fi network, or if the Smart Device **901** is in communication with at least one of a specified Node or Nodes. Communication with a Node may include, for example, wireless communication via a wireless modality, such as, one or more of: UWB; Bluetooth, infrared, sonic, or other modality discussed. In other embodiments, interactive area **904** may display on any user interface **902** (if the RTA **905** includes transmitter **907**), but further functionality may be based upon successfully responding to a security challenge. A security challenge may include, for example, a biometric measurement, inputting a password, correctly input an answer to a question, a gesture made with the Smart Device, a gesture made in communication with a sensor integrated within or with a Smart Device (such as, for example, motion of hand(s) in front of a camera, or motion of a hand wearing a Smart Ring or a wrist wearing a Smart Wristband).

In some embodiments, the appearance of interactive area **904** may change based upon variables, such as, one or more of: the position of the Smart Device **901**; the identity of user or Agent; if the interactive area **904** is related to a query and/or query response; if the interactive area **904** is within an RTA **905** or based upon some other dynamic. For example, if the user has a certain UUID, and the database includes a message specifically intended for a user with that UUID, then the interactive area **904** or an icon may flash to indicate the presence of a message. This message may be displayed textually, visually, audibly, or by a hologram. Similarly, the database may record one or more instances in which the Smart Device **901** is accessed via a query from a Smart Device. Access via a query may be associated with a time stamp. If data related to structure **906** has changed since a previous time stamp, then interactive area **904** may be presented in a highlighting color (such as, for example be presented in red or other color) to indicate that a change has been detected. In addition, digital content may be appended to any content already in the database, such as additional alphanumeric annotation, an audio file, an image file, a video file, or a story file.

In some embodiments, in response to activation of an interactive area **904** (such as a click, screen touch, voice command, gesture, etc.), additional functionality may be provided via the Smart Device **901** or other item of equipment. For example, selecting interactive area **904** may display digital content related to Structure **906**. Alternatively, activating the interactive user device associated with interactive area **904** may generate a control panel, which may allow the user to control aspects relating to sensors or other electronics within structure **906**. For example, upon confirmation that Smart Device **901** has the appropriate permissions, selecting interactive area **904** (or other activa-

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tion of the interactive area **904**) may allow the user to turn off the lights within structure **906**. Still other embodiments allow for activation of an interactive area **904** to be a prerequisite to operation of equipment located within the RTA **905** or other defined area.

An interactive area **904**, may be incorporated into a user interface in almost any manner conducive to a user activating the interactive area **904**. For example, a user interface that recreates a visual of a physical area, such as, by way of non-limiting example: an image (or video) based upon a CCD sensor array; a two dimensional representation of a physical area (such as a floorplan, site plan or architectural drawing); and a three dimensional representation (such as a CAD model, AVM; or Augmented Reality model) may include interactive areas that include areas of the image data, integration of one or more of: an icon, an outlined image area, a highlighted image area, an image area with a changed appearance (e.g. change in hue or color), integration or overlay of an image (e.g. a logo, emoticon, or other device).

The Smart Device **901** may also display other functional buttons on its user interface **902**. In some examples, one such function may be to show displays of the sensor in the context of the universal RTA **905** surrounding the user. By activating the functional button, the user may be presented with a set of options to display the universal RTA **905**.

According to the present invention, an interactive area **904** may be used to retrieve digital content, and/or to store digital content for subsequent retrieval. Digital content may be associated with one or more sets of position coordinates (e.g., cartesian coordinates, polar coordinates, and/or cylindrical coordinates). A user interface **902**, AVM and/or two dimensional representation of a structure or geospatial area may be produced that allows the digital content to be accessed based upon the associated coordinates. The retrieval of the digital content is persistent in the sense that it may be retrieved, and new digital content may be added for so long as an underlying infrastructure enabling determination of coordinates used to access and/or place digital content within a coordinate framework exists. Referring to FIG. **9B**, nonlimiting and exemplary apparatus and methods for presenting a user interface including an interactive area **904** is illustrated. The display screen of the Smart Device **901** may display a number of informational digital content components. In some embodiments, a similar illustration as FIG. **8G** may be included as an inset **923** of the user interface **920**. In addition, in some embodiments, a user interface **920** may include a representation of the interactive area **904** may be formed by flattening the surface of the illustrated sphere **925** into a flat depiction with surface regions flattened into a segment **921**. The interactive area **904** may be illustrated on the flat segments. A user interactive area **920** (which may be an icon, highlighted area, outline, portion of an image, or other defined area) may be located within the user interface representing the interactive area **904** and structure **906A**. The interactive area may also be included in the real time display of a representation of data generated by an energy-receiving Sensor **922**. Tags may be located within or outside of the interactive area **904** such that an Agent may move the Smart Device **901** to redirect an interactive area **904** to align the interactive area **904** into a position that encompasses Tag **924** the Agent wishes to include.

Referring to FIG. **9C**, a nonlimiting exemplary user interface **937** generated on a display screen **930** for Smart Device **901** is illustrated. The user interface **937** may be displayed, for example, when a user selects an interactive area associated with a Tag **931-932**.

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The Tag **931-932** may be located at a physical location within or outside of the RTA **932** (is illustrated). Selection of the Tag (sometimes referred to activating the Tag **931-932**), a menu **933** may display. Amongst the various information such as text, imagery, video content and the like that may be displayed an identification of the Tag **934**, associated textual information and data **935** as well as functional buttons **936** may be displayed on the user interface and may be used by the user to activate additional function including new display layers, content integration and control function such as in a non-limiting sense a control to revert to a previous menu display.

In some examples, a Smart Device may function as a Tag. The Tag functionality may include providing location-related information as broadcasted digital content. In providing such broadcasted digital content, the Smart Device tab may employ numerous forms of security protocols for the protection of the information and authorization of its use which may include sign-in/password protocols, sharing of encryption keys and the like. In similar methods, a central server may provide content related to a Tag and may manage security protocols and the like where a Smart Device acting as a Tag may merely share an identification phrase that a user could use with applications running or connecting with the central server could use to be authorized for additional content. Location may be determined by the various means as described herein including wireless communication with position Nodes by GPS, Cellular, Wi-Fi, Ultrawideband, Bluetooth and the like. If the Smart Device is operating in a mesh Node, the mesh could communicate within Nodes relative and absolute location information which the Smart Device may share as its role as a Tag. In addition to location, other sensor data at the Smart Device such as temperature, vibration, sensor imagery, LiDAR scan imagery, sound sensing.

In addition to real-world data, the Smart Device Tag may also provide virtual content associated with itself and its connected environment. The Smart Device may provide content stored within its memory devices and may provide dynamically calculated results of processing on content stored in its memory devices. The virtual content may also correspond to a user interface of the Smart Device Tag that may be used to initiate or authorize function of the Smart Device including real-world activities such as a communication via internet protocol, text, phone, or video.

In some embodiments, an energy-receiving Sensor may receive energy associated with a LiDAR transmission and/or other functionality involved in LiDAR scanning which can be used to interrogate the local environment for physical shapes. In a Smart Device Tag function, the Smart Device may stream its video and scanning data directly or through a server model. Some Smart Devices may be configured to operate as a smart security monitoring systems and may provide the video, topographic, audio, and other sensor streams as Tag related content. There may be numerous manners that a Smart Device could function as a Tag in an environment.

A Smart Device with either a single- or multiple-sensor system may also have a LiDAR scanning capability or other three-dimensional scanning capability. The Smart Device may utilize a number of systems to refine and improve its accuracy in determining the location that it is at. In an example, a Smart Device may utilize a GPS or cellular system to get an approximate location of the device. In a next step, a user may initiate the Smart Device to take a series of image and scanning data acquisitions of its environment. For example, the user may move the phone by

hand to different directions while maintaining their feet in a fixed location. The phone may use one of the orientation methods as have been discussed to determine its orientation as it is moved to different vantage points. The Smart Device may either process those images and compare against a database in its memory, or it may communicate the data to a server to do the comparison. With an approximate location, the orientation information, and the streams of video and/or topographic information, a calculation may be performed to match the image/topographic information to a more exact positional location. In alternative examples, the device may use the image and/or topographic information to determine the orientation of the device itself.

In some examples, the Smart Device may act as a receiver of one or multiple types of wireless energy input. For example, the acquisition of data based upon a visual light spectrum (approximately 380 to 700 nm wavelength) may be modelled as spatially-characterized electromagnetic energy. Electromagnetic energy in the visible band may enter a focusing lens and be focused up an array of devices. The devices may be CMOS-active pixel sensors, CMOS back-illuminated sensors, or CCDs, as non-limiting examples, to receive the energy and convert it into spatially-arrayed pixel data.

In some examples, the Smart Device may have an energy-receiving Sensor incorporated or attached which may quantify energy levels for frequencies outside the visible spectrum. Any optics employed in such sensors may be different from the previously discussed CMOS and CCD Sensors since some of those energy receiving devices may have filters or lenses that absorb wavelengths outside of the visible spectrum. Sensors with infrared capabilities may have specialized optics and may use different materials for the CMOS and CCD elements—such as indium gallium arsenide-based sensors for wavelengths in the regime of 0.7-2.5 μm .

Alternatively, entirely different sensing elements, such as bolometers, which sense temperature differences of the incoming radiation, may be employed for longer wavelengths in the regime of 7-14 μm and may include filters that remove other wavelengths. A display of an infrared Sensor, which senses incoming energy in the infrared band, may be rendered on a typical visual display, but the colors of such displays may have no direct physical meaning. Instead, a color scheme may be instituted to represent different infrared wavelengths with different visible colors. Alternatively, the colors may be used to represent different intensities of infrared energy received across bands of infrared wavelengths.

In some examples, a Smart Device may both project and receive energy. For example, a Smart Device may scan the topography of its surroundings by use of LiDAR. In LiDAR, a laser may be used to emit energy into the environment. The energy may be emitted as pulses or continuous trains, and the light source may be scanned across the environment. Light emitted from the Smart Device may proceed into the environment until it is absorbed or reflected. When it is reflected and subsequently received at the Sensor, the transit time can be converted to distance measurements of the environment. Many different wavelengths of light may be used to scan an environment, but numerous factors may favor certain choices such as invisibility to human/animal eyes, safety, absorption by the airspace surrounding the user and the like. Atmospheric gases may absorb significant amounts of infrared transmissions at certain frequencies; therefore, for LiDAR to be effective in the infrared spectral region, certain bands of emitted frequencies may be favored.

A standard LiDAR system may operate at a band from 900-1100 nm infrared wavelength or at a band centered at approximately 1550 nm. As discussed previously, select optic components and materials may be useful for these wavelengths and the detectors may have improved function based on materials such as “black” silicon, germanium, indium phosphide, gallium arsenide, and indium gallium arsenide as exemplary detector materials.

In an example, a laser light source may be rastered across a dimension of forward looking positions of a Smart Device, which may be represented by a conic section or Radio Target Area in front of the Smart Device. As the light is raster across the surface it can address, it may be pulsed on or off. As the light travels out along a collimated path, it may interact with a surface and a portion of the intensity may be reflected backwards.

A resulting reflected ray may come back to the Smart Device and be received by a Sensor in the device. Since the emitted light source may be orders of magnitude more intense than the surroundings, reflected light may dominate a background intensity and the signal detected may be compared with the time of the leading edge of the laser pulse. The repeated acquisition of the timing signals in the various directions can be used to form a point cloud that represents the distance to reflective features from the Smart Device.

As mentioned previously sound may be reflected off of surfaces and the transit time may be used to characterize a distance between a focused ultrasonic transducer and a reflective surface. In similar manners, points or lines of focused sound emissions may be pulsed at the environment and a sensor or array of sensors may detect the reflected signals and feed the result to a controller which may calculate point cloud representation or other or topographic line representations of the measured surface topography. In some examples, ultrasonic focused and scanned soundwaves in the frequency range of hundreds of megahertz may result in small, focused sources whose reflections may be detected by magnetic or piezoelectric sound transducers as non-limiting examples.

A Smart Device may have numerous different types of energy-collection devices which may characterize data values with spatial relevance. As mentioned before, infrared imaging may be performed on some Smart Devices, and a user may desire to view a spatial representation of the infrared imaging that represents the data as it may appear if the user’s eyes could perceive the energy. In some examples, data values for the wireless energy sensing of infrared energy may be assigned color values and displayed in an image format. For examples, low levels of infrared energy, which may relate to colder temperatures in the imaged regions, may be assigned blue color values, and high levels of infrared energy, which may relate to warmer temperatures, may be assigned red color values. Other color assignments to data values may be used. A legend for the conversion of the color values to the data values may be provided.

In some examples, the data descriptive of spatially descriptive energy levels quantified by an energy-receiving Sensor data may be portrayed in a user interface. In some user interfaces, representations based upon spatially representative energy levels of different wavelengths may be aggregated or otherwise combined in one or more related user interfaces. Such a combination may allow a user to understand the regional nature of various quantified energy.

In some examples, a user interface may allow for display of the positional location image points. In some examples, a location of a pixel element chosen by a user may be

converted to a real-world location within the RTA which may be represented in Cartesian coordinates (X, Y, Z) or in other coordinate systems such as polar coordinate systems involving angles and distances as discussed previously. In some examples, topographic data obtained by scanning an area with an RTA may be used quantify topography within the RTA. A user interface based upon such quantified energy levels may include virtual presentations of the quantified energy levels from different perspectives and may allow for coordinate grids (Cartesian or other) to coordinate placement of facets of a user interface based upon combinations of energy level data, Tag locations and perspective relevance.

In some examples, distinct structures within the RTA may be highlighted and assigned positional coordinates. In some examples, this may occur by image processing directly, in other examples a user interface may allow for a user to pick items/regions of interest in an RTA presentation.

In other examples, real and virtual Tags may exist within the RTA. A physical Tag may include a position Node, another Smart Device, or any device with communication capability that can communicate with either a position Node or with the Smart Device of the user directly. Such physical Tags may be located in numerous manners. In some examples, the physical Tag may have a direct determination of its location either because it is stationary and has been programmed with its location or because it has the capability of determining its own position with the various methods as have been described herein. In other examples, a physical Tag may be able to communicate with Nodes such as Reference Point Transceivers and a location may be determined based upon an exchange of data, such as timing values, in the wireless communications. A Node may also be functional to determine, store and communicate a location of other Tags. The Smart Device of the user may gain access to the locations of Tags, either because they are publicly available or because the user has established rights digitally to obtain the information from some or all of these physical Tags.

There may also be virtual Tags that are associated with positional coordinates. The distinction of these Tags over physical Tags is that there may be no physical presence to the virtual Tag. It may be a digital or virtual-world entity that has an association with a real-world positional coordinate. Except for this distinction, a virtual Tag and a real-world Tag may behave similarly with respect to their association with a physical coordinate.

In these examples, an interactive user interface based upon energy levels and Tags located with an RTA may have icons associated with the placement of Tags. The user interface may include an icon positional designation and a graphic to indicate the presence of a Tag. It may be apparent that, in some cases, multiple Tags may lay along a single direction from a given Smart Device location and RTA, and thus multiple icons may be included within a user interface in close proximity. The user interface may indicate multiple Tag icons by color changes, blinking or other indicators. As an RTA is changed, Tags along a same perspective may resolve into different directions for Tags with different positional coordinates.

The Tag icon may indicate to the user a digital functionality associated with a real-world or virtual Tag. For example, the icon may allow a user to choose the functionality of the icon by moving a cursor over the icon and making a keystroke or mouse click or for touch screens by pressing the display location of the Tag icon. The choosing of the Tag icon may activate user interface dialogs to allow

the user to control subsequent functionality. In cases of superimposed Tag icons on a same pixel location in a user display, a first functionality may allow the user to choose one of the multiple Tag icons to interact with. In some examples, a Tag icon may be displayed with an associated ID/name and a user may select the icon with voice commands rather than physically selecting the icon as described previously. Displays of these Tags may follow similar protocols as have been discussed in reference to FIGS. 9A-9C.

Referring now to FIG. 10A, a method for generating an augmented-reality Radio Target Area for a Smart Device is shown. At step 1001, wireless energy of a first wavelength is received into a wireless receiver. In exemplary embodiments, this step may include receiving image data based on visible light into a sensor of the Smart Device. The wireless energy may be dispersed over a one-, two-, or three-dimensional space in a defined physical area, and may be received into a one-, two-, or three-dimensional array in the receiver. The wireless energy may take the form of electromagnetic radiation, such as light in the human-visible light spectrum (generally having a wavelength between 380 nm-740 nm), ultraviolet light (generally having a wavelength between 10.0 nm-400 nm), or infrared light (generally having a wavelength between 740 nm-2.00 mm) as examples. The set of wireless energy available to the wireless receiver is the Smart Device's Radio Target Area.

The wireless receiver may be a Smart Device sensor, including a CMOS active pixel sensor, a CMOS back illuminated sensors, CCD, or a LIDAR apparatus, including a solid-state/MEMS-based LIDAR. The wireless receiver may comprise an array or other plurality of other wireless receivers. The wireless receiver may be operative to receive the wireless energy into an array of an appropriate dimension for subsequent display (possibly after processing) on the Smart Device. For example, where the wireless receiver is a Sensor, the Sensor may be operative to translate the wireless energy into a two-dimensional array.

At step 1002, a pattern of digital values is generated based upon receipt of wireless energy into the wireless receiver. This pattern of digital values may be based on one or more qualities of the received wireless energy, including its intensity, spatial dispersion, wavelength, or angle of arrival. The pattern may be placed into an appropriate array. For example, if the display of the Smart Device is a two-dimensional display, then the pattern of digital values may comprise a two-dimensional representation of the image data received. In some embodiments, the pattern of digital values may be based on an aggregated set of values from an array of receivers. For example, if the basis of the digital values is the intensity of the wireless energy received into the receiver, then the digital value assigned to a given entry in the array may be based on a weighted average of intensity of wireless energy received at a plurality of the receivers in the array. Optionally, at step 1003, the wireless receiver may receive the wireless energy as an analog signal (for example, if the wireless receiver is a black-and-white sensor or an unfiltered CCD), and convert the analog signal to digital values through filtration or other analog-to-digital conversion. The set of digital values within the Radio Target Area is the Digital Radio Target Area.

With the Smart Device wireless receiver's Radio Target Area determined, the Smart Device's position should be determined as well, along with the positions of any items of interest in a given space. Collectively, the Smart Device and the item of interest may comprise wireless Nodes. Accordingly, at step 1004, coordinates representative of a wireless Node may be determined relative to a base Node. These

coordinates may be determined in any appropriate coordinate system (such as Cartesian, polar, spherical polar, or cylindrical polar) and may be determined via RTLS or the orienteering-triangulation methods with various wavelengths or modalities, such as ultra-wideband, Bluetooth, etc. Additionally, the coordinates may be determined using an angle of arrival or angle of departure of a signal to or from the base Node, along with the distance from the base Node. By way of non-limiting example, this could produce a dataset that correlates the coordinates of three elements with the identities of those elements: $\{(0,0,0), \text{BaseNode}; (1,1,1), \text{SmartDevice}; (2,2,2), \text{ItemOfInterest}\}$. While this example may be used throughout the following discussion, it is understood to be non-limiting, as a given space may include a plurality of items of interest. Note that, in some embodiments, the Smart Device itself may become a dynamic database entry with a continuously (or periodically) updating set of coordinates. This may be useful in allowing a plurality of Smart Devices engaged with the system at the same time to interact with one another.

At step **1005**, the position of the Base Node is determined relative to the defined physical area. In exemplary embodiments, this may include establishing the Base Node as an origin in the coordinate system and determining vectors from the Base Node to boundaries and items of interest (i.e., the distance from the Base Node and the direction from the Base Node to the boundaries and items of interest). In some examples, the Base Node may have an established reference relative to a global coordinate system established.

At step **1006**, a Target Area is generated within a controller of the Smart Device. The Target Area may be the set of coordinates (relative to the Base Node) within the Radio Target Area of the wireless receiver. The Target Area may be limited by physical boundaries of the given space, such as walls, floors, ceilings, occlusions, etc. The Target Area may also be limited by distances that various types of signals may travel. For example, a sensor of audio signals may not be able to practically pickup signals over a background noise level that originate more than 1000 feet from a user position, purely as an example. In such a case, the Target Area for such signal types may be limited to that dimension.

At step **1007**, respective positions of one or more wireless Nodes within the Target Area are determined. These positions may be determined relative to the physical Target Area or the Radio Target Area. The determination may be made with reference to the dataset discussed at step **1005**, or it may be made dynamically based upon one or more Base Nodes and/or the Radio Target Area. Moreover, the determination may additionally be based on receipt of a wireless signal into the Smart Device from the wireless Node. This signal may indicate a position using the orienteering methods described herein.

At step **1008**, a user interface may be generated on the Smart Device based upon the pattern of digital values generated at step **1002**. The user interface may comprise a plurality of pixels, wherein each pixel comprises a visible color based upon the pattern of digital values generated at step **1002**. For example, if the digital values were based upon receipt of visible light into the wireless receiver (e.g., a sensor), then the display may reflect a reasonably accurate color photograph of the Radio Target Area of the wireless receiver. If the digital values were based upon an intensity of received light from, for example, LIDAR, then the display may reflect a scan of the Radio Target Area. In some embodiments, the pixel may include an intensity of energy received into the receiver. In this way, aspects of the Radio Target Area characterized by an intensity of energy may be

emphasized. For example, this may produce a LIDAR relief of an area or a heatmap of an area.

At step **1009**, an icon may be generated in the user interface. Preferably the icon will be placed at a position relative to data quantifying received energy levels. In some embodiments, the icon location in a user interface will be indicative of a position of a Tag (Virtual or Physical). This position may be quantified via positional coordinates, such as Cartesian Coordinates, Polar Coordinates, Spherical Coordinates, and the like. The icon may be based upon an input from a user, stored data, quantified environmental conditions or other criteria related to an aspect of the Radio Target Area.

For example, an icon may indicate information about an Item of Interest located at a given set of coordinates within the Radio Target Area or Digital Radio Target Area. In another embodiment, the user may indicate on the display a position in which the user wishes to place an icon and add information about an Item of Interest (thus creating a new entry in the database, which may be populated with the coordinates of the indicated position). Moreover, the icon may change colors based upon the pattern of digital values. The icon may be overlaid on top of the display. The icon may resemble the letter "i", a question mark, a thumbnail, or any other suitable image from a library. In some embodiments, the icon may change depending on one or more attributes of its corresponding database entry. For example, if the icon located at (4,4,4) relates to a restaurant menu, then the icon may resemble the letter "i" or a thumbnail of a menu. On the other hand, if this database entry is modified so that the corresponding database entry is a message, then the icon may update to a picture of an envelope.

In some embodiments, the icon-generation step may be based upon an inquiry to a database that uses the Digital Radio Target Area as an input. For example, upon generation of the Digital Radio Target Area, an associated set of coordinates in one or more dimensions may be generated. This may then be submitted to a database. An associated display may be as illustrated in FIG. 9A. In some embodiments, the icon-generation step may be based upon an inquiry to a database that uses the user's position coordinates as an input. In these embodiments, both the Digital Radio Target Area based on an RTA as well as the universal Radio Target Area may be included in an inquiry submitted to the database. An associated display may be as illustrated in FIG. 9C. In some examples, the user may have an option to limit or filter the types of database entries that may be queried for, such as in a non-limiting sense, the existence of real-world Tags, virtual Tags, sensor data values and streams from a particular class of sensors and the like.

Continuing with the example from step **1004**, the Digital Radio Target Area may comprise the set of coordinates: $[(1.5,10), [1.5,10], [1.5,10]]$. In this example, the database may return information about the Item Of Interest, but not about the Base Node. The Digital Radio Target Area may update when the Smart Device position changes, or by user input, the Digital Radio Target Area may remain static after a certain instance in time.

Continuing with FIG. 10B, at step **1010**, the icon may be positioned in the user interface at a given position based upon coordinates representative of the position of the wireless Node or Tag in the Target Area. This may comprise a selection of a multitude of pixels related to the position of the wireless Node or Tag and changing those pixels from the digital values determined at step **1002** (check ref #) to a second set of pixels to indicate the presence of an icon. In some embodiments, the icon may be dynamically updated

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based upon movement of the Smart Device (and, accordingly, the wireless receiver). In some embodiments, the icon may be permanently associated with a set of coordinates. In such embodiments, the icon may be generated whenever a Smart Device with appropriate permissions includes in its Radio Target Area the set of coordinates of Nodes or Tags associated with the icon.

At step **1011**, user-interactive functionality may be associated with the pixels comprising the icon. This may allow the user to “select” the icon by means of an input device (e.g., mouse, touchpad, keyboard), touchscreen, digital input, etc. Upon selection, the icon may be operative to interact with the user in one or more ways, including: displaying a message intended for the user (by text, audio, video, hologram, etc.); requesting credentials from the user to verify permissions (e.g., a password), displaying information about an item associated with the icon, prompting the user to update information about an item associated with the icon, etc. The user-interactive functionality may display static information (e.g., dimensions of the item), display dynamic information (e.g., an alarm state or sensor information relating to the item; for example, if the item is a refrigerator, internal temperature may be displayed), or produce a control panel that allows the user to issue control commands (e.g., remotely operating an automated apparatus by resetting an alarm state, taking remedial action based upon a sensor state as described herein, etc.) or to issue menu control commands such as to invoke a different user interface or screen of a user interface.

This may be useful in geospatial applications, or in procedurally generated activities. For example, a first user may generate a positional designation on a user interactive device, such as, for example an augmented-reality display to leave a narrative, icon or other input associated with the first use. Additionally, the same or another user may log positional coordinates and upload an image that could be displayed submitting a database query including those coordinates. Entry of the coordinates and essential credentials may provide access to the content associated with the positional coordinates.

At step **1012**, the preceding steps may be integrated by generating a display comprising the user interface, the icon, and at least some of the associated user-interactive functionality. In embodiments, in which a plurality of Smart Devices are themselves part of the database, this may allow various users to send messages, images, etc. to each other.

At step **1013**, detection of movement of the Smart device may cause a branch back to step **1005**. Based upon that movement of the Smart Device, a defined physical area from which wireless energy is received (i.e., the Radio Target Area based upon the Target Area) may be changed. The movement may be detected using input from wireless communications, magnetic field sensors, an accelerometer, feature-recognition software, or other similar apparatus and algorithms. In other examples, the position of the Smart Device may be dynamically obtained using any of the techniques of position determination, such as triangulation with reference nodes. Here, too, a change of position detected in this manner may cause a branch back to step **1005**. The Target Area may be based upon the position of the Base Node, the relative positions of the wireless Nodes, and the Smart Device.

Referring now to FIG. **11**, an exemplary database structure usable in conjunction with the present disclosure is shown. In this non-limiting example, the database has five sets of information: coordinates **1101** associated with an action, permissions **1102** associated with the action, the

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action type **1103**, attributes **1104** for the action, and notes **1105**. The example shown in FIG. **11** may suppose the following: the augmented-reality system is deployed in an enclosed space, definable by a coordinate system set relative to a Base Node having an origin point (0,0,0); the enclosed space spans, in that coordinate system, ([0, 10], [0, 10], [0, 10]) (using traditional set notation; in other words, each coordinate can take on any number between 0 and 10, inclusive); and the Radio Target Area is ([1, 10], [1, 10], [1, 10]).

The bolded entries in the database shown in FIG. **11** represent the database responses to the query given by the Radio Target Area of the Smart Device, i.e., all entries having a Coordinate value within the Radio Target Area. In some embodiments, the database may sort through all coordinates within the Radio Target Area and then return any entries for which the Smart Device has appropriate permissions. In other embodiments, the database may sort through all entries for which the Smart Device has appropriate permissions and then return any entries with coordinates within the Radio Target Area. The latter approach may be beneficial in circumstances in which there are numerous database entries with varying permissions; for example, if a database has 10,000,000 entries, but a given user might only have access to five of those entries, sorting by permissions first may be more beneficial.

The ActionType variable may include any action for which interactivity with an icon may be desirable. In FIG. **11**, the ActionType variables shown are Information, Message, Action, and Directions. Each of these represents functionalities within the scope of this disclosure. For example, Information may relate to information that the Smart Device user may find helpful. Continuing with the shop example from FIG. **9A** (check Fig Ref #), Information may include store hours, discounts, reviews, etc. Similarly, Message may be a message to the general public (e.g., an announcement), or a message tailored to a specific user. In the latter case, permissions may operate to ensure that only the specific user (or set of users) may access the Message.

Action may relate to any action that a sensor, electronic device, or other apparatus connected to the database may take. For example, Action may include changing a temperature, measuring a temperature, turning off lights, activating an emergency sprinkler system, opening a door, etc. In some embodiments, prior to taking the Action, a password may be requested as part of the permission check.

Directions may show a user how to navigate (using, in exemplary embodiments, orienteering methods) from the current position to a desired position. For example, upon scanning an entry on a map, virtual arrows may be generated to guide the user to a chosen store.

The ActionAttributes may have attributes based on the ActionType. For example, if the ActionType is Information or Message, then the ActionAttributes may be a text string or a stored audiovisual file containing the message. Similarly, if the ActionType requires a sensor or other electronic device to take an Action, then the ActionAttributes may include a command or subroutine to affect such an Action. In the example shown here, the ActionType Directions comprises an ActionAttribute that includes a command to the Smart Device (i.e., show directions in the form of green arrows).

Referring to FIG. **12**, an illustration of alternative methods for display of information relating to RTA is provided. At the beginning of the process, a system of components which may include a smart device with a user of the smart device may be established. Amongst the various components

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a Home Position may be established for all the components at step **1201**. The system may proceed by establishing and initiating transceiving of data and information at step **1202**.

In some examples, the user may be prompted to choose a desired coordinate system for the display at step **1203**. In other examples, a user interface of the system may have a setpoint function which the user may invoke to gain access to user settable parameters which may include they type of coordinate system to use, such as for example Cartesian or spherical coordinates.

In still further examples, the system may decide to default to a particular coordinate system depending on the nature of the type of data its positional reference devices may be obtaining or providing.

At step **1204**, if the coordinate system was chosen as Cartesian coordinates, the system may utilize triangulation amongst multiple reference point transceivers. Alternatively, at step **1205** if the coordinate system was chosen as polar coordinates, the system may utilize positioning systems that utilize angles and distances involved in transceiving and location. In either event, at step **1206**, the position of a Sensor attached to the smart device of the user may be determined. In some examples, the system may have multiple and redundant location system. A combination of such position determinations may result in superior accuracy of an aggregated position result. Accordingly, at optional step **1207**, a wireless position determination may be performed with the smart device to establish a verification of the position of the Smart Device and the Sensor attached. Referring now to step **1208**, a direction that the sensor is facing in may be determined. Although there may be a number of different manners of determining orientation as have been described herein, in an example, the orientation may be determined based upon wireless transmission and/or wireless verification.

Referring now to step **1209**, an energy-receiving Sensor included in the Smart Device or in logical communication with the Smart Device may be used to quantify energy levels perceivable at the position and in the direction of the Smart Device. The resulting quantification may depend on aspects of the Sensor device, but the resulting data will quantify a characteristic for the RTA.

In some embodiments, an optional step **1210** may be performed by an element of the system such as the smart device or a server in communication with the Smart Device. The element of the system may compare one or more of position information, orientation information and the image data itself to calculate an estimate of whether the RTA angle has changed for the sensing element.

In general, at step **1211**, the RTA of the Sensor device used to capture the image in step **1209** may be quantified. In an optional step **1212**, coordinates relating to the instant RTA of the image may be established. In some examples, this may relate to a range of three-dimensional coordinates that are addressed by the RTA of the Sensor element. In general, at step **1213**, the system may look up, or in some cases generate, location coordinates for Tags that are determined to be within the quantified RTA. In some database systems that the system may have access to, real-world or virtual-world tags may be tracked in a coordinate system with a certain origin.

If the current origin established at step **1201** is offset from a particular database related origin, then one or both the coordinate system values may be converted to each other to align their respective origins. At step **1214**, the Tags in an aligned coordinate system may have their positions com-

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pared to the current RTA and a selection for the set of Tags that are within the RTA may be made.

In some alternative examples, a display of all Tags that are authorized for access to the user regardless of whether they are in the RTA may be made using associated aligned coordinates as discussed in reference to step **1213**.

Referring now to step **1215**, in an example, the Smart Device of the user may be used to generate and display a user interface to the user based upon the captured image and the associated tag icons within the RTA. These associated Tag icons may have at least the functionality as has been discussed in reference to FIGS. **10A** and **10B**.

Referring now to FIG. **13**, a Smart Device **1301** is illustrated within a wireless communication area (WCA) **1302**. The extent of the particular WCA **1302** may be defined according to a select bandwidth and/or a particular modality of the wireless communication the Smart Device **1301** uses to transmit and receive information.

For example, bandwidths may include those associated with UWB, Wi-Fi, Bluetooth, ANT, ultrasonic, infrared, and cellular modalities of communication. In general, unless otherwise constrained by physical modification such as the use of a directional antenna, or the presence of radio frequency interference from a physical object (such as objects with significant metallic content; objects with high water content; electrical fields; etc.), a WCA **1302** may include spherical area(s) emanating from one or more transceivers and/or transceiver antennas operated by the Smart Device **1301**.

As discussed extensively herein, and in patent applications referenced by this application, the location of the Smart Device **1301** may be determined based upon wireless communication to and/or from the Smart Device **1301**; and described via a coordinate system, such as via generation of Cartesian coordinates, or other coordinates such as: polar coordinates, spherical coordinates, and cylindrical coordinates. Modalities of wireless communications that may be referenced to generate location coordinates may include one or more of: RTT (round trip time), time of flight, RSSI (received signal strength indicator); angle of arrival, angle of departure, and other methods, equipment and modalities as have been described herein.

With the location of the Smart Device **1301** determined, a location of the WCA **1302** may be extrapolated based upon the location of the Smart Device and a range or transceiving distance the Smart Device may be capable of.

According to the present invention, a portion of the WCA **1302** may be selected as a radio target area (RTA) **1312** from which the Smart Device **1301** may receive specific bandwidths of electromagnetic radiation. In preferred embodiments, the RTA **1312** may include a frustum expanding outward in a conical shape from one or more energy-receiving Sensors **1309** included in the Smart Device **1301**. The frustum shaped RTA **1312** may overlap with a portion of the generally spherically shaped WCA **1302**. Other shapes for an RTA **1312** are also within the scope of this specification.

In some embodiments, a shape of the RTA **1312** may be based upon receiving capabilities of the one or more energy-receiving Sensors **1309** incorporated into or in logical communication with the Smart Device **1301**. For example, an energy-receiving Sensors **1309** with a charge coupled device (CCD) or complementary metal oxide semiconductor (CMOS) receiver may have a single plane receiving surface and be best matched with a frustum of a generally pyramidal or conical shape. Whereas an energy-receiving Sensors **1309**

with multiple receiving surfaces (e.g., with multiple CCD and/or CMOS devices) may be arranged to enable a more complex shaped RTA **1312**.

In some preferred embodiments, a direction of interest **1311** may intersect the RTA **1312**. As discussed herein, the direction of interest **1311** may be represented by a ray or vector **1311A** and **1311B**. In addition, the direction of interest **1311** may be represented as a direction of interest area, such as a frustum defined by multiple rays or vectors, **1311A** and **1311B**. In various embodiments, the direction of interest **1311** area may encompass the RTA **1312** or be a subset of the RTA **1312**.

A direction of interest **1311** may be determined for example via the methods and devices described herein and in referenced patent applications and may be associated with a direction based upon a ray or vector indicative of a direction of interest **1311**, a direction based upon a magnetic field sensor, an accelerometer, a light beam, correlation between two Tags or Nodes, Agent gestures, or other Smart Device recognized apparatus and/or method.

One or more transceivers **1303-1305** (typically included within a Smart Device, Tag, or Node) may be located within an area defined by the RTA **1312**. According to the present disclosure, a position of the transceiver **1303-1305** may be determined and a user interactive mechanism may be generated at a position of the transceiver **1303-1305** within a graphical user interface emulating aspects of the RTA **1312** on the Smart Device **1301** or another user interactive interface screen (not shown, and perhaps at a site remote to the RTA **1312**).

According to the present disclosure, some portion of the RTA **1312** (which may include the entirety of the RTA **1312**) may be portrayed on an Agent interface **1310**, including, in some embodiments, a human-readable graphical user interface (GUI). The interface **1310** may include a representation **1313** of a particular level of electromagnetic energy received via the energy-receiving Sensors **1309** and associated with a particular area of the RTA **1312**. For example, energy levels of an infrared wavelength that has emanated from or reflected off of an item in the RTA **1312** and received via an infrared receiver in the Smart Device **1301** may be used to generate a heat map type interface display. Similarly, energy that has emanated from or reflected off of an item in the RTA **1312** in the 400 nm to 700 nm range and been received via a charge-coupled/or CMOS image sensing device in the Smart Device **1301** may be portrayed as a human visible image of items in the area included in the RTA **1312**.

Other embodiments may include a point cloud derived from electromagnetic energy bouncing off of or emanating from items included in the RTA **1312** or a series of polygons generated based upon a LIDAR receiver in the Smart Device **1301**. An Agent interface **1310** may be presented in a modality understandable to an Agent type. For example, an interface presented to a UAV or UGV may include a digital pattern and an interface presented to a human Agent may include multiple pixels or voxels generating a pattern visible to a human being.

The wireless location methods and apparatus described herein may be deployed in conjunction with one or more Transceivers **1303-1305** or Tags and/or Nodes **1306-1308** located with the WCA **1302** to generate location coordinates for the one or more Transceivers **1303-1305** or Tags and/or Nodes **1306-1308**. A controller or other device operating a processor may determine which one or more Transceivers **1303-1305** or Tags and/or Nodes **1306-1308** located within the three-dimensional space included in the RTA **1312** based upon a) the location of the one or more Transceivers

1303-1305 or Tags and/or Nodes **1306-1308**; and b) the location of area included in the RTA **1312**.

In another aspect of the present disclosure, in some embodiments, some energy levels may not be represented in the Agent interface **1310**. For example, in some embodiments, energy levels reflected off of a particular item may not be included in the Agent interface **1310**. Other embodiments may only represent energy levels that have reflected off of selected items within the RTA **1312** thereby emphasizing the presence of the selected items and ignoring the presence of other items within the RTA **1312**.

As described above, some portion of the RTA **1312** may be portrayed on an Agent interface **1310**, including, in some embodiments, a human readable graphical user interface (GUI), as a point cloud derived from electromagnetic energy bouncing off of or emanating from items included in the RTA **1312** or a series of polygons generated based upon a LIDAR receiver in the Smart Device **1301**. An example of such a representation is shown in FIG. **14**. In this example, the GUI includes a human visual image **1401** of an RTA **1400** overlaid with a series of polygons **1402** generated based upon a LIDAR receiver in the Smart Device. The LIDAR sensor illuminates the RTA **1400** with laser light and then measures the reflection with a sensor. The resulting polygons **1402** represent differences in laser return times, which provides a topographical representation of objects in the RTA **1400**.

In this example, Virtual Tags **1403** and **1404** are created by the Smart Device by methods described herein and icons may be present on the GUI to identify the position of the Virtual Tags **1403** and **1404**. The Virtual Tags **1403** and **1404** may, for example, represent various locations of interest in the RTA **1400**, such as an object of interest or an exit or entrance. The icons associated with the Virtual Tags **1403** and **1404** may be engaged or “clicked” or otherwise activated to be made operational; for the Smart Device to receive (e.g., retrieved from a database) additional information associated with the object or location of interest.

For example, if the object of interest is a statue, clicking on the icon associated with the Virtual Tag **1403** associated therewith may provide information regarding the statue, such as the history, origin, and the like. If, for example, the Virtual Tag **1404** is associated with an exit of the room, clicking the Virtual Tag may provide information on what is present in the adjacent room, or where the Smart Device is in relation to exiting the building, or any other desired information.

In some embodiments, mathematical data associated with a LIDAR rendering, such as parameters of triangles formed by various LIDAR points **1405-1406** within an associated RTA may be stored and a relative position of a smart device with the RTA **1400** may be determined based upon the recognition of similarities of the LIDAR point **1405-1406** patterns. A resolution of laser scanning involved in generation of data based upon LIDAR techniques may influence a number of data points within a selected RTA, but in general, pattern recognition and determination of an orientation of a smart device based upon LIDAR data may be much more efficient, than, for example image data based pattern recognition. In addition, the LIDAR based patterns may be formed in a “fingerprint” of an RTA, wherein it would be very rare, if not impossible to replicate the LIDAR point patterns at two disparate locations. Therefore, recognition of a point pattern may be used to identify a location of a particular RTA.

Referring now to FIG. **15** a multi-modal tag (MMT) **1500** with multiple electronic sensors is illustrated. The MMTS

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1500 will include a processor, such as a processor in a controller 1510 that will in turn include storage for executable code and input/output communication means. The controller 1510 will have the ability for logical communication with multiple wireless communication transceivers 1501-1506 and one or more electronic sensors 1507 and a storage device 1508 for storing digital data. The controller 1510 may also manage power allocated to devices or components 1501-1509 included in the MMTS 1500. The power may be supplied via a power source 1511.

The controller may cause one or more transceivers 1501-1506 and/or a GPS receiver 1509 to be operational during a given time period. Operation of the one or more transceivers 1501-1506 and/or a GPS receiver 1509 may be based upon a function being undertaken, such as, for example a function being undertaken may include one or more of: communication to provide variables useful for real time location services or operation of a sensor to quantify a physical condition. The block diagram of FIGS. 15 and 16A-16C illustrate components that may be included in some preferred embodiments of an integrated multi-mode tag (MMT). The MMT combines components for conducting wireless communication suitable for real time location service functionality such as transceivers 1501-1506, with components functional to quantify one or more environmental conditions as digital content (such as, for example, IoT sensors), and with data aggregation functionality (such as, for example, onboard digital memory and/or cloud based data aggregation), to enable improved levels of real time location based augmented reality.

Referring now to FIGS. 17A-17C, method steps are illustrated that may be performed in some implementations of the present invention. During operation, executable software may be operative with the controller to cause the MMT to perform one or more of the following method steps, the order of the steps is not limited by the listing and may be performed in various sequences. In some or all of the steps may be performed and still be within the scope of the invention. By way of non-limiting example, an illustrative scenario of how an MMTS may be deployed includes:

- 1) An MMTS may be included with a shipment of items being assembled in a warehouse to be delivered to a work site. The MMTS power will be turned on. A power on mechanism may include a wireless communication, such as, via Bluetooth, Zigby, UWB, magnetic interaction, cellular or another wireless modality, or a mechanical and/or electronic switch.
- 2) A MMT may include a primary communication modality, such as a modality enabling relatively more accurate location services than a secondary modality. Examples of a primary modality may include: UWB, Bluetooth, Zigby, ANT, ultrasonic and other relatively shorter range bandwidth, and communication protocols. A secondary modality may include GPS, cell tower triangulation, subGHz and other relatively longer range bandwidths and communication protocols.
- 3) Power management for an MMT may follow a life cycle of an item being tracked. For example, an item of equipment to be included in a structure being constructed (or upgraded or maintained) at a worksite may have an onboard power supply, such as a battery that is powered on as the item is included in a bill of material (sometimes referred to as a "BOM"). The battery will power wireless communication receivers, transceivers, and transmitters prior to installation of the items at the worksite. In some embodiments, the battery may also provide power to sensors used to quantify conditions

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encountered by the item of equipment during shipment and storage before installation at the worksite. Upon installation of the item of equipment, the MMT may be connected to a more permanent power source, such as inhouse AC or DC power, power over ethernet, or other hardwired power connection.

- 4) The MMT may seek to communicate via a primary communication modality and for so long as the primary modality is operational and functioning, a secondary (or tertiary) form of wireless communication may remain powered off. This is in part based upon the secondary modality will be redundant and may not provide as accurate location determination.
- 5) As a primary communication modality fails, such as when an item is shipped out of a warehouse and beyond the range of a warehouse based UWB wireless communication, a secondary communication modality, such as GPS, may be engaged, including the provision of power. A GPS receiver may be provided with power to receive GPS signals from multiple satellites. So long as GPS communications are received, the GPS receiver may be provided with power on a constant or periodic basis. However, if an obstruction, such as an interior of a warehouse location blocks GPS reception, power to the GPS receiver may be discontinued until an "event" causes the power to be restored. An event may include, for example, one or more of: an onboard accelerometer indicating movement of the MMTS; a cellular transceiver be involved in transceiving that indicated a change in position of the MMTS, such as a change in values for triangulation between cell towers and/or a transition of cell towers in communication, change in a time of travel from a cell tower or other change.
- 6) During storage, while en route, while located at a work site but not installed, and during an installed state, digital content may be associated with the MMTS and stored for tracking and historical purposes, as well as for AI analysis. In some embodiments, the digital content will additionally be associated with location coordinates of the MMTS at the time of association of the digital content or the generation of the digital content (such as, via a Virtual Tag and/or Hybrid Tag). In some embodiments, location coordinates and/or sensor data associated with other MMTSs within a specified distance of a primary MMTS (such as within wireless communication range, or within a distance indicative of an interior of a shipping box or other shipping container) may also be associated with the Primary MMTS and stored. In this manner, it may be determined whether a BOM is complete within a single shipping container or box or if other items are in additional boxes.
- 7) Upon occurrence of an event, such as arrival at a destination (e.g., within a geofence associated with a worksite) or a sensor reading that exceeds a threshold, a Virtual Tag and/or Hybrid Tag may be manually or automatically generated and associated with the MMTS thereby generating historical data of what an item associated with the MMTS has experienced. By way of non-limiting example, upon opening of a shipping box, a user may take a picture of the contents of the box and an environment in which the box was opened, and the pictures will become a part of a Virtual Tag that indicates that this box was opened at these RTLS coordinates and this image captured the contents of the opened box. Sensor readings or other data, such

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as an audio message, a video or narrative may also be included in the Virtual Tag.

- 8) An icon may be generated for a Physical Tag, Virtual Tag and/or Hybrid Tag according to the teaching contained herein and related patent applications. In some embodiments, icons may indicate a communication modality referenced to determine the location coordinates associated with the icon. For example, an icon based upon coordinates generated by receipt of GPS communications may have a first visual appearance, and an icon based upon location coordinates generated via UWB communication may have a second visual appearance. In another aspect, selecting the icon may bring a use to an interface that lists one or more communication modalities involved in determining location coordinated associated with the icon. As such, a single communication modality or multiple communication modalities may be involved in generation of a set of location coordinates.
- 9) Power management may also reference the availability of a type of power source such that if the equipment is installed at the worksite and the MMTS is supplied with a hardwired power source, the MMTS may provide power to multiple communication modality modules and sensors and increase the frequency of communications and sensor readings.
- 10) Location coordinates of the MMTS may also be supplied to an A/R interface for generation of icons within the A/R interface.

Referring again to FIG. 17A exemplary method steps may include the following. At step 1701 a timing sequence may be registered. At step 1702 one or more sensors may have power supplied to it. Next at step 1703 the apparatus may be used to quantify a condition present to the MMT as digital content. Continuing to step 1704 the apparatus may periodically repeat the quantification of a condition(s) present to the MMT. Proceeding to step 1705 a conditional processing may determine if an equipped GPS receiver is receiving GPS signals and then store the GPS signals that are received. At step 1706, a conditional processing may determine if the GPS is not receiving signals and if not then cease polling until an event occurs. At step 1707, if a local array receiver is receiving local array communications, then the system may receive and store local array data that has been transceived. Proceeding to step 1708, the step of receiving and storing local array communication data on a periodic basis may be repeated. At step 1709 a conditional processing may determine if internet communications are available, and if so then transmit some or all of the stored data. At step 1710, the apparatus may operate a set of wireless transceivers to transmit and receive wireless communications. Next, at step 1711, the system may calculate a distance between two or more sets of transceivers. Continuing at step 1712, the system may generate a position of at least one of the transceivers relative to the other based upon the wireless communications.

The exemplary method may continue in FIG. 17B. At step 1713, the system may be set up to co-locate one or more wireless transceivers and/or receivers (GPS) with one or more IOT sensors. Proceeding to step 1714, the system may generate location coordinates of the transceiver/receiver collated with the sensor. At step 1715, the system may quantify physical condition with the sensor. At step 1716, the system may transmit digital quantification of condition generated by the sensor. Furthermore, at step 1717, the system may associate digital quantification of the condition with the location coordinates. And at step 1718, the system

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may generate an icon associated with the location coordinates. At step 1719, the system may generate a user interface with spatial recognition. At step 1720, the systems may locate icons in an interface in spatial congruity with the location coordinates. And at step 1721, the system may include image data descriptive of the area including location coordinates.

Exemplary method steps are further illustrated in FIG. 17C. At step 1722, the system may specify location coordinates for one or more virtual tags. Next, at step 1723, the system may identify digital content for the one or more virtual tags. At step 1724, the system may display an icon or icons for the one or more virtual tags integrated into an A/R interface. Further, at step 1725 the system may integrate the icon into 2D site plan. At step 1726, the system may detect whether a user selects the icon to display digital content. At step 1727, the system may query digital content included in area specified. The area specified may be within a given radius of the user's location. In some examples, the area specified may be chosen in meeting search criteria. In other examples, the area specified may be in response to an alert status.

Referring now to FIG. 18A an exemplary user interactive interface 1800A is illustrated with features that are conducive to enabling the methods, processes and apparatus deployment described herein. The user interactive interface 1800A includes various user interactive areas 1801-1809. Each interactive area 1801-1809 may be activated to have an associated controller become operative to perform a function. The interactive areas 1801-1809 may be integrated into image data. Image data may be from a file, such as a two dimensional image of a site plan, floor design, or room layout, and/or energy levels received by a sensor in a smart device. For example, activation of an icon 1801 on a video conferencing screen may be integrated with image data of the video conferencing screen 1807 and allow a user to control functionality included in the video conferencing screen, such as, by way of example, powering on/off; connecting to a virtual meeting app; adjusting volume; adjust a display option; or other functionality. Similar user interactive areas are linked to control of other items of equipment and functionality appropriate for an associated item of equipment.

IoT sensors may be associated with transceivers and combined into a unit for quantifying conditions present at location, such as, for example a Shade™ Multi Sensor unit 1802 and/or a Shade Action Box 1803. A Shade Action Box 1803 will be associated with interactive controls linked to the user interface that allow for wireless of a structure aspect, such as a water turn off valve, a door lock, HVAC control, electrical on/off and the like.

Specific icons each associated with a disparate IoT sensor may be displayed with associated logos in a specified user interactive area 1809. Icons 1809a, 1809b may be associated with one or more of: fire, power, power surge, power outage, temperature, water, humidity, vibration and almost any condition quantifiable via an electronic or electromechanical sensor.

Interactive areas 1804-1806 may provide user control of aspects of the user interactive interface. For example, a calibration interactive area 1804 provides a user with control of functionality that allows the user to align A/R sensor data, such a visible light wave data and/or infrared image data to be aligned with an icon 1801-1803 or other designated user interactive area.

Another control area 1808 allows a user to which assets are displayed in the user interactive interface 1800A.

Options illustrated include all assets, fixed assets, and mobile assets. Fixed assets may be associated with an IoT sensor that is not combined with a transceiver and/or a virtual tag **1802** with fixed coordinates. A mobile asset may include an IoT sensor combined with a transceiver.

Referring now to FIG. **18B** illustrates a user interface on smart device **1810** on a Smart Device **1812**. The user interface includes image data **1810B** representative of a physical environment **1800B** in a direction of interest **1811** generated via the orientation of the smart device **1810** by the user **1824**. The user interface **1810A** also includes an icon **1801** associated with a physical asset in the area, as illustrated, the asset is a video screen **1814**, other assets may be similarly associated with an icon. Some embodiments may also include an interactive area **1813** that is congruent with an area and/or inclusive of an area on the interactive interface of a physical environment **1800B** that corresponds with an asset or other feature. As illustrated, the asset is shown as a chair and interactive area **1813** allows a user **1824** to select (e.g., click or touch a touch screen) the chair and activate presentation of digital data associated with the chair.

Similarly, the user may select the icon **1801** and activate presentation of digital data associated with the video screen **1814**.

Referring now to FIG. **18C**, the step of activating an icon **1821** via a user **1824** touch the interactive area **1822** including the icon **1821** is illustrated.

Referring now to FIG. **18D**, an action screen **1823** that is generated in response to the user selecting the interactive area **1822** is illustrated. The action screen includes additional user interactive areas that are operative with the controller to monitor a temperature sensor **1823A**, monitor humidity **1823B**, open and/or close an alarm state **1823C**, and/or monitor a CO2 meter **1823D**.

Referring now to FIG. **19**, a basic illustration **1900** is provided for description of the principles involved in determining the location of a radio receiver in three dimensional space based on its interaction with radio transceivers whose location are precisely known. In principle, to determine a location in three dimensional space at least three radio transceivers located at different locations is required. As will be described in more detail following, it is possible to obtain improvements in operations by use of additional transceivers. For illustration purposes, four transceivers are included in the illustration of FIG. **19**. A first transceiver **1901** is illustrated as a "base position" with exemplary cartesian position coordinates of 0, 0, 3000. In an example, the home position of 0, 0 may be a relative position along a floor of a structure and as the transceiver may be located upon a ceiling of a structure, the height of 3000 mm may represent a 3 meter elevation of the first transceiver **1901**. Other transceivers may also be located on a ceiling such as the second transceiver **1902** at position 5000, 0, 3000. The third transceiver **1903** may be located at position 5000, 15000, 3000 in an example. And a fourth transceiver **1904** may be located at position 0, 15000, 3000. A radio position tag may be located at a first position location **1905**.

There may be numerous techniques that the system of transceivers and a radio position tag may be used with to determine positions by triangulation. In some examples, a distance from a transceiver to the radio tag may be determined by the measurement of the strength of the signal which is related to the distance the signal has travelled. In other examples, timing signals may be used to determine the precise time it takes for a signal to transit the distance. In still further examples, the angles of arrival of signals from or to

the transponder may be used in combination to determine a location. Thus, the first position location **1905** may be determined by making at least three determinations of distance or angles between a radio position tag at first position location **1905** and the four transceivers (**1901-1904**).

It may be appreciated, that all four of the determinations may be made and can be used to calculate a position. In a real world environment, there may be a number of factors that lead to difficulties in obtaining a position or angle measurement with precision. In a first example, the positioning system may be installed in an interior location with walls, equipment, occupants, and other elements formed of materials that may interact with the radio signals (or other position signals such as infrared, laser, and ultrasonic signals as non-limiting examples. Scattered signals from walls may cause multiple signal paths which may cause confused signal determination. Depending on the location of the first position within the space of interest, one or more of the signal paths from the transceivers to the first position tag may experience signal degradation from one of the causes.

The positioning system may utilize all four of the signals to determine that one of the three signals could be omitted to improve the "goodness" of a position determination. Both the instantaneous signal path determination as well as the time progression of a signal path determination may factor into the rejection of a particular signal path measurement. Accuracy of position determination may be improved by sampling numerous times. In some examples, ultrawide band transmission protocols may be used to make position determination with relatively low energy consumption and good signal to noise aspects. Ultrawideband protocols may use extremely narrow (nanosecond) pulses which may allow for improved discrimination of multipath signal arrivals. Many different frequencies may be utilized for high degrees of sampling which can improve the accuracy.

Referring again to FIG. **19**, a second position location may include an agent with two tags located at locations **1906** and **1907**. The accuracy of determination of a second position may improve by the use of multiple tags at the different locations **1906** and **1907** whose aggregate position determination may be used to determine a second position location. As may be observed in the dashed lines and the complex dot dashed line there are individual signal paths from each of the four transceivers to the two tags. Here too the increased number of position signal paths allow for statistical improvement of an average location and also allow for the ability to statistically process signals to eliminate poor measurements or to average out poor measurements by acquiring collections of signals over time.

In some examples, advantages may be obtained by establishing systems with larger numbers of transceivers.

Referring now to FIG. **19A** an example **1900A** including an assembly of eight reference point transceivers **1901A-1907A** are operative to transceive in an area proximate to (within wireless communication distance to) a position tag **1910A**. Each of the eight transceivers **1901A-1907A** are located at respective discrete locations may again be at a height of 3000 mm such as a system mounted on a ceiling where a floor might be located at 0. A base coordinate location may be defined as the base position of reference point transceiver **1901A**. The other reference point transceivers (**1902A-1908A**) may be located at other discrete locations also indicated at a height of 3000 mm for example, although disparate heights are also within the scope of the invention.

Position tag **1910A** may transceive a wireless communication to and/or from the 8 transponders and each of these paths may be used to determine the location. The statistical combination of additional transponder paths will implicitly improve the accuracy of measurements. In some embodiments, a system including the reference point transceivers **1901A-1907A** is trained with known positions from time to time or a set periodic basis to remove static or systematic errors such as may occur in the location of a given position reference point transceivers **1901A-1907A**.

In some embodiments, dynamic aspects of an environment in which the reference point transceivers **1901A-1907A** are located may affect performance and accuracy of transceiving of wireless communications, such as for example changing a timing aspect associated with such transceiving.

For example, a human occupant of a space or an object that is placed transiently into a location may create an obstruction **1920A** to line of sight path measurements such as the illustrated obstruction **1920A**. The location determination system may detect the presence of an obstruction by a number of means such as a reduced signal strength over historical levels relative to other signal strengths at other transceivers. In some examples, interference by an obstruction **1920A** may result in a distance or angle determination of the signal path **D5** that is entirely inconsistent with the determination of position from some or all of the other paths **D1-D4** and **D6-D8**.

Referring now to FIG. 20, in some embodiments, wireless position devices **2102A-2109A** may be incorporated into a smart device **2101A** and not require a smart receptacle to house wireless position devices **2102A-2109A**. Wireless position devices **2102A-2109A** that are incorporated into a smart device, such as a smart phone or smart tablet, will include internal power and logic connections and therefore not require wireless communication between the controller in the smart device **2101A**.

According to the present invention, the smart device **2101A** may include integrated wireless position devices **2102A-2109A** and/or wireless position devices **2102A-2109A** in a smart receptacle **2100B** may provide a directional indication, such as a directional vector **2110A**, without needing to move the smart device **2101A** from a first position to a second position since a directional vector may be determined from a relative position of a first wireless position devices **2102A-2109A** and a second wireless positional device wireless position devices **2102A-2109A**.

In preferred embodiments, a position of the Smart Device **2101A** may be determined based upon wireless communication with one or more Reference point transceivers **2111-2114**. As described herein, a position may be determined based upon timing of wireless transmissions, such as via TDOA methodologies understood in the industry, and/or a time involved in a wireless communication and an angle of arrival and/or angle of departure for the wireless transmissions.

In other exemplary embodiments, distances may be triangulated based on measurements of wireless strength at two or more points, such as a strength of a wireless WiFi transmission. In the case of a WiFi transmission, a wireless signal may propagate outward as a wave, ideally according to an inverse square law. Ultimately, the crucial feature of the present invention relies on measuring relative distances at two points. In light of the speed of WiFi waves and real-time computations involved in orienteering, these computations need to be as computationally simple as possible. Thus, depending upon the specific application and means for

taking the measurements, various coordinate systems may be desirable. In particular, if the smart device moves only in a planar direction while the elevation is constant, or only at an angle relative to the ground, the computation will be simpler.

A position may be described in reference to coordinates descriptive of a location, one such an exemplary coordinate system includes a cartesian coordinate system including an X, Y, Z position and other embodiments, may include a polar coordinate system. One example of a three-dimensional polar coordinate system is a spherical coordinate system. A spherical coordinate system typically comprises three coordinates: a radial coordinate, a polar angle, and an azimuthal angle (r , θ , and φ , respectively, though a person of ordinary skill in the art will understand that θ and φ are occasionally swapped).

By way of non-limiting example, suppose Point 1 is considered the origin for a spherical coordinate system (i.e., the point (0, 0, 0)). Each WiFi emitter e_1 , e_2 , e_3 can be described as points $(r_1, \theta_1, \varphi_1)$, $(r_2, \theta_2, \varphi_2)$, and $(r_3, \theta_3, \varphi_3)$, respectively. Each of the r_i 's ($1 \leq i \leq 3$) represent the distance between the WiFi emitter and the WiFi receiver on the smart device.

In some embodiments, the orienteering occurs in a multi-story building, in which WiFi emitters may be located above and/or below the technician. In these embodiments, a cylindrical coordinate system may be more appropriate. A cylindrical coordinate system typically comprises three coordinates: a radial coordinate, an angular coordinate, and an elevation (r , θ , and z , respectively). A cylindrical coordinate system may be desirable where, for example, all WiFi emitters have the same elevation.

Referring now to FIG. 21, in some embodiments, one or both of a smart device **2101B** and a smart receptacle **2100B** may be rotated in a manner (such as, for example in a clockwise or counterclockwise movement **2115** **2116** relative to a display screen) that repositions one or more wireless position devices **2102B-2109B** from a first position to a second position relative to reference point transceivers **2111-2114**. A vector **2117** may be generated at an angle that is perpendicular **2118** or some other designated angle in relation to the smart device **2101B**. In some embodiments, an angle in relation to the smart device **2101B** is perpendicular **2118** and thereby viewable via a forward looking camera on the smart device **2101B**.

A user may position the smart device **2101B** such that an object in a direction of interest is within in the camera view. The smart device **2101B** may then be moved to reposition one or more of the wireless position devices **2102B-2109B** from a first position to a second position and thereby capture the direction of interest via a generation of a vector in the direction of interest.

Referring now to FIG. 21A, as illustrated, a vector in a direction of interest **2130** may be based upon a rocking motion **2131-2132** of the smart device **2101C**, such as a movement of an upper edge **2119C** in a forward arcuate movement **2131**. The lower edge **2120C** may also be moved in a complementary arcuate movement **2132** or remain stationary. The movement of one or both the upper edge and the lower edge **2119C-2120C** also results in movement of one or more wireless position devices **2102C-2109C** as shown in FIG. 21B). The movement of the wireless position devices **2102C-2109C** as shown in FIG. 21B) will be a sufficient distance to register to geospatial positions based upon wireless transmissions. A required distance will be contingent upon a type of wireless transmission referenced to calculate the movement; for example, an infrared beam

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may require less distance than a WiFi signal, and a WiFi transmission may require less distance than a cell tower transmission which in turn may require less distance than a GPS signal.

Referring to FIG. 22A, an illustration of the six degrees of freedom is provided. In FIG. 22A, a set of sensing and/or transmitting devices within a defined area 2200A are illustrated at a first point 2201, and at a second point 2202. The positional coordinates of the first point 2201 may be represented in a Cartesian Coordinate system as P_x , P_y , and P_z . Accordingly, the second point 2202 may be represented in a Cartesian Coordinate system P'_x , P'_y , and P'_z . A direction from the first point 2201 to the second point 2202 may be represented by a vector 2203 as V_x , V_y , and V_z . An Agent 2204 may be located at the first point 2201 in an example. In representing the orientation and direction of an element of interest there may be a number of possible position references that may be associated with the element.

In some examples, a controller determining a position may default to either a first point 2201 or a second point 2202 (additional points may also be calculated and used according to the methods presented in this disclosure) or a mathematical combination of the first point, second point (and any additional points) locations. A vector 2203 may be generated and a direction of the vector may be used as an Agent 2204 defined direction of interest.

A hierarchy of the first point 2201 to the second point 2202 may be specified to generate a starting point of the vector (e.g., first point 2201) and an intersecting point (e.g., second point 2202), a magnitude may be generated based upon a model of a position of the Agent 2204. A generated direction may be inverted by swapping the hierarchy of the first point 2201 and the second point 2202.

One or more of radio frequency and sound frequency transmissions, emissions, reflections, absorption, and detections may be used as input into a controller for determining a location of the first point 2201 and the second point 2202 and generation of a direction of interest.

Referring now to FIG. 22B, a defined area 2200B may be equipped with fixed reference point transceivers 2210-2213, each transceiver capable of one or both of transmitting and receiving one or both of radiofrequency encoded data and soundwave encoded data. Numerous frequency bandwidths are within the scope of the invention, including radio waves that are sometimes referred to as Ultra-Wideband (UWB) technology which focusses radio wave emissions of low power consumption to achieve high bandwidth connections, WiFi bandwidths, including WiFi RTT and frequencies compliant with 802.11 specifications, ultrasonic bandwidths, infrared bandwidths, and Bluetooth bandwidths, including Bluetooth 5.1.

In some embodiments, each transceiver 2210-2213 may in turn include multiple transmitters and/or receivers 2214-2217. The multiple transmitters and receivers 2214-2217 may operate on a same or different frequencies. Different frequencies may be within a same bandwidth, such as for example UWB bandwidth, or the different frequencies may be across different bandwidths, such as, for example an UWB and a WiFi bandwidth. In some embodiments a single transceiver 2210-2213 may thereby operate on multiple different frequencies. In other embodiments, different transceivers 2210-2213 may operate on a same or different frequencies. The multiple transceivers 2210-2213 may be operative to implement simultaneous or sequenced transmitting and receiving.

In some embodiments, some, or all of the multiple transmitters and/or receivers 2214-2217 may be incorporated into

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a transceiver device. The multiple transmitters and/or receivers 2214-2217 may also include an antenna 2220 with a same or different physical characteristics. For example, different antenna may be tuned to a same or different frequencies. In some embodiments, tens, hundreds or more antennae may be incorporated into a device in order to enable redundant communications and improve quality of a wireless communication.

Wireless communication may be accomplished for example via bandwidths associated with one or more of: Bluetooth; UWB; WiFi (including RTT Wi-Fi); Ultrasonic; and infrared communications. Transceivers used in transceiving may include directional and/or omni-directional antennas 2220. Antennae 2220 may be tuned similarly or tuned differently. Transceiving may be accomplished simultaneously, in timed sequence and/or based upon occurrence of an event. For example, a sensor may transceive on a predetermined timed schedule and also transceive following the occurrence of an event, such as a sensor reading that exceeds a threshold.

As illustrated in FIG. 22B, at least three Reference Point Transceivers 2210-2213 are mounted in different reference locations within or proximate to the defined area 2200B. Preferably each Reference Point Transceivers 2210-2213 has a relatively clear line of sight to an Agent Transceiver 2218-2219 supported by an Agent (not shown in FIG. 22B) and the line of sight is conducive to successful wireless communications.

In some examples, mounting (either permanent or temporary) will include fixedly attaching a Reference Point Transceiver 2210-2213 to a position and may be made to one or more of: a ceiling within the defined area, a wall mount; a stand mount; a pole; or integrated into or otherwise connected to an electrical receptacle. In some examples, a reference transceiver 2210-2213 may be mounted at a calibrated location within the defined area 2200B and act as a coordinate reference location.

The Reference Point Transceivers 2210-2213 may be placed in logical communication with a controller (such as via a distributed communications system, wireless or hard-wired), the controller may cyclically receive logical communication from one or more transceivers supported by an Agent located within the defined area 2200B while simultaneously monitoring the reference location. A Reference Point Transceiver 2210-2213 may be useful for calibrating various aspects of wireless communication between a Reference Point Transceiver 2210-2213 and an Agent supported Transceiver 2218-2219, aspects may include, for example variables in communication relating to one or more of: environmental condition such as humidity, temperature, and the like; as well as a variation in transceiver power levels, noise levels, amplification aspects and the like.

There may be numerous sources and causes of noise in a radiofrequency environment and/or a sound frequency environment (such as ultrasonic) that may come into play when using a Reference Point Transceivers 2210-2213 and an Agent supported Transceiver 2218-2219 that operate in one or more of: WiFi bandwidth; Bluetooth bandwidth; ultra-wide band; ultrasonic or similar technology. For example, in an indoor environment walls, structures, furniture, occupants, HVAC settings; particulate in the air (such as smoke or steam) human traffic; machinery movement; and the like may create a complex and dynamic environment where radiofrequency logical communications reflect and are absorbed. Reflections, particularly multiple reflections, may

cause spurious logical communications where the time for the logical communication transmission may be inappropriately long.

Accordingly, in some embodiments, a controller may benefit from receiving many data from multiple closely sequenced logical communications included in transmissions/receptions between Reference Point Transceivers and Transceivers supported by an Agent. Examples of multiple logical communications include less than ten samples to billions of samples per second. A large number of logical communications may be averaged or otherwise mathematically processed to determine a localization. Mathematical processing may include less consideration (e.g., weight, or removal) of logical communications outside of an otherwise closely associated data set. Other mathematical processing may include a mean, an average and a median of data included in the logical communications.

Systems with Transceiver counts of as few as six and sampling frequencies in the hundreds of billions of samples per second have been demonstrated to localize Transceiver locations with sub-millimeter accuracy. High sampling rates may require specialized data acquisition capabilities including advanced filtering systems, ultrafast digital to analog converters and the like. Fundamentally, the more samples that are collected per unit of time a more accurate a position determination may be.

A wireless positioning system (such as, WiFi, UWB, Ultrasonic and Bluetooth) with high positioning accuracy may be used for determination of a direction of interest using transceivers sized to be unobtrusive in a defined area **2200B** and/or to be able to be supported by a human Agent or an automation Agent capable of traversing the defined area **2200B**.

Referring now to FIGS. **22C-22E**, an example of Agent supported Transceivers **2218-2219** (in FIG. **22B**) may include a combination of an Agent's smart phone **2230** and an ancillary position determining device **2231**, **2232-2233**, **2234** linked to the smart phone **2230**. An ancillary position determining device **2231**, **2232-2233**, **2234** may provide one location position, such as for example, a first location position (**P1**), and the smart phone **2230** may provide another location position, such as for example a second location position (**P2**). A vector may be generated based upon **P1** and **P2**. For example, a generated vector may intersect **P1** and **P2**, or the vector may be in a calculated direction, such as an angle, with reference to one or both of **P1** and **P2**.

Linking between a smart device, such as a smart phone **2230**, and an ancillary position determining device **2231**, **2232-2233**, **2234** may be accomplished, for example via a hardwire connection, such as a lightning port or USB, mini USB type connector, Bluetooth, ANT, near field communications and the like. A smart watch (ancillary position determining device **2231**) that may be worn on an Agent's arm, a wand **2234** may be held in an Agent's hand, similarly, a ring **2232** may be worn on a finger and a tag **2233** may be incorporated into a badge, a button, an adhesive backed patch, a clip, or a wide variety of attachment mechanisms. Each ancillary position determining device **2231**, **2232-2233**, **2234** may include one or more Transceivers capable of communicating with a Reference Point Transceiver to generate logical communications from which a position may be calculated.

The Agent's smart phone **2230** and an ancillary position determining device **2231**, **2232-2233**, **2234**, may each have one or more Transceivers and may be used with the methods and apparatus described herein to determine a first point and

a second point. The first point and the second point may be used for generating a vector indicating a direction of interest (as discussed above). Other combinations of devices may be used, such as those illustrated in FIG. **22D** where a smart ring **2232** and a smart tag **2233** may be used to determine the multiple position location samples.

Referring to FIG. **22E** in some embodiments, a single ancillary position determining device may be able to employ multiple transceivers **2235-2237** on its body. For example, a wand (ancillary position determining device **2234**) may include a tip Transceiver **2235** and a base Transceiver **2236**. A wand (ancillary position determining device **2234**) may be used in conjunction with a smart device, such as a smart phone **2230**, where the phone **2230** is in a first position in close proximity to an Agent (such as in a pocket or holster worn by the Agent). The wand (ancillary position determining device **2234**) may be extended out from a handle portion of the wand.

The devices shown as examples may allow a single hand to be used to indicate position and direction. Various other devices that may include transceiver capability may be used in similar manners. A user may have one hand occupied holding a tool or sensor or may be otherwise occupied and can still indicate a desired direction of focus. In the example of a wand (ancillary position determining device **2234**), the user may press a button, switch, or engage other activation mechanism, such as a capacitive discharge device on the wand to indicate that the orientation of the wand is at a desired input condition.

Transceiver devices may be operative to employ various methods to improve accuracy of location determination, including but not limited to varying a frequency of transmission and reception of logical communications; varying a pulse pattern transmission and reception of logical communications, and varying intensity of emissions used in transmitting a logical communication.

In some embodiments of the present invention, Agent supported Transceivers and ancillary position determining devices **2231**, **2232-2233**, **2234** may communicate bursts of logical communications that include timing information. A delay of logical communications between the transmitter and the receiver may be converted to a distance measurement and a combination of a number of such logical communications may be used to triangulate the position. In some examples, the smart phone **2230** and an ancillary position determining device **2231**, **2232-2233**, **2234** may transmit the timing logical communications which the mounted transceivers receive and process for a distance determination. In other examples, an ancillary position determining device **2231**, **2232-2233**, **2234** smart phone **2230** may receive logical communications and determine timing delays and associated distances. Results of distance determinations may be communicated to controller, such as processing devices located at a smart device. A suitable controller may be located at one or more of the Transceivers or at a location remote to the Transceivers and connected by a communication network.

There may be many physical properties that may be used to make localization measurements/determinations. In an example of another type of sensing system an Infrared based sensor and camera system may be used to determine localization and orientation.

Referring to FIG. **23A**, in some embodiments, one or several wireless communications modalities may be operational during a same time period. For example, one or more of UWB; Bluetooth; WiFi; ultrasonic and infrared transmitters may be included in a system in a defined area. The

system may include three Reference Point Transceivers **2301-2303** that are positioned to transmit to a portion of the defined area. Some Reference Point Transceivers operate without movement of the Reference Point Transceivers **2301-2303**. Additional embodiments may include one or more of the Reference Point Transceivers **2301-2303** sweeping the defined area, or otherwise changing a field of view associated with the respective Reference Point Transceivers **2301-2303**. For systems that include Reference Point Transceivers **2301-2303** that change a field of view, a timing sequence may be generated and used to correlate with a logical communication such that the logical communication is associated with both the Reference Point Transceivers **2301-2303** and a particular field of view.

Some particular embodiments will include Reference Point Transceivers **2301-2303** that include one or more infrared cameras, each camera will have a defined field of view. Other directional transceivers may operate similarly.

A Transceiver may be located at a position **2305** and wirelessly communicate with a multitude of the Reference Point Transceivers **2301-2303**. As mentioned in reference to FIGS. **17A-17C** a user may wear one or more Transceivers that include transmitters, such as infrared emitting LEDs, laser or lights that emanate logical communications; WiFi, UWB; Bluetooth and/or Ultrasonic Transceivers. One or more of the Reference Point Transceivers **2301-2303** receive logical communications transmitted via an Agent supported Transceiver. The infrared transmitters may change intensity, cycle on and off, and/or vary in patterns to enable logical communication, such as information that allows for the extraction of location information. In some examples, Transceivers that include infrared transmitters have calibrated maximum intensities, and the logical communication levels received may be used to determine additional confirming information related to location. Various smart devices and/or position determining devices described herein may include or be equipped with an infrared emission element that may serve as a Transceiver supported by an Agent and used to indicate position and direction orientation.

In some examples, the aspects of FIG. **23A** may represent a virtual viewing environment that a user, such as an architect or engineer may be immersed in with a viewing apparatus, such as a Virtual Reality headset. The user may utilize localization and direction orientation aspects of the camera systems to drive the content of a virtual display view plane and a virtual location within a virtual model being displayed. In some examples, a user may be located at a building site configured with a camera system such as illustrated in FIG. **23A** while an architect may be located in a separate room configured with a similar camera system as illustrated in FIG. **23A**. In some of these examples, the architect may observe the view perspective of the user in the real-world location. In some other examples, the architect may occupy a virtual location and observe, through camera output of the real-world location, both the real-world location, the user in the real-world location and a display of the virtual model.

Referring to FIG. **23B**, in some specific embodiments, ancillary position determining devices may include an extension apparatus **2306** supported by an Agent. The extension apparatus **2306** may include, for example a pointer **2300**. The pointer **2300** may include a fixed length of rigid or semi-rigid material, or a telescopic combination of multiple lengths of rigid and/or semi-rigid materials. The pointer **2300** may be configured with areas of one or more wireless transceivers **2301-2303** at various distances from a first

point **2310** of the pointer **2300**. A location of the first point **2310** may essentially be the tip, or other convenient area.

A second area containing one or more transceivers **2301** and **2302** may be used as indicators that will be detected by directional apparatus, such as an infrared camera. A user may direct a pointer **2300** in a direction of interest and engage an activation mechanism, such as a switch, or engage in a motion to indicate the time to obtain the position and direction orientation. For example, an agent may activate a switch to activate a Transceiver **2301-2303** and partake in logical communication with a Reference Point Transceiver **2307**. In some embodiments, the logical communication may be manifested as a pattern of light. A controller may be supplied with the pattern of light transmitted as well as Reference Position information and generate a direction of interest.

According to the methods of the present invention, position points **P1-P4** may be generated based upon the logical communications between the Reference Point Transceiver **2307** and the Transceivers **2301-2303** supported by an Agent. A vector **2308** may be generated based upon the position points **P1-P4**. In addition, a smart device **2304** may also communicate with the Reference Point Transceiver **2307** and a position point **P5** associated with the smart device **2304** may be generated.

In some embodiments, a vector **2309** may be generated based upon the position point **P5** of the smart device **2304** and a position point **P1-P4** generated based upon logical communications with Transceivers **2301-2303** located on or within the extension apparatus **2306**.

Referring now to FIG. **23C** as discussed further herein, a sensor that includes a microelectromechanical system (MEMS) accelerometer may be used to track vibration patterns. In some embodiments, a MEMS accelerometer **2344** may be included within a smart device **2343**, such as a tablet or a smart phone. Other embodiments include a sensor independent of a smart device. Still other embodiments include a sensor packaged with a controller for executing software specific to the sensor, such as the Fluke™ 3561 FC Vibration Sensor. A structural component **2340** of a structure for which conditions will be monitored with sensors may include a vibration integrator **2341** with an attachment fixture **2342** that establishes vibrational integrity between an accelerometer **2344** in a smart phone **2343** and the structural component **2340**. The vibration integrator **2341** may be matched via its shape and material to accurately convey vibrations present in the structural component to the accelerometer **2344** in the smart device **2343**. In some embodiments a vibration integrator may include a damper or filter to exclude certain frequencies that may be considered noise to some applications. A damper may be directional such that only vibration frequencies in a particular direction are excluded.

It is understood that an accelerometer **2344** does not need to be incorporated into a smart phone and may be directly fixed to an attachment fixture **2342** or fixed to a vibration integrator **2341** or fixed to a structural component **2340**.

Vibrations present in the structural component may be indicative of a state of functioning of equipment included in the structure (not shown in FIG. **22C**). For example, a first pattern of vibrations, which may include frequency and/or amplitude and variations of one or both of frequency and amplitude may indicate a proper functioning of a piece of equipment. Patterns of equipment installed in a setting in a structure may be recorded under proper operating conditions to set up an initial proper state of functioning. Patterns derived from a subsequent sensor reading, such as an

accelerometer **2344** reading may indicate a variation from the initial pattern of sufficient magnitude to indicate a malfunction or wear present in the equipment.

In some embodiments, a user, such as a service technician, may install an accelerometer into the attachment fixture for the specific purpose of taking an accelerometer reading. A smart phone **2343** may run an app that records a time and place and vibration pattern received. The vibration pattern may be compared with a known set of vibration patterns and a condition of the structure may be ascertained from the comparison. The time date and vibration pattern may be transmitted to a server and aggregated with other sensor readings.

In another aspect, in some embodiments, a second accelerometer **2344A** may be used to introduce a vibration pattern into the structural component **2340**. The second device may include a second attachment fixture **2342A** that establishes vibrational integrity between the second accelerometer **2344A** in a second smart device **2343A** and a second vibration integrator **2341A**. The vibration pattern introduced may include a known frequency and amplitude. In some embodiments, the vibration pattern will include a sequence of frequencies and amplitudes, wherein different frequencies and amplitudes will be effective in diagnosing or otherwise indicating an underlying causation for a pattern of vibration. The second accelerometer **2344A** and the first accelerometer **2344** may be synchronized via executable software such that the first accelerometer will detect the vibrations introduced by the second accelerometer **2344A**. Any discrepancies between what was introduced by the first accelerometer **2344** and the first accelerometer **2344** may be indicative of a state of the structure.

For example, introduction of a frequency pattern into a beam that is sound may transmit well through the beam and be detected with minimal variations from the frequency pattern that was introduced. However, a beam that is cracked or has rot within it may not convey the frequency pattern to the first accelerometer or convey the frequency pattern with significant distortion and/or diminutions in amplitude.

A history of sensor readings associated with a particular structure and/or group of structures may be stored and referenced to assist in interpreting a cause for a particular vibration pattern.

Vibration sensors may be installed and recorded in as built data or added to a structure in a retrofit. Some commercial sensors (such as the Fluke 3561 FC Vibration Sensor) may be associated with vendor supplied software for ease of retrofit implementation.

According to the present invention, accelerometers or other vibration sensors are deployed in specific locations and tracked in a structure according to the respective sensor location. In addition, a relative position of a particular sensor position is tracked relative to other sensors (vibration sensors or sensors for monitoring different modalities of ambient conditions). The present system includes an AVM that may store and make available to a user and/or to AI applications which structural components are in vibrational communication with a particular sensor. Various sensors include underlying piezoelectric, accelerometers of other technologies.

Embodiments also include a sensor programmed to reside in a lower power states and to periodically “wake itself up” (enter a higher powered state) to take a reading and transmit the reading. Sensor readings may be correlated with different types of wear, damage, failure, or proper operation of components included in a structure. The AVM may track location and may rank a likelihood of a component respon-

sible for a particular vibration pattern detected by a sensor. The ranking may be based upon proximity, mediums available for communicating the vibration pattern (such as a beam traversing a significant portion of a structure, but which provides excellent mechanical communication for the vibration).

Some embodiments also associate a sensor reading of vibration with a type of motion likely to cause such a reading. For example, some readings may include a linear component and a rotational component (such as operation of a washing machine during certain cycles). Patterns of normal and abnormal operation may be recorded and deciphered via programmable software on a controller.

In another aspect, a pattern of sensor data that denotes spikes of linear data may be associated with a human being walking. Overtime, a controller may track sensor reading patterns and associate a particular pattern with the walk of a particular person.

It is also within the scope of the invention to track and analyze a set of data associated with a primary signal and additional sets of data (secondary, tertiary etc.) tracking harmonics of the primary signal. The AVM may also track sets of data associated with simultaneous, and/or closely timed readings received from multiple sensors and associate an amplitude, sequence, delay, or other attribute of the data sets relative to each other to provide input as to a location of a source of the vibration. Additionally, vibration sensors may include axis within the sensor. For example, two axis and three axis sensors may have a direction of each axis included in the AVM and used in analysis of a vibration pattern.

The present invention also provides simple and fast procedures for the provision of directions of a User or a sensor to a source of vibration based upon analysis of readings of one or more sensors via the X, Y and Z location determination and directional ray or vector generation methods described herein.

Disparate types of sensor may also provide disparate data types that are useful in combination to determine a source of sensor readings. For example, a vibration sensor reading indicating erratic motion may be combined with an increased temperature reading from a sensor proximate to an item of equipment. The combined sensor readings may assist in an analysis of a cause of the sensor readings.

In still another aspect, one or more sensor readings may be correlated to a life expectancy of an item of equipment, such as for example a heating Ventilation and Air Conditioning (HVAC) unit. By way of non-limiting example, an ammeter sensor reading measuring an electrical draw of an HVAC unit may be quantified upon deployment of the unit. The initial readings may act as a baseline of a unit in excellent operational condition. A similar baseline reading may be taken via an accelerometer measuring a vibration generated by the HVAC unit. Still other sensor readings may include airflow, temperature, humidity, or other condition. Over time, a change in one or more sensor reading values may indicate some wear and move the HVAC equipment item into a “normal wear but operational” status.

Still further along a time continuum, one or more sensor readings may indicate a pending failure. For example, a current required to run the unit may be measured by the ammeter sensor and indicate an increased draw in electrical current. Likewise, airflow may decrease, and temperature may increase, and other sensors may provide additional evidence of a pending failure. Finally, a failed unit may generate a very high temperature reading and ammeter

readings may increase to a level of sufficient electrical current draw to trip an electrical breaker, thereby indicating a failure.

According to the present invention, any of the sensor readings (or all, or some subset of all sensor readings) may be referenced to generate an alert. Following the alert, remedial action may be taken.

Referring now to FIG. 24A, a method for generating an augmented-reality Radio Target Area for a Smart Device is shown. At step 2401, wireless energy of a first wavelength is received into a wireless receiver. In exemplary embodiments, this step may include receiving image data based on visible light into a sensor of the Smart Device. The wireless energy may be dispersed over a one-, two-, or three-dimensional space in a defined physical area, and may be received into a one-, two-, or three-dimensional array in the receiver. The wireless energy may take the form of electromagnetic radiation, such as light in the human-visible light spectrum (generally having a wavelength between 380 nm-740 nm), ultraviolet light (generally having a wavelength between 10.0 nm-400 nm), or infrared light (generally having a wavelength between 740 nm-2.00 mm) as examples. The set of wireless energy available to the wireless receiver is the Smart Device's Radio Target Area.

The wireless receiver may be a Smart Device sensor, including a CMOS active pixel sensor, a CMOS back illuminated sensors, CCD, or a LIDAR apparatus, including a solid-state/MEMS-based LIDAR. The wireless receiver may comprise an array or other plurality of other wireless receivers. The wireless receiver may be operative to receive the wireless energy into an array of an appropriate dimension for subsequent display (possibly after processing) on the Smart Device. For example, where the wireless receiver is a Sensor, the Sensor may be operative to translate the wireless energy into a two-dimensional array.

At step 2402, a pattern of digital values is generated based upon receipt of wireless energy into the wireless receiver. This pattern of digital values may be based on one or more qualities of the received wireless energy, including its intensity, spatial dispersion, wavelength, or angle of arrival. The pattern may be placed into an appropriate array. For example, if the display of the Smart Device is a two-dimensional display, then the pattern of digital values may comprise a two-dimensional representation of the image data received. In some embodiments, the pattern of digital values may be based on an aggregated set of values from an array of receivers. For example, if the basis of the digital values is the intensity of the wireless energy received into the receiver, then the digital value assigned to a given entry in the array may be based on a weighted average of intensity of wireless energy received at a plurality of the receivers in the array. Optionally, at step 2403, the wireless receiver may receive the wireless energy as an analog signal (for example, if the wireless receiver is a black-and-white sensor or an unfiltered CCD), and convert the analog signal to digital values through filtration or other analog-to-digital conversion. The set of digital values within the Radio Target Area is the Digital Radio Target Area.

With the Smart Device wireless receiver's Radio Target Area determined, the Smart Device's position should be determined as well, along with the positions of any items of interest in a given space. Collectively, the Smart Device and the item of interest may comprise wireless Nodes. Accordingly, at step 2404, coordinates representative of a wireless Node may be determined relative to a base Node. These coordinates may be determined in any appropriate coordinate system (such as Cartesian, polar, spherical polar, or

cylindrical polar) and may be determined via RTLS or the orienteering-triangulation methods with various wavelengths or modalities, such as ultra-wideband, Bluetooth, etc. Additionally, the coordinates may be determined using an angle of arrival or angle of departure of a signal to or from the base Node, along with the distance from the base Node. By way of non-limiting example, this could produce a dataset that correlates the coordinates of three elements with the identities of those elements: $\{(0,0,0), \text{BaseNode}; (1,1,1), \text{SmartDevice}; (2,2,2), \text{ItemOfInterest}\}$. While this example may be used throughout the following discussion, it is understood to be non-limiting, as a given space may include a plurality of items of interest. Note that, in some embodiments, the Smart Device itself may become a dynamic database entry with a continuously (or periodically) updating set of coordinates. This may be useful in allowing a plurality of Smart Devices engaged with the system at the same time to interact with one another.

At step 2405, the position of the Base Node is determined relative to the defined physical area. In exemplary embodiments, this may include establishing the Base Node as an origin in the coordinate system and determining vectors from the Base Node to boundaries and items of interest (i.e., the distance from the Base Node and the direction from the Base Node to the boundaries and items of interest). In some examples, the Base Node may have an established reference relative to a global coordinate system established.

At step 2406, a Target Area is generated within a controller of the Smart Device. The Target Area may be the set of coordinates (relative to the Base Node) within the Radio Target Area of the wireless receiver. The Target Area may be limited by physical boundaries of the given space, such as walls, floors, ceilings, occlusions, etc. The Target Area may also be limited by distances that various types of signals may travel. For example, a sensor of audio signals may not be able to practically pickup signals over a background noise level that originate more than 1000 feet from a user position, purely as an example. In such a case, the Target Area for such signal types may be limited to that dimension.

At step 2407, respective positions of one or more wireless Nodes within the Target Area are determined. These positions may be determined relative to the physical Target Area or the Radio Target Area. The determination may be made with reference to the dataset discussed at step 2405, or it may be made dynamically based upon one or more Base Nodes and/or the Radio Target Area. Moreover, the determination may additionally be based on receipt of a wireless signal into the Smart Device from the wireless Node. This signal may indicate a position using the orienteering methods described herein.

At step 2408, a user interface may be generated on the Smart Device based upon the pattern of digital values generated at step 2402. The user interface may comprise a plurality of pixels, wherein each pixel comprises a visible color based upon the pattern of digital values generated at step 2402. For example, if the digital values were based upon receipt of visible light into the wireless receiver (e.g., a sensor), then the display may reflect a reasonably accurate color photograph of the Radio Target Area of the wireless receiver. If the digital values were based upon an intensity of received light from, for example, LIDAR, then the display may reflect a scan of the Radio Target Area. In some embodiments, the pixel may include an intensity of energy received into the receiver. In this way, aspects of the Radio Target Area characterized by an intensity of energy may be emphasized. For example, this may produce a LIDAR relief of an area or a heatmap of an area.

At step **2409**, an icon may be generated in the user interface. Preferably the icon will be placed at a position relative to data quantifying received energy levels. In some embodiments, the icon location in a user interface will be indicative of a position of a Tag (Virtual or Physical). This position may be quantified via positional coordinates, such as Cartesian Coordinates, Polar Coordinates, Spherical Coordinates, and the like. The icon may be based upon an input from a user, stored data, quantified environmental conditions or other criteria related to an aspect of the Radio Target Area.

For example, an icon may indicate information about an Item of Interest located at a given set of coordinates within the Radio Target Area or Digital Radio Target Area. In another embodiment, the user may indicate on the display a position in which the user wishes to place an icon and add information about an Item of Interest (thus creating a new entry in the database, which may be populated with the coordinates of the indicated position). Moreover, the icon may change colors based upon the pattern of digital values. The icon may be overlaid on top of the display. The icon may resemble the letter “i”, a question mark, a thumbnail, or any other suitable image from a library. In some embodiments, the icon may change depending on one or more attributes of its corresponding database entry. For example, if the icon located at (4,4,4) relates to a restaurant menu, then the icon may resemble the letter “i” or a thumbnail of a menu. On the other hand, if this database entry is modified so that the corresponding database entry is a message, then the icon may update to a picture of an envelope.

In some embodiments, the icon-generation step may be based upon an inquiry to a database that uses the Digital Radio Target Area as an input. For example, upon generation of the Digital Radio Target Area, an associated set of coordinates in one or more dimensions may be generated. This may then be submitted to a database.

In some embodiments, an icon-generation step may be based upon an inquiry to a database that uses the user’s position coordinates as an input. In these embodiments, both the Digital Radio Target Area based on an RTA as well as the universal Radio Target Area may be included in an inquiry submitted to the database.

In some examples, the user may have an option to limit or filter the types of database entries that may be queried for, such as in a non-limiting sense, the existence of real-world Tags, virtual Tags, sensor data values and streams from a particular class of sensors and the like.

Continuing with the example from step **2404**, the Digital Radio Target Area may comprise the set of coordinates: ([1.5,10], [1.5,10], [1.5,10]). In this example, the database may return information about the Item Of Interest, but not about the Base Node. The Digital Radio Target Area may update when the Smart Device position changes, or by user input, the Digital Radio Target Area may remain static after a certain instance in time.

Continuing with FIG. **24B**, at step **2410**, the icon may be positioned in the user interface at a given position based upon coordinates representative of the position of the wireless Node or Tag in the Target Area. This may comprise a selection of a multitude of pixels related to the position of the wireless Node or Tag and changing those pixels from the digital values determined at step **2402** to a second set of pixels to indicate the presence of an icon. In some embodiments, the icon may be dynamically updated based upon movement of the Smart Device (and, accordingly, the wireless receiver). In some embodiments, the icon may be permanently associated with a set of coordinates. In such

embodiments, the icon may be generated whenever a Smart Device with appropriate permissions includes in its Radio Target Area the set of coordinates of Nodes or Tags associated with the icon.

At step **2411**, user-interactive functionality may be associated with the pixels comprising the icon. This may allow the user to “select” the icon by means of an input device (e.g., mouse, touchpad, keyboard), touchscreen, digital input, etc. Upon selection, the icon may be operative to interact with the user in one or more ways, including: displaying a message intended for the user (by text, audio, video, hologram, etc.); requesting credentials from the user to verify permissions (e.g., a password), displaying information about an item associated with the icon, prompting the user to update information about an item associated with the icon, etc. The user-interactive functionality may display static information (e.g., dimensions of the item), display dynamic information (e.g., an alarm state or sensor information relating to the item; for example, if the item is a refrigerator, internal temperature may be displayed), or produce a control panel that allows the user to issue control commands (e.g., remotely operating an automated apparatus by resetting an alarm state, taking remedial action based upon a sensor state as described herein, etc.) or to issue menu control commands such as to invoke a different user interface or screen of a user interface.

This may be useful in geospatial applications, or in procedurally generated activities. For example, a first user may generate a positional designation on a user interactive device, such as, for example an augmented-reality display to leave a narrative, icon or other input associated with the first use. Additionally, the same or another user may log positional coordinates and upload an image that could be displayed submitting a database query including those coordinates. Entry of the coordinates and essential credentials may provide access to the content associated with the positional coordinates.

At step **2412**, the preceding steps may be integrated by generating a display comprising the user interface, the icon, and at least some of the associated user-interactive functionality. In embodiments, in which a plurality of Smart Devices are themselves part of the database, this may allow various users to send messages, images, etc. to each other.

At step **2413**, detection of movement of the Smart device may cause a branch back to step **2405**. Based upon that movement of the Smart Device, a defined physical area from which wireless energy is received (i.e., the Radio Target Area based upon the Target Area) may be changed. The movement may be detected using input from wireless communications, magnetic field sensors, an accelerometer, feature-recognition software, or other similar apparatus and algorithms. In other examples, the position of the Smart Device may be dynamically obtained using any of the techniques of position determination, such as triangulation with reference nodes. Here, too, a change of position detected in this manner may cause a branch back to step **2405**. The Target Area may be based upon the position of the Base Node, the relative positions of the wireless Nodes, and the Smart Device.

Referring to FIG. **25**, an illustration of alternative methods for display of information relating to RTA is provided. At the beginning of the process, a system of components which may include a smart device with a user of the smart device may be established. Amongst the various components a Home Position may be established for all the components at step **2501**. The system may proceed by establishing and initiating transceiving of data and information at step **2502**.

In some examples, the user may be prompted to choose a desired coordinate system for the display at step **2503**. In other examples, a user interface of the system may have a setpoint function which the user may invoke to gain access to user settable parameters which may include they type of coordinate system to use, such as for example Cartesian or spherical coordinates.

In still further examples, the system may decide to default to a particular coordinate system depending on the nature of the type of data its positional reference devices may be obtaining or providing.

At step **2504**, if the coordinate system was chosen as Cartesian coordinates, the system may utilize triangulation amongst multiple reference point transceivers. Alternatively, if the coordinate system was chosen as polar coordinates, at step **2505**, the system may be operative to execute software and reference positioning devices that utilize angles and distances involved in transceiving and location.

In either case, at step **2506**, the position of a Sensor attached to the smart device of the user may be determined. In some examples, the system may have multiple and redundant location system. A combination of such position determinations may result in superior accuracy of an aggregated position result. Accordingly, at optional step **2507**, a wireless position determination may be performed with the smart device to establish a verification of the position of the Smart Device and the Sensor attached. Referring now to step **2508**, a direction that the sensor is facing in may be determined. Although there may be a number of different manners of determining orientation as have been described herein, in an example, the orientation may be determined based upon wireless transmission and/or wireless verification.

Referring now to step **2509**, an energy-receiving Sensor included in the Smart Device or in logical communication with the Smart Device may be used to quantify energy levels perceivable at the position and in the direction of the Smart Device. The resulting quantification may depend on aspects of the Sensor device, but the resulting data will quantify a characteristic for the RTA.

In some embodiments, an optional step **2510** may be performed by an element of the system such as the smart device or a server in communication with the Smart Device. The element of the system may compare one or more of position information, orientation information and the image data itself to calculate an estimate of whether the RTA angle has changed for the sensing element.

In general, at step **2511**, the RTA of the Sensor device used to capture the image in step **1209** may be quantified. In an optional step **2512**, coordinates relating to the instant RTA of the image may be established. In some examples, this may relate to a range of three-dimensional coordinates that are addressed by the RTA of the Sensor element. In general, at step **2513**, the system may look up, or in some cases generate, location coordinates for Tags that are determined to be within the quantified RTA. In some database systems that the system may have access to, real-world or virtual-world tags may be tracked in a coordinate system with a certain origin.

If the current origin established at step **2501** is offset from a particular database related origin, then one or both of the coordinate system values may be converted to each other to align their respective origins. At step **2514**, the Tags in an aligned coordinate system may have their positions compared to the current RTA and a selection for the set of Tags that are within the RTA may be made.

In some alternative examples, a display of all Tags that are authorized for access to the user regardless of whether they are in the RTA may be made using associated aligned coordinates as discussed in reference to step **2513**.

Referring now to step **2515**, in an example, the Smart Device of the user may be used to generate and display a user interface to the user based upon the captured image and the associated tag icons within the RTA. These associated Tag icons may have at least the hardware and be operative to perform the functionality as has been discussed herein.

Referring now to FIG. **26A**, an automated controller is illustrated that may be used to implement various aspects of the present invention in various embodiments, and for various aspects of the present invention. Controller **2600** may be included in one or more of: a wireless tablet or handheld smart device, a server, an integrated circuit incorporated into a Node, appliance, equipment item, machinery, or other automation. The controller **2600** includes a processor unit **2602**, such as one or more semiconductor based processors, coupled to a communication device **2601** configured to communicate via a communication network (not shown in FIG. **26A**). The communication device **2601** may be used to communicate, for example, with one or more online devices, such as a smart device, a Node, personal computer, laptop, or a handheld device.

The processor unit **2602** is also in communication with a storage device **2603**. The storage device **2603** may comprise any appropriate information storage device, including combinations of digital storage devices (e.g., an SSD), optical storage devices, and/or semiconductor memory devices such as Random Access Memory (RAM) devices and Read Only Memory (ROM) devices.

The storage device **2603** can store a software program **2604** with executable logic for controlling the processor unit **2602**. The processor unit **2602** performs instructions of the software program **2604**, and thereby operates in accordance with the present invention. The processor unit **2602** may also cause the communication device **2601** to transmit information, including, in some instances, timing transmissions, digital data and control commands to operate apparatus to implement the processes described above. The storage device **2603** can additionally store related data in a database **2605** and database **2606**, as needed.

Referring now to FIG. **26B**, an illustration of an exemplary wireless Node (transceiver module **2610**) configured with a transceiver **2624** to wirelessly communicate via one or more wireless communication Modalities, including a bandwidth and protocol, such as the Bluetooth 5.1; BLE5.1; Wi-Fi RT and/or GPS standard is illustrated. As discussed, many different Modalities of wireless technology may be utilized with the content presented herein, but a BLE5.1 "radio" module is an interesting example since its standards provide for angle of arrival (AoA) capability as well as angle of departure (AoD) and a distance determination based upon a timing signal. With AoA/AoD a designed antenna array **2625** can be used by a transceiver **2624** to measure a phase shift amongst multiple antenna elements to estimate distance differences between the antennas and to extract an angle from the antenna array to the source of radiation. A BLE5.1-consistent multichip transceiver **2624** may include circuitry and software code to perform the acquisition of data and determine the angle of arrival in some examples. In other examples, a BLE5.1-consistent multichip transceiver **2624** may control the acquisition of data from an antenna array while streaming the data to off module processing capabilities. The BLE5.1-consistent Node **2610** may contain functional blocks of circuitry for peripheral **2620** control. The

peripherals may include a connection to external host controllers/MCUs **2621**. The peripheral **2620** control may also interact with peripheral and IoT Sensors and other devices **2622**.

The BLE5.1-consistent Node **2610** may include a processing element **2623** which may have its own memory of different types as well as capabilities for encryption of data. The BLE5.1 consistent Node (transceiver module **2610**) may also have Transceiver **2624**. This circuitry may include Baseband and RF functions as well as control the AoA functions and the self-verifying array functions. The Bluetooth transceiver **2624** may receive signals through an on-module antenna **2625** or an external antenna or array of antennas may provide external RF input **2626**. The BLE5.1-consistent Node **2610** may include functional circuitry blocks for control of Security functions **2627**, crypto-generations, random number generation and the like. The BLE5.1-consistent Node **2610** may include functional blocks for power management **2628**.

The BLE5.1-consistent Node **2610** may be operative for quantification of temperature aspects of the Node (transceiver module **2610**), battery-control functions and power-conversion functions. An external power source **2633** may be included to provide electrical energy to a power management unit **2628** which, in some examples, may be from a battery unit, or a grid connected power supply source in other examples. The BLE5.1-consistent Node **2610** may include functions for control of timing and triggering **2629**. In a related sense, the BLE5.1-consistent Node **2610** may include functions for clock management **2630** within the module. The BLE5.1-consistent Node **2610** may also include circuit elements that are always-on **2631** to allow external connections **2632** to interact with the device and perhaps awake it from a dormant state. There may also be other customized and/or generic functions that are included in a BLE5.1-consistent Node **2610** and/or multichip module.

In some embodiments, functionality performed by a Node (transceiver module **2610**) may be executed via interaction with one or more external controller and/or MCU **2621**.

Referring now to FIG. **26C**, a Node **2650** included in a higher order deployment assembly is illustrated. A deployment Node **2650** may be in logical communication with one or more of: sensors, customized control commands, antenna array designs and the like.

A Node **2650** may include multiple antennas or antenna arrays **2651-2656**. As described previously, the Node **2650** may include a transceiver module **2610**, and in some examples, the transceiver module may include Bluetooth-adherent aspects. Communications received via an antenna **2651-2656** may be directly ported into the transceiver module **2610**. Embodiments may also include routing particular antenna/antenna array outputs to the transceiver module **2610** in a controlled and timed sequence. A processing Module **2670** may coordinate a connection of the Node **2650** to external peripherals.

In some examples, circuitry **2680** to logically communicate with one or more of: a Peripheral, a data Connection, Cameras and Sensors controllers, and components to perform data and image acquisition of various kinds, or it may interface external components with the Node **2650**.

The Node **2650** may also include its own power management unit **2660** which may take connected power or battery power or both and use it to prove the various power needs of the components of the assembly. The Node **2650** may have its own processing modules **2670** or collections of different types of processing functions which may have dedicated memory components **2671**. In some examples,

specialized processing chips of various kinds such as Graphical Processing Units and fast mathematics function calculators as well as dedicated artificial intelligence processing chips may be included to allow the Node **2650** to perform various computational functions including location determination of wirelessly connected devices amongst other functions. There may be numerous other functions to include in a Node **2650** and alternative types of devices to perform the functions presented herein.

In some examples as illustrated in FIG. **26D** antenna arrays **2690**, **2691** may be assembled into a "Puck" shown as Node **2650** wherein the antenna arrays are configured with antenna designs which have directional aspects to them. Directional aspects may mean that the antennas may be sensitive to incident radiation coming from a certain direction but not sensitive to radiation coming from a different direction. Antenna arrays **2690**, **2691** may include antennas that may have maximized signals for a particular incident waveform, the identification of which antenna may provide or supplement angle of incidence calculations.

A directional antenna may include, for example, an antenna with RF shielding over some portion of an antenna's circumference. For example, 270° (or some other subset of a 360° circumference of an antenna), or an antenna array may have RF shielding to block and/or reflect back an RF signal towards the antenna-receiving portion. Other directional antennas may include a shield blocking less than 360° of RF transmissions that rotates around a receiving portion of an antenna and only receives RF communications from a direction of an opening in the shield. Shielded antennas may provide improved determination of a direction from which a wireless transmission is being received from, since RF noise is blocked from a significant portion of a reception sphere.

Referring now to FIG. **27**, a block diagram of an exemplary Smart Device **2702** is shown. Smart Device **2702** comprises an optical capture device **2708** to capture an image and convert it to machine-compatible data, and an optical path **2706**, typically a lens, an aperture, or an image conduit to convey the image from the rendered document to the optical capture device **2708**. The optical capture device **2708** may incorporate a CCD, a Complementary Metal Oxide Semiconductor (CMOS) imaging device, or an optical Sensor **2724** of another type.

A microphone **2710** and associated circuitry may convert the sound of the environment, including spoken words, into machine-compatible signals. Input facilities may exist in the form of buttons, scroll wheels, or other tactile Sensors such as touch-pads. In some embodiments, input facilities may include a touchscreen display.

Visual feedback to the user is possible through a visual display, touchscreen display, or indicator lights. Audible feedback **2734** may come from a loudspeaker or other audio transducer. Tactile feedback may come from a vibrate module **2736**.

A magnetic force sensor **2737** such as a Hall Effect Sensor, solid state device, MEMS device or other silicon based or micro-electronic apparatus.

A motion Sensor **2738** and associated circuitry converts motion of the smart device **2702** into a digital value or other machine-compatible signals. The motion Sensor **2738** may comprise an accelerometer that may be used to sense measurable physical acceleration, orientation, vibration, and other movements. In some embodiments, motion Sensor **2738** may include a gyroscope or other device to sense different motions.

A location Sensor **2740** and associated circuitry may be used to determine the location of the device. The location

Sensor **2740** may detect Global Position System (GPS) radio signals from satellites or may also use assisted GPS where the mobile device may use a cellular network to decrease the time necessary to determine location. In some embodiments, the location Sensor **2740** may use radio waves to determine the distance from known radio sources such as cellular towers to determine the location of the smart device **2702**. In some embodiments these radio signals may be used in addition to GPS.

Smart Device **2702** comprises logic **2726** to interact with the various other components, possibly processing the received signals into different formats and/or interpretations. Logic **2726** may be operable to read and write data and program instructions stored in associated storage or memory **2730** such as RAM, ROM, flash, SSD, or other suitable memory. It may read a time signal from the clock unit **2728**. In some embodiments, Smart Device **2702** may have an on-board power supply **2732**. In other embodiments, Smart Device **2702** may be powered from a tethered connection to another device or power source.

Smart Device **2702** also includes a network interface **2716** to communicate data to a network and/or an associated computing device. Network interface **2716** may provide two-way data communication. For example, network interface **2716** may operate according to the internet protocol. As another example, network interface **2716** may be a local area network (LAN) card allowing a data communication connection to a compatible LAN. As another example, network interface **2716** may be a cellular antenna and associated circuitry which may allow the mobile device to communicate over standard wireless data communication networks. In some implementations, network interface **2716** may include a Universal Serial Bus (USB) to supply power or transmit data. In some embodiments, other wireless links may also be implemented.

As an example of one use of Smart Device **2702**, a reader may scan some coded information from a location marker in a facility with Smart Device **2702**. The coded information may include for example, a hash code, bar code, RFID, or other data storage device. In some embodiments, the scan may include a bit-mapped image via the optical capture device **2708**. Logic **2726** causes the bit-mapped image to be stored in memory **2730** with an associated time-stamp read from the clock unit **2728**. Logic **2726** may also perform optical character recognition (OCR) or other post-scan processing on the bit-mapped image to convert it to text. Logic **2726** may optionally extract a signature from the image, for example by performing a convolution-like process to locate repeating occurrences of characters, symbols, or objects, and determine the distance or number of other characters, symbols, or objects between these repeated elements. The reader may then upload the bit-mapped image (or text or other signature if post-scan processing has been performed by logic **2726**) to an associated computer via network interface **2716**.

As an example of another use of Smart Device **2702**, a reader may recite words to create an audio file by using microphone **2710** as an acoustic capture port. Logic **2726** causes audio file to be stored in memory **2730**. Logic **2726** may also perform voice recognition or other post-scan processing on the audio file to convert it to text. As above, the reader may then upload the audio file (or text produced by post-scan processing performed by logic **2726**) to an associated computer via network interface **2716**.

A directional sensor **2741** may also be incorporated into Smart Device **2702**. The directional device may be a compass and be based upon a magnetic reading or based upon

network settings. The magnetic sensor may include three axes of magnetic sensitive elements and may also be coupled with an accelerometer in the directional sensor **2741**.

A LiDAR sensing system **2751** may also be incorporated into Smart Device **2702**. The LiDAR system may include a scannable laser light (or other collimated) light source which may operate at nonvisible wavelengths such as in the infrared. An associated sensor device, sensitive to the light of emission may be included in the system to record time and strength of returned signal that is reflected off of surfaces in the environment of Smart Device **2702**. Aspects relating to capturing data with LiDAR and comparing it to a library of stored data (which may be obtained at multiple angles to improve accuracy) are discussed above.

Physical world and virtual-world based imagery related to the environment of a user may be presented via a user interface that may display on a Smart Device screen or other interactive mechanism, or in some embodiments, be presented in an augmented or virtual environment, such as via a VR or AR headset. The imagery displayed upon these devices may represent a composite of image data reflective of a real-world data stream as well as digitally added/superimposed image data from a virtual or digital source data stream. A user may be presented a typical image as it would look to the user's eyes physically, upon which digital shapes representing virtual "Tags" may be superimposed to represent the presence of digital information that may be accessed by a user. In other examples, the digital information may be directly displayed as a superposition. In some examples, the real-world and virtual-world environments may be displayed separately on a screen or separately in time.

In some examples, the "physical world image" may also be digitally formed or altered. For, example, an imaging device may obtain images where the sensing elements of the imaging device are sensitive to a different frequency of electromagnetic radiation, such as in a non-limiting sense infrared radiation. The associated "real-world image" may be a color scale representation of the images obtained in the infrared spectrum. In still further examples, two different real-world images may be superimposed upon each other with or without additional virtual elements. Thus, a sensor image may have an IR sensor image superimposed over part or all of the image and a digital shape representing a virtual Tag may be superimposed.

In some implementations, a virtual reality headset may be worn by a user to provide an immersive experience from a vantage point such that the user may experience a virtual representation of what it would be like to be located at the vantage point within an environment at a specified point in time. The virtual representation may include a combination of simulated imagery, textual data, animations and the like and may be based on scans, image acquisition and other Sensor inputs, as examples. A virtual representation may therefore include a virtual representation of image data via the visual light spectrum, image data representing image scans obtained via infrared light spectrum, noise, and vibration reenactment. Although some specific types of exemplary sensor data have been described, the descriptions are not meant to be limiting unless specifically claimed as a limitation and it is within the scope of this disclosure to include a virtual representation based upon other types of captured sensor data may also be included in the AVM virtual reality representation.

It should be noted that although a Smart Device is generally operated by a human Agent, some embodiments of

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the present disclosure include a controller, accelerometer, data storage medium, Image Capture Device, such as a CCD capture device and/or an infrared capture device being available in an Agent that is an unmanned vehicle, including for example an unmanned ground vehicle (“UGV”) such as a unit with wheels or tracks for mobility and a radio control unit for communication, or an unmanned aerial vehicle (“UAV”) or other automation.

In some embodiments, multiple unmanned vehicles may capture data in a synchronized fashion to add depth to the image capture and/or a three-dimensional and four-dimensional (over time) aspect to the captured data. In some implementations, UAV position may be contained within a perimeter and the perimeter may have multiple reference points to help each UAV (or other unmanned vehicle) determine a position in relation to static features of a building within which it is operating and also in relation to other unmanned vehicles. Still other aspects include unmanned vehicles that may not only capture data, but also function to perform a task, such as paint a wall, drill a hole, cut along a defined path, or other function. As stated throughout this disclosure, the captured data may be incorporated into a virtual model of a space or Structure.

Referring now to FIG. 28A, an A/R interface **2810** is shown with a physical world environment image **2800A** displayed on a smart device screen. The physical world environment image **2800A** may be updated on an interval basis, such as at video camera speed, or as a still image. Integrated into the physical world environment image **2800A** are user interactive areas, illustrated in FIG. 28A as interactive icons **2801-2803** representing one or both of Virtual Tags (associated with icon **2801**), and Physical Tags (associated with icons **2802-2803**).

As presented in FIG. 28A, the Virtual Tag icon **2801** is not associated with a physical tag transceiver or other transceiver, and is associated with location coordinates. A Virtual Tag (such as the virtual tag associated with icon **2801**) may include one or more of: contain digital content, link to digital content, poll data from an IoT sensor, receiving storage for digital content, and distribution channel for digital content. IoT sensors may include stand-alone sensors or sensors integrated into a device or equipment item. Exemplary sensors those included in a Shade Multi-Sensor aggregation co-located with Physical Tag **2802**, and sensors in a Shade Action Sensor aggregation co-located with Physical Tag **2802**.

A/R interfaces according to the present invention thereby create a reciprocal relationship of a user or other Agent with its environment, including a real time view of the physical world that is augmented with the power of the cyber world. The power of the cyber world may include one or more of: IoT sensing capabilities that far exceed those of human sensory capabilities; location specific data (or other digital content) storage; location specific data (or other digital content) access; location tracking of Agents and/or equipment items; condition tracking of Agents, structures, or equipment items; location specific ecommerce; and almost any functionality available via a distributed network. Such as the Internet.

Additional non-limiting exemplary aspects of an A/R interface according to the present invention include one or more of: an interactive portion indicating RTLS coordinates of a Tag **2804**; RTLS coordinates of an image capture device, such as a CCD camera **2805**; bearings of the image capture device (or other environment mapping device, such as a laser device, a sonic device etc.) **2806**; and calibration controls **2807**.

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Referring now to FIG. 28B, a smart device **2812** (in this example an Apple iPad) is supported with an image capture device included in the smart device **2812** directed in a direction of interest **2811**. The smart device **2812** is positioned in a physical world environment **2800**, and includes an A/R interface **2810** which may include an image of the physical world environment **2800A**.

According to the present invention, an additional layer of security may be provided by requiring that a user **2824** or other Agent requesting access to digital content accessible via the Virtual Tags (represented by icon **2801**) and Physical Tags (represented by icons **2802-2803**) must be in a qualified area to access the Tags. An accessible area may include an area from which the icons **2801-2803** associated with the respective Tags may be viewed, or another physical area designated as a qualified access area.

Referring now to FIG. 28C, an interactive icon may be accessed via most forms of user or Agent interaction, such as a touchscreen (illustrated in FIG. 28C) at **2821**; voice control; user “click”; user gesture; eye control or other biometric control; keystroke, accelerometer pattern registration; etc.

Referring now to FIG. 28D, a Tag (either Virtual Tag or Physical Tag) icon may be activated to provide access to associated digital content. As illustrated in FIG. 28D, digital content may include data **2822**, a device control **2823** for action, text, documents, video, or other content capable of memorializing and/or communicating information, images, data, concepts, and the like.

A qualified access area may include, by way of non-limiting example, an area around a workstation in an office, an area within a structure (such as an office building, manufacturing facility, distribution warehouse, consulate etc.); an area proximate to or within an infrastructure item (such as a bridge; tunnel; power station; cell tower, etc.); a home office area; on a ship; or other quantifiable area that has been designated as authorized to access designated digital content. The designated digital content may or may not be associated with a Virtual Tag, Physical Tag and/or an icon in an A/R or VR interface.

Although FIGS. 28A-28D illustrate a smart device **2810** capturing an image of a physical area **2800** as being supported by a user **2824**, the present invention also includes a smart device, Node, image capture device, energy capture device for wavelengths other than the visual light bandwidths, or other device capable of generating data for use in an interface that is supported by an Agent, fixture, architectural aspect, vehicle, bot, UGV, UAV or other support item. Using an Tags in a remote area and also interact with those tags through a user interface as if the remote user were on site.

In some embodiments, remote or on site interaction may include, for example, monitoring digital content that is deposited into a Virtual Tag and/or retrieved from a Virtual Tag, and/or deleted from a Virtual Tag.

For example, by way of illustration, an administrator may be able to view cyber activity associated with a remote worker by monitoring work product generated by the remote worker in an authorized remote work area. The remote worker may have an onsite work area within a virtual fence around a company facility and another authorized work area designated as a home office. The two areas may be the only areas from which the worker is able to access protected digital content. A respective Virtual Tag may be associated with each of the onsite work area and the at homework area and work product produced by the worker will be deposited into an appropriate Virtual Tag based upon where the worker

is located while working during a specified time period. Virtual Tags may be protected in that they are only visible to the worker and to the worker's administrator.

In this example, an unauthorized person would need to know normal login credentials, and have a Node associated with the worker (the node may be a smart device, employee badge, dongle etc.); and also know what areas are authorized for access to specified digital content and also be at an authorized area, in order to gain access to protected digital content.

Location coordinates may also make up a portion of an enhanced PKI key. In such embodiments, instead of simply having a private key, the private key portion is replaced by a value generated via location coordinates; or a range of values for location coordinates of an Agent position. Still other embodiments include a private key comprising a value generated by an algorithm that processes location coordinates for one or both of: location coordinates at a time of request of access to digital content; and a series of values for locations coordinates associated with the user at specified time periods (or instances) prior to the request for access to the digital content. For example, if the user generates location coordinates indicating that they are in their office during a particular hour of the day, and a request for access to protected digital data is received with coordinates indicating that the user is at an authorized home office at the same hour of the day, the VTI may deny access to the digital content until the conflict is resolved. Such embodiments may include a private key portion that is not known to the user and is generated by a smart device or other Node designating the user's position.

Referring now to FIG. 29, beacon transceivers (sometimes referred to herein as Reference Point Transceivers) **290-2904** are positioned in an area **2900** that may be authorized for access to specified digital content being protected. As illustrated, if an authorized area position **2900** is designated as being with a rectangular area formed by beacons **2901-2904**, then an Agent **2905** will be in a position authorized to have access to the specified digital content being protected; and Agents **2906-2907** are at positions that are not within the authorized area and a request from Agent **2906-2907** to access specified digital content being protected will be denied.

Referring now to FIG. 30, a 2D floorplan **3000** may also be used in a user interface to designate an area **3005** for locating a Tag (Virtual Tag, Physical Tag or Hybrid Tag) **3006**. The authorized area **3005**, may or may not have beacons **3001-3004** for wireless location of the Virtual Tag **3004**. IN the embodiments illustrated in FIG. 30, a remote user, such as an administrator, may select a Tag **3006** and monitor digital content being stored within the Tag and/or retrieved from the Tag.

In another aspect of the present invention, a user may retrieve digital content based upon knowledge of where an icon associated with storage of the digital content is located. Users sometimes have difficulty remembering which file folder, or disk drive, or other file storage mechanism holds particular content, however, humans often associate particular knowledge with a particular location which may aid in remembering where to retrieve information.

For example, a family may have a "digital safe" located at a particular position within the home. The digital safe will in essence be a Virtual Tag. The digital safe may contain sensitive documents, passwords, photos, videos, or other personal items that are important and private to the family members. The digital safe will only be accessible from within specified areas, e.g., the room in the house corre-

sponding to the location coordinates of the digital safe. A physical hard drive, SSD or other storage device for the device may be located on site of the home, or remotely, such as in the cloud. If on site, the storage device need not be accessible from any location outside of the home (or other designated location). Access to the digital safe may require an authorized user, with an authorized device, making an access request from an authorized location (location coordinates or area of coordinates), as well as knowledge of the existence of the digital safe (Virtual Tag) which is essentially invisible to unauthorized users.

Glossary

"Agent" as used herein refers to a person or automation capable of supporting a Smart Device at a geospatial location relative to a Ground Plane.

"Area of digital content interaction" as used herein refers to an area with multiple additional sets of positional coordinates that may be used to access digital content with a first set of positional coordinates.

"Augmented Virtual Model" (sometimes referred to herein as "AVM"): as used herein is a digital representation of a real property parcel including one or more three-dimensional representations of physical structures suitable for use and As Built data captured descriptive of the real property parcel. An Augmented Virtual Model includes As Built Features of the structure and may include improvements and features contained within a Processing Facility.

"Bluetooth" as used herein means the Wireless Personal Area Network (WPAN) standards managed and maintained by Bluetooth SIG. Unless otherwise specifically limited to a subset of all Bluetooth standards, the Bluetooth will encompass all Bluetooth standards (e.g., Bluetooth 4.0; 5.0; 5.1 and BLE versions).

"Digital Content" as used herein refers to any artifact that may be quantified in digital form. By way of non-limiting example, digital content may include, one or more of: alphanumeric text; audio files; image data; video data; digital stories and media.

"Energy-Receiving Sensor" as used herein refers to a device capable of receiving energy from a Radio Target Area and quantifying the received energy as a digital value.

"Ground Plane" as used herein refers to horizontal plane from which a direction of interest may be projected.

"Hybrid Tag" as used herein means digital content associated with a location coordinates of a position previously occupied by a Physical Tag. In some embodiments, a Hybrid Tag may include digital content based upon data generated by a sensor co-located with a Physical Tag or while the sensor was within a specified distance of a position described by location coordinates.

"Image Capture Device" or "Scanner" as used herein refers to apparatus for capturing digital or analog image data, an Image capture device may be one or both of: a two-dimensional sensor (sometimes referred to as "2D") or a three-dimensional sensor (sometimes referred to as "3D"). In some examples an Image Capture Device includes a charge-coupled device ("CCD") sensor. "Intelligent Automation" as used herein refers to a logical processing by a device, system, machine, or equipment item (such as data gathering, analysis, artificial intelligence, and functional operation) and communication capabilities.

"IoT Tag" as used herein refers to a Node co-located with an IoT Sensor.

"Multi-Modal" as used herein refers to the ability of a device to communication using multiple protocols and/or

bandwidths. Examples of multimodal may include being capable of communication using two to more of: Ultra-Wideband, Bluetooth; Bluetooth Low Energy; Wi-Fi; Wi-Fi RT; GPS; ultrasonic; infrared protocols and/or mediums.

“Multi-Modal Tag” as used herein refers to a device including multiple wireless transceivers operating in different bandwidths and according to different communication parameters.

“Node” as used herein means a device including at least a processor, a digital storage, and a wireless transceiver.

“Physical Tag” as used here shall mean a physical device with a transceiver capable of wireless communication sufficient to determine a geospatial position of the device. The Physical Tag may also be associated with a data set that is not contingent upon the geospatial location of the physical device.

“Radio Target Area” an area from which an energy-receiving Sensor will receive energy of a type and bandwidth that may be quantified by the energy-receiving Sensor.

“Ray” as used herein refers to a straight line including a starting point and extending indefinitely in a direction.

“Sensor” (sometimes referred to as an IoT sensor) as used herein refers to one or more of a solid state, electro-mechanical, and mechanical device capable of transducing a physical condition or property into an analogue or digital representation and/or metric.

“Smart Device” as used herein includes an electronic device including, or in logical communication with, a processor and digital storage and capable of executing logical commands.

“Smart Receptacle” as used herein includes a case or other receiver of a smart device with components capable of receiving wireless transmissions from multiple wireless positional reference transceivers. In some embodiments, the smart receptacle will include a wireless transmitter and/or a physical connector for creating an electrical path for carrying one or both of electrical power and logic signals between an associated Smart Device and the Smart Receptacle.

“Structure” as used herein refers to a manmade assembly of parts connected in an ordered way. Examples of a Structure in this disclosure include a building; a sub-assembly of a building; a bridge, a roadway, a train track, a train trestle, an aqueduct; a tunnel a dam, and a retainer berm.

“Tag” as used herein refers to digital content and access rights associated with a geospatial position

“Transceive” as used herein refers to an act of transmitting and receiving data.

“Transceiver” as used herein refers to an electronic device capable of one or both of wirelessly transmitting and receiving data.

“Vector” as used herein refers to a magnitude and a direction as may be represented and/or modeled by a directed line segment with a length that represents the magnitude and an orientation in space that represents the direction.

“Virtual Tag” as used here shall mean digital content associated with a location identified via positional coordinates.

“Wireless Communication Area” (sometimes referred to as “WCA”) as used herein means an area through which wireless communication may be completed. A size of a WCA may be dependent upon a specified modality of wireless communication and an environment through which the wireless communication takes place.

In discussion (and as illustrated), a WCA may be portrayed as being spherical in shape, however in a physical

environment a shape of a WCA may be amorphous or of changing shape and more resemble a cloud of thinning density around the edges.

A number of embodiments of the present disclosure have been described. While this specification contains many specific implementation details, there should not be construed as limitations on the scope of any disclosures or of what may be claimed, but rather as descriptions of features specific to particular embodiments of the present disclosure. While embodiments of the present disclosure are described herein by way of example using several illustrative drawings, those skilled in the art will recognize the present disclosure is not limited to the embodiments or drawings described. It should be understood the drawings and the detailed description thereto are not intended to limit the present disclosure to the form disclosed, but to the contrary, the present disclosure is to cover all modification, equivalents and alternatives falling within the spirit and scope of embodiments of the present disclosure as defined by the appended claims.

The headings used herein are for organizational purposes only and are not meant to be used to limit the scope of the description or the claims. As used throughout this application, the word “may” is used in a permissive sense (i.e., meaning having the potential to), rather than the mandatory sense (i.e., meaning must). Similarly, the words “include”, “including”, and “includes” mean including but not limited to. To facilitate understanding, like reference numerals have been used, where possible, to designate like elements common to the figures.

The phrases “at least one”, “one or more”, and “and/or” are open-ended expressions that are both conjunctive and disjunctive in operation. For example, each of the expressions “at least one of A, B and C”, “at least one of A, B, or C”, “one or more of A, B, and C”, “one or more of A, B, or C” and “A, B, and/or C” means A alone, B alone, C alone, A and B together, A and C together, B and C together, or A, B and C together.

The term “a” or “an” entity refers to one or more of that entity. As such, the terms “a” (or “an”), “one or more” and “at least one” can be used interchangeably herein. It is also to be noted the terms “comprising”, “including”, and “having” can be used interchangeably.

Certain features that are described in this specification in the context of separate embodiments can also be implemented in combination in a single embodiment. Conversely, various features that are described in the context of a single embodiment can also be implemented in combination in multiple embodiments separately or in any suitable sub-combination. Moreover, although features may be described above as acting in certain combinations and even initially claimed as such, one or more features from a claimed combination can in some cases be excised from the combination, and the claimed combination may be directed to a sub-combination or variation of a sub-combination.

Similarly, while method steps may be depicted in the drawings in a particular order, this should not be understood as requiring that such operations be performed in the particular order shown or in a sequential order, or that all illustrated operations be performed, to achieve desirable results.

Certain features that are described in this specification in the context of separate embodiments can also be implemented in combination in a single embodiment. Conversely, various features that are described in the context of a single embodiment can also be implemented in combination in multiple embodiments separately or in any suitable sub-

combination. Moreover, although features may be described above as acting in certain combinations and even initially claimed as such, one or more features from a claimed combination can in some cases be excised from the combination, and the claimed combination may be directed to a sub-combination or variation of a sub-combination.

Moreover, the separation of various system components in the embodiments described above should not be understood as requiring such separation in all embodiments, and it should be understood that the described program components and systems can generally be integrated together in a single software product or packaged into multiple software products.

Thus, particular embodiments of the subject matter have been described. Other embodiments are within the scope of the following claims. In some cases, the actions recited in the claims can be performed in a different order and still achieve desirable results. In addition, the processes depicted in the accompanying figures do not necessarily require the particular order show, or sequential order, to achieve desirable results. In certain implementations, multitasking and parallel processing may be advantageous. Nevertheless, it will be understood that various modifications may be made without departing from the spirit and scope of the claimed disclosure.

What is claimed is:

1. A method for accessing digital content associated with a position in a building, the method comprising the steps of:

- a) determining a location of a smart device based upon operation of an inertial measurement unit included in the smart device while the smart device traverses a known orienteering area in the building;
- b) generating an area of interest comprising the location of the Smart Device;
- c) associate positional coordinates of a tag with digital content associated with a position in the building;
- d) determining that the positional coordinates of the tag are within the area of interest;
- e) determining that the Smart Device has access rights to the digital content;
- f) receiving a user input into a dynamic portion of a user interactive interface on the Smart Device, the user input operative to cause the Smart Device to display the digital content; and
- g) based upon the user input received into the dynamic portion of the user interactive interface, displaying the digital content in the user interactive interface.

2. The method of claim 1 additionally comprising the step of including in the user interactive interface a dynamic portion representative of the tag.

3. The method of claim 1, additionally comprising the steps of:

- establishing a radio target area for an energy receiving sensor;
- receiving energy into the energy receiving sensor from the radio target area;
- generating a digital representation of the energy received into the energy receiving sensor at an instance in time; and
- including in the user interactive interface the digital representation of the energy received into the energy receiving sensor at the instance in time and a dynamic portion representative of the tag.

4. The method of claim 3, additionally comprising the steps of: receiving multiple disparate energy wavelengths into the energy receiving sensor at the instance in time, the multiple disparate energy wavelengths received from differ-

ent geospatial locations; and indicating the multiple disparate energy wavelengths in the user interactive interface.

5. The method of claim 3 wherein the radio target area comprises the position of the energy receiving sensor and an area providing electromagnetic energy to the energy receiving sensor.

6. The method of claim 5 wherein the radio target area further comprises a frustum of a generally conical shape with the energy receiving sensor located at a relatively narrow end of the frustum.

7. The method of claim 6 additionally comprising the steps of changing a direction of the Smart Device; and redefining the radio target area based upon the change in the direction of the Smart Device.

8. The method of claim 7 wherein a structure comprises the radio target area.

9. The method of claim 5 additionally comprising the steps of co-locating the tag with an IoT sensor; with the IoT sensor, quantifying a condition present in a structure as digital data; and including the digital data quantifying the condition present in the structure in the digital content included in the user interactive interface.

10. The method of claim 1, wherein the tag comprises a virtual tag associated with a location identified via the positional coordinates of the tag.

11. The method of claim 1, wherein the tag comprises a physical tag comprising a transceiver capable of wireless communication with multiple reference point transceivers and the method additionally comprises the steps of: transceiving a wireless communication between the tag and reference point transceivers; and generating the positional coordinates for the tag based upon the wireless communication between the tag and the multiple reference point transceivers.

12. The method of claim 11, wherein the wireless communication between the Smart Device and the multiple reference point transceivers comprises transceiving using an Ultra-Wideband modality.

13. The method of claim 12, wherein the tag comprises the transceiver capable of wireless communication with the multiple reference point transceivers and the method additionally comprises the steps of: transceiving the wireless communication between the tag and the multiple reference point transceivers using the Ultra-Wideband modality; and generating the positional coordinates for the tag based upon the wireless communication between the tag and the reference point transceivers.

14. The method of claim 13 additionally comprising the step of identifying a wireless communication area comprising a radio target area of an energy receiving sensor, the wireless communication area based upon a communication distance of the Ultra-Wideband modality in an area encompassing the energy receiving sensor.

15. The method of claim 14, wherein the wireless communication between the Smart Device and the multiple reference point transceivers comprises transceiving using a Bluetooth modality.

16. The method of claim 15, wherein the tag comprises the transceiver capable of wireless communication with the multiple reference point transceivers and the method additionally comprises the steps of: transceiving the wireless communication between the tag and the multiple reference point transceivers using the Bluetooth modality; and generating the positional coordinates for the tag based upon the wireless communication between the tag and the multiple reference point transceivers.

17. The method of claim 15, additionally comprising the step of identifying the wireless communication area comprising the radio target area of the energy receiving sensor, the wireless communication area based upon a communication distance of the Bluetooth modality in an area encompassing the energy receiving sensor. 5

18. The method of claim 1 wherein the positional coordinates comprise Cartesian Coordinates.

19. The method of claim 1 wherein the positional coordinates comprise at least one of: an angle of arrival and an angle of departure. 10

20. The method of claim 1 additionally comprising the step of including in the dynamic portion of the user interactive interface an icon indicative of the digital content associated with the tag. 15

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